

most abundant omega-3 fatty acid in the brain and an important component of breast milk. Draw the structure of this fatty acid.

3. Melting Points of Lipids The melting points of a series of 18-carbon fatty acids are stearic acid, 69.6 °C; oleic acid, 13.4 °C; linoleic acid, -5 °C; and linolenic acid, -11 °C.

(a) What structural aspect of these 18-carbon fatty acids can be correlated with the melting point?

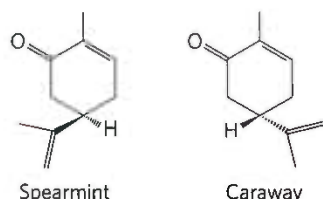
(b) Draw all the possible triacylglycerols that can be constructed from glycerol, palmitic acid, and oleic acid. Rank them in order of increasing melting point.

(c) Branched-chain fatty acids are found in some bacterial membrane lipids. Would their presence increase or decrease the fluidity of the membrane (that is, give the lipids a lower or a higher melting point)? Why?

4. Catalytic Hydrogenation of Vegetable Oils Catalytic hydrogenation, used in the food industry, converts double bonds in the fatty acids of the oil triacylglycerols to $-\text{CH}_2-\text{CH}_2-$. How does this affect the physical properties of the oils?

5. Impermeability of Waxes What property of the waxy cuticles that cover plant leaves makes the cuticles impermeable to water?

6. Naming Lipid Stereoisomers Carvone, a member of the terpenoid family of chemicals, forms two enantiomers with quite different properties. One enantiomer, abundant in spearmint, smells sweet and minty. The other enantiomer, abundant in caraway seeds, smells spicy and of rye bread. Name the compounds abundant in spearmint and caraway seeds using the RS system.



7. Chemical Reactivity of Lipids Soaps are salts of fatty acids and can be made by mixing triacylglycerols with a strong base such as NaOH. This saponification reaction produces glycerol and fatty acid salts. In a lab experiment, students saponify the triacylglycerol tristearin in the presence of ^{18}O -labeled water. What saponification reaction products will contain the ^{18}O label?

8. Hydrophobic and Hydrophilic Components of Membrane Lipids A common structural feature of membrane lipids is their amphipathic nature. For example, in phosphatidylcholine, the two fatty acid chains are hydrophobic and the phosphocholine head group is hydrophilic. Name the components that serve as the hydrophobic and hydrophilic units for each membrane lipid: (a) phosphatidylethanolamine (b) sphingomyelin (c) galactosylcerebroside (d) ganglioside (e) cholesterol.

9. Deducing Lipid Structure from Composition A biochemist completely digests a glycerophospholipid with a mixture of phospholipases A and D. HPLC and MS analysis reveals the presence of an amino acid of 105.09 Da, a saturated fatty acid of 256.43 Da, and an omega-3 monounsaturated fatty acid of 282.45 Da. Which amino acid does the glycerophospholipid contain? Draw the most likely structure of this glycerophospholipid.

10. Deducing Lipid Structure from Molar Ratio of Components Complete hydrolysis of a glycerophospholipid yields glycerol, two fatty acids (16:1(Δ^9) and 16:0), phosphoric acid,

and serine in the molar ratio 1:1:1:1:1. Name this lipid and draw its structure.

11. Lipids in Blood Group Determination We note in Figure 10-13 that the structure of glycosphingolipids determines the blood groups A, B, and O in humans. It is also true that glycoproteins determine blood groups. How can both statements be true?



12. The Action of Phospholipases The venom of the Eastern diamondback rattler and the Indian cobra contains phospholipase A_2 , which catalyzes the hydrolysis of fatty acids at the C-2 position of glycerophospholipids. The phospholipid breakdown product of this reaction is lysolecithin, which is derived from phosphatidylcholine. At high concentrations, this and other lysophospholipids act as detergents, dissolving the membranes of erythrocytes and lysing the cells. Extensive hemolysis may be life-threatening.

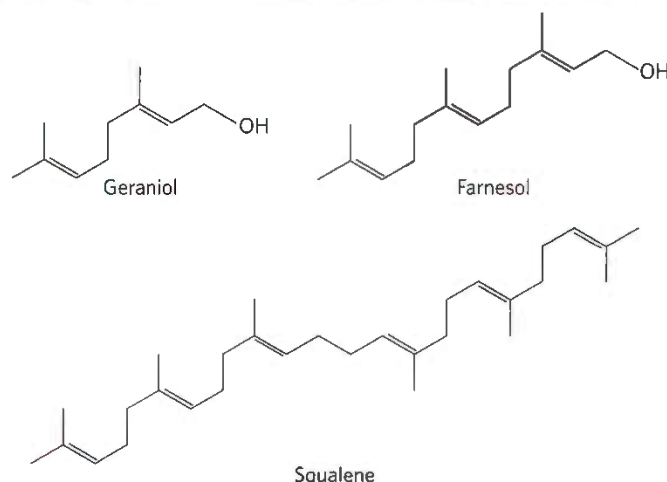
(a) All detergents are amphipathic. What are the hydrophilic and hydrophobic portions of lysolecithin?

(b) The pain and inflammation caused by a snake bite can be treated with certain steroids. What is the basis of this treatment?

(c) Though the high levels of phospholipase A_2 in venom can be deadly, this enzyme is necessary for a variety of normal metabolic processes. What are these processes?

13. Intracellular Messengers from Phosphatidylinositols The hormone vasopressin is an extracellular signal that activates a specific phospholipase C in the membrane. Cleavage of PIP_2 by phospholipase C generates two products. What are they? Compare their properties and their solubilities in water, and predict whether either would diffuse readily through the cytosol.

14. Isoprene Units in Isoprenoids Geraniol, farnesol, and squalene are called isoprenoids because they are synthesized from five-carbon isoprene units. In each compound, circle the five-carbon units representing isoprene units (see Fig. 10-22).



15. Hydrolysis of Lipids Name the products of mild hydrolysis with dilute NaOH of (a) 1-stearoyl-2,3-dipalmitoylglycerol (b) 1-palmitoyl-2-oleoylphosphatidylcholine.

16. Effect of Polarity on Solubility Rank a triacylglycerol, a diacylglycerol, and a monoacylglycerol in order of decreasing solubility in water. Assume that each acylglycerol contains only palmitic acid.

17. Chromatographic Separation of Lipids Suppose that you apply a mixture of lipids to a silica gel column and then

wash the column with increasingly polar solvents. The mixture consists of phosphatidylserine, phosphatidylethanolamine, phosphatidylcholine, cholesteryl palmitate (a sterol ester), sphingomyelin, palmitate, *n*-tetradecanol, triacylglycerol, and cholesterol. In what order will the lipids elute from the column? Explain your reasoning.

18. Identification of Unknown Lipids Johann Thudichum, who practiced medicine in London about 100 years ago, also dabbled in lipid chemistry in his spare time. He isolated a variety of lipids from neural tissue and characterized and named many of them. His carefully sealed and labeled vials of isolated lipids were rediscovered many years later.

(a) How would you confirm, using techniques not available to Thudichum, that the vials labeled “sphingomyelin” and “cerebroside” actually contain these compounds?

(b) How would you distinguish sphingomyelin from phosphatidylcholine by chemical, physical, or enzymatic tests?

BIOCHEMISTRY ONLINE

19. Using the LIPID MAPS Database to Find Solubility Information Lipidomics has identified thousands of cellular lipids. LIPID MAPS is an online database containing over 40,000 unique lipid structures, as well as information on the chemical and physical properties of each lipid (www.lipidmaps.org). One important parameter when working with lipids is log *P*, where *P* is the octanol:water partition coefficient, an indicator of lipophilicity.

(a) Look up cholesterol, sphingosine, linoleic acid, and stearic acid in LIPID MAPS and use the reported log *P* values to place them in order of increasing solubility in octanol.

(b) Pharmacologists often study log *P* values when developing new drugs. Why would knowing a drug's log *P* value be informative?

20. Characteristics of Lipid Transport Proteins Often when lipids are transported between different tissues, they are carried by proteins. In this exercise, you will explore the interactions between a lipid and a protein using the PDB (www.rcsb.org). Use the PDB identifier 2YG2 and study the structure of the complex between HDL-associated apolipoprotein M and sphingosine-1-phosphate. Navigate to 3D View: Structure to answer the following questions.

(a) What protein motif is adopted by apolipoprotein M?

(b) Which amino acid residues do you find lining the sphingosine binding pocket? What do they have in common?

(c) The phosphoryl group of sphingosine-1-phosphate is exposed on the surface of the protein. Why do you suppose it is important that the transport protein binds the hydrocarbon tail of sphingosine-1-phosphate but not necessarily the polar head group?

DATA ANALYSIS PROBLEM



21. Determining the Structure of the Abnormal Lipid in Tay-Sachs Disease Box 10-1, Figure 1, shows the pathway of breakdown of gangliosides in healthy (normal) individuals and in individuals with certain genetic diseases.

Some of the data on which the figure is based were presented in a paper by Lars Svennerholm (1962). Note that the sugar Neu5Ac, *N*-acetylneuraminic acid, represented in the Box 10-1 figure as $\blacklozenge$, is a sialic acid.

Svennerholm reported that “about 90% of the monosialogangliosides isolated from normal human brain” consisted of a compound with ceramide, hexose, *N*-acetylgalactosamine, and *N*-acetylneuraminic acid in the molar ratio 1:3:1:1.

(a) Which of the gangliosides (GM1 through GM3 and globoside) in Box 10-1, Figure 1, fits this description? Explain your reasoning.

(b) Svennerholm reported that 90% of the gangliosides from a patient with Tay-Sachs disease had a molar ratio (of the same four components given above) of 1:2:1:1. Is this consistent with the Box 10-1 figure? Explain your reasoning.

To determine the structure in more detail, Svennerholm treated the gangliosides with neuraminidase to remove the *N*-acetylneuraminic acid. This resulted in an asialoganglioside that was much easier to analyze. He hydrolyzed it with acid, collected the ceramide-containing products, and determined the molar ratio of the sugars in each product. He did this for both the normal gangliosides and the Tay-Sachs gangliosides. His results are shown below.

Ganglioside	Ceramide	Glucose	Galactose	Galactosamine
<i>Normal</i>				
Fragment 1	1	1	0	0
Fragment 2	1	1	1	0
Fragment 3	1	1	1	1
Fragment 4	1	1	2	1
<i>Tay-Sachs</i>				
Fragment 1	1	1	0	0
Fragment 2	1	1	1	0
Fragment 3	1	1	1	1

(c) Based on these data, what can you conclude about the structure of the normal ganglioside? Is this consistent with the structure in Box 10-1? Explain your reasoning.

(d) What can you conclude about the structure of the Tay-Sachs ganglioside? Is this consistent with the structure in Box 10-1? Explain your reasoning.

Svennerholm also reported the work of other researchers who “permethylated” the normal asialoganglioside. Permethylation is the same as exhaustive methylation: a methyl group is added to every free hydroxyl group on a sugar. They found the following permethylated sugars: 2,3,6-trimethylglycopyranose; 2,3,4,6-tetramethylgalactopyranose; 2,4,6-trimethylgalactopyranose; and 4,6-dimethyl-2-deoxy-2-aminogalactopyranose.

(e) To which sugar of GM1 does each of the permethylated sugars correspond? Explain your reasoning.

(f) Based on all the data presented so far, what pieces of information about normal ganglioside structure are missing?

Reference

Svennerholm, L. 1962. The chemical structure of normal human brain and Tay-Sachs gangliosides. *Biochem. Biophys. Res. Comm.* 9:436–441.

BIOLOGICAL MEMBRANES AND TRANSPORT

11.1 The Composition and Architecture of Membranes 367

11.2 Membrane Dynamics 377

11.3 Solute Transport across Membranes 385

The first cell probably came into being when a membrane formed, enclosing a small volume of aqueous solution and separating it from the rest of the universe. Membranes define the external boundaries of cells and control the molecular traffic across that boundary; in eukaryotic cells, they also divide the internal space into discrete compartments to segregate processes and components. Proteins embedded in and associated with membranes organize complex reaction sequences and are central to both biological energy conservation and cell-to-cell communication.

In this chapter we focus on these principles that underlie the structure and function of biological membranes:

P1 **The biological membrane is a lipid bilayer with proteins of various functions (enzymes, transporters) embedded in or associated with the bilayer.** The hydrophobic effect stabilizes structures (lipid bilayers and vesicles) in which lipids with some polar and some nonpolar regions can protect their nonpolar regions from interaction with the very polar solvent, water. Membrane proteins are associated with the lipid bilayer more-or-less tightly, and proteins and lipids are both allowed limited lateral motion in the plane of the bilayer.

P2 **All of the internal membranes of cells are part of an interconnected, functionally specialized, and dynamic endomembrane system.** Proteins and lipids synthesized in the endoplasmic reticulum move through the Golgi apparatus, and they are targeted to the membranes of organelles or to the plasma membrane, where they provide the essential properties of these structures. During this

membrane trafficking, some proteins are covalently altered, and some lipids are segregated into different organelles or are concentrated in one of the bilayer leaflets. Rafts are functionally specialized regions with unique lipid and protein compositions.

P3 **Although the lipid bilayer is impermeable to charged or polar solutes, cells of all kinds have many membrane transporters and ion channels that catalyze transmembrane movement of specific solutes.** Some transporters merely speed the movement of solutes in the direction that simple diffusion takes them, whereas others use an energy source to move solutes against a concentration gradient.

11.1 The Composition and Architecture of Membranes

We begin our discussion of biological membranes by looking at the stable bilayer structures formed spontaneously by phospholipids in water. In biological membranes we see this basic lipid bilayer combined with a wide array of membrane proteins specialized for one of the many roles that membranes play in cells. Each intracellular compartment has membranes with specific protein and lipid content, uniquely disposed across the two leaflets of the bilayer. The intracellular membranes together make up a dynamic endomembrane system that synthesizes and distributes membrane components to create functionally specialized cell compartments. Depending on its function, a protein may form one or several transmembrane segments, or it may adhere to the membrane bilayer through lipids attached to the protein, or it may alternate between membrane-associated and free forms.

The Lipid Bilayer Is Stable in Water

Glycerophospholipids, sphingolipids, and sterols are virtually insoluble in water. Recall that glycerophospholipids contain two long-chain fatty acids (which are

hydrophobic) and a head group consisting of glycerol and one of several polar or charged substituents such as phosphocholine or phosphoethanolamine (see Fig. 10-8). Sphingolipids are constructed from a long-chain alkyl amine (sphingosine), a saturated long-chain fatty acid, and a polar head group that may be as simple as phosphocholine or may be a complex oligosaccharide (see Fig. 10-11). Sterols (cholesterol in animal membranes) have a very nonpolar steroid nucleus of four fused rings, and a polar hydroxyl group at one end of the ring system (see Fig. 10-15). Each type of membrane lipid has its particular effect on membrane structure and function. When mixed with water, these lipids spontaneously form microscopic lipid aggregates, clustering together, with their hydrophobic moieties in contact with each other and their hydrophilic groups in contact with the surrounding water. The clustering reduces the amount of hydrophobic surface exposed to water. This minimizes the number of molecules in the shell of ordered water at the lipid-water interface (see Fig. 2-7), resulting in an increase in entropy. This hydrophobic effect provides the thermodynamic driving force for the formation and maintenance of these clusters of lipid molecules. The term **hydrophobic interactions** is sometimes used to describe the clustering of hydrophobic molecular surfaces in an aqueous environment, but the molecules are not interacting chemically; they are simply finding the lowest-energy environment by reducing the hydrophobic, or nonpolar, surface area exposed to water.

Depending on the precise conditions and the nature of the lipids, several types of lipid aggregate can form when amphipathic lipids are mixed with water (Fig. 11-1). **Micelles** are spherical structures that contain anywhere from a few dozen to a few thousand amphipathic molecules. These molecules are arranged with their hydrophobic

regions aggregated in the interior, where water is excluded, and their hydrophilic head groups at the outer surface, in contact with water. Micelle formation is favored when the cross-sectional area of the head group is greater than that of the acyl side chain(s), as in free fatty acids, lysophospholipids (phospholipids lacking one fatty acid), and many detergents, such as sodium dodecyl sulfate (SDS; p. 88).

A second type of lipid aggregate in water is the **bilayer**, in which two lipid monolayers (leaflets) form a two-dimensional sheet. Bilayer formation is favored if the cross-sectional areas of the head group and acyl side chain(s) are similar, as in glycerophospholipids and sphingolipids. The hydrophobic portions in each monolayer, excluded from water, make contact with each other. The hydrophilic head groups interact with water at one or the other surface of the bilayer. Because the hydrophobic regions at its edges (Fig. 11-1b) are in contact with water, a bilayer sheet is relatively unstable and spontaneously folds back on itself to form a hollow sphere, called a **vesicle** or liposome (Fig. 11-1c). The continuous surface of vesicles eliminates exposed hydrophobic regions, allowing bilayers to achieve maximal stability in their aqueous environment. Vesicle formation also creates a separate internal aqueous compartment (the vesicle lumen). It is likely that the antecedents to the first living cells resembled lipid vesicles, their aqueous contents segregated from their surroundings by a lipid bilayer.

Studies of lipid bilayers in vitro (Fig. 11-2) show that the hydrocarbon core of the bilayer, made up of the $-\text{CH}_2-$ and $-\text{CH}_3$ of the fatty acyl groups, is about as nonpolar as decane, and about 3 nm (30 Å) thick, roughly the width of two extended fatty acyl chains. Biological membranes, to which we turn next, are 50–80 Å thick, when proteins protruding on either side are included.

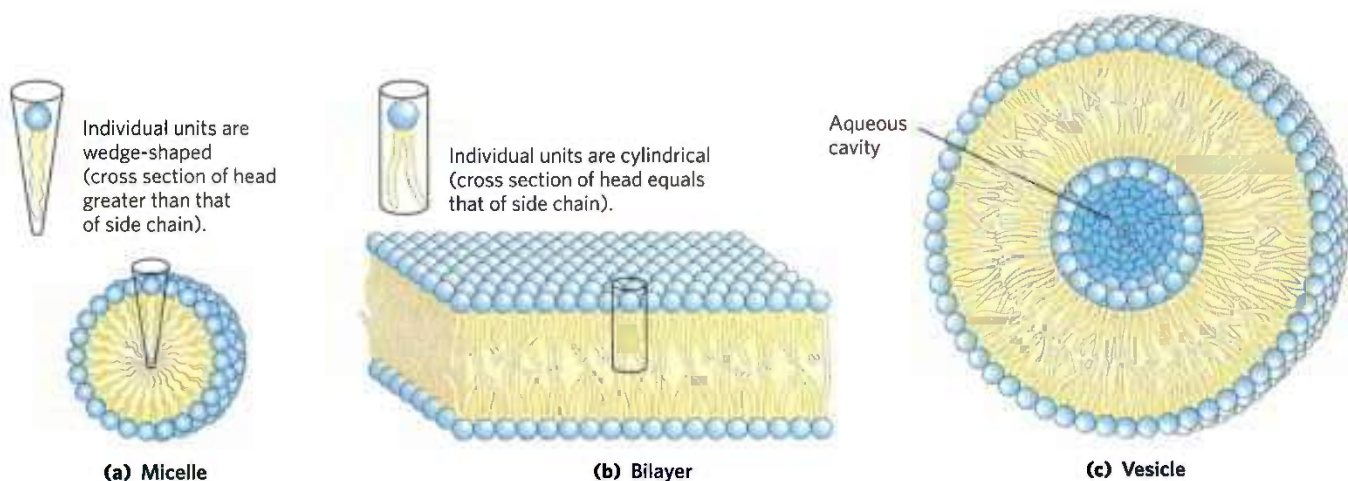


FIGURE 11-1 Amphipathic lipid aggregates that form in water. (a) In micelles, the hydrophobic chains of the fatty acids are sequestered at the core of the sphere. There is virtually no water in the hydrophobic interior. (b) In an open bilayer, all acyl side chains except

those at the edges of the sheet are protected from interaction with water. (c) When a two-dimensional bilayer folds on itself, it forms a closed bilayer, a three-dimensional hollow vesicle (liposome) enclosing an aqueous cavity.

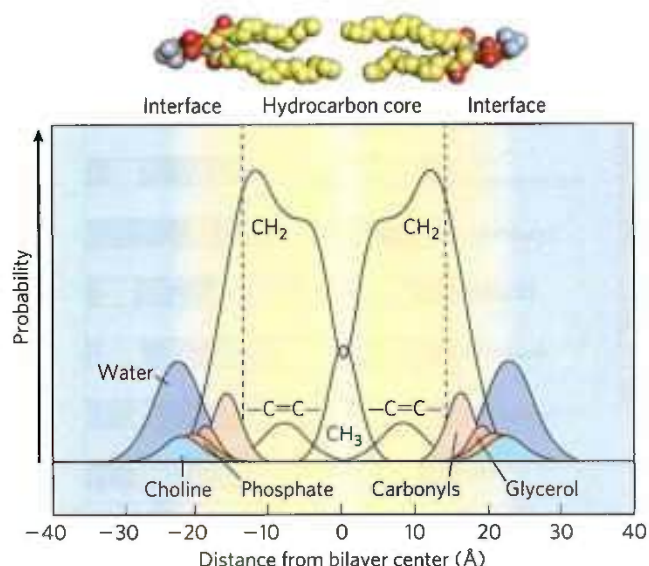


FIGURE 11-2 Distribution of membrane lipids across an artificial membrane. The membrane, composed of pure phosphatidylcholine in which both fatty acids are oleic acid (18:1 Δ^9), was studied with x-ray and neutron diffraction. The y axis shows the probability that one of the chemical moieties will be found at the position indicated on the x axis. The phosphatidylcholine molecules are shown at the same scale as the x axis. Fatty acyl chains in the hydrocarbon core are fluid and are about as nonpolar as decane. At either side, a polar region about 15 Å from the center of the bilayer contains the fatty acyl carbonyls. The glycerol, choline, and phosphate that make up the polar head group are at 18–20 Å, outside the hydrocarbon core and in contact with water. [Information from A. Rath and C. M. Deber, *Annu. Rev. Biophys.* 41:135, 2012, Fig. 5a, based on M. C. Wiener and S. H. White, *Biophys. J.* 61:434, 1992.]

Bilayer Architecture Underlies the Structure and Function of Biological Membranes

In the generalized membrane in **Figure 11-3**, phospholipids form a bilayer in which proteins are embedded, their hydrophobic domains in contact with the fatty acyl chains of membrane lipids. Some proteins protrude from only one side of the membrane; others have domains exposed on both sides. The orientation of both lipids and proteins in the bilayer is asymmetric, giving the membrane structural and functional “sidedness.” **P1** The individual lipid and protein units in a membrane form a **fluid mosaic** with a pattern that, unlike a mosaic of ceramic tile and mortar, is able to change as lipids and proteins move in the plane of the membrane, while maintaining the permeability barrier to polar and charged solutes.

Membranes are not merely passive barriers. They are flexible, self-repairing, and selectively permeable. Their flexibility permits the shape changes that accompany cell growth and movement (such as amoeboid movement). With their ability to break and reseal, two membranes can fuse, as in exocytosis, or a single membrane-enclosed compartment can undergo fission to yield two sealed compartments, as in endocytosis or cell division, without creating gross leaks through cellular surfaces. Their selective permeability allows them to serve as molecular gatekeepers.

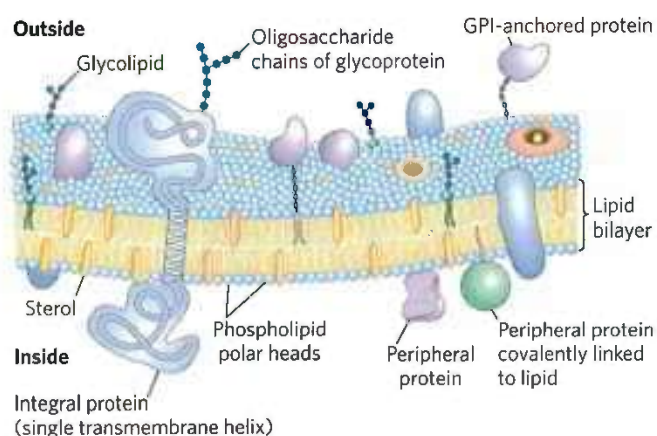


FIGURE 11-3 Fluid mosaic model for plasma membrane structure. In this simplified and general model, the fatty acyl chains in the interior of the membrane form a fluid, hydrophobic region. Integral proteins float in this sea of lipid, with their nonpolar amino acid side chains protected from interaction with water.

These membrane functions are accomplished primarily by a broad array of proteins and enzymes in and on membranes. At the cell surface, transporters move specific organic solutes and inorganic ions across the membrane; receptors sense extracellular signals and trigger molecular changes in the cell; ion channels mediate electrical signaling between cells; and adhesion molecules hold neighboring cells together. The relative proportions of protein and lipid vary with the type of membrane, reflecting the diversity of biological roles. For example, certain neurons have a myelin sheath—an extended plasma membrane that wraps around the cell many times and acts as a passive electrical insulator. The myelin sheath consists primarily of lipids (good insulators). In contrast, the plasma membranes of bacteria and the membranes of mitochondria and chloroplasts, the sites of many enzyme-catalyzed processes, contain more protein than lipid (in mass per total mass).

Within the cell, membranes organize cellular processes such as the synthesis of lipids and certain proteins and the energy transductions that produce ATP in mitochondria and chloroplasts. Proteins associated with membranes are essentially confined to two-dimensional space, where intermolecular collisions of membrane proteins and lipids are far more probable than in the three-dimensional cytoplasm, so the efficiency of enzyme-catalyzed processes organized within membranes can be vastly increased.

The Endomembrane System Is Dynamic and Functionally Differentiated

P2 In eukaryotic cells, the endoplasmic reticulum (ER), Golgi apparatus, lysosomes, and various small vesicles that carry lipids and proteins throughout the cell are each surrounded and defined by a single membrane.

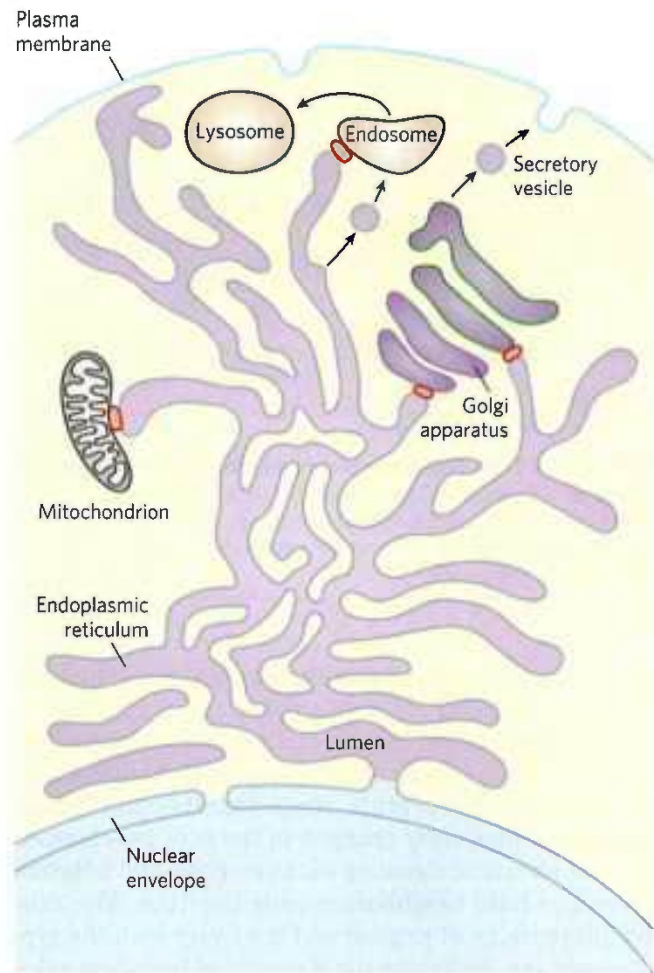


FIGURE 11-4 Trafficking in the endomembrane system of an animal cell. Lipids and proteins move from the site of their synthesis (ER) through the Golgi apparatus to the cell surface (or to organelles such as lysosomes). Mitochondria (and in plants, chloroplasts) are also part of the endomembrane system. Small vesicles bud off the ER, move to and fuse with the cis Golgi apparatus, exit the trans Golgi apparatus as secretory or transport vesicles, and fuse with the plasma membrane or with endosomes, giving rise to lysosomes. Individual lipid molecules can be carried throughout the cell by lipid transfer proteins (LTPs), which act at membrane contact points, shown here in red.

This endomembrane system (Fig. 11-4) also includes three organelles with double membranes: the nucleus, mitochondrion, and (in plants) chloroplast. (Recall the likely origin of mitochondria and chloroplasts by an endosymbiont mechanism shown in Fig. 1-37.) The convoluted and tubular ER is dynamic and extends throughout the cell, touching the other compartments at specific membrane contact points. Although each compartment is closed, they actively exchange proteins and lipids in several ways.

Cells have mechanisms to target specific lipids to particular organelles. Each species, each tissue or cell type, and the organelles of each cell type have a characteristic set of membrane lipids. Plasma membranes, for example, are enriched in cholesterol and sphingolipids, but they contain no detectable cardiolipin (Fig. 11-5); mitochondrial

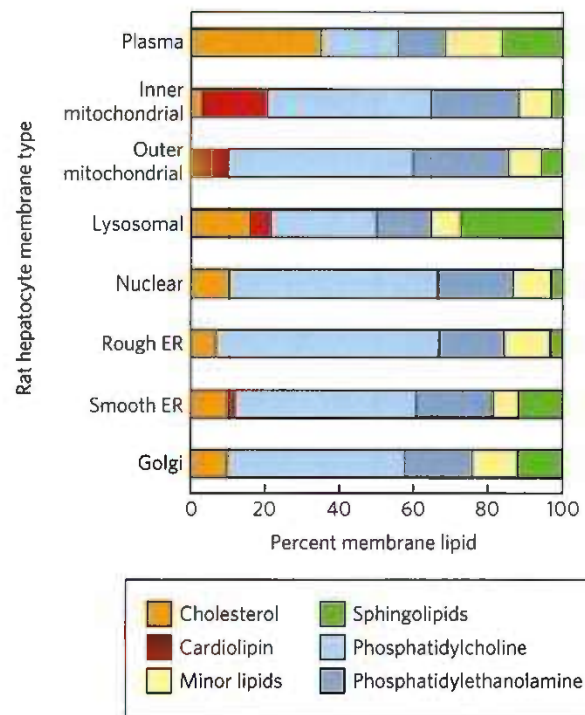


FIGURE 11-5 Lipid composition of the plasma and organelle membranes of a rat hepatocyte. The functional specialization of each membrane type is reflected in its unique lipid composition. Cholesterol is prominent in plasma membranes but barely detectable in mitochondrial membranes. Cardiolipin is a major component of the inner mitochondrial membrane but not of the plasma membrane. Phosphatidylserine, phosphatidylinositol, and phosphatidylglycerol are relatively minor components of most membranes but serve critical functions; phosphatidylinositol and its derivatives, for example, are important in signal transductions triggered by hormones. Sphingolipids, phosphatidylcholine, and phosphatidylethanolamine are present in most membranes but in varying proportions. Glycolipids, which are major components of the chloroplast membranes of plants, are virtually absent in animal cells.

membranes are very low in cholesterol and sphingolipids, but they contain most of the cell's phosphatidylglycerol and cardiolipin, which are synthesized within the mitochondria. Cardiolipin is specifically required for correct assembly of the respiratory complexes of mitochondria. In all but a few cases, the functional significance of these different combinations of lipids is not yet known.

Most membrane lipids and proteins are synthesized in the ER, and from there they move to their destination organelles or to the plasma membrane. In this process of membrane trafficking, small membrane vesicles bud from the ER, then move to and fuse with the cis Golgi apparatus. As lipids and proteins move across the Golgi apparatus to its trans side, they undergo covalent modifications required for their cellular function. In many cases, these covalent alterations act as molecular ZIP codes, dictating the eventual location of the mature protein.

P2 Membrane trafficking achieves striking changes in lipid composition and disposition across the bilayer (Fig. 11-6). Phosphatidylcholine is the principal phospholipid in the luminal leaflet of the Golgi membrane,

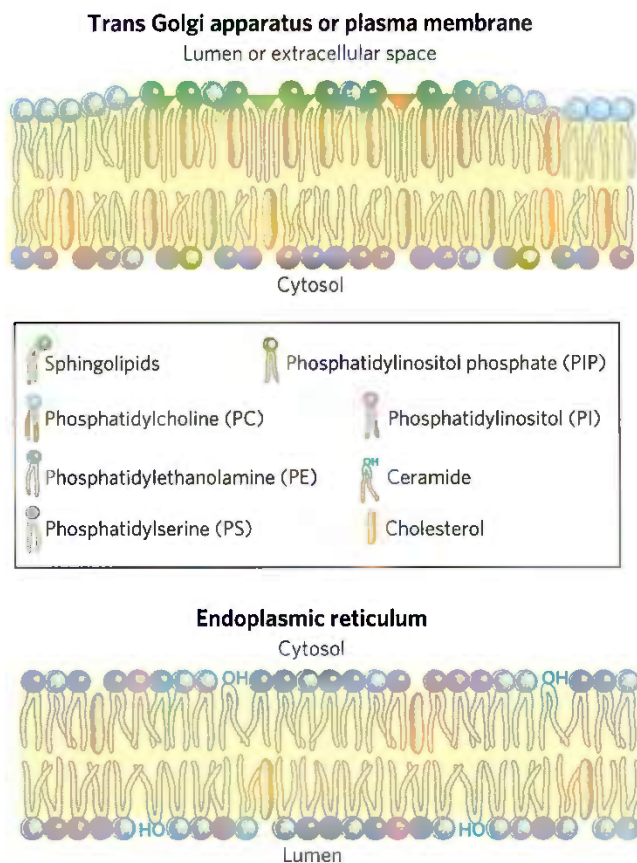


FIGURE 11-6 Change in lipid composition of secretory vesicles with their passage through the Golgi apparatus. Both the lipid composition of the bilayer and the disposition of specific lipids between inner and outer leaflets change remarkably as a result of vesicle trafficking. In the trans Golgi apparatus and plasma membrane, sphingolipids are greatly enriched in the leaflet facing the lumen or extracellular space, and phosphatidylserine, phosphatidylethanolamine, and the inositol phospholipids are almost exclusively in the cytosolic leaflet. The plasma membrane is also enriched with cholesterol. [Information from G. Drin, *Annu. Rev. Biochem.* 83:51, 2014, Fig. 1.]

but in transport vesicles leaving the trans Golgi apparatus, phosphatidylcholine has largely been replaced by sphingolipids and cholesterol, which, following fusion of the transport vesicles with the plasma membrane, make up the majority of the lipids in the outer leaflet of the cell's plasma membrane. Plasma membrane lipids are asymmetrically distributed between the two leaflets of the bilayer. Choline-containing lipids (phosphatidylcholine and sphingomyelin) are typically found in the outer (extracellular, or exoplasmic) leaflet, whereas phosphatidylserine, phosphatidylethanolamine, and the phosphatidylinositols are almost exclusively in the inner (cytoplasmic) leaflet. Here the negatively charged serine and inositol phosphate head groups can interact electrostatically with positively charged regions of peripheral or amphitropic membrane proteins (described below).

A second route for redistributing lipids from their site of synthesis to their destination membrane is via **lipid transfer proteins (LTPs; Fig. 11-7)**, which occur universally in various forms. LTPs are soluble in water, but they

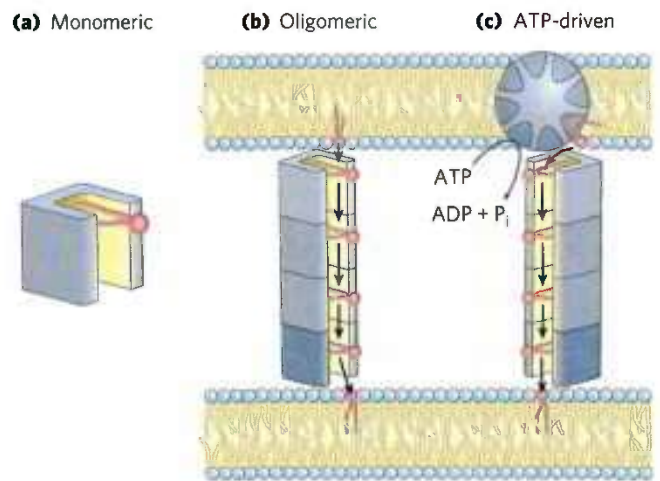


FIGURE 11-7 Lipid transfer protein (LTP) action. (a) Lipid transfer proteins move lipids and other nonpolar solutes through the very polar cytosol by providing a hydrophobic groove or pocket into which the hydrophobic solute can fit, with its nonpolar regions masked from the surrounding water. (b) Some LTPs form multisubunit structures to generate a hydrophobic groove long enough to connect two separate membranes at contact points (see Fig. 11-5). (c) To move lipid molecules against a concentration gradient (say, from a leaflet poor in phosphatidylserine to one relatively rich in it), the LTP is coupled to an ATP-dependent pump. [Information from L. H. Wong et al., *Nat. Rev. Mol. Cell Biol.* 20:85, 2019, Fig. 2.]

have a hydrophobic lipid-binding pocket in which they carry a lipid from one membrane to another through the cytosol. Some LTPs are bispecific: they carry, for example, a molecule of cholesterol to the plasma membrane, and they return with a phosphatidylinositol molecule. One class of lipid transfer protein forms a hydrophobic tunnel between two membranes at their contact points. In some cases, ATP is needed to drive the process, supplied by an ATP-binding cassette (ABC) transporter (described below). The rate of lipid movement from one membrane to another in vivo is significantly greater than the rate of vesicle budding and fusion.

Finally, once phospholipids reach their destination membrane, they can undergo enzymatic remodeling of their fatty acid constituents, a subject discussed in Chapter 21.

Membrane Proteins Are Receptors, Transporters, and Enzymes

P2 The protein composition of membranes reflects each membrane's functional specialization. One large group of membrane proteins are the receptors for extracellular signals such as hormones and changes in membrane potential. Hundreds of membrane proteins are transporters that carry specific polar or charged compounds—sugars, amino acids, vitamins, a wide variety of metabolic intermediates, minerals (Mg^{2+} , Ca^{2+} , Na^+ , K^+), and trace metal ions (Mn^{2+} , Ni^{2+} , Co^{2+})—across the plasma membrane or between organelles. All transporters and many receptors must span the membrane at least once to play their roles. Some membrane-associated enzymes also span the membrane, but many carry out their catalytic

functions on one side or the other of their membrane setting, partially penetrating just one leaflet of the bilayer.

The synthesis and delivery to the site of function are more complicated for membrane proteins than for typical cytosolic proteins. Many of the proteins destined for the plasma membrane undergo extensive posttranslational modification as they pass through the ER and the Golgi apparatus. One covalent modification is the glycosylation (attachment of oligosaccharides) of proteins bound for the plasma membrane. In glycophorin, a glycoprotein of the erythrocyte plasma membrane, 60% of the mass consists of complex oligosaccharides covalently attached to specific amino acid residues (**Fig. 11-8**). For

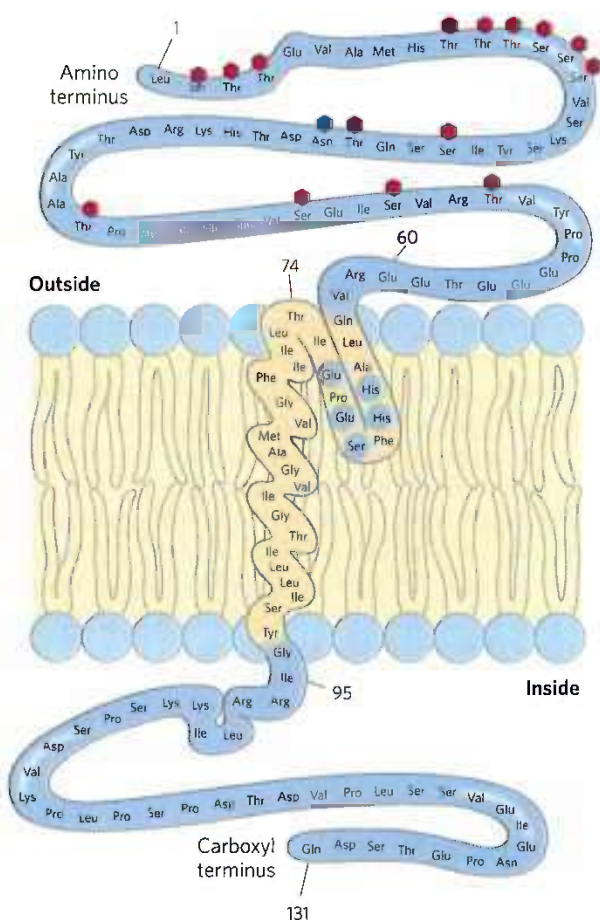


FIGURE 11-8 Transbilayer disposition of glycophorin in an erythrocyte. One hydrophilic domain, containing all the sugar residues, is on the outer surface, and another hydrophilic domain protrudes from the inner face of the membrane. Each red hexagon represents a tetrasaccharide (containing two Neu5Ac (sialic acid), Gal, and GalNAc) O-linked to a Ser or Thr residue; the blue hexagon represents an oligosaccharide N-linked to an Asn residue. The oligosaccharides are much larger than the hexagons used to represent them. A segment of 19 hydrophobic residues (residues 75 to 93) forms an α helix that traverses the membrane bilayer (see Fig. 11-13a). The segment from residues 64 to 74 has some hydrophobic residues and probably penetrates the outer face of the lipid bilayer, as shown. The functional unit of glycophorin is a homodimer; for clarity we show only one of the subunits here. [Information from V. T. Marchesi et al., *Annu. Rev. Biochem.* 45:667, 1976.]

such glycoproteins, Ser, Thr, and Asn residues are the most common points of carbohydrate attachment (see Fig. 7-27), and the carbohydrates are invariably on the outer face of the plasma membrane. Another posttranslational modification of some proteins is the covalent attachment of one or more lipids, which serve as hydrophobic anchors to hold a protein to the membrane or as tags to target the protein to a specific location.

Membrane Proteins Differ in the Nature of Their Association with the Membrane Bilayer

Integral membrane proteins are firmly embedded within the lipid bilayer and are removable only by agents such as detergents or organic solvents that overcome hydrophobic interactions (**Fig. 11-9**). **Peripheral membrane proteins** associate with the membrane through electrostatic interactions and hydrogen bonding with hydrophilic domains of integral proteins and with the charged head groups of membrane lipids. In the laboratory, they can be released from their membrane association by relatively mild treatments that interfere with electrostatic interactions or that break hydrogen bonds. **Amphitropic proteins** associate reversibly with membranes and are therefore found both in the membrane and in the cytosol. Their affinity for membranes results in some cases from the protein's noncovalent interaction with another membrane protein or lipid, and in other cases from the presence of one or more covalently attached lipids (see Fig. 11-16).

The firm attachment of integral proteins to membranes is the result of the hydrophobic interaction between the nonpolar core of the lipid bilayer and nonpolar side chains of the protein. **Monotopic** integral proteins have small hydrophobic domains that interact with only a single leaflet of the membrane (**Fig. 11-10**). These domains, which make up as little as a few percent of the protein's total mass, provide enough hydrophobic surface to hold the proteins firmly to the membrane. **Bitopic** proteins span the bilayer once, extending on either surface. They have a single hydrophobic sequence somewhere in the molecule. Glycophorin (**Fig. 11-8**) is bitopic. **Polytopic** proteins cross the membrane several times. They have multiple hydrophobic sequences, each of which, when in the α -helical conformation, is long enough (about 20 residues) to span the lipid bilayer. (Recall from Worked Example 4-1 that each residue in an α helix adds 1.5 Å to its length.)

One of the best-studied polytopic proteins, bacteriorhodopsin, has seven very hydrophobic internal sequences and spans the lipid bilayer seven times. Bacteriorhodopsin is a light-driven proton pump densely packed in regular arrays in the purple membrane of the bacterium *Halobacterium salinarum*. X-ray crystallography and cryo-electron microscopy reveal a structure in which each of the seven α -helical segments crosses the

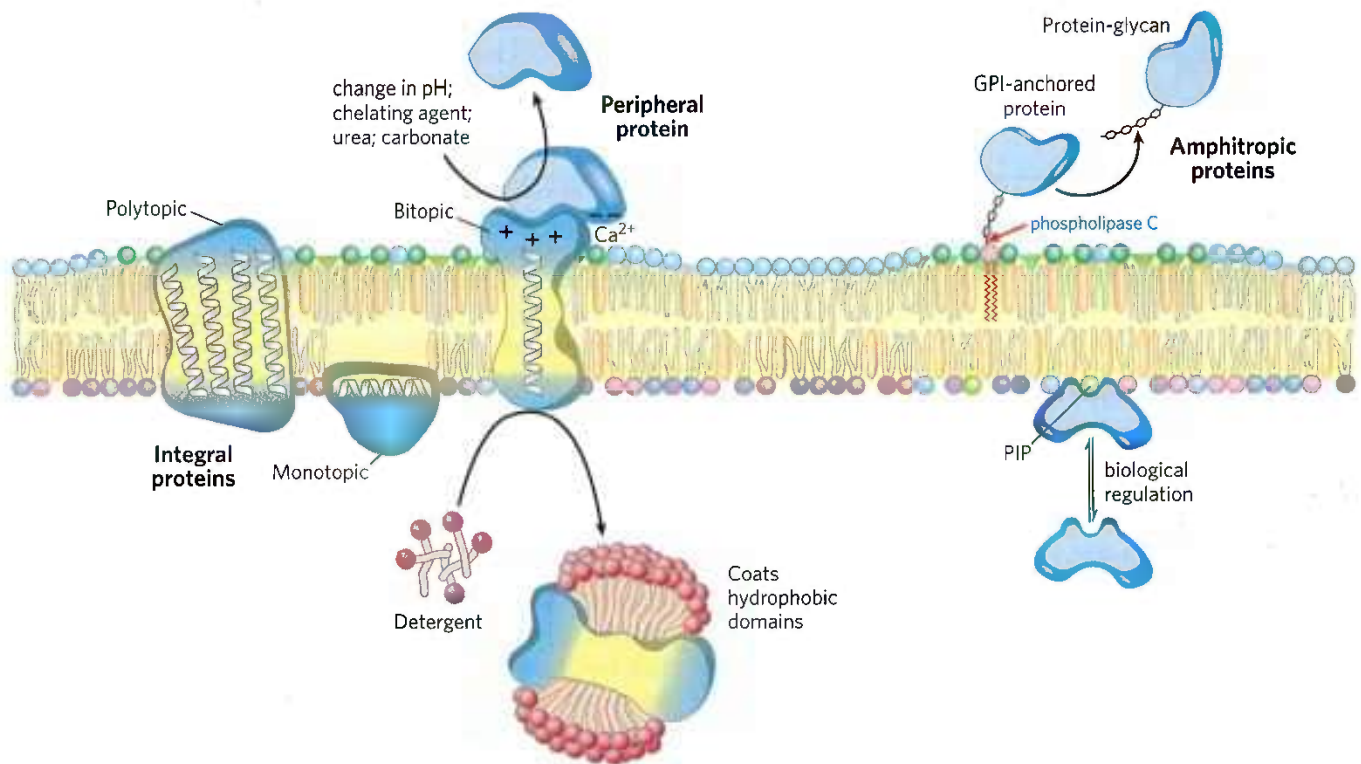


FIGURE 11-9 Integral, peripheral, and amphotropic proteins. Membrane proteins can be operationally distinguished by the conditions required to release them from the membrane. Integral proteins can only be removed from membranes with harsh treatment such

as detergent. Peripheral proteins are more loosely associated. Amphotropic proteins have a reversible association with membranes that depends on a regulatory process, such as reversible palmitoylation or phosphorylation.

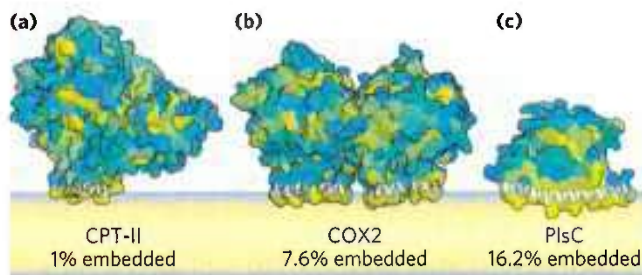


FIGURE 11-10 Monotopic proteins penetrate only one leaflet. Three typical monotopic membrane proteins have relatively little of their structure (1% to 16%) associated with lipids in one leaflet of the bilayer. They are colored to show surface hydrophobicity, with yellow hydrophobic, blue hydrophilic, and hues of green in between. Each of these proteins catalyzes a reaction that involves a hydrophobic substrate. (a) CPT-II is a mammalian carnitine palmitoyltransferase II; (b) COX2 is a mammalian prostaglandin H2 synthase-1; (c) PlsC is a bacterial acyltransferase (1-acyl-*sn*-glycerol-6-phosphate acyltransferase). [Data from (a) PDB ID 2H4T, Y. S. Hsiao et al., *Biochem. Biophys. Res. Commun.* 346:974, 2006; (b) PDB ID 1CQE, D. Picot et al., *Nature* 367:243, 1994; (c) PDB ID 5KYM, R. M. Robertson et al., *Nat. Struct. Mol. Biol.* 24:666, 2017.]

lipid bilayer, and all are connected by nonhelical loops at the inner and outer faces of the membrane (Fig. 11-11). In the amino acid sequence of bacteriorhodopsin, seven segments of about 20 hydrophobic residues can be identified, each forming an α helix that spans the bilayer. The seven helices are clustered together and oriented not quite perpendicular to the bilayer plane, a motif that (as we shall see in Chapter 12) is very common in membrane

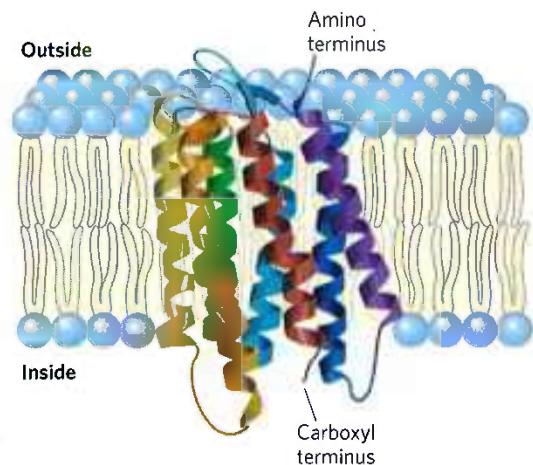


FIGURE 11-11 Bacteriorhodopsin, a membrane-spanning protein. The single polypeptide chain folds into seven hydrophobic α helices, each of which traverses the lipid bilayer roughly perpendicular to the plane of the membrane. The seven transmembrane helices are clustered, and the space around and between them is filled with the acyl chains of membrane lipids. The light-absorbing pigment retinal (see Fig. 10-20) is buried deep within the membrane in contact with several of the helical segments (not shown). The helices are colored to correspond with the hydropathy plot in Figure 11-13. [Data from PDB ID 2AT9, K. Mitsuoka et al., *J. Mol. Biol.* 286:861, 1999.]

proteins involved in signal reception. The hydrophobic interaction keeps the nonpolar amino acid residues firmly anchored among the fatty acyl groups of the membrane lipids in the membrane's lipid core.

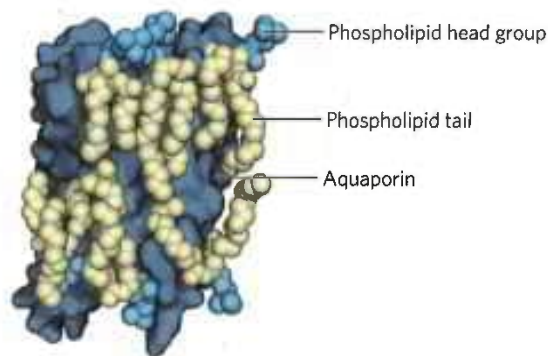


FIGURE 11-12 Lipid annuli associated with an integral protein.

The crystal structure of sheep aquaporin, a transmembrane water channel, includes a shell of phospholipids positioned with their head groups (blue) at the expected positions on the inner and outer membrane surfaces and their hydrophobic acyl chains (gold) intimately associated with the surface of the protein exposed to the bilayer. The lipid forms a “grease seal” around the protein, which is depicted by a dark blue surface representation. [Data from PDB ID 2B6O, T. Gonen et al., *Nature* 438:633, 2005.]

Many membrane proteins co-crystallize with phospholipid molecules, which are presumed to be positioned in the native membranes as they are in the protein crystals. Many of these phospholipid molecules lie on the protein surface, their head groups interacting with polar amino acid residues at the inner and outer membrane-water interfaces and their side chains associated with nonpolar residues. Other phospholipids are found at the interfaces between monomers of multisubunit membrane proteins, where they form a “grease seal” (Fig. 11-12).

The Topology of an Integral Membrane Protein Can Often Be Predicted from Its Sequence

Determination of the location of a membrane protein relative to the bilayer—that is, its topology—is generally much more difficult than determining its amino acid sequence, either directly or by gene sequencing. The amino acid sequences of many thousands of membrane proteins are known, but fewer three-dimensional structures have been established by x-ray crystallography or cryo-electron microscopy. The presence of unbroken sequences of more than 20 hydrophobic residues in a membrane protein is commonly taken as evidence that these sequences traverse the lipid bilayer, acting as hydrophobic anchors or forming transmembrane channels. All known integral proteins have at least one such sequence. Application of this logic to entire genomic sequences leads to the conclusion that in many organisms, 20% to 30% of all proteins are integral proteins.

What can we predict about the secondary structure of the membrane-spanning portions of integral proteins? At 1.5 Å (0.15 nm) per amino acid residue, an α -helical sequence of 20 to 25 residues is just long enough to span the thickness (30 Å) of the lipid bilayer. A polypeptide

chain surrounded by lipids, having no water molecules with which to hydrogen-bond, will tend to form α helices or β sheets in which intrachain hydrogen bonding is maximized. If the side chains of all amino acids in a helix are nonpolar, the helix is further stabilized in the surrounding lipids by the hydrophobic effect.

Several simple methods of analyzing amino acid sequences yield reasonably accurate predictions of secondary structure for transmembrane proteins. The relative polarity of each amino acid has been determined experimentally by measuring the free-energy change accompanying the movement of that amino acid side chain from a hydrophobic environment into water. This free energy of transfer, which can be expressed as a **hydropathy index** (see Table 3-1), ranges from highly exergonic for charged or polar residues to highly endergonic for amino acids with aromatic or aliphatic hydrocarbon side chains. The overall hydropathy index (hydrophobicity) of a sequence of amino acids is estimated by summing the free energies of transfer for the residues in the sequence. To scan a polypeptide sequence for potential membrane-spanning segments, an investigator calculates the hydropathy index for successive segments (called windows) of a given size, from 7 to 20 residues. For a window of seven residues, for example, the average indices for residues 1 to 7, 2 to 8, 3 to 9, and so on, are plotted as in Figure 11-13 (plotted for the middle residue in each window—residue 4 for residues 1 to 7, for example). A region with more than 20 residues of high hydropathy index is presumed to be a transmembrane segment. When the sequences of membrane proteins of known three-dimensional structure are scanned using simple online bioinformatics tools, we find a reasonably good correspondence between predicted and known membrane-spanning segments. Hydropathy analysis predicts a single hydrophobic helix for glycoporphin (Fig. 11-13a) and seven transmembrane segments for bacteriorhodopsin (Fig. 11-13b)—in agreement with structures known from x-ray crystallography.

Not all integral proteins are composed of transmembrane α helices. Another structural motif common in bacterial and mitochondrial membrane proteins is the **β barrel** (see Fig. 4-16b), in which 20 or more transmembrane segments form β sheets that line a cylinder (Fig. 11-14). The same factors that favor α helix formation in the hydrophobic interior of a lipid bilayer also stabilize β barrels: when no water molecules are available to hydrogen-bond with the carbonyl oxygen and nitrogen of the peptide bond, maximal intrachain hydrogen bonding gives the most stable conformation. Planar β sheets do not maximize these interactions and are generally not found in the membrane interior; β barrels allow all possible hydrogen bonds and are common among membrane proteins. **Porins**, proteins that allow certain polar solutes to cross the outer membrane of gram-negative bacteria such as *E. coli*, have many-stranded β barrels lining the polar transmembrane

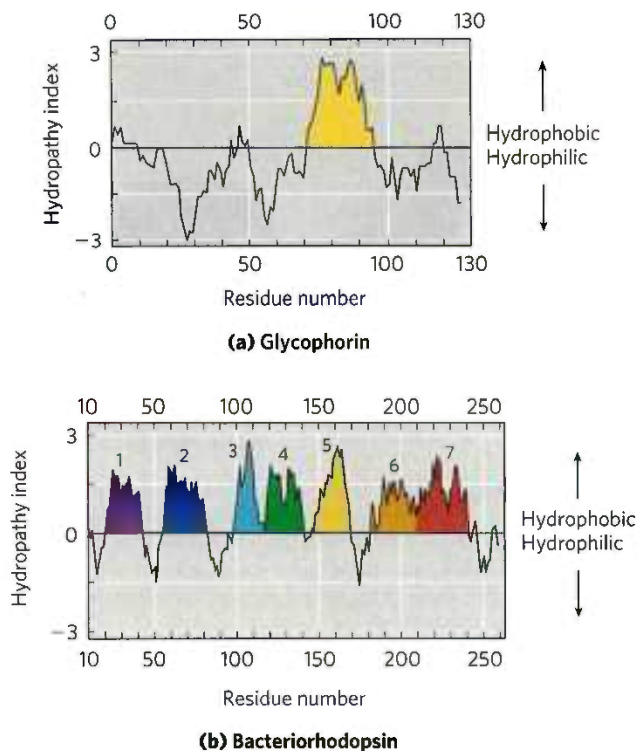


FIGURE 11-13 Hydropathy plots. Average hydropathy index (see Table 3-1) is plotted against residue number for two integral membrane proteins. The hydropathy index for each amino acid residue in a sequence of defined length, or “window,” is used to calculate the average hydropathy for that window. The horizontal axis shows the residue number in the middle of the window. (a) Glycophorin from human erythrocytes has a single hydrophobic sequence between residues 75 and 93 (yellow); compare this with Figure 11-8. (b) Bacteriorhodopsin, known from independent physical studies to have seven transmembrane helices (see Fig. 11-11), has seven hydrophobic regions. Note, however, that the hydropathy plot is ambiguous in the region of segments 6 and 7. X-ray crystallography has confirmed that this region has two transmembrane segments.

passage. The outer membranes of mitochondria and chloroplasts also contain a variety of β barrels, as do the outer membranes of the modern bacteria descended from those believed to have evolved to mitochondria and chloroplasts.

A polypeptide is more extended in the β conformation than in an α helix; just seven to nine residues of β conformation are needed to span a membrane. Recall that in the β conformation, alternating side chains project above and below the sheet (see Fig. 4-5). In β strands of membrane proteins, every second residue in the membrane-spanning segment is hydrophobic and interacts with the lipid bilayer; aromatic side chains are commonly found at the lipid-protein interface. The other residues may or may not be hydrophilic.

A further remarkable feature of many transmembrane proteins of known structure is the presence of Tyr and Trp residues at the interface between lipid and water (Fig. 11-15). The side chains of these residues seem to serve as membrane interface anchors, able to interact simultaneously with the central lipid phase and the

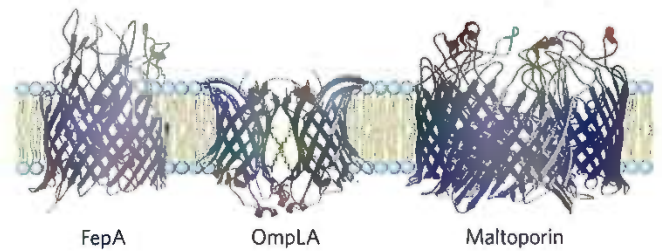


FIGURE 11-14 Polytopic integral proteins with β -barrel structure.

Three proteins of the *E. coli* outer membrane are shown, viewed in the plane of the membrane. FepA, involved in iron uptake, has 22 membrane-spanning β strands. Outer membrane phospholipase A, or OmpLA, is a 12-stranded β barrel that exists as a dimer in the membrane. Maltoporin, a maltose transporter, is a trimer; each monomer consists of 16 β strands. [Data from FepA, PDB ID 1FEP; S. K. Buchanan et al., *Nat. Struct. Biol.* 6:56, 1999; OmpLA, PDB ID 1QD5; H. J. Snijder et al., *Nature* 401:717, 1999; maltoporin, PDB ID 1MAL; T. Schirmer et al., *Science* 267:512, 1995.]

aqueous phases on either side of the membrane. Another generalization about amino acid location relative to the bilayer is described by the **positive-inside rule**: the positively charged Lys and Arg residues in the extramembrane loop of membrane proteins occur more commonly on the cytoplasmic face of plasma membranes.

Covalently Attached Lipids Anchor or Direct Some Membrane Proteins

Some membrane proteins are covalently linked to one or more lipids, which may be of several types: long-chain fatty acids, isoprenoids, sterols, or glycosylated derivatives of phosphatidylinositol (GPIs; Fig. 11-16). The attached lipid provides a hydrophobic anchor that inserts into the lipid bilayer and holds the protein at the membrane surface. The strength of the hydrophobic interaction between a bilayer and a single hydrocarbon chain linked to a protein is barely enough to anchor the protein securely, but many proteins have more than one attached lipid moiety. Furthermore, other interactions, such as ionic attractions between positively charged Lys residues in the protein and negatively charged lipid head groups, can add to the anchoring effect of a covalently bound lipid. For example, the plasma membrane protein MARCKS (myristoylated alanine-rich C-kinase substrate), which interacts with actin filaments in the process of cell motility, has a covalently attached myristoyl moiety, but it also contains the sequence

KKKKKRFSFKKSFKLSGFSFKNNKK
151 175

which adds to the protein's affinity for the membrane. Three clusters of positively charged Lys and Arg residues (screened blue) interact with the negatively charged head group of phosphatidylinositol 4,5-bisphosphate (PIP₂) on the cytoplasmic face of the plasma membrane; five aromatic residues (screened yellow) insert into the lipid bilayer. When the head-group phosphates of PIP₂ are enzymatically removed, MARCKS loses its hold on

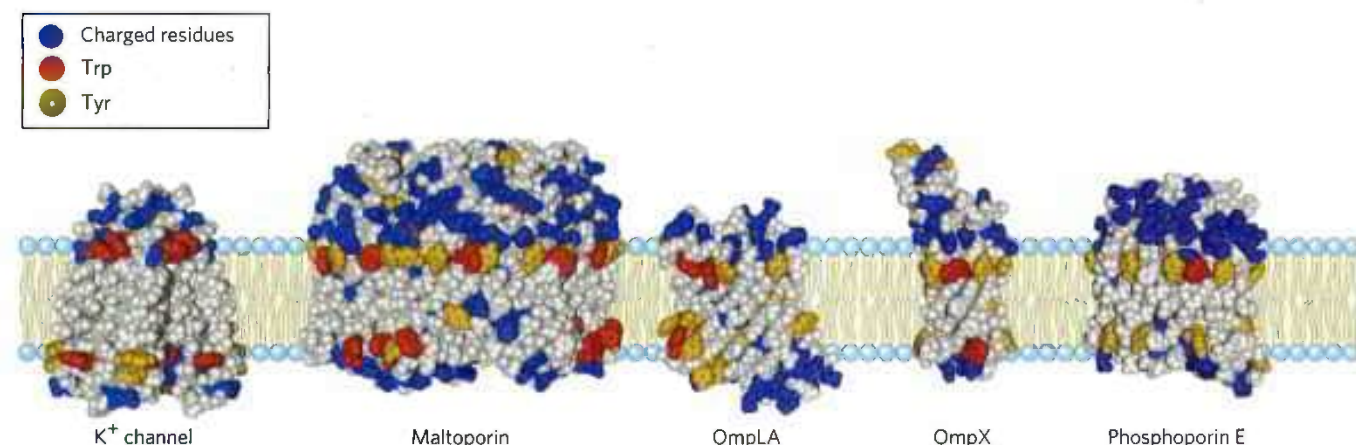
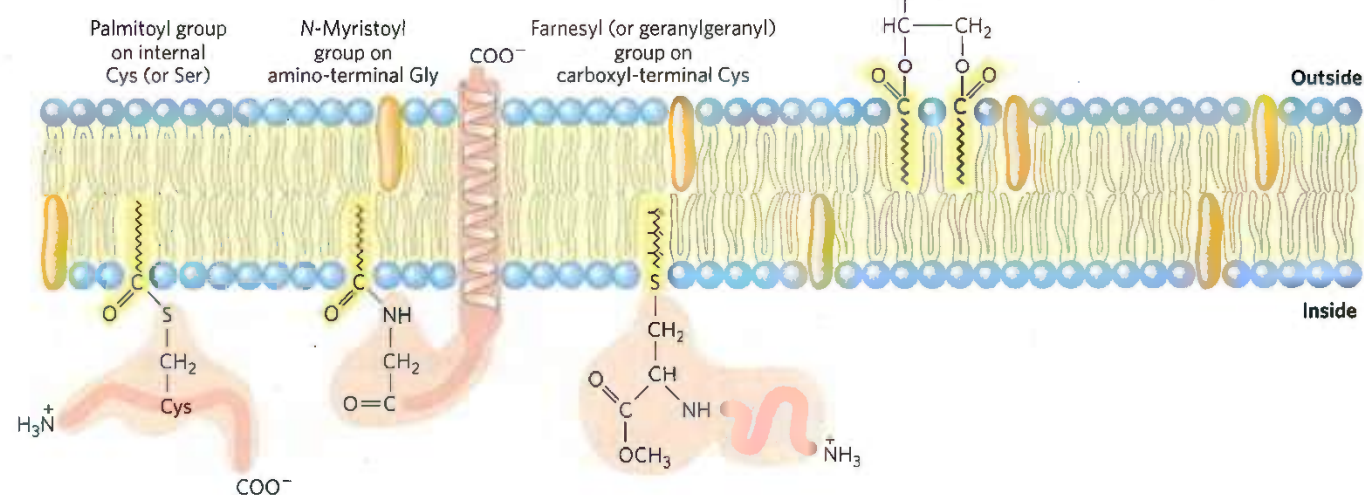


FIGURE 11-15 Tyr and Trp residues of integral proteins clustering at the water-lipid interface. The detailed structures of these five integral proteins are known from crystallographic studies. The K^+ channel is from the bacterium *Streptomyces lividans* (see Fig. 11-45); maltoporin, OmpLA, OmpX, and phosphoporin E are proteins of the outer membrane of *E. coli*. Residues of Tyr and Trp are found predominantly where the nonpolar region

of acyl chains meets the polar head-group region. Charged residues (Lys, Arg, Glu, Asp) are found almost exclusively in the aqueous phases. [Data from K^+ channel, PDB ID 1BL8, D. A. Doyle et al., *Science* 280:69, 1998; maltoporin, PDB ID 1AF6, Y. F. Wang et al., *J. Mol. Biol.* 272:56, 1997; OmpLA, PDB ID 1QD5, H. J. Snijder et al., *Nature* 401:717, 1999; OmpX, PDB ID 1QJ9, J. Vogt and G. E. Schulz, *Structure* 7:1301, 1999; phosphoporin E, PDB ID 1PHO, S. W. Cowan et al., *Nature* 358:727, 1992.]

FIGURE 11-16 Lipid-linked membrane proteins. Covalently attached lipids anchor membrane proteins to the lipid bilayer. A palmitoyl group is shown attached by thioester linkage to a Cys residue; an *N*-myristoyl group is generally attached to an amino-terminal Gly residue, typically of a protein that also has a hydrophobic transmembrane segment; the farnesyl and geranylgeranyl groups attached to carboxyl-terminal Cys residues are isoprenoids of 15 and 20 carbons, respectively. The carboxyl-terminal Cys residue is invariably methylated. Glycosyl phosphatidylinositol (GPI) anchors are derivatives of phosphatidylinositol in which the inositol bears a short oligosaccharide covalently joined to the carboxyl-terminal residue of a protein through phosphoethanolamine. GPI-anchored proteins are always on the extracellular face of the plasma membrane. Farnesylated and palmitoylated membrane proteins are found on the inner surface of the plasma membrane, and myristoylated proteins have domains both inside and outside the plasma membrane. For some lipid-linked proteins, the lipid attachment process is reversible, so the protein is amphitropic: membrane-bound when lipid-linked, soluble when not.



the plasma membrane and dissociates. The reversible nature of the association of MARCKS with the membrane makes it an amphitropic protein.

Beyond merely anchoring a protein to the membrane, the attached lipid may have a more specific role.

In the plasma membrane, **GPI-anchored proteins** are exclusively on the outer face and are clustered in certain regions, as discussed later in the chapter (p. 380), whereas other types of lipid-linked proteins (with farnesyl or geranylgeranyl groups attached; Fig. 11-16) are exclusively

on the inner face. In polarized epithelial cells (such as intestinal epithelial cells), in which apical and basal surfaces have different roles, GPI-anchored proteins are directed specifically to the apical surface. Attachment of a specific lipid to a newly synthesized membrane protein therefore has a targeting function, directing the protein to its correct cellular location.

SUMMARY 11.1 *The Composition and Architecture of Membranes*

- Biological phospholipids and sterols spontaneously form a lipid bilayer to protect their hydrophobic hydrocarbon chains from energetically unfavorable interaction with water.

- In the fluid mosaic model, a lipid bilayer is the basic structural unit, and proteins associate with or span the bilayer. Biological membranes are flexible, self-repairing, and selectively permeable. They regulate the traffic of small molecules into and out of the cell and among the organelles, and they provide a scaffold on which proteins assemble into catalytic and structural aggregates that function in two-dimensional space.

- Proteins and lipids are trafficked through the dynamic endomembrane system of a eukaryote from their point of synthesis to their appropriate cellular locations. Small vesicles and soluble transporter proteins ensure that each membrane leaflet has its unique complement of lipids and proteins to achieve its specialized function.

- Cells have hundreds of membrane transporters that carry polar solutes and ions in and out of cells and throughout the endomembrane system. The plasma membrane contains protein receptors that sense signals from outside the cell and carry their message into the cell. Many enzymes are associated with membranes, where they interact with each other in essentially two-dimensional space.

- Integral proteins are embedded within membranes, their nonpolar amino acid side chains stabilized by contact with the lipid bilayer. Peripheral proteins associate with membranes through electrostatic interactions and hydrogen bonding with membrane phospholipids and integral proteins. Amphitropic proteins associate reversibly with the membrane.

- Many integral membrane proteins span the lipid bilayer several times, with hydrophobic sequences of about 20 amino acid residues forming transmembrane α helices. Multistranded β barrels are also common in integral proteins of bacterial and mitochondrial membranes. A hydropathy plot of the amino acid sequence identifies segments that are likely to cross the lipid bilayer as helices or barrels.

- Covalent attachment of a hydrophobic molecule such as a fatty acid serves to anchor some membrane proteins to the bilayer.

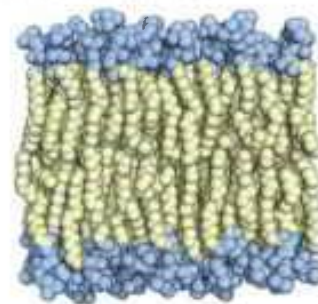
11.2 Membrane Dynamics

One remarkable feature of all biological membranes is their plasticity—their ability to change shape without losing their integrity and becoming leaky. The bases for this property are the noncovalent interactions among lipids in the bilayer and the mobility allowed to individual lipids because they are not covalently anchored to one another. We turn now to the dynamics of membranes: the motions that occur and the transient structures that are allowed by these motions.

Acyl Groups in the Bilayer Interior Are Ordered to Varying Degrees

Although the lipid bilayer structure is stable, its individual phospholipid molecules have much freedom of motion (Fig. 11-17), depending on the temperature and the lipid composition. Below normal physiological temperatures, **P1** the lipids in a bilayer form a gel-like **liquid-ordered (L_o) state**, in which all types of motion of individual lipid molecules are strongly constrained (Fig. 11-17a). Above physiological temperatures, individual hydrocarbon chains of fatty acids are in constant motion produced by rotation about the carbon–carbon bonds of the long acyl side chains and by lateral diffusion of individual lipid molecules in

(a) Liquid-ordered state L_o



Heat produces thermal motion of side chains ($L_o \rightarrow L_d$ transition).

(b) Liquid-disordered state L_d

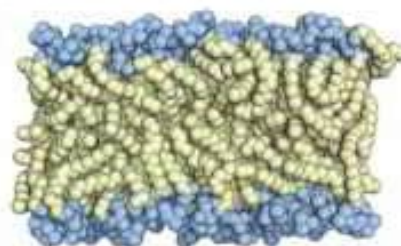


FIGURE 11-17 Two extreme states of bilayer lipids. (a) In the liquid-ordered (L_o) state, polar head groups are uniformly arrayed at the surface, and the acyl chains are nearly motionless and packed with regular geometry. (b) In the liquid-disordered (L_d) state, or fluid state, acyl chains undergo much thermal motion and have no regular organization. The state of membrane lipids in biological membranes is maintained somewhere between these extremes. [Information from H. Heller et al., *J. Phys. Chem.* 97:8343, 1993.]

the plane of the bilayer. This is the **liquid-disordered (L_d) state** (Fig. 11-17b). In the transition from the L_o state to the L_d state, the general shape and dimensions of the bilayer are maintained; what changes is the degree of motion (lateral and rotational) allowed to individual lipid molecules.

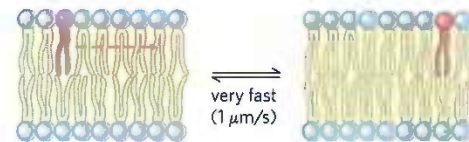
At temperatures in the physiological range for a mammal (about 20 to 40 °C), long-chain saturated fatty acids (such as 16:0 and 18:0) tend to pack into an L_o gel phase, but the kinks in unsaturated fatty acids interfere with packing, favoring the L_d state (see Fig. 10-1). Shorter-chain fatty acyl groups are more mobile than longer-chain fatty acyl groups and thus also favor the L_d state. The sterol content of a membrane, which varies greatly with organism and organelle, is another important determinant of lipid state. Sterols (such as cholesterol) have paradoxical effects on bilayer fluidity: they interact with phospholipids containing unsaturated fatty acyl chains, compacting them and constraining their motion in bilayers. In contrast, their association with sphingolipids and phospholipids having long, saturated fatty acyl chains tends to make a bilayer fluid that would otherwise, without cholesterol, adopt the L_o state. In biological membranes composed of a variety of phospholipids and sphingolipids, cholesterol tends to associate with sphingolipids and to form regions in the L_o state surrounded by cholesterol-poor regions in the L_d state (see the discussion of membrane rafts below).

Transbilayer Movement of Lipids Requires Catalysis

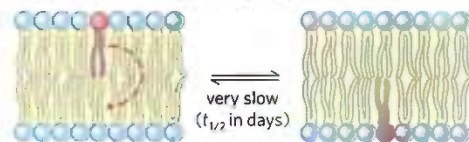
At physiological temperatures, lateral diffusion *in the plane* of the bilayer is very rapid, but the movement of a lipid molecule from one leaflet of the bilayer to the other occurs very slowly, if at all, in most membranes (Fig. 11-18). Transbilayer—or “flip-flop”—movement requires that a polar or charged head group leave its aqueous environment and move into the hydrophobic interior of the bilayer, a process with a large, positive free-energy change. There are, however, situations in which such movement is essential. For example, in the ER, membrane glycerophospholipids are synthesized on the cytosolic face, whereas sphingolipids are synthesized or modified on the luminal surface. To get from their site of synthesis to their eventual point of deposition, these lipids must undergo flip-flop diffusion.

P2 Membrane proteins called flippases, floppases, and scramblases (Fig. 11-18c) facilitate the transbilayer movement (translocation) of individual lipid molecules. Like enzymes, these translocators act by providing a path that is energetically more favorable and therefore much faster than the uncatalyzed movement. The combination of asymmetric biosynthesis of membrane lipids, no uncatalyzed flip-flop diffusion, and the presence of selective, energy-dependent lipid translocators accounts for the transbilayer asymmetry in lipid composition discussed in Section 11.1.

(a) Uncatalyzed lateral diffusion



(b) Uncatalyzed transbilayer (“flip-flop”) diffusion



(c) Catalyzed transbilayer translocations

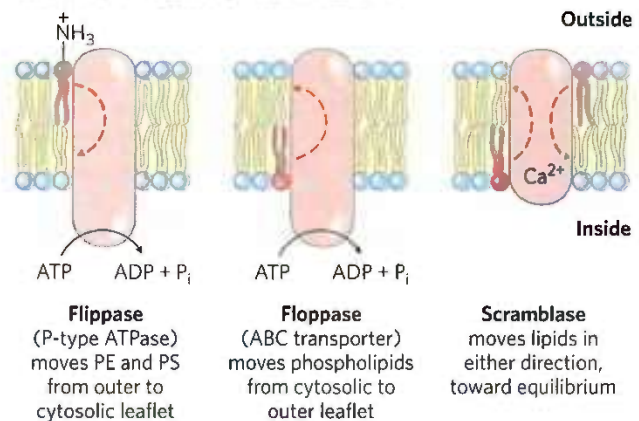


FIGURE 11-18 Motion of single phospholipids in a bilayer.

(a) Lateral diffusion within a leaflet is very rapid, but (b) uncatalyzed movement from one leaflet to the other is very slow. (c) Three types of phospholipid translocators in the plasma membrane. PE is phosphatidylethanolamine; PS is phosphatidylserine.

Flippases catalyze translocation of the *amino*-phospholipids phosphatidylethanolamine and phosphatidylserine from the extracellular to the cytoplasmic leaflet of the plasma membrane, contributing to the asymmetric distribution of phospholipids: phosphatidylethanolamine and phosphatidylserine primarily in the cytoplasmic leaflet, and the sphingolipids and phosphatidylcholine in the outer leaflet. Keeping phosphatidylserine out of the extracellular leaflet is important: its exposure on the outer surface triggers apoptosis (programmed cell death; see Chapter 12) and engulfment by macrophages that carry phosphatidylserine receptors. Flippases also act in the ER, where they move newly synthesized phospholipids from their site of synthesis in the cytosolic leaflet to the luminal leaflet. Flippases consume about one ATP per molecule of phospholipid translocated, and they are structurally and functionally related to the P-type ATPases (active transporters) described in Section 11.3.

There are three other types of lipid-translocating activities: floppases, scramblases, and phosphatidylinositol transfer proteins. **Floppases** move plasma membrane phospholipids and sterols from the cytoplasmic leaflet

to the extracellular leaflet and, like flippases, are ATP-dependent. Floppases are members of the ABC transporter family, described on page 395; all ABC transporters actively transport hydrophobic substrates outward across the plasma membrane. Each floppase specializes in movement of specific lipids: cholesterol, phosphatidylcholine, sphingomyelin, and phosphatidylserine. **Scramblases** are proteins that move any membrane phospholipid across the bilayer down its concentration gradient (from the leaflet where it has a higher concentration to the leaflet where it has a lower concentration); their activity is not dependent on ATP, although some require Ca^{2+} . Scramblase activity leads to controlled randomization of the head-group composition on the two faces of the bilayer. The activity of some scramblases rises sharply with an increase in cytosolic Ca^{2+} concentration, which may result from cell activation, cell injury, or apoptosis. Finally, a group of proteins that act primarily to move phosphatidylinositol lipids across lipid bilayers, the **phosphatidylinositol transfer proteins**, are believed to have important roles in lipid signaling and membrane trafficking.

Lipids and Proteins Diffuse Laterally in the Bilayer

P2 Individual lipid molecules can move laterally in the plane of the membrane by changing places with neighboring lipid molecules; that is, they undergo Brownian movement within the bilayer (Fig. 11-18a). This rapid lateral diffusion in the plane of the bilayer tends to randomize the positions of individual molecules in a few seconds.

Lateral diffusion can be shown experimentally by attaching fluorescent probes to the head groups of lipids and using fluorescence microscopy to follow the probes over time (Fig. 11-19). In one technique, a small region ($5 \mu\text{m}^2$) of a cell surface with fluorescence-tagged lipids is bleached by intense laser radiation so that the irradiated patch no longer fluoresces when viewed with less-intense (nonbleaching) light in the fluorescence microscope. However, within milliseconds, the region recovers its fluorescence as unbleached lipid molecules diffuse into the bleached patch and bleached lipid molecules diffuse away from it. The rate of fluorescence recovery after photobleaching, or **FRAP**, is a measure of the rate of lateral diffusion of the lipids. Using the FRAP technique, researchers have shown that some membrane lipids diffuse laterally at rates of up to $1 \mu\text{m/s}$. At this rate, a lipid molecule could move from one end of a eukaryotic cell to the other in a few seconds.

Another technique, single particle tracking, has allowed researchers to follow the movement of a *single* lipid molecule in the plasma membrane on a much shorter time scale. Results from these studies confirm that lipid molecules diffuse laterally with rapidity within small, discrete regions of the cell surface but that movement from one such region to a nearby region ("hop diffusion") is rarer; membrane lipids behave as though corralled by fences that they can occasionally cross by hop diffusion (Fig. 11-20).

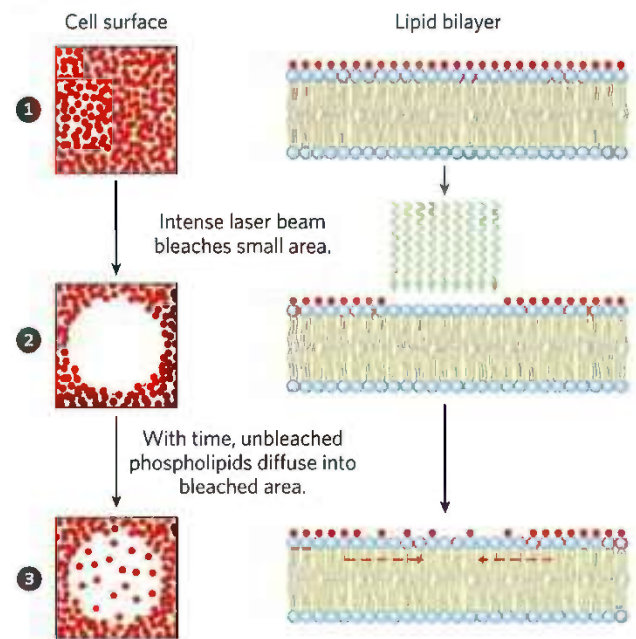


FIGURE 11-19 Measurement of lateral diffusion rates of lipids by fluorescence recovery after photobleaching (FRAP).

Lipids in the outer leaflet of the plasma membrane are labeled by reaction with a membrane-impermeant fluorescent probe (red) so that the surface is uniformly labeled when viewed with a fluorescence microscope. A small area is bleached with a laser, then recovers its fluorescence. With time, unbleached lipid molecules diffuse into the bleached region, and it again becomes fluorescent. The FRAP method can also be used to measure lateral diffusion of membrane proteins labeled with a fluorescent tag.

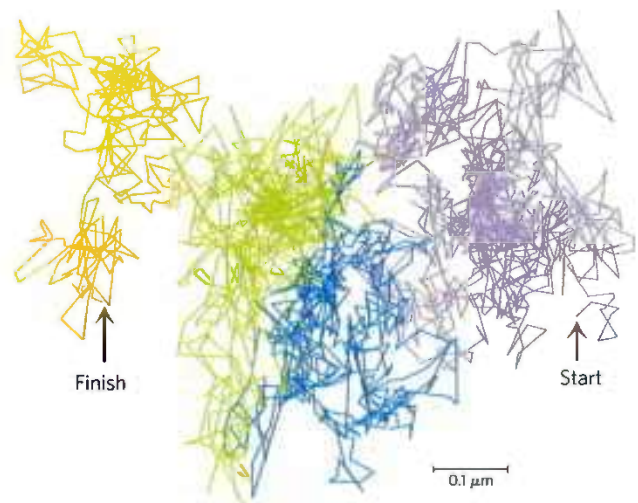


FIGURE 11-20 Hop diffusion of individual lipid molecules.

The motion of a single fluorescently labeled lipid molecule in a cell surface is recorded on video by fluorescence microscopy, with a time resolution of $25 \mu\text{s}$ (equivalent to 40,000 frames/s). The track shown here represents a molecule followed for 56 ms (2,250 frames); the trace begins in the purple area and continues through blue, green, and orange. The pattern of movement indicates rapid diffusion within a confined region (about 250 nm in diameter, shown by a single color), with occasional hops into an adjoining region. This finding suggests that the lipids are corralled by molecular fences that they occasionally jump. [Data from Takahiro Fujiwara, Ken Ritchie, Hideji Murakoshi, Ken Jacobson, and Akihiro Kusumi.]

form more stable associations with the cholesterol and long acyl groups in rafts than they form with the surrounding phospholipids. (It is notable that other lipid-linked proteins, those with covalently attached isoprenyl groups such as farnesyl, are *not* preferentially associated with the outer leaflet of sphingolipid rafts; see Fig. 11-22.) The “raft” and “sea” domains of the plasma membrane are not rigidly separated; membrane proteins can move into and out of lipid rafts in a fraction of a second. But in the shorter time scale (microseconds) more relevant to many membrane-mediated biochemical processes, many of these proteins reside primarily in a raft.

The approximate fraction of the cell surface occupied by rafts can be as high as 50% in some cases; the rafts cover half of the ocean. Indirect measurements in cultured fibroblasts suggest a diameter of roughly 50 nm for an individual raft, which corresponds to a patch containing a few thousand sphingolipids and perhaps 10 to 50 membrane proteins. Because most cells express more than 50 different kinds of plasma membrane proteins, it is likely that a single raft contains only a subset of membrane proteins and that this segregation of membrane proteins is functionally significant. Their presence in a single raft would hugely increase the likelihood of their collision. Certain membrane receptors and signaling proteins, for example, seem to be segregated together in membrane rafts. Experiments show that signaling through these proteins can be disrupted by manipulations that deplete the plasma membrane of cholesterol and destroy lipid rafts.

Plasma membranes of many cells have specialized rafts called **caveolae** (“little caves”), which can represent about half of the total area of the plasma membrane (Fig. 11-23a). Closely associated with caveolae is **caveolin**, an integral protein with two globular domains connected by a hairpin-shaped hydrophobic domain that binds the protein to the cytoplasmic leaflet of the plasma membrane (Fig. 11-23b). Three palmitoyl groups attached to the carboxyl-terminal globular domain further anchor the protein to the membrane. Caveolins form dimers and associate with cholesterol-rich regions in the membrane. The presence of caveolin dimers forces the associated lipid bilayer to curve inward, forming caveolae. Caveolae involve *both* leaflets of the bilayer—the cytoplasmic leaflet, from which the caveolin globular domains project, and the extracellular leaflet, a cholesterol and sphingolipid raft with associated GPI-anchored proteins. Caveolae are implicated in a variety of cellular functions, including membrane trafficking within cells and the transduction of external signals into cellular responses.

Caveolae may also provide a means of expanding the cell surface. The lipid bilayer itself is not elastic, but if existing caveolae lose their associated caveolin as the result of a regulatory signal, the caveolae flatten into

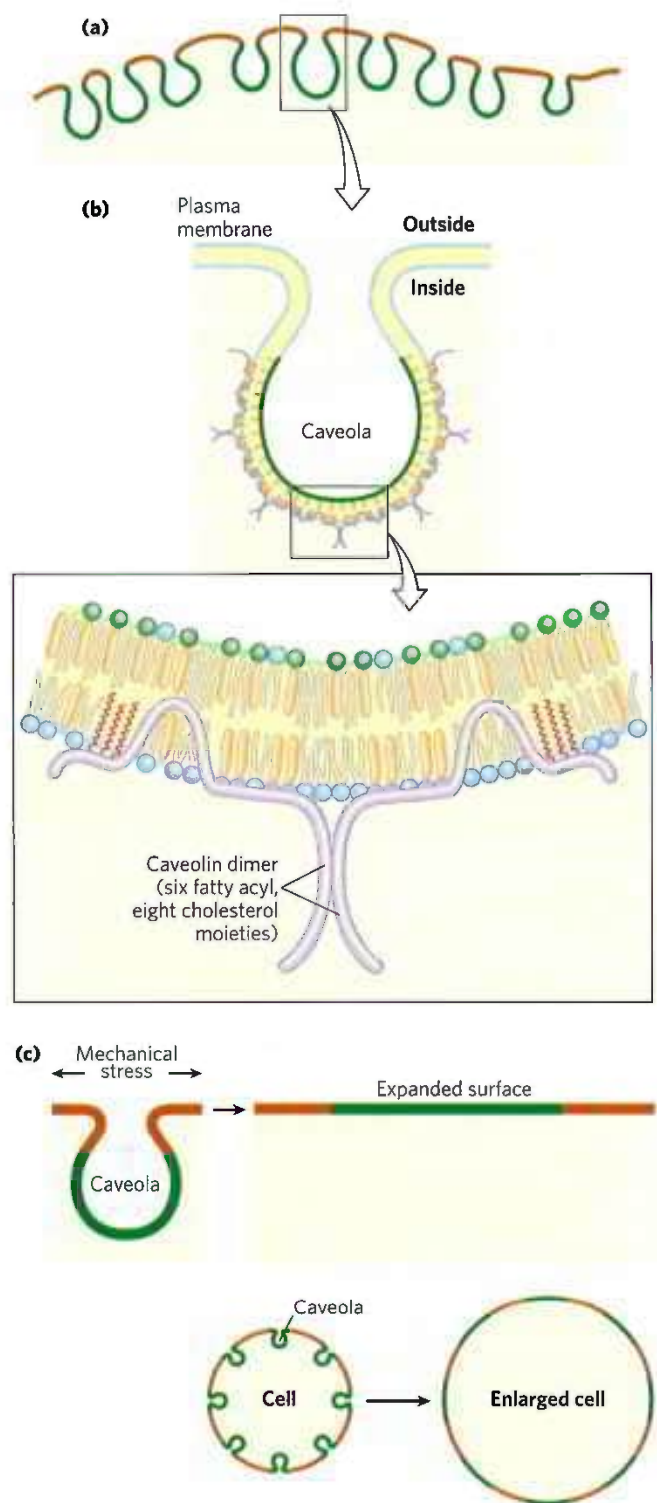


FIGURE 11-23 Caveolin forces inward curvature of a membrane. (a) Caveolae are small invaginations in the plasma membrane. (b) Cartoon showing the location and role of a caveolin dimer in causing inward membrane curvature. Each caveolin monomer has a central hydrophobic domain and three long-chain acyl groups (red), which hold the molecule to the inside of the plasma membrane. When several caveolin dimers are concentrated in a small region (a raft), they force a curvature in the lipid bilayer, forming a caveola. Cholesterol molecules in the bilayer are shown in orange. (c) Flattening of caveolae allows the plasma membrane to expand in response to various stresses. [Information from R. G. Parton, *Nat. Rev. Mol. Cell Biol.* 8:185–194, 2007.]

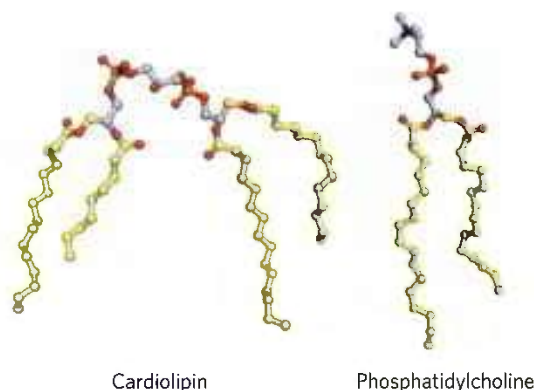
the plasma membrane (Fig. 11-23c). The effect is to add surface area, allowing the cell to expand without bursting in response to osmotic or other stress.

Membrane Curvature and Fusion Are Central to Many Biological Processes

Caveolins are not unique in their ability to induce curvature in membranes. **P2** Changes of curvature are central to one of the most remarkable features of biological membranes: their ability to undergo fusion with other membranes without losing their continuity. Within the eukaryotic endomembrane system, the membranous compartments constantly reorganize. Vesicles bud from the ER to carry newly synthesized lipids and proteins to other organelles and to the plasma membrane. Exocytosis, endocytosis, cell division, fusion of egg and sperm cells, and entry of a membrane-enveloped virus into a host cell all involve a membrane reorganization that requires the fusion of two membrane segments without loss of continuity (Fig. 11-24). Most of these processes begin with a local increase in membrane curvature.

Cardiolipin, present primarily in mitochondrial membranes of eukaryotes, but also a major component of the membrane lipids of bacteria, can create or recognize membrane curvature. It is cone-shaped—its head group

is small relative to its four fatty acyl chains—so it can act as a wedge in the monolayer, contracting that monolayer relative to the other.



Membrane curvature is the result. In *E. coli*, cardiolipin is highly localized at the two poles of the rod-shaped cell, where its curvature is sharpest. It seems very likely that other membrane lipids will be found to influence local curvature of the bilayer.

A protein that is intrinsically curved may force a bilayer to curve by binding to it (Fig. 11-25); the binding energy provides the driving force for the increase in bilayer curvature. Alternatively, multiple subunits of a scaffold protein may assemble into curved supra-molecular complexes and stabilize curves that spontaneously form in the bilayer. For example, a superfamily of proteins containing **BAR domains** (named for the first three members of the family that were identified: BIN1, amphiphysin, and RVS167) can assemble into a crescent-shaped scaffold that binds to the membrane surface, forcing or favoring membrane curvature. BAR domains consist of coiled coils that form long, thin, curved dimers with a positively charged concave surface that tends to form ionic interactions with the negatively charged head groups of membrane lipids PIP₂ and PIP₃. The enzymatic formation of these inositol lipids can tag a plasma membrane area for creation of inward curvature by a BAR protein (Fig. 11-25). Some of these BAR proteins also have an amphipathic helix (having one polar face and one hydrophobic face; see Fig. 11-32) that inserts like a wedge into one leaflet of the bilayer, expanding its area relative to the other leaflet and thereby forcing curvature. Such a protein might also serve as a detector of preexisting membrane curvature due to differences in lipid composition of two leaflets of a bilayer.

Septins make up a family of GTP-binding proteins (14 genes in humans) that polymerize at curved regions of the plasma membrane and participate in cellular processes such as cell division, exocytosis, phagocytosis, and apoptosis. All septins have an amphipathic helix that can submerge its hydrophobic side in one leaflet of the bilayer, forcing one leaflet of the bilayer to expand laterally, either causing or sensing local curvature in the membranes. Studies of cells with mutations in this helix

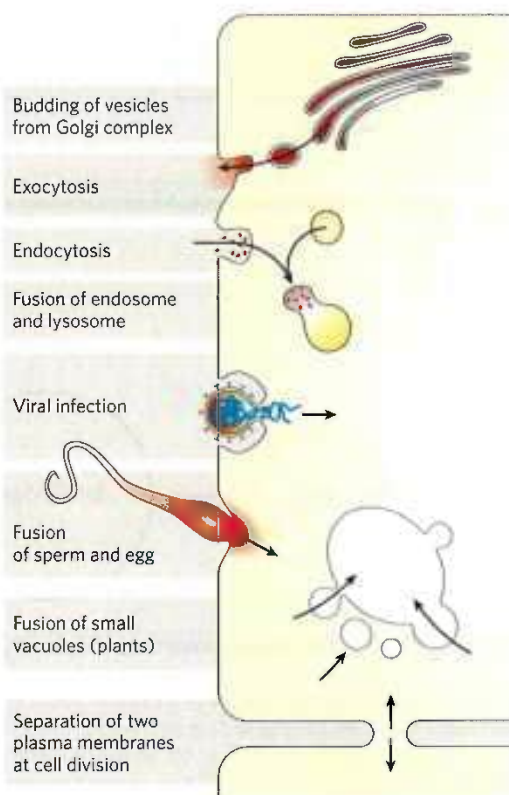


FIGURE 11-24 Membrane fusion. The fusion of two membranes is central to a variety of cellular processes involving organelles and the plasma membrane.

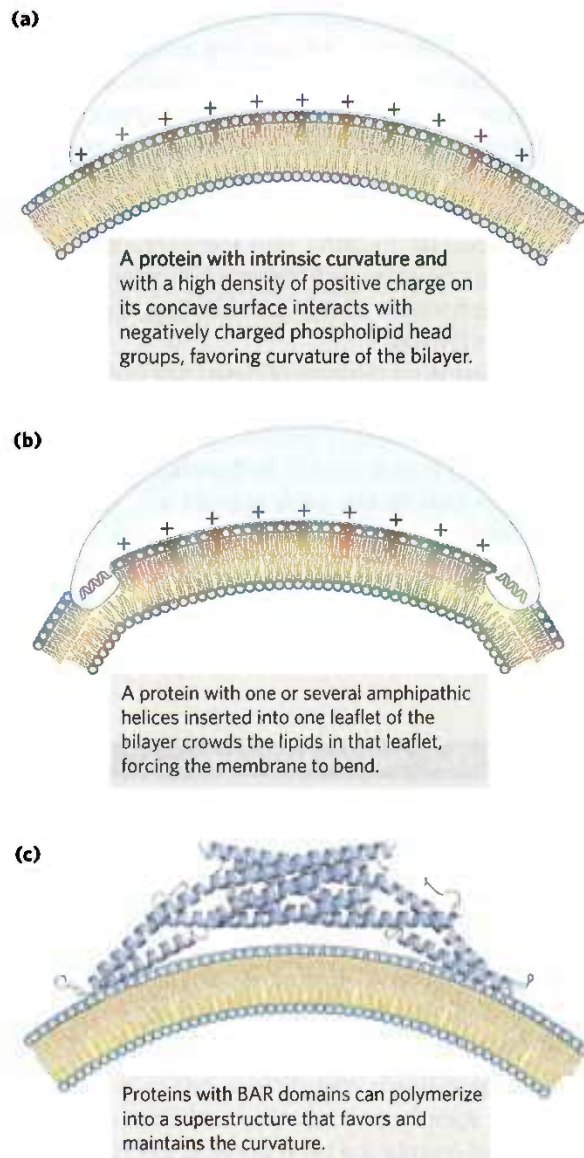


FIGURE 11-25 Three models for protein-induced curvature of membranes. [Information from (a, b) B. Qualmann et al., *EMBO J.* 30:3501, 2011, Fig. 1; (c) B. J. Peter et al., *Science* 303:495, 2004, Fig. 1A.]

show its biological importance in vesicle trafficking and neurotransmitter release.

Specific fusion of two membranes requires mediation by **fusion proteins**. Fusion proteins ensure (1) that the two membranes recognize each other; (2) that their surfaces become closely apposed, which requires removal of the water molecules normally associated with the polar head groups of lipids; (3) that their bilayer structures become locally disrupted, resulting in fusion of the outer leaflets of the two membranes (hemifusion); and (4) that their bilayers fuse to form a single continuous bilayer. The fusion occurring in receptor-mediated endocytosis, or regulated secretion, also requires (5) that the process is triggered at the appropriate time or in response to a specific signal. Fusion proteins mediate these events,

bringing about specific recognition and a transient local distortion of the bilayer structure that favors membrane fusion. (Note that these fusion proteins are unrelated to the products encoded by two fused genes, also called fusion proteins, discussed in Chapter 9.)

A well-studied example of membrane fusion occurs at synapses, when intracellular (neuronal) vesicles loaded with neurotransmitter fuse with the plasma membrane. Yeast cells provide another experimentally accessible system in which vesicles fuse with the plasma membrane, releasing their secretion products. Both processes involve a family of proteins called **SNAREs** (*snap* receptors; **Fig. 11-26**). SNAREs in the cytoplasmic face of the intracellular vesicle are called **v-SNAREs** (*v* for vesicle); those in the target membrane with which the vesicle fuses (the plasma membrane during exocytosis) are **t-SNAREs** (*t* for target). The protein NSF (*N*-ethylmaleimide sensitive factor) regulates the interactions among SNAREs. During fusion, a v-SNARE and a t-SNARE bind to each other and undergo a structural change that produces a bundle of long, thin rods made up of helices from both SNAREs and two helices from the protein SNAP25 (**Fig. 11-26**). The two SNAREs initially interact at their ends, then zip up into the bundle of helices. This structural change pulls the two membranes into contact and initiates the fusion of their lipid bilayers. An alternative designation of SNARE types is based on structural features of the proteins: R-SNAREs have an Arg residue critical to their function, and Q-SNAREs have a critical Gln residue. Typically, R-SNAREs act as v-SNAREs, and Q-SNAREs act as t-SNAREs.



The complex of SNAREs and SNAP25 is the target of several powerful neurotoxins. *Clostridium botulinum* toxin is a bacterial protease that cleaves specific bonds in SNARE proteins, preventing neurotransmission and causing paralysis and death. Because of its very high specificity for these proteins, purified botulinum toxin has served as a powerful tool for dissecting the mechanism of neurotransmitter release in vivo and in vitro. Used in small amounts, botulinum toxin (Botox) is used in medicine to treat disorders of eye and neck muscles, as well as cosmetically for the removal of skin wrinkles. Tetanus toxin, produced by the bacterium *Clostridium tetani*, also is a protease with high specificity for SNARE proteins. It causes painful muscle spasms and rigidity of voluntary muscles—hence the characteristic symptom “lockjaw.” ■

Integral Proteins of the Plasma Membrane Are Involved in Surface Adhesion, Signaling, and Other Cellular Processes

Several families of integral proteins in the plasma membrane provide specific points of attachment between cells or between a cell and proteins of the extracellular matrix. **Integrins** are surface adhesion proteins that mediate a cell's interaction with the extracellular matrix and with other cells, including some pathogens.

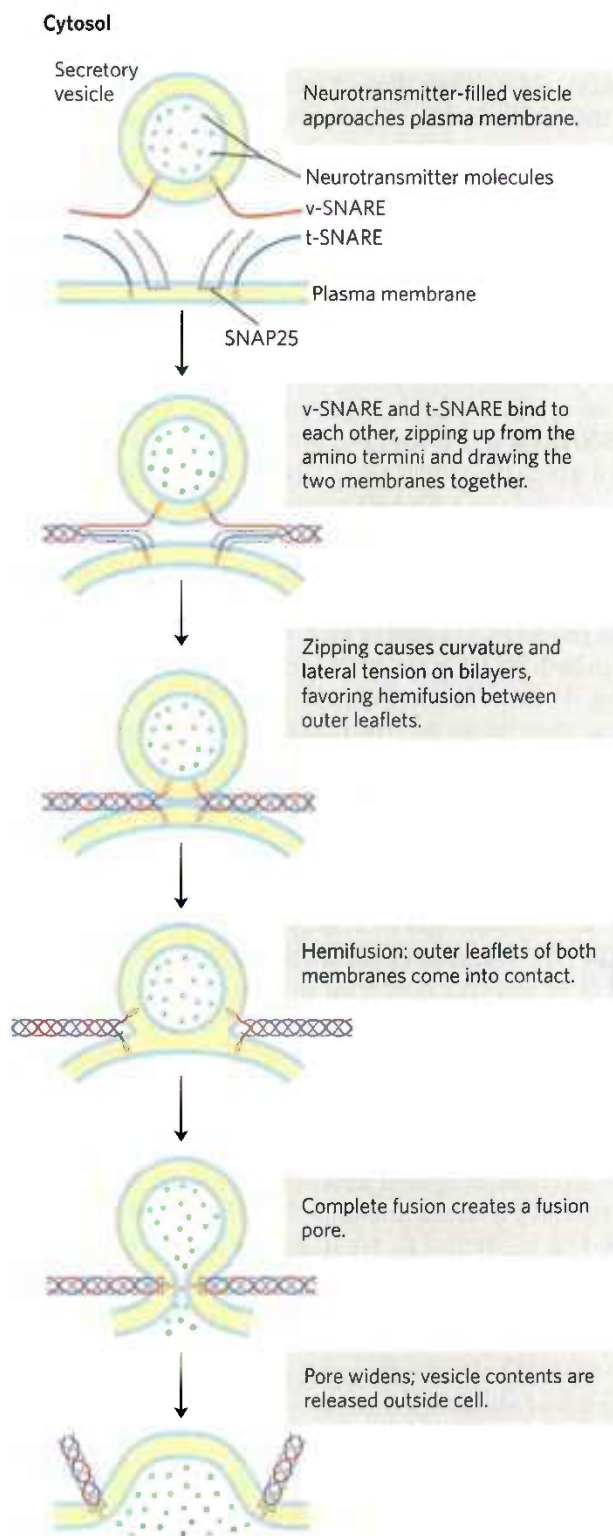


FIGURE 11-26 Membrane fusion during neurotransmitter release at a synapse. The secretory vesicle membrane contains the v-SNARE synaptobrevin (red). The target (plasma) membrane contains the t-SNAREs syntaxin (blue) and SNAP25 (violet). When a local increase in $[Ca^{2+}]$ signals release of neurotransmitter, the v-SNARE, SNAP25, and t-SNARE interact, forming a coiled bundle of four α helices, pulling the two membranes together and disrupting the bilayer locally. This leads first to hemifusion, joining the outer leaflets of the two membranes, then to complete membrane fusion and neurotransmitter release. When fusion is complete, the SNARE complex is disassembled. [Information from Y. A. Chen and R. H. Scheller, *Nat. Rev. Mol. Cell Biol.* 2:98, 2001.]

Integrins also carry signals in both directions across the plasma membrane, integrating information about the extracellular and intracellular environments. All integrins are heterodimeric proteins composed of two unlike subunits, α and β , each anchored to the plasma membrane by a single transmembrane helix. The large extracellular domains of the α and β subunits combine to form a specific binding site for extracellular proteins such as collagen and fibronectin, which contain a common determinant of integrin binding, the sequence Arg-Gly-Asp (RGD).

Other plasma membrane proteins involved in surface adhesion are the **cadherins**, which undergo homophilic ("with the same kind") interactions with identical cadherins in an adjacent cell. **Selectins** have extracellular domains that, in the presence of Ca^{2+} , bind specific polysaccharides on the surface of an adjacent cell. Selectins are present primarily in the various types of blood cells and in the endothelial cells that line blood vessels (see Fig. 7-29). They are an essential part of the blood-clotting process.

SUMMARY 11.2 Membrane Dynamics

- Lipids in a biological membrane can exist in liquid-ordered or liquid-disordered states. Membrane fluidity is affected by temperature, fatty acid composition, and sterol content.
- Flip-flop diffusion of lipids between the inner and outer leaflets of a membrane occurs only when the diffusion is specifically catalyzed by flippases, floppases, scramblases, or PI transporters.
- Proteins and lipids can diffuse laterally within the plane of the membrane, but this mobility is limited by interactions of membrane proteins with internal cytoskeletal structures and interactions of lipids with lipid rafts.
- One class of lipid rafts is enriched for sphingolipids and cholesterol with a subset of membrane proteins that are GPI-linked or attached to several long-chain fatty acyl moieties. Caveolins are integral proteins that associate with the inner leaflet of the plasma membrane, forcing it to curve inward to form caveolae, which are involved in membrane transport, signaling, and the expansion of plasma membranes.
- Proteins containing BAR domains cause local membrane curvature and mediate the fusion of two membranes, which accompanies processes such as endocytosis, exocytosis, and viral invasion. Septins are proteins that sense or cause membrane curvature. Membrane fusion depends on SNARE proteins, which draw two membranes close together and favor fusion.
- Integrins, cadherins, and selectins are transmembrane proteins of the plasma membrane that act both to attach cells to each other and to carry messages between the extracellular matrix and the cytoplasm.

11.3 Solute Transport across Membranes

Every living cell must acquire from its surroundings the raw materials for biosynthesis and for energy production, and must release the byproducts of metabolism to its environment; both processes require that small compounds or inorganic ions cross the plasma membrane. Within the eukaryotic cell, different compartments have different concentrations of ions and of metabolic intermediates and products, and these, too, must move across intracellular membranes in tightly regulated processes.

P3 A few nonpolar compounds can dissolve in the lipid bilayer and cross a membrane unassisted, but for any polar compound or ion, a specific membrane protein carrier is essential. Approximately 2,000 genes in the human genome encode proteins that function in transporting solutes across membranes. In some cases, a membrane protein simply facilitates the diffusion of a solute down its concentration gradient; but transport can also occur against a gradient of concentration, electrical potential, or both, and in these cases, as we shall see, the transport process requires energy. Ions may also diffuse across membranes via ion channels formed by proteins, or they may be carried across by ionophores, small molecules that mask the charge of ions and allow them to diffuse through the lipid bilayer. **Figure 11-27** summarizes the various types of transport mechanisms discussed in this section.

Transport May Be Passive or Active

When two aqueous compartments containing unequal concentrations of a soluble compound or ion are separated by a permeable divider (membrane), the solute moves by **simple diffusion** from the region of higher concentration, through the membrane, to the region of lower concentration, until the two compartments have equal solute concentrations (**Fig. 11-28a**). When ions of opposite charge are separated by a permeable membrane, there is a transmembrane electrical gradient, a **membrane potential**, V_m (expressed in millivolts). This membrane potential produces a force opposing ion movements that increase V_m and driving ion movements that reduce V_m (**Fig. 11-28b**). **P3** Thus, the direction

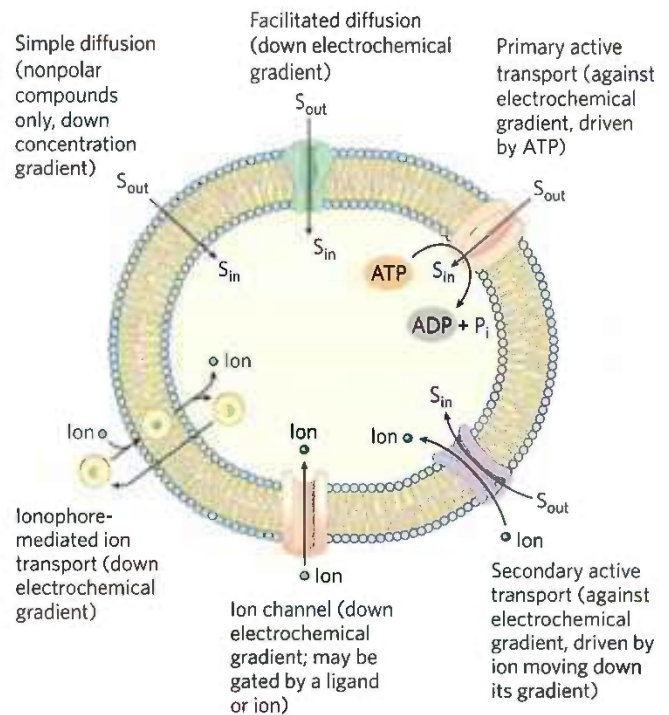
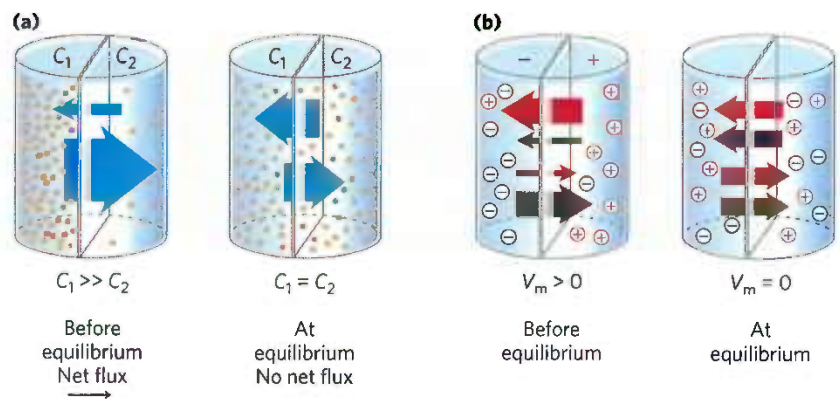


FIGURE 11-27 Summary of transporter types. Some types (ionophores, ion channels, and passive transporters) simply speed transmembrane movement of solutes (S) down their electrochemical gradients, whereas others (active transporters) can pump solutes against a gradient, using ATP or a gradient of a second solute to provide the energy.

in which a charged solute tends to move spontaneously across a membrane depends on both the chemical gradient (the difference in solute concentration) and the electrical gradient (V_m) across the membrane. Together these two factors are referred to as the **electrochemical gradient** or **electrochemical potential**. This behavior of solutes is in accord with the second law of thermodynamics: molecules tend to spontaneously assume the distribution of greatest randomness and lowest energy.

Membrane proteins that act by increasing the rate of solute movement across membranes are called transporters or carriers. **P3** Transporters are of two general types: passive and active. **Passive transporters** simply facilitate movement down a concentration gradient, increasing the transport rate. This process is called **passive transport** or

FIGURE 11-28 Movement of solutes across a permeable membrane. (a) Net movement of an electrically neutral solute is toward the side of lower solute concentration until equilibrium is achieved. The solute concentrations on the left and right sides of the membrane, as shown here, are designated C_1 and C_2 . The rate of transmembrane solute movement (indicated by the arrows) is proportional to the concentration ratio. (b) Net movement of an electrically charged solute is dictated by a combination of the electrical potential (V_m) and the ratio of chemical concentrations (C_2/C_1) across the membrane; net ion movement continues until this electrochemical potential reaches zero.



facilitated diffusion. Active transporters (sometimes called pumps) can move substrates across a membrane against a concentration gradient or an electrical potential, a process called **active transport**. **Primary active transporters** use energy provided directly by a chemical reaction; **secondary active transporters** couple uphill transport of one substrate with downhill transport of another.

Transporters and Ion Channels Share Some Structural Properties but Have Different Mechanisms

To pass through a lipid bilayer, a polar or charged solute must first give up its interactions with the water molecules in its hydration shell, then diffuse about 3 nm (30 Å) through a substance (lipid) in which it is poorly soluble (Fig. 11-29a). The energy used to strip away the hydration shell and to move the polar compound from water into lipid, then through the lipid bilayer, is regained as the compound leaves the membrane on the other side and is rehydrated. However, the intermediate stage of transmembrane passage is a high-energy state comparable to the transition state in an enzyme-catalyzed chemical

reaction. In both cases, an activation barrier must be overcome to reach the intermediate stage (Fig. 11-29; compare with Fig. 6-3). The energy of activation ($\Delta G^\ddagger$) for translocation of a polar solute across the bilayer is so large that pure lipid bilayers are virtually impermeable to polar and charged species on time scales relevant to cell growth and division.

P3 Membrane proteins lower the activation energy for transport of polar compounds and ions by providing an alternative path across the membrane for specific solutes. Lowering the activation energy greatly increases the rate of transmembrane movement (recall Eqn 6-6, p. 182). Transporters are not enzymes in the usual sense; their “substrates” are moved from one compartment to another but are not chemically altered. Like enzymes, however, transporters bind their substrates with stereochemical specificity through multiple weak, noncovalent interactions. The negative free-energy change associated with these weak interactions, $\Delta G_{\text{binding}}$, counterbalances the positive free-energy change that accompanies loss of the water of hydration from the substrate, $\Delta G_{\text{dehydration}}$, thereby lowering $\Delta G^\ddagger$ for transmembrane passage (Fig. 11-29b). Transporter proteins span the lipid bilayer several times, forming a transmembrane pathway lined with hydrophilic amino acid side chains. The pathway provides an alternative route for a specific substrate to move across the lipid bilayer without its having to dissolve in the bilayer, further lowering $\Delta G^\ddagger$ for transmembrane diffusion. The result is an orders-of-magnitude increase in the substrate’s rate of passage across the membrane.

Ion channels use a different mechanism than transporters to move inorganic ions across membranes.

P3 Ion channels speed the passage of ions across membranes by providing an aqueous path across the membrane through which inorganic ions can diffuse at very high rates. Most ion channels have a “gate” (Fig. 11-30a) regulated by a biological signal. When the gate is open, ions move across the membrane, through the channel, in the direction dictated by the ion’s charge and the electrochemical gradient. Movement occurs at rates approaching the limit of unhindered diffusion (tens of millions of ions per second per channel—much higher than typical transporter rates). Ion channels typically show some specificity for an ion, but they are not saturable with their ion substrate. Flow through a channel stops either when the gating mechanism is closed (again, by a biological signal) or when there is no longer an electrochemical gradient providing the driving force for the movement. In contrast, transporters, which bind their “substrates” with high stereospecificity, catalyze transport at rates well below the limits of free diffusion, and they are saturable in the same sense as are enzymes: there is some substrate concentration above which further increases will not produce a greater rate of transport. Transporters have a gate on either side of the membrane, and the two gates are never open at the same time (Fig. 11-30b).

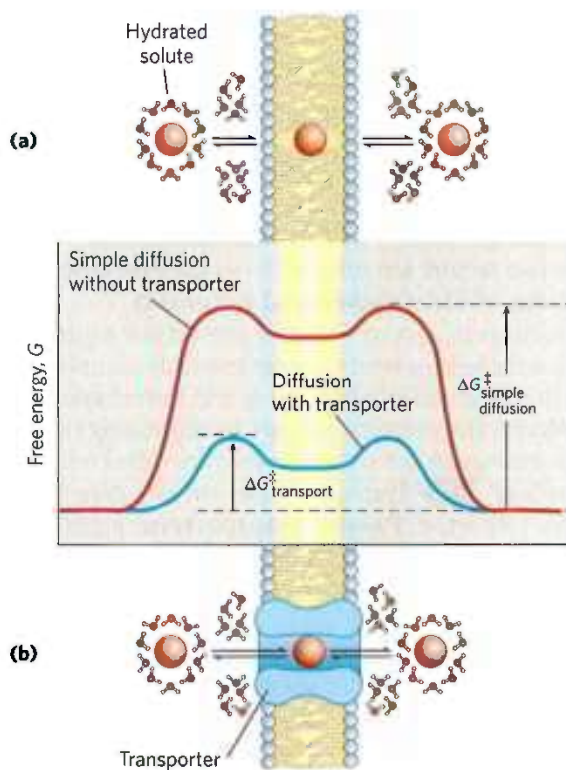
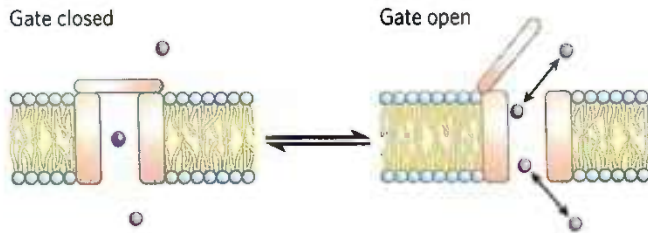
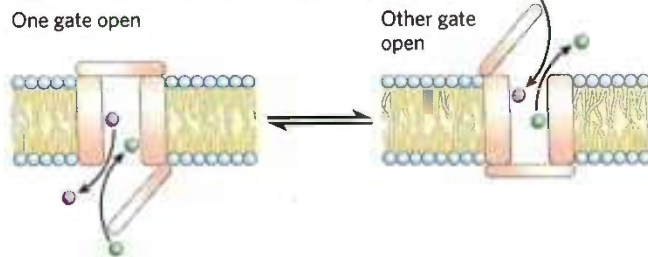


FIGURE 11-29 Energy changes accompanying passage of a hydrophilic solute through the lipid bilayer of a biological membrane. (a) In simple diffusion, removal of the hydration shell is highly endergonic, and the energy of activation ($\Delta G^\ddagger$) for diffusion through the bilayer is very high. (b) A transporter protein reduces the $\Delta G^\ddagger$ for transmembrane diffusion of the solute. It does this by forming noncovalent interactions with the dehydrated solute to replace the hydrogen bonding with water and by providing a hydrophilic transmembrane pathway.

(a) Ion channel: single gate**(b) Transporter (pump): alternating gates****FIGURE 11-30 Differences between channels and transporters.**

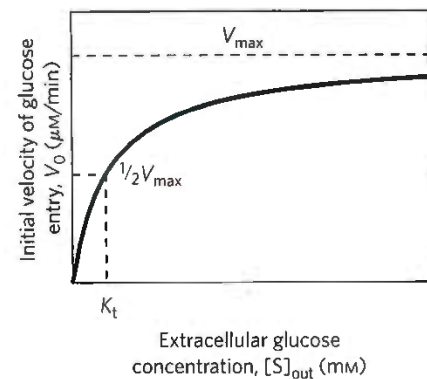
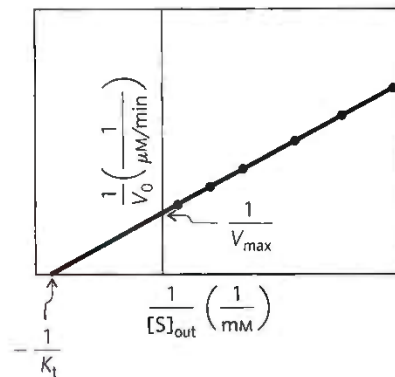
(a) Ion channels have a transmembrane pore that is either open or closed, depending on the position of the single gate. When the gate is open, ions move through at a rate limited only by the maximum rate of diffusion. (b) Transporters have two gates, and both are never open at the same time. Movement of a substrate (an ion or a small molecule) through the membrane is therefore limited by the time needed for one gate to open and close (on one side of the membrane) and the second gate to open. Rates of movement through ion channels can be orders of magnitude greater than rates through transporters, but channels simply allow the ion to flow down the electrochemical gradient, whereas active transporters (pumps) can move a substrate against its concentration gradient. [Information from D. C. Gadsby, *Nat. Rev. Mol. Cell Biol.* 10:344, 2009, Fig. 1]

Both transporters and ion channels constitute large families of proteins, defined not only by their primary sequences but also by their secondary structures. We next consider some well-studied representatives of the main transporter and channel families. You will also encounter some of these in Chapter 12 when we discuss transmembrane signaling, and some in later chapters in the context of the metabolic pathways in which they participate.

The Glucose Transporter of Erythrocytes Mediates Passive Transport

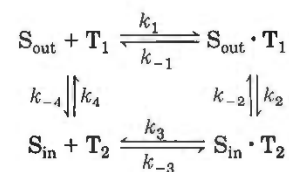
Energy-yielding metabolism in erythrocytes depends on a constant supply of glucose from the blood plasma, where the glucose concentration is maintained at about 4.5 to 5 mM. Glucose enters the erythrocyte by passive transport via a specific glucose transporter called GLUT1, at a rate about 50,000 times greater than it could cross the membrane unassisted.

P3 The process of glucose transport can be described by analogy with an enzymatic reaction in which the “substrate” is glucose outside the cell (S_{out}), the “product” is glucose inside the cell (S_{in}), and the “enzyme” is the transporter, T. When the initial rate of glucose uptake is measured as a function of external glucose concentration (Fig. 11-31), the resulting plot is hyperbolic: at high external

**(a)****(b)****FIGURE 11-31 Kinetics of glucose transport into erythrocytes.**

(a) The initial rate of glucose entry into an erythrocyte, V_0 , depends on the initial concentration of glucose on the outside, $[S]_{\text{out}}$. (b) Double-reciprocal plot of the data in (a). The kinetics of passive transport is analogous to the kinetics of an enzyme-catalyzed reaction. (Compare these plots with Figs 6-12 and 6-14.) K_t is analogous to K_m , the Michaelis constant.

glucose concentrations, the rate of uptake approaches V_{max} . Formally, such a transport process can be described by the set of equations



in which k_1 , k_{-1} , and so forth are the forward and reverse rate constants for each step; T_1 is the transporter conformation in which the glucose-binding site faces outward (in contact with blood plasma), and T_2 is the conformation in which it faces inward. Given that every step in this sequence is reversible, the transporter is, in principle, equally able to move glucose into or out of the cell. As for enzyme assays, this transporter assay measures the initial rate of uptake, when the product concentration (glucose concentration inside the cell) is zero, while the substrate concentration (glucose on the outside) is varied. In the living cell, GLUT1 accelerates the movement of glucose down its concentration gradient, which

normally means *into* the cell. Glucose that enters a cell is generally metabolized immediately, and the intracellular glucose concentration is thereby kept low relative to its concentration in the blood.

The rate equations for glucose transport can be derived exactly as for enzyme-catalyzed reactions (Chapter 6), yielding an expression analogous to the Michaelis-Menten equation

$$V_0 = \frac{V_{\max}[S]_{\text{out}}}{K_t + [S]_{\text{out}}} \quad (11-1)$$

in which V_0 is the initial velocity of accumulation of glucose inside the cell when its concentration in the surrounding medium is $[S]_{\text{out}}$, and K_t ($K_{\text{transport}}$) is a constant analogous to the Michaelis constant, a combination of rate constants that is characteristic of each transport system. This equation describes the *initial* velocity, the rate observed when $[S]_{\text{in}} = 0$. As is the case for enzyme-catalyzed reactions, the slope-intercept form of the equation describes a linear plot of $1/V_0$ against $1/[S]_{\text{out}}$, from which we can obtain values of K_t and $V_{\max}$ (Fig. 11-31b). When $[S]_{\text{out}} = K_t$, the rate of uptake is $\frac{1}{2}V_{\max}$; the transport process is half-saturated. The concentration of glucose in blood, as noted above, is 4.5 to 5 mM, which is close enough to the K_t to ensure that GLUT1 is half-saturated with substrate and operates near one-half $V_{\max}$.

Because no chemical bonds are made or broken in the conversion of S_{out} to S_{in} , neither “substrate” nor “product” is intrinsically more stable, and the process of entry is therefore fully reversible. As $[S]_{\text{in}}$ approaches $[S]_{\text{out}}$, the rates of entry and exit become equal. Such a system is therefore incapable of accumulating glucose within a cell

at concentrations above that in the surrounding medium; it simply equilibrates glucose on the two sides of the membrane much faster than would occur in the absence of a specific transporter. GLUT1 is specific for D-glucose, with a K_t of about 6 mM. For the close analogs D-mannose and D-galactose, which differ only in the position of one hydroxyl group, the values of K_t are 20 mM and 30 mM, respectively, and for L-glucose, K_t exceeds 3,000 mM. Thus, GLUT1 shows the three hallmarks of passive transport: high rates of diffusion down a concentration gradient, saturability, and stereospecificity.

GLUT1 is an integral protein with 12 hydrophobic segments, each forming a membrane-spanning helix (Fig. 11-32a). The helices that line the transmembrane path for glucose are **amphipathic**; for each helix, the residues along one side are predominantly nonpolar, and those on the other side are mainly polar. This amphipathic structure is evident in a helical wheel diagram (Fig. 11-32b). A cluster of amphipathic helices are arranged so that their polar sides face each other and line a hydrophilic pore through which glucose can pass (Fig. 11-32c), while their hydrophobic sides interact with the surrounding membrane lipids such that the hydrophobic effect stabilizes the entire transporter structure.

Structural studies of mammalian GLUT1 and other GLUT transporters suggest that the protein cycles through a series of conformational changes, interconverting a form (T_1) with its glucose-binding site accessible only from the extracellular side, through a form in which the bound glucose is sequestered and inaccessible from either side, to a form (T_2) with the glucose-binding site open only to the intracellular side (Fig. 11-33).

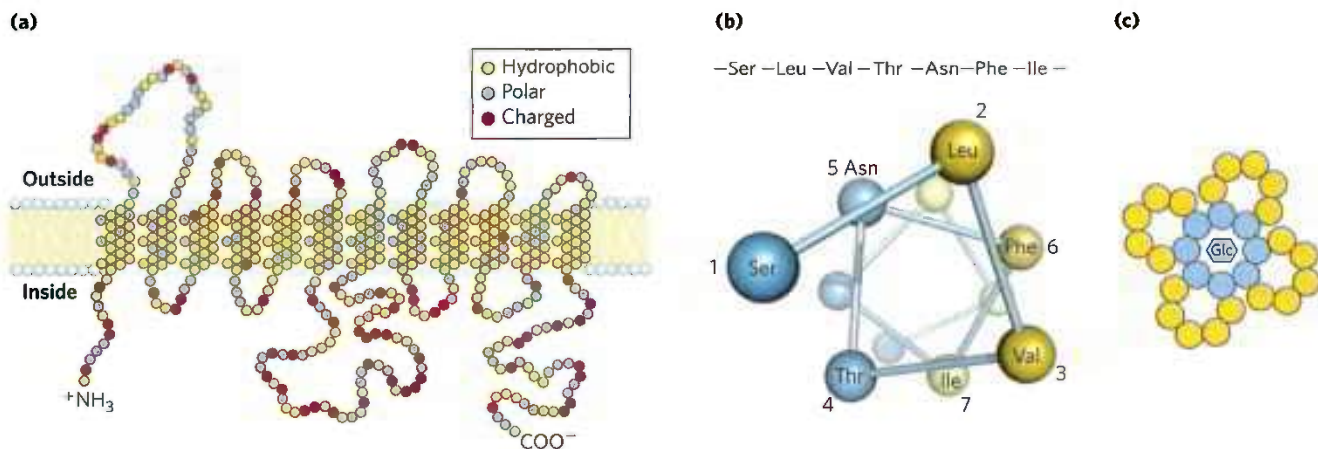


FIGURE 11-32 Membrane topology of the glucose transporter GLUT1. (a) Transmembrane helices are represented here as oblique (angled) rows of three or four amino acid residues, each row depicting one turn of the α helix. Of the 12 helices, 9 contain three or more polar or charged residues (blue or red), often separated by several hydrophobic residues (yellow). (b) A helical wheel diagram shows the distribution of polar and nonpolar residues on the surface of a helical segment. The helix is diagrammed as though observed along its axis from the amino terminus. Adjacent residues in the linear sequence are

connected, and each residue is placed around the wheel in the position it occupies in the helix; recall that 3.6 residues are required to make one complete turn of the α helix. In this example, the polar residues (blue) are on one side of the helix, and the hydrophobic residues (yellow) are on the other. This is, by definition, an amphipathic helix. (c) Side-by-side association of amphipathic helices, each with its polar face oriented toward the central cavity, produces a transmembrane channel lined with polar (and charged) residues, available for interaction with glucose. [Information from (a, c) M. Mueckler, *Eur. J. Biochem.* 219:713, 1994.]

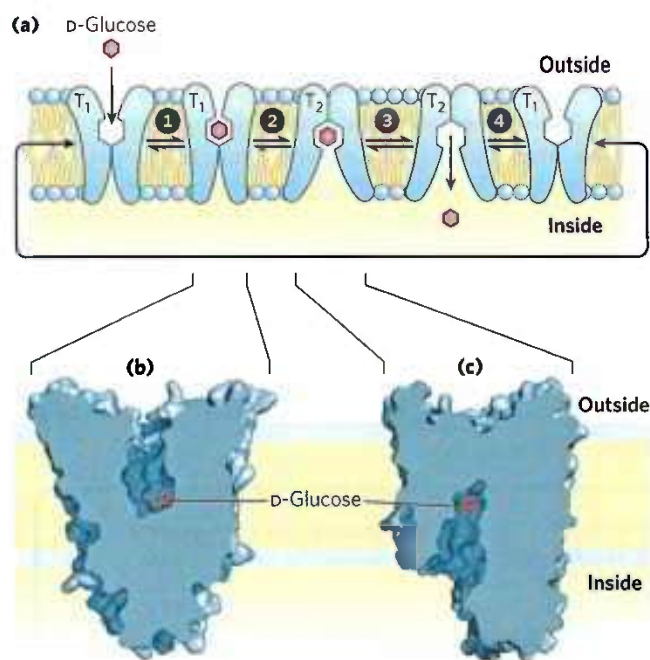


FIGURE 11-33 Model of glucose transport into erythrocytes by GLUT1. (a) The transporter exists in two extreme conformations: T_1 , with the glucose-binding site exposed on the outer surface of the plasma membrane, and T_2 , with the binding site exposed on the inner surface. Glucose transport occurs in four steps. ① Glucose in blood plasma binds to a stereospecific site on T_1 ; this lowers the activation energy for ② a conformational change from glucose_{out} • T_1 to glucose_{in} • T_2 , effecting transmembrane passage of the glucose. ③ Glucose is released from T_2 into the cytoplasm, and ④ the transporter returns to the T_1 conformation, ready to transport another glucose molecule. Between the forms T_1 and T_2 , there is an intermediate form (not shown here) in which glucose is sequestered within the transporter, with access to neither side. The structures of (b) human GLUT3 in the T_1 conformation and (c) human GLUT1 in the T_2 conformation, determined by x-ray crystallography, support the model shown in (a). [Data from (b) PDB ID 4ZWC, D. Deng et al., *Nature* 526:391, 2015; (c) PDB ID 4PYP, D. Deng et al., *Nature* 510:121, 2014.]



Twelve passive glucose transporters are encoded in the human genome, each with its unique kinetic properties, patterns of tissue distribution, and function (Table 11-1). GLUT1, in addition to supplying glucose to erythrocytes, also transports glucose across the blood-brain barrier, supplying the glucose that is essential for normal brain metabolism. The very rare individuals with defects in GLUT1 have a variety of brain-related symptoms, including seizures, movement and language disorders, and developmental delays. Standard care for such individuals includes a ketogenic diet, which provides the ketones that can serve as an alternative energy source for the brain. In the liver, GLUT2 transports glucose out of hepatocytes when liver glycogen is broken down to replenish blood glucose. GLUT2 has a large K_t (≥ 17 mM) and can therefore respond to increased levels of intracellular glucose (produced by glycogen breakdown) by increasing outward transport. Skeletal and heart muscle and adipose tissue have yet another glucose transporter, GLUT4 ($K_t = 5$ mM), which is distinguished by its response to insulin: its activity increases when insulin signals a high blood glucose concentration, thus increasing the rate of glucose uptake into muscle and adipose tissue. Box 11-1 describes the effect of insulin on this transporter. ■

The Chloride-Bicarbonate Exchanger Catalyzes Electroneutral Cotransport of Anions across the Plasma Membrane

The erythrocyte contains another passive transport system, an anion exchanger that is essential in CO_2 transport to the lungs from tissues such as skeletal muscle and liver. Waste CO_2 released from respiring tissues

TABLE 11-1 Glucose Transporters in Humans

Transporter	Tissue(s) where expressed	K_t (mM)	Role/characteristics ^a
GLUT1	Erythrocytes, blood-brain barrier, placenta, most tissues at a low level	3	Basal glucose uptake; defective in De Vivo disease
GLUT2	Liver, pancreatic islets, intestine, kidney	17	In liver and kidney, removal of excess glucose from blood; in pancreas, regulation of insulin release
GLUT3	Brain (neuron), testis (sperm)	1.4	Basal glucose uptake; high turnover number
GLUT4	Muscle, fat, heart	5	Activity increased by insulin
GLUT5	Intestine (primarily), testis, kidney	6 ^b	Primarily fructose transport
GLUT6	Spleen, leukocytes, brain	>5	Possibly no transporter function
GLUT7	Small intestine, colon, testis, prostate	0.3	—
GLUT8	Testis, sperm acrosome	~2	—
GLUT9	Liver, kidney, intestine, lung, placenta	0.6	Urate and glucose transporter in liver, kidney
GLUT10	Heart, lung, brain, liver, muscle, pancreas, placenta, kidney	0.3 ^c	Glucose and galactose transporter
GLUT11	Heart, skeletal muscle	0.16	Glucose and fructose transporter
GLUT12	Skeletal muscle, heart, prostate, placenta	—	—

Information on localization from M. Mueckler and B. Thorens, *Mol. Aspects Med.* 34:121, 2013. K_t values for glucose from R. Augustin, *IUBMB Life* 62:315, 2010.

^aDash indicates role uncertain.

^b K_m for fructose.

^c K_m for 2-deoxyglucose.

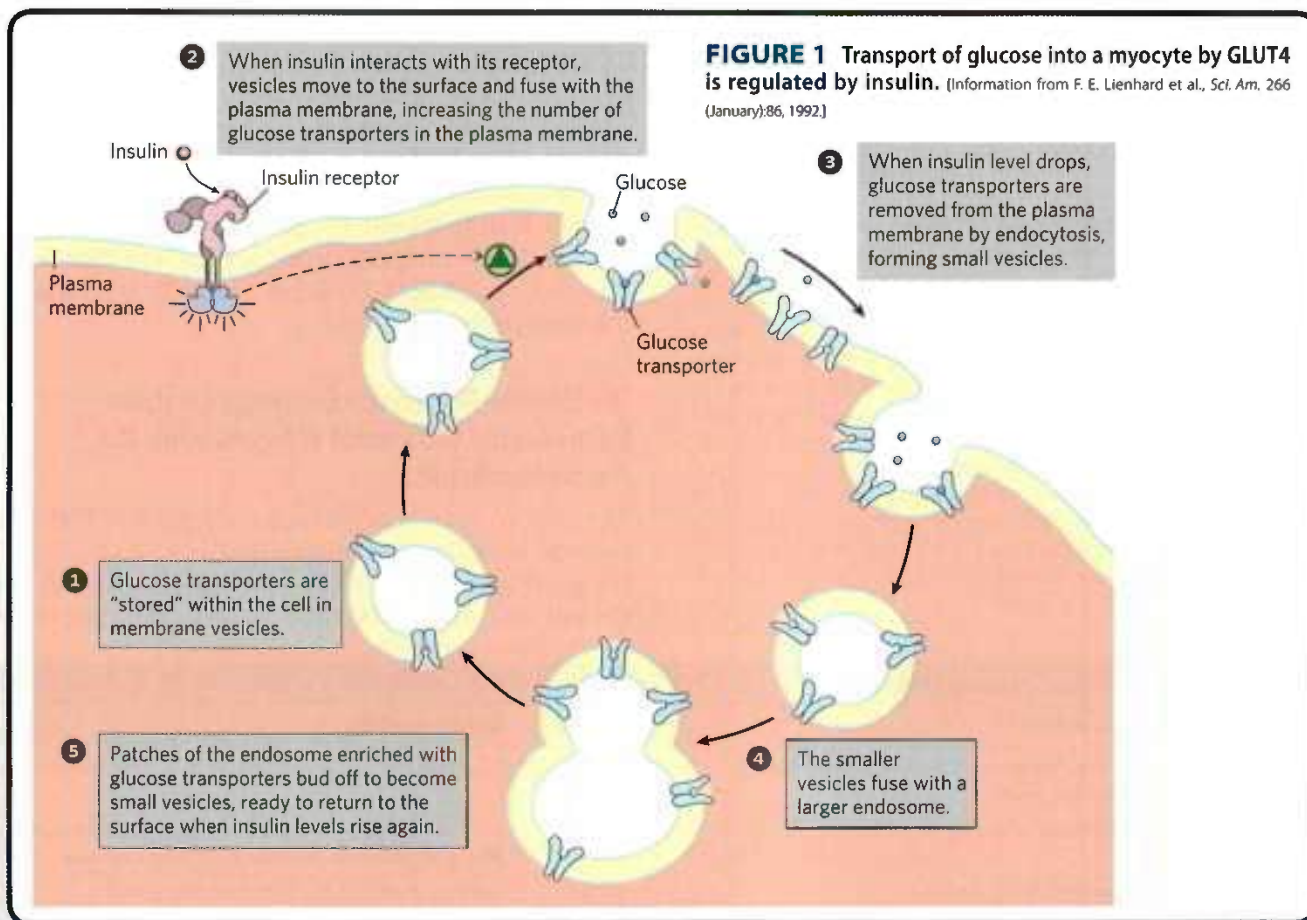


Defective Glucose Transport in Diabetes

When ingestion of a carbohydrate-rich meal causes blood glucose to exceed the usual concentration between meals (about 5 mM), excess glucose is taken up by the myocytes of cardiac and skeletal muscle (which store it as glycogen) and by adipocytes (which convert it to triacylglycerols). Glucose uptake into myocytes and adipocytes is mediated by the glucose transporter GLUT4. Between meals, some GLUT4 is present in the plasma membrane, but most (90%) is sequestered in the membranes of small intracellular vesicles (Fig. 1). Insulin is released from the pancreas in response to high blood glucose, and it triggers, within minutes, the movement of these vesicles to the plasma membrane. The vesicles fuse with the membrane,

bringing most of the GLUT4 molecules to the cell surface (see Fig. 12-23) and increasing the rate of glucose uptake 15-fold or more. When blood glucose levels return to normal, insulin release slows and most GLUT4 molecules are removed from the plasma membrane by endocytosis and stored in vesicles.

In type 1 (insulin-dependent) diabetes mellitus, the inability to release insulin (and thus to mobilize glucose transporters) results in low rates of glucose uptake into muscle and adipose tissue. One consequence is a prolonged period of high blood glucose after a carbohydrate-rich meal. This condition is the basis for the glucose tolerance test used to diagnose diabetes (Chapter 23).



into the blood plasma enters the erythrocyte, where it is converted to bicarbonate HCO_3^- by the enzyme carbonic anhydrase. (Recall that HCO_3^- is the primary buffer of blood pH; see Fig. 2-20.) The HCO_3^- reenters the blood plasma for transport to the lungs (Fig. 11-34). Because HCO_3^- is much more soluble in blood plasma than is CO_2 , this roundabout route increases the capacity of the blood to carry carbon dioxide from the tissues to the lungs. In the lungs, HCO_3^- reenters the erythrocyte and is converted to CO_2 , which is eventually released into the lung space and exhaled. To be effective, this shuttle requires very

rapid movement of HCO_3^- across the erythrocyte membrane. As described in Chapter 5 (pp. 160–161), there is a second mechanism for moving CO_2 from tissue to lung, involving reversible binding of CO_2 to hemoglobin.)

The chloride-bicarbonate exchanger, also called the anion exchange (AE) protein, increases the rate of HCO_3^- transport across the erythrocyte membrane more than a millionfold. Like the glucose transporter, it is a dimeric integral protein that spans the membrane 14 times. This protein mediates the simultaneous movement of two anions: for every HCO_3^- ion that moves in one direction,

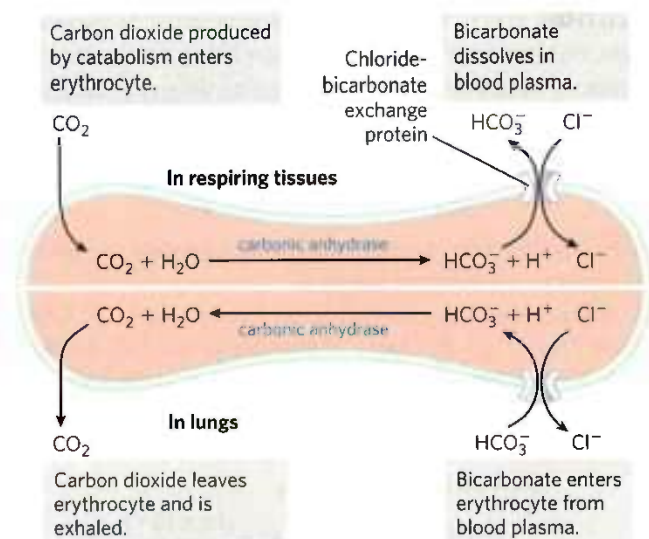


FIGURE 11-34 Chloride-bicarbonate exchanger of the erythrocyte membrane. This cotransport system allows the entry and exit of HCO_3^- without changing the membrane potential. Its role is to increase the CO_2 -carrying capacity of the blood. The top half of the figure illustrates the events that take place in respiring tissues; the bottom half illustrates the events in the lungs.

one Cl^- ion moves in the opposite direction, with no net transfer of charge: the exchange is **electroneutral**. The coupling of Cl^- and HCO_3^- movements is obligatory; in the absence of chloride, bicarbonate transport stops. In this respect, the anion exchanger is typical of those systems, called **cotransport** systems, that simultaneously carry two solutes across a membrane (Fig. 11-35). When, as in this case, the two substrates move in opposite directions, the process is **antiport**. In **symport**, two substrates are moved simultaneously in the same direction. Transporters that carry only one substrate, such as the erythrocyte glucose transporter, are known as **uniport** systems.

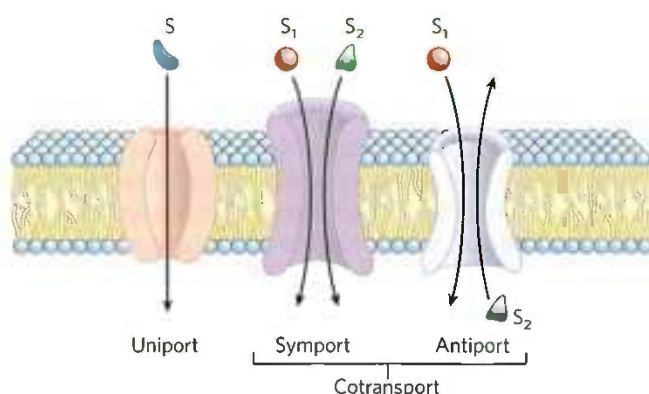


FIGURE 11-35 Three general classes of transport systems. Transporters differ in the number of solutes (substrates) transported and the direction in which each solute moves. Examples of all three types of transporter are discussed in the text. Note that this classification tells us nothing about whether these are energy-requiring (active transport) or energy-independent (passive transport) processes.

Active Transport Results in Solute Movement against a Concentration or Electrochemical Gradient

P3 In passive transport, the transported species always moves down its electrochemical gradient and is not accumulated above the equilibrium concentration. Active transport, by contrast, results in the accumulation of a solute above the equilibrium point. Active transport is essential when cells function in an environment in which key substrates are present outside the cell only at very low concentrations. For example, the bacterium *E. coli* can grow in a medium containing only $1 \mu\text{M}$ P_i (inorganic phosphate), but the cell must maintain internal P_i levels in the millimolar range. Worked Example 11-2 describes another such situation, which requires cells to pump Ca^{2+} outward across the plasma membrane. Active transport is thermodynamically unfavorable (endergonic) and takes place only when coupled, directly or indirectly, to an exergonic process such as the absorption of sunlight, an oxidation reaction, the breakdown of ATP, or the concomitant flow of some other chemical species down its electrochemical gradient. In **primary active transport**, solute accumulation is coupled directly to an exergonic chemical reaction, such as conversion of ATP to $\text{ADP} + \text{P}_i$ (Fig. 11-36). **Secondary active transport** occurs when endergonic (uphill) transport of one solute is coupled to the exergonic (downhill) flow of a different solute that was originally pumped uphill by primary active transport.

The amount of energy needed for the transport of a solute against a gradient can be calculated from the initial concentration gradient. The general equation for the free-energy change in the chemical process that converts substrate (S) to product (P) is

$$\Delta G = \Delta G^\circ + RT \ln ([\text{P}]/[\text{S}]) \quad (11-2)$$

where ΔG° is the standard free-energy change, R is the gas constant ($8.315 \text{ J/mol} \cdot \text{K}$), and T is the absolute

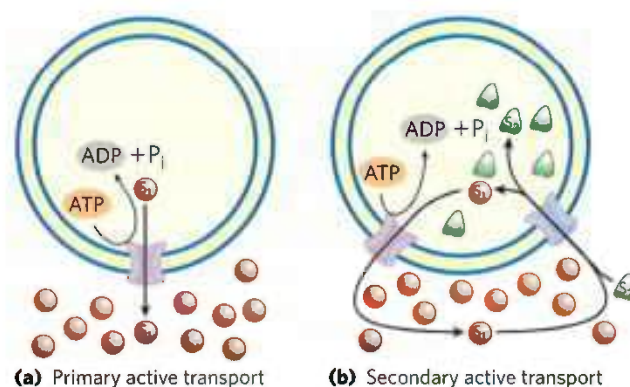


FIGURE 11-36 Two types of active transport. (a) In primary active transport, the energy released by ATP hydrolysis drives solute (S_1) movement against an electrochemical gradient. (b) In secondary active transport, a gradient of an ion (designated S_1 ; often Na^+) has been established by primary active transport. Movement of S_1 down its electrochemical gradient now provides the energy to drive cotransport of a second solute, S_2 , against its electrochemical gradient.

temperature. When the “reaction” is simply transport of a solute from a region where its concentration is C_1 to a region where its concentration is C_2 , no bonds are made or broken and ΔG° is zero. The free-energy change for transport, ΔG_t , is then

$$\Delta G_t = RT \ln (C_2/C_1) \quad (11-3)$$

If there is, say, a 10-fold difference in concentration between two compartments, the cost of moving 1 mol of an uncharged solute at 25 °C uphill across a membrane separating the compartments is

$$\begin{aligned} \Delta G_t &= (8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K}) \ln (10/1) = 5,700 \text{ J/mol} \\ &= 5.7 \text{ kJ/mol} \end{aligned}$$

Equation 11-3 holds for all uncharged solutes.

WORKED EXAMPLE 11-1 Energy Cost of Pumping an Uncharged Solute

Calculate the energy cost (free-energy change) of pumping an uncharged solute against a 10^4 -fold concentration gradient at 25 °C.

SOLUTION: Begin with Equation 11-3. Substitute 1.0×10^4 for (C_2/C_1) , $8.315 \text{ J/mol} \cdot \text{K}$ for R , and 298 K for T :

$$\begin{aligned} \Delta G_t &= RT \ln (C_2/C_1) \\ &= (8.315 \text{ J/mol} \cdot \text{K})(298 \text{ K})(1.0 \times 10^4) \\ &= 23 \text{ kJ/mol} \end{aligned}$$

When the solute is an *ion*, its movement without an accompanying counterion results in the endergonic separation of positive and negative charges, producing an electrical potential; such a transport process is said to be **electrogenic**. The energetic cost of moving an ion depends on the electrochemical potential (Fig. 11-25), the sum of the chemical and electrical gradients:

$$\Delta G_t = RT \ln (C_2/C_1) + ZF \Delta \psi \quad (11-4)$$

where Z is the charge on the ion, F is the Faraday constant ($96,480 \text{ J/V} \cdot \text{mol}$), and $\Delta \psi$ is the transmembrane electrical potential (in volts). Eukaryotic cells typically have plasma membrane potentials of about 0.05 V (with the inside negative relative to the outside), so the second term on the right side of Equation 11-4 can make a significant contribution to the total free-energy change for transporting an ion. Most cells maintain more than a 10-fold difference in ion concentrations across their plasma or intracellular membranes, and for many cells and tissues active transport is therefore a major energy-consuming process.

WORKED EXAMPLE 11-2 Energy Cost of Pumping a Charged Solute

Calculate the energy cost (free-energy change) of pumping Ca^{2+} from the cytosol, where its concentration is about $1.0 \times 10^{-7} \text{ M}$, to the extracellular fluid, where its concentration is about 1.0 mM . Assume a temperature of 37 °C (body temperature in a mammal) and a standard transmembrane potential of 50 mV (inside negative) for the plasma membrane.

SOLUTION: This is a case in which energy must be expended to counter two forces acting on the ion being transported: the membrane potential and the concentration difference across the membrane. These forces are expressed in the two terms on the right side of Equation 11-4:

$$\Delta G_t = RT \ln (C_2/C_1) + ZF \Delta \psi$$

in which the first term describes the chemical gradient and the second describes the electrical potential.

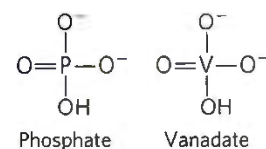
In Equation 11-4, substitute $8.315 \text{ J/mol} \cdot \text{K}$ for R , 310 K for T , 1.0×10^{-3} for C_2 , 1.0×10^{-7} for C_1 , $+2$ (the charge on a Ca^{2+} ion) for Z , $96,500 \text{ J/V} \cdot \text{mol}$ for F , and 0.050 V for $\Delta \psi$. Note that the transmembrane potential is 50 mV (inside negative), so the change in potential when an ion moves from inside to outside is 50 mV .

$$\begin{aligned} \Delta G_t &= RT \ln (C_2/C_1) + ZF \Delta \psi \\ &= (8.315 \text{ J/mol} \cdot \text{K})(310 \text{ K}) \ln \frac{(1.0 \times 10^{-3})}{(1.0 \times 10^{-7})} \\ &\quad + 2(96,500 \text{ J/V} \cdot \text{mol})(0.050 \text{ V}) \\ &= 33 \text{ kJ/mol} \end{aligned}$$

The mechanism of active transport is of fundamental importance in biology. As we shall see in Chapters 19 and 20, ATP is formed in mitochondria and chloroplasts by a mechanism that is essentially ATP-driven ion transport operating in reverse. The energy made available by the spontaneous flow of protons across a membrane is calculable from Equation 11-4; remember that ΔG for flow *down* an electrochemical gradient has a negative value, and ΔG for transport of ions *against* an electrochemical gradient has a positive value.

P-Type ATPases Undergo Phosphorylation during Their Catalytic Cycles

The family of active transporters called **P-type ATPases** are cation transporters that are reversibly phosphorylated by ATP (thus the name P-type) as part of the transport cycle. Phosphorylation forces a conformational change that is central to movement of the cation across the membrane. The human genome encodes at least 70 P-type ATPases that share similarities in amino acid sequence and topology, especially near the Asp residue that undergoes phosphorylation. All are integral proteins with 8 or 10 predicted membrane-spanning regions in a single polypeptide, and all are sensitive to inhibition by the transition-state analog **vanadate**, which mimics phosphate when under nucleophilic attack by a water molecule.



The P-type ATPases are widespread in eukaryotes and bacteria. The Na^+K^+ ATPase of animal cells (an antiporter for Na^+ and K^+ ions) and the plasma membrane H^+ ATPase of plants and fungi set the transmembrane

electrochemical potential in cells by establishing ion gradients across the plasma membrane. These gradients provide the driving force for secondary active transport and are also the basis for electrical signaling in neurons. In animal tissues, the **sarcoplasmic/endoplasmic reticulum Ca^{2+} ATPase (SERCA) pump** and the plasma membrane Ca^{2+} ATPase pump, which are uniporters for Ca^{2+} ions, together maintain the cytosolic level of Ca^{2+} below $1\ \mu\text{M}$. The SERCA pump moves Ca^{2+} from the cytosol into the lumen of the sarcoplasmic reticulum. Parietal cells in the lining of the mammalian stomach have a P-type ATPase that pumps H^+ and K^+ out

of the cells and into the stomach, thereby acidifying the stomach contents. Lipid flippases, as we noted earlier, are structurally and functionally related to P-type transporters. Bacteria and eukaryotes use P-type ATPases to pump toxic heavy metal ions such as Cd^{2+} and Cu^{2+} out of cells.

All P-type pumps have similar structures (Fig. 11-37) and similar mechanisms. The mechanism postulated for P-type ATPases takes into account the large conformational changes and the phosphorylation-dephosphorylation of the critical Asp residue in the P (phosphorylation) domain that is known to occur during a catalytic cycle. For the SERCA pump (Fig. 11-38), each catalytic cycle moves two Ca^{2+} ions across the membrane and converts an ATP to ADP and P_i . The role of ATP binding and phosphoryl transfer to the enzyme is to bring about the interconversion of two conformations, E1 and E2, of the transporter. In the E1 conformation, the two Ca^{2+} -binding sites are exposed on the cytosolic side of the ER or sarcoplasmic reticulum and bind Ca^{2+} with high affinity. ATP binding and Asp phosphorylation drive a conformational change from E1 to E2 that exposes the Ca^{2+} -binding sites to the lumen and greatly reduces their affinity for Ca^{2+} , which releases the Ca^{2+} ions into the lumen. By this mechanism, the

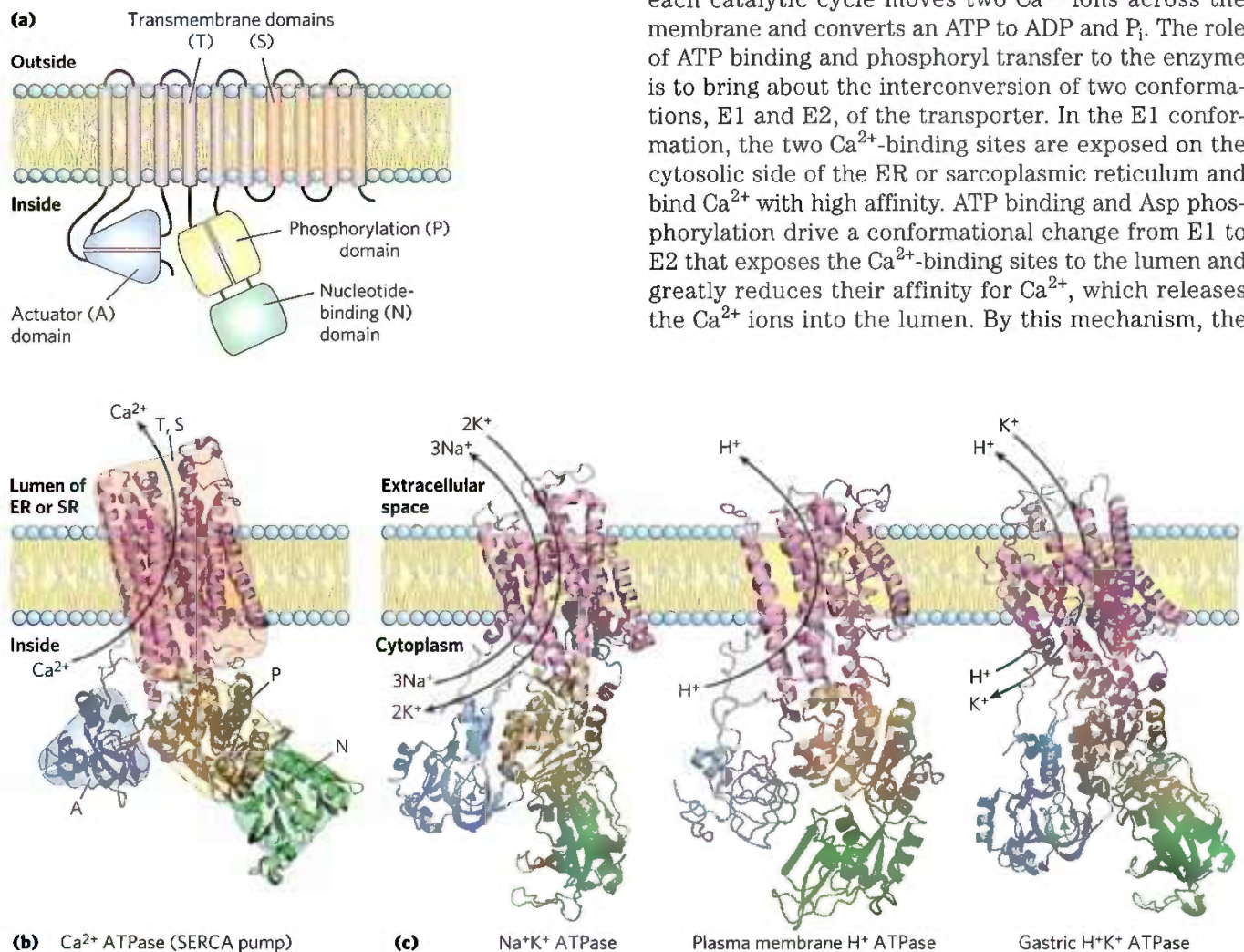
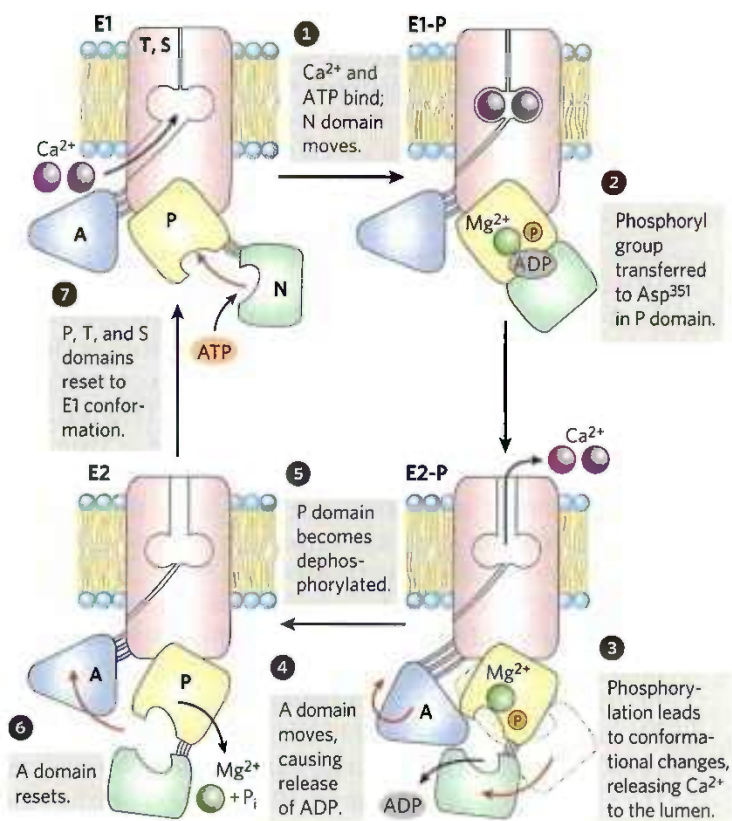


FIGURE 11-37 The general structure of the P-type ATPases.

(a) P-type ATPases have three cytoplasmic domains (A, N, and P) and two transmembrane domains (T and S) consisting of multiple helices. The N (nucleotide-binding) domain binds ATP and Mg^{2+} and has protein kinase activity that phosphorylates a specific Asp residue in the P (phosphorylation) domain of all P-type ATPases. The A (actuator) domain has protein phosphatase activity and removes the phosphoryl group from the Asp residue with each catalytic cycle of the pump. A transport (T) domain with six transmembrane helices includes the ion-transporting structure, and four more transmembrane helices make up the support (S) domain, which provides physical support to the transport domain and may have other specialized functions in certain P-type ATPases. The binding sites for the ions to be transported are near the middle

of the membrane, 40 to 50 Å from the phosphorylated Asp residue — thus, Asp phosphorylation-dephosphorylation does not directly affect ion binding. The A domain communicates movements of the N and P domains to the ion-binding sites. (b) A ribbon representation of the Ca^{2+} ATPase (SERCA pump). ATP binds to the N domain, and the Ca^{2+} ions to be transported bind to the T domain. (c) Other P-type ATPases have domain structures, and presumably mechanisms, like those of the SERCA pump; shown here are Na^+K^+ ATPase, the plasma membrane H^+ ATPase, and the gastric H^+K^+ ATPase. [(a) Information from M. Bublitz et al., *Curr. Opin. Struct. Biol.* 20:431, 2010, Fig. 1. Data from (b) PDB ID 1SU4, C. Toyoshima et al., *Nature* 405:647, 2000; (c) Na^+K^+ ATPase, PDB ID 3KDP, J. Preben Morth et al., *Nature* 450:1043, 2007; H^+ ATPase, PDB ID 3B8C, B. P. Pedersen et al., *Nature* 450:1111, 2007; H^+K^+ ATPase, PDB ID 3IXZ, K. Abe et al., *EMBO J.* 28:1637, 2009, and PDB ID 3B8E, J. Preben Morth et al., *Nature* 450:1043, 2007.]



energy released by hydrolysis of ATP during one phosphorylation-dephosphorylation cycle drives Ca^{2+} across the membrane against a large electrochemical gradient.

A variation on this basic mechanism is seen in the **Na^+K^+ ATPase** of the plasma membrane. This cotransporter couples phosphorylation-dephosphorylation of the critical Asp residue to the simultaneous movement of both Na^+ and K^+ against their electrochemical gradients. The Na^+K^+ ATPase is responsible for maintaining low Na^+ and high K^+ concentrations in the cell relative to the extracellular fluid (Fig. 11-39). For each molecule of ATP converted to ADP and P_i , the transporter moves two K^+ ions inward and three Na^+ ions outward across the plasma membrane. Cotransport is therefore electrogenic, creating a net separation of charge across the membrane; in animals, this produces the membrane potential of -50 to -70 mV (inside negative relative to outside) that is characteristic of most cells and is essential to the conduction of action potentials in neurons. The central role of the Na^+K^+ ATPase is reflected in the energy invested in this single reaction: about 25% of the total energy consumption of a human at rest.

V-Type and F-Type ATPases Are ATP-Driven Proton Pumps

V-type ATPases, a class of proton-transporting ATPases, are responsible for acidifying intracellular compartments in many organisms (thus, *V* for *vacuolar*). Proton pumps of this type maintain the vacuoles of fungi and higher plants at a pH between 3 and 6, well below that of the

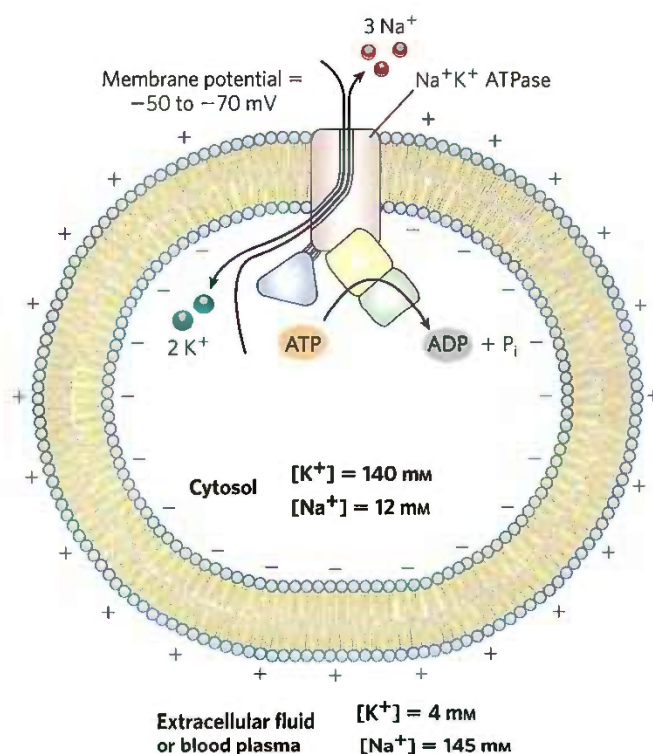


FIGURE 11-39 Role of the Na^+K^+ ATPase in animal cells. This active transport system is primarily responsible for setting and maintaining the intracellular concentrations of Na^+ and K^+ in animal cells and for generating the membrane potential. It does this by moving three Na^+ ions out of the cell for every two K^+ ions it moves in. The electrical potential across the plasma membrane is central to electrical signaling in neurons, and the gradient of Na^+ is used to drive the uphill cotransport of solutes in many cell types.

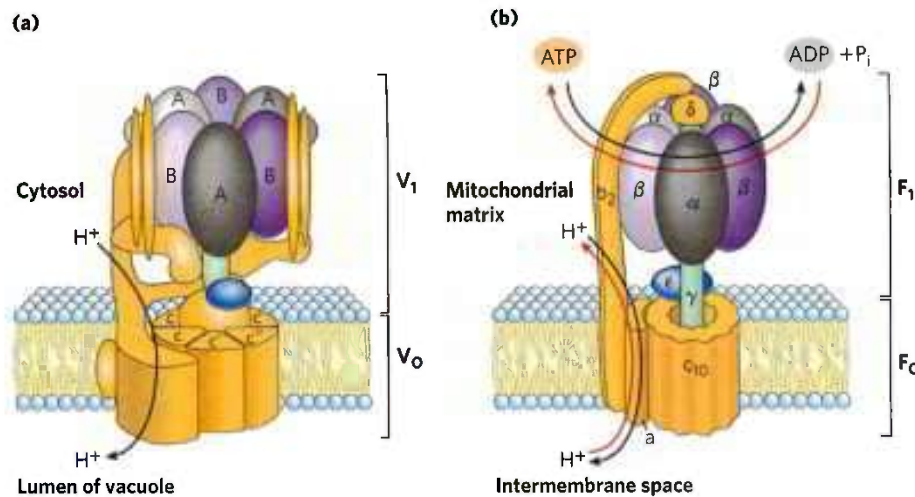


FIGURE 11-40 Two proton pumps with similar structures. (a) The V_0V_1 H^+ ATPase uses ATP to pump protons into vacuoles and lysosomes, creating their low internal pH. It has an integral (membrane-embedded) domain, V_0 , that includes multiple identical c subunits, and a peripheral domain that projects into the cytosol and contains the ATP-hydrolyzing sites, on three identical B subunits (purple). (b) The F_0F_1 ATPase/ATP synthase of mitochondria has an integral domain, F_0 , with multiple copies of the c subunit, and a

peripheral domain, F_1 , consisting of three α subunits, three β subunits, and a central shaft joined to the integral domain. F_0 and V_0 provide transmembrane channels through which protons are pumped as ATP is hydrolyzed on the β subunits of F_1 (B subunits of V_1). An ATP-driven proton transporter also can catalyze ATP synthesis (red arrows) as protons flow down their electrochemical gradient. This is the central reaction in the processes of oxidative phosphorylation and photophosphorylation.

surrounding cytosol (pH 7.5). V-type ATPases are also responsible for the acidification of lysosomes, endosomes, the Golgi complex, and secretory vesicles in animal cells. All V-type ATPases have a similar complex structure, with an integral (transmembrane) domain (V_0) that serves as a proton channel and a peripheral domain (V_1) that contains the ATP-binding site and the ATPase activity (Fig. 11-40a). The structure is similar to that of the well-characterized F-type ATPases.

F-type ATPase transporters catalyze the uphill transmembrane passage of protons, driven by ATP hydrolysis. The “F-type” designation derives from the identification of these ATPases as energy-coupling factors. The F_0 integral membrane protein complex (Fig. 11-40b; subscript “o” denotes its inhibition by the drug oligomycin) provides a transmembrane pathway for protons, and the peripheral protein F_1 (subscript “1” indicating that this was the first of several factors isolated from mitochondria) uses the energy of ATP to drive protons uphill (into a region of higher H^+ concentration). The F_0F_1 organization of proton-pumping transporters must have developed very early in evolution. Bacteria such as *E. coli* use an F_0F_1 ATPase complex in their plasma membrane to pump protons outward, and archaea have a closely homologous proton pump, the A_0A_1 ATPase.

P3 Like all enzymes, F-type ATPases catalyze their reactions in both directions. Therefore, a sufficiently large proton gradient can supply the energy to drive the reverse reaction, ATP synthesis (Fig. 11-40b). When functioning in this direction, the F-type ATPases are more appropriately named **ATP synthases**. ATP synthases are central to ATP production in mitochondria during oxidative phosphorylation and in chloroplasts

during photophosphorylation, as well as in bacteria and archaea. The proton gradient needed to drive ATP synthesis is produced by other types of proton pumps powered by substrate oxidation or sunlight. We provide a detailed description of these processes in Chapters 19 and 20.

ABC Transporters Use ATP to Drive the Active Transport of a Wide Variety of Substrates

ABC transporters constitute a large family of ATP-driven transporters that pump amino acids, peptides, proteins, metal ions, various lipids, bile salts, and many hydrophobic compounds, including drugs, across a membrane against a concentration gradient. Many ABC transporters are located in the plasma membrane, but some are also found in the ER and in the membranes of mitochondria and lysosomes. All members of this family have two ATP-binding domains (“cassettes”) that give the family its name—*ATP-binding cassette* transporters—and two transmembrane domains, each containing six transmembrane helices. In some cases, all these domains are in a single, long polypeptide; other ABC transporters have two subunits, each contributing a nucleotide-binding domain (NBD) and a domain with six transmembrane helices. The transport mechanism is believed to involve two forms of the transporter, one with its substrate-binding site facing the outside of the cell, the other open to substrate on the inside (Fig. 11-41). Substrates move across the membrane when the two forms interconvert, driven by ATP hydrolysis. The NBDs of all ABC proteins are similar in sequence and presumably in three-dimensional structure. They constitute

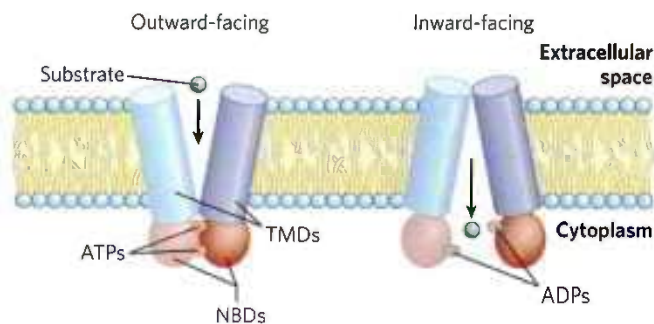


FIGURE 11-41 ABC transporters. The protein has two homologous halves, each with a six-helix transmembrane domain (TMD; blue), and a cytoplasmic nucleotide-binding domain (NBD; red). In the mechanism proposed for the coupling of ATP hydrolysis to transport, with ATP bound to the NBD sites, substrate binds to the transporter on the cytoplasmic side. Upon substrate binding and ATP hydrolysis to ADP, a conformational change exposes the substrate to the outside surface and lowers the transporter's affinity for its substrate; substrate diffuses away from the transporter into the extracellular space. This mechanism for the coupling of ATP hydrolysis to transport is based on structures of a number of ABC transporters crystallized under different conditions. Compare this process with the model of glucose transport in Figure 11-33.

the conserved molecular motor that can be coupled to a wide variety of transmembrane domains, each capable of pumping one specific substrate across a membrane. When coupled this way, the ATP-driven motor moves solutes against a concentration gradient, with a stoichiometry of about one ATP hydrolyzed per molecule of substrate transported.



The human genome contains at least 48 genes that encode ABC transporters; a number of these are presented in Table 11-2. Some of these transporters have very high specificity for a single substrate; others are more promiscuous, able to transport drugs that cells presumably did not encounter during their evolution. Many ABC transporters are involved in maintaining the composition of the lipid bilayer, such as the floppases that move membrane lipids from one leaflet of the bilayer to the other. Many others are needed to move sterols, sterol derivatives, and fatty acids into the bloodstream for transport throughout the body. For example, the cellular

machinery for exporting excess cholesterol includes an ABC transporter (see Fig. 21-47). Mutations in the genes that encode some of these proteins contribute to genetic diseases, including liver failure, retinal degeneration, and Tangier disease. The cystic fibrosis transmembrane conductance regulator protein (CFTR) of the plasma membrane is an interesting case of an ABC protein that is an ion channel (for Cl^-), regulated by ATP hydrolysis, but without the pumping function characteristic of an active transporter (Box 11-2).

One human ABC transporter with very broad substrate specificity is the **multidrug transporter (MDR1)**, encoded by the *ABCB1* gene. MDR1 in the placental membrane and in the blood-brain barrier ejects toxic compounds that would damage the fetus or the brain. But it is also responsible for the striking resistance of certain tumors to some generally effective antitumor drugs. For example, MDR1 pumps the chemotherapeutic drugs doxorubicin and vinblastine out of cells, thus preventing their accumulation within a tumor and blocking their therapeutic effects. Overexpression of MDR1 is often associated with treatment failure in cancers of the liver, kidney, and colon. A related ABC transporter, BCRP (breast cancer resistance protein, encoded by the *ABCG2* gene), is overexpressed in breast cancer cells, also conferring resistance to anticancer drugs. Highly selective inhibitors of these multidrug transporters are expected to enhance the effectiveness of antitumor drugs and are the objects of current drug discovery and design.

ABC transporters are also present in simpler animals and in plants and microorganisms. Yeast has 31 genes that encode ABC transporters, *Drosophila* has 56, and *E. coli* has 80, representing 2% of its entire genome. ABC transporters that are used by *E. coli* and other bacteria to import essentials such as vitamin B_{12} are the presumed evolutionary precursors of the MDRs of animal cells. The presence of ABC transporters that confer antibiotic resistance in pathogenic microbes (*Pseudomonas aeruginosa*, *Staphylococcus aureus*, *Candida albicans*, *Neisseria gonorrhoeae*, and *Plasmodium falciparum*) is a serious public health concern and makes these transporters attractive targets for drug design. ■

TABLE 11-2 Some ABC Transporters in Humans

Gene(s)	Role/characteristics	Text reference
<i>ABCA1</i>	Reverse cholesterol transport; defect causes Tangier disease	Fig. 21-47
<i>ABCA4</i>	Only in visual receptors, recycling of all- <i>trans</i> -retinal	Fig. 12-19
<i>ABCB1</i>	Multidrug resistance P-glycoprotein 1; transport across blood-brain barrier	—
<i>ABCB4</i>	Multidrug resistance; transport of phosphatidylcholine in bile	—
<i>ABCB11</i>	Transports bile salts out of hepatocytes	Fig. 17-1
<i>ABCC6</i>	Sulfonylurea receptor; targeted by the drug glipizide in type 2 diabetes	Fig. 23-27
<i>ABCG2</i>	Breast cancer resistance protein (BCRP); major exporter of anticancer drugs	p. 396
<i>ABCC7</i>	CFTR (Cl^- channel); defect causes cystic fibrosis	Box 11-2



A Defective Ion Channel in Cystic Fibrosis

Cystic fibrosis (CF) is a serious hereditary disease. In the United States, the frequency of CF is about 1 in 3,200 live births, and 1% to 4% (depending on ethnicity) are carriers, having one defective copy of the gene and one normal copy. Only individuals with two defective copies show the severe symptoms of the disease: obstruction of the gastrointestinal and respiratory tracts, commonly leading to bacterial infection of the airways.

The defective gene underlying CF was discovered in 1989. It encodes a membrane protein called cystic fibrosis transmembrane conductance regulator, or CFTR. This protein has two segments, each containing six transmembrane helices, two nucleotide-binding domains (NBDs), and a regulatory region that connects them (Fig. 1). CFTR is therefore very similar to other ABC transporter proteins, except that it functions as an *ion channel* (for Cl^-), not as a pump. The channel conducts Cl^- across the plasma membrane when both NBDs have bound ATP, and it closes when the ATP on one of the NBDs is broken down to ADP and P_i . The Cl^- channel is further regulated by phosphorylation of several Ser residues in the regulatory domain, catalyzed by cAMP-dependent protein kinase. When the regulatory domain is not phosphorylated, the Cl^- channel is closed.

The mutation responsible for CF in up to 90% of cases results in deletion of a Phe residue at position 508 (a mutation denoted F508del). The mutant protein folds incorrectly,

causing it to be degraded in proteasomes. As a result, Cl^- movement is reduced across the plasma membranes of epithelial cells that line the airways, digestive tract, exocrine glands (pancreas, sweat glands), bile ducts, and vas deferens. Less-common mutations, such as G551D (Gly⁵⁵¹ changed to Asp), lead to production of CFTR that is correctly folded and inserted into the membrane but is defective in Cl^- transfer.

Diminished export of Cl^- in individuals with CF is accompanied by diminished export of water from cells, causing the mucus on cell surfaces to become dehydrated, thick, and excessively sticky. In normal circumstances, cilia on the epithelial cells lining the inner surface of the lungs constantly sweep away bacteria that settle in this mucus, but the thick mucus in individuals with CF hinders this process, providing a haven in the lungs for pathogenic bacteria. Frequent infections by bacteria such as *Staphylococcus aureus* and *Pseudomonas aeruginosa* cause progressive damage to the lungs and reduce respiratory efficiency, eventually resulting in death due to inadequate lung function.

Advances in therapy have raised the average life expectancy for people who have CF from just 10 years in 1960 to more than 40 years today. CFTR potentiators such as ivacaftor (VX-770) increase the function of the mutant G551D protein that is properly folded and in place in the plasma membrane. For individuals with the folding defect, F508del, CFTR

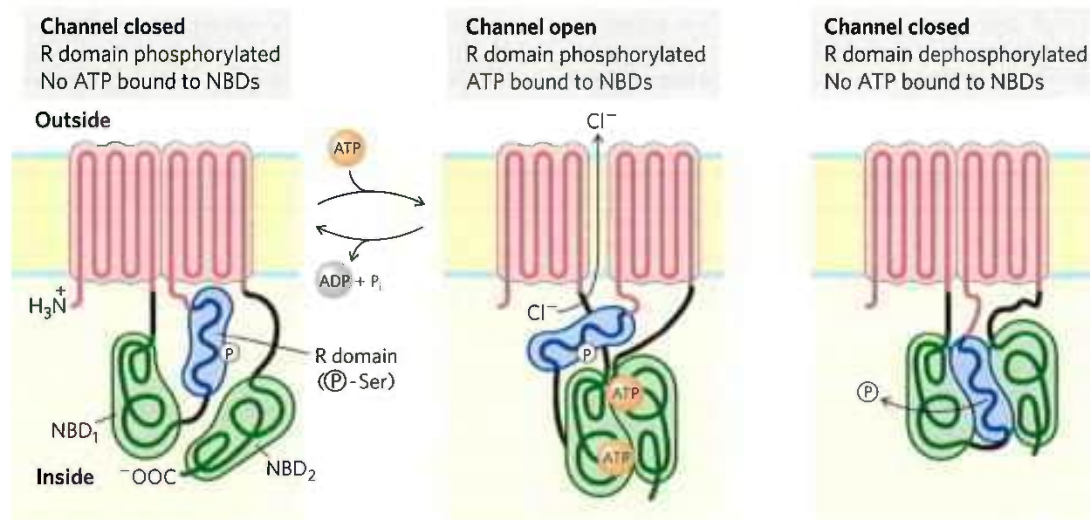


FIGURE 1 Three states of the CFTR protein. The protein has two segments, each with six transmembrane helices, and three functionally significant domains extend from the cytoplasmic surface: NBD₁ and NBD₂ (green) are nucleotide-binding domains that bind ATP, and the regulatory R domain (blue) is the site of phosphorylation by cAMP-dependent protein kinase. When this R domain is phosphorylated but no ATP is bound to the NBDs

(left), the channel is closed. The binding of ATP opens the channel (middle) until the bound ATP is hydrolyzed. When the R domain is unphosphorylated (right), it binds the NBD domains and prevents ATP binding and channel opening. CFTR is a typical ABC transporter in all but two respects: most ABC transporters lack the regulatory domain, and CFTR acts as an ion channel (for Cl^-), not as a typical transporter.

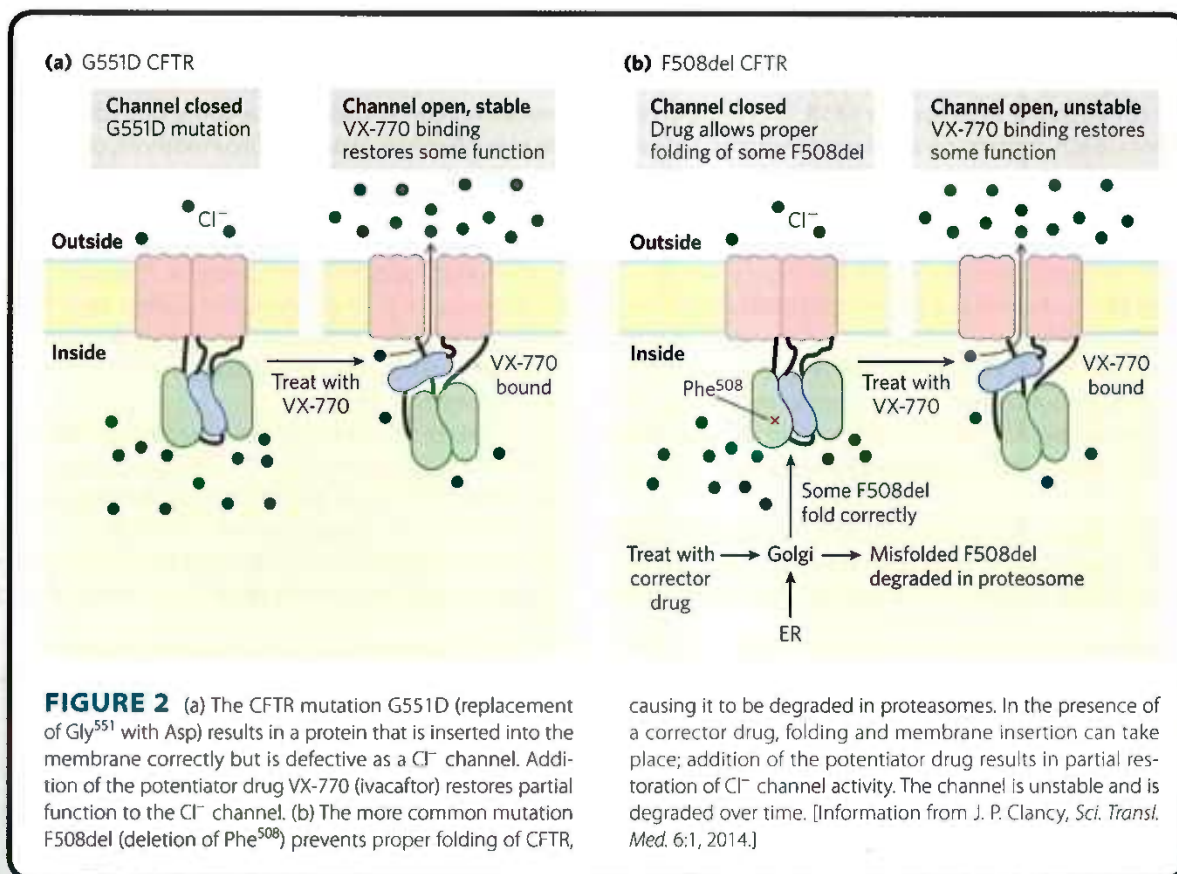
(Continued on next page)



A Defective Ion Channel in Cystic Fibrosis (Continued)

correctors improve the processing and delivery of the mutant protein to the cell surface; a combination of potentiator and corrector drugs is more effective than the corrector drug alone for these patients (Fig. 2). In 2019, clinical trials of a combi-

nation of three drugs (ivacaftor, tezacaftor, and elexacaftor), acting as both potentiators and correctors, showed dramatic improvements in individuals with the most common mutation (F508del).



causing it to be degraded in proteasomes. In the presence of a corrector drug, folding and membrane insertion can take place; addition of the potentiator drug results in partial restoration of Cl⁻ channel activity. The channel is unstable and is degraded over time. [Information from J. P. Clancy, *Sci. Transl. Med.* 6:1, 2014.]

Ion Gradients Provide the Energy for Secondary Active Transport

The ion gradients formed by primary transport of Na⁺ or H⁺ can, in turn, provide the driving force for cotransport of other solutes. Many cell types have transport systems that couple the spontaneous, downhill flow of these ions to the simultaneous uphill pumping of another ion, sugar, or amino acid.

In intestinal epithelial cells, glucose and certain amino acids are accumulated by symport with Na⁺, down the Na⁺ gradient established by the Na⁺K⁺ ATPase of the plasma membrane (Fig. 11-42). The apical surface of the intestinal epithelial cell (the surface that faces the intestinal contents) is covered with microvilli—long, thin projections of the plasma membrane that greatly increase the surface area exposed to the intestinal contents. The

Na⁺-glucose symporter in the apical plasma membrane takes up glucose from the intestine in a process driven by the downhill flow of Na⁺:



The energy required for this process comes from two sources: the greater concentration of Na⁺ outside than inside the cell (the chemical potential) and the membrane (electrical) potential, which is inside negative and therefore draws Na⁺ inward. **P3** The strong thermodynamic tendency for Na⁺ to move into the cell provides the energy needed for the transport of glucose into the cell, against its concentration gradient. An ion gradient created and sustained by energy-dependent ion pumping serves as the potential energy for cotransport of another species against its concentration gradient.

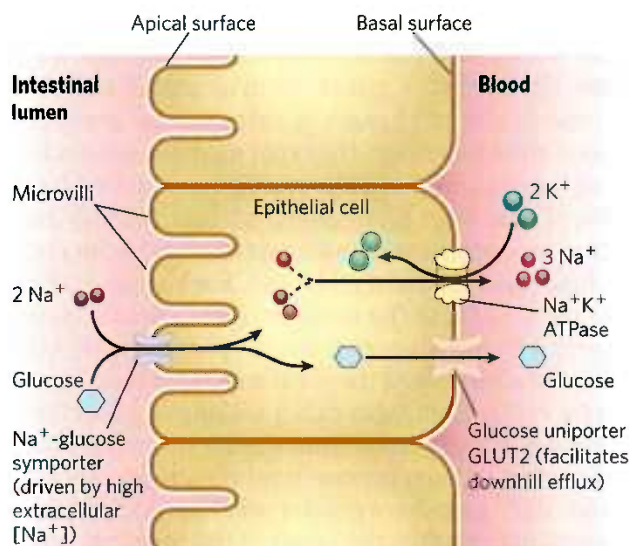


FIGURE 11-42 Glucose transport in intestinal epithelial cells.

Glucose is cotransported with Na^+ across the apical plasma membrane into the epithelial cell. It moves through the cell to the basal surface, where it passes into the blood via GLUT2, a passive glucose uniporter. The Na^+K^+ ATPase continues to pump Na^+ outward to maintain the Na^+ gradient that drives glucose uptake.

WORKED EXAMPLE 11-3 Energetics of Pumping by Symport

Calculate the maximum $\frac{[\text{glucose}]_{\text{in}}}{[\text{glucose}]_{\text{out}}}$ ratio that can be achieved by the plasma membrane Na^+ -glucose symporter of an epithelial cell when $[\text{Na}^+]_{\text{in}}$ is 12 mM, $[\text{Na}^+]_{\text{out}}$ is 145 mM, the membrane potential is -50 mV (inside negative), and the temperature is 37°C .

SOLUTION: Using Equation 11-4 (p. 392), we can calculate the energy inherent in an electrochemical Na^+ gradient—that is, the cost of moving one Na^+ ion up this gradient:

$$\Delta G_t = RT \ln \frac{[\text{Na}^+]_{\text{out}}}{[\text{Na}^+]_{\text{in}}} + ZF \Delta \psi$$

We then substitute standard values for R , T , and F ; the given values for $[\text{Na}^+]$ (expressed as molar concentrations); $+1$ for Z (because Na^+ has a positive charge); and 0.050 V for $\Delta \psi$. Note that the membrane potential is -50 mV (inside negative), so the change in potential when an ion moves from inside to outside is 50 mV.

$$\begin{aligned} \Delta G_t &= (8.315 \text{ J/mol}\cdot\text{K})(310 \text{ K}) \ln \frac{(1.45 \times 10^{-1})}{(1.2 \times 10^{-2})} \\ &\quad + 1(96,500 \text{ J/V}\cdot\text{mol})(0.050 \text{ V}) \\ &= 11.2 \text{ kJ/mol} \end{aligned}$$

When Na^+ reenters the cell, it releases the electrochemical potential created by pumping it out; ΔG for reentry is -11.2 kJ/mol of Na^+ . This is the potential energy per mole of Na^+ that is available to pump glucose. Given that two Na^+ ions pass down their electrochemical gradient and into the cell for each glucose carried

in by symport, the energy available to pump 1 mol of glucose is $2 \times 11.2 \text{ kJ/mol} = 22.4 \text{ kJ/mol}$. We can now calculate the maximum concentration ratio of glucose that can be achieved by this pump (from Eqn 11-3, p. 392):

$$\Delta G_t = RT \ln \frac{[\text{glucose}]_{\text{in}}}{[\text{glucose}]_{\text{out}}}$$

Rearranging, then substituting the values of ΔG_t , R , and T , gives

$$\begin{aligned} \ln \frac{[\text{glucose}]_{\text{in}}}{[\text{glucose}]_{\text{out}}} &= \frac{\Delta G_t}{RT} = \frac{22.4 \text{ kJ/mol}}{(8.315 \text{ J/mol}\cdot\text{K})(310 \text{ K})} = 8.69 \\ \frac{[\text{glucose}]_{\text{in}}}{[\text{glucose}]_{\text{out}}} &= e^{8.69} = 5.94 \times 10^3 \end{aligned}$$

Thus, the cotransporter can pump glucose inward until its concentration inside the epithelial cell is about 6,000 times the concentration outside (in the intestine). (This is the maximum theoretical ratio, assuming a perfectly efficient coupling of Na^+ reentry and glucose uptake.)

As glucose molecules are pumped from the intestine into the epithelial cell at the apical surface, glucose is simultaneously moved from the cell into the blood by passive transport through a glucose transporter (GLUT2) in the basal surface (Fig. 11-42). The crucial role of Na^+ in symport and antiport systems such as this requires the continued outward pumping of Na^+ to maintain the transmembrane Na^+ gradient.



In the kidney, a different Na^+ -glucose symporter (SGLT2) is the target of drugs used to treat type 2 diabetes. Gliflozins are specific inhibitors of this Na^+ -glucose symporter. They lower blood glucose by inhibiting glucose reabsorption in the kidney, thus preventing the damaging effects of elevated blood glucose. Glucose not reabsorbed in the kidney is cleared in the urine. This class of drugs, taken orally in combination with diet and exercise, lowers blood glucose significantly in individuals with type 2 diabetes.

Because of the essential role of ion gradients in active transport and energy conservation, compounds that collapse ion gradients across cellular membranes are effective poisons, and those that are specific for infectious microorganisms can serve as antibiotics. One such substance is valinomycin, a small cyclic peptide that neutralizes the K^+ charge by surrounding the ion with six carbonyl oxygens (Fig. 11-43). The hydrophobic peptide then acts as a shuttle, carrying K^+ across the membrane down its concentration gradient and deflating that gradient. Compounds that shuttle ions across membranes in this way are called **ionophores** (“ion bearers”). Both valinomycin and monensin (a Na^+ -carrying ionophore) are antibiotics; they kill microbial cells by disrupting secondary transport processes and energy-conserving reactions. Monensin is widely used as an antifungal and antiparasitic agent. ■

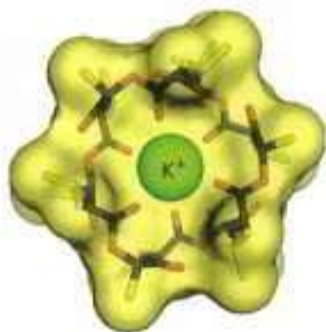


FIGURE 11-43 Valinomycin, a peptide ionophore that binds K^+ .

The central K^+ ion is surrounded by inward-facing polar and charged amino acid side chains, and the outside surface is covered with nonpolar side chains that make the whole structure hydrophobic enough to diffuse through the lipid bilayer, carrying K^+ down its concentration gradient. The resulting dissipation of the transmembrane ion gradient kills microbial cells, making valinomycin a potent antibiotic. [Information from P. Barak and E. A. Nater, <http://virtual-museum.soils.wisc.edu>, and K. Neupert-Laves and M. Dobler, *Helv. Chim. Acta* 58:432, 1975.]

Aquaporins Form Hydrophilic Transmembrane Channels for the Passage of Water

P3 A family of integral membrane proteins, the **aquaporins (AQPs)**, provide channels for rapid movement of water molecules across all plasma membranes. Aquaporins are found in all organisms, and multiple aquaporin genes are generally present, encoding similar but not identical proteins. Eleven aquaporins are known in mammals, each with a specific location and

role (Table 11-3). The exocrine glands that produce sweat, saliva, and tears secrete water through aquaporins. Erythrocytes, which swell or shrink rapidly in response to abrupt changes in extracellular osmolarity as blood travels through the renal medulla, have a high density of aquaporin in their plasma membrane (2×10^5 copies of AQP1 per cell). Seven aquaporins play roles in urine production and water retention in the nephron (the functional unit of the kidney). Each renal AQP has a specific location in the nephron, and each has specific properties and regulatory features. For example, AQP2 in the epithelial cells of the renal collecting duct is regulated by vasopressin (also called antidiuretic hormone): more water is reabsorbed from the duct into the kidney tissues when the vasopressin level is high. Mutant mice with no AQP2 gene have greater urine output (polyuria) and more dilute urine, the result of the proximal tubule becoming less permeable to water. In humans, genetically defective AQPs are known to be responsible for a variety of diseases.

Water molecules flow through an AQP1 channel at a rate of about 10^9 s^{-1} . For comparison, the highest known turnover number for an enzyme is that for catalase, $4 \times 10^7 \text{ s}^{-1}$, and many enzymes have turnover numbers between 1 s^{-1} and 10^4 s^{-1} (see Table 6-7). The low activation energy for passage of water through aquaporin channels ($\Delta G^\ddagger < 15 \text{ kJ/mol}$) suggests that water moves through the channels in a continuous stream, in the direction dictated by the osmotic gradient. (For a

TABLE 11-3 Permeability Characteristics and Predominant Distribution of Known Mammalian Aquaporins

Aquaporin	Permeant (permeability)	Tissue distribution	Primary subcellular distribution ^a
AQP0	Water (low)	Lens	Plasma membrane
AQP1	Water (high)	Erythrocyte, kidney, lung, vascular endothelium, brain, eye	Plasma membrane
AQP2	Water (high)	Kidney, vas deferens	Apical plasma membrane, intracellular vesicles
AQP3	Water (high), glycerol (high), urea (moderate)	Kidney, skin, lung, eye, colon	Basolateral plasma membrane
AQP4	Water (high)	Brain, muscle, kidney, lung, stomach, small intestine	Basolateral plasma membrane
AQP5	Water (high)	Salivary gland, lacrimal gland, sweat gland, lung, cornea	Apical plasma membrane
AQP6	Water (low), anions ($\text{NO}_3^- > \text{Cl}^-$)	Kidney	Intracellular vesicles
AQP7	Water (high), glycerol (high), urea (high)	Adipose tissue, kidney, testis	Plasma membrane
AQP8 ^b	Water (high)	Testis, kidney, liver, pancreas, small intestine, colon	Plasma membrane, intracellular vesicles
AQP9	Water (low), glycerol (high), urea (high)	Liver, leukocyte, brain, testis	Plasma membrane
AQP10	Water (low), glycerol (high), urea (high)	Small intestine	Intracellular vesicles

Information from L. S. King et al., *Nat. Rev. Mol. Cell Biol.* 5:688, 2004.

^aThe apical plasma membrane faces the lumen of the gland or tissue; the basolateral plasma membrane is along the sides and base of the cell, not facing the lumen of the gland or tissue.

^bAQP8 might also be permeated by urea.

discussion of osmosis, see p. 52.) Aquaporins do not allow passage of protons (hydronium ions, H_3O^+), which would collapse membrane electrochemical gradients.

Ion-Selective Channels Allow Rapid Movement of Ions across Membranes

Ion-selective channels—first recognized in neurons and now known to be present in the plasma membranes of all cells, as well as in the intracellular membranes of eukaryotes—provide another mechanism for moving inorganic ions across membranes. Ion channels, together with ion pumps such as the Na^+K^+ ATPase, determine a plasma membrane's permeability to specific ions and regulate the cytosolic concentration of ions and the membrane potential. In neurons, very rapid changes in the activity of ion channels cause the changes in membrane potential (action potentials) that carry signals from one end of a neuron to the other. In myocytes, rapid opening of Ca^{2+} channels in the sarcoplasmic reticulum releases the Ca^{2+} that triggers muscle contraction. We discuss the signaling functions of ion channels in Chapter 12. Here we describe the structural basis for ion-channel function, using as our example a well-studied K^+ channel.

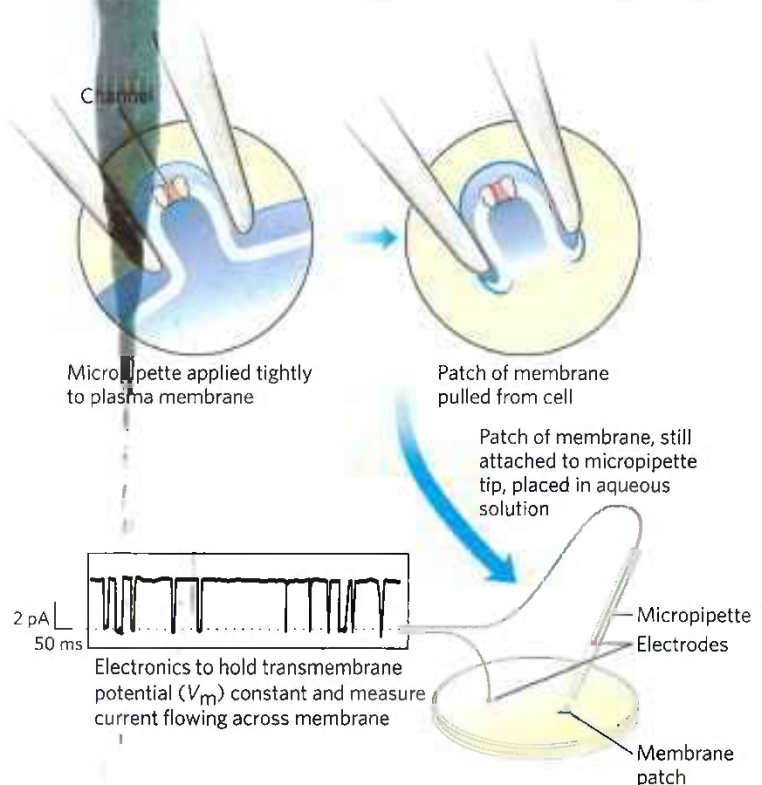
Ion channels are distinct from ion transporters in at least three ways. First, the rate of flux through channels can be orders of magnitude greater than the turnover number for a transporter— 10^7 to 10^8 ions/s for an ion channel, approaching the theoretical maximum for unrestricted diffusion. By contrast, the turnover rate of

the Na^+K^+ ATPase is about 100 s^{-1} . Second, ion channels are not saturable: rates do not approach a maximum at high substrate concentration. Third, they are gated in response to some type of cellular event. In **ligand-gated channels** (which are generally oligomeric), binding of an extracellular or intracellular small molecule forces an allosteric transition in the protein, which opens or closes the channel. In **voltage-gated ion channels**, a change in transmembrane electrical potential (V_m) causes a charged protein domain to move relative to the membrane, opening or closing the channel. Both types of gating can be very fast. A channel typically opens in a fraction of a millisecond and may remain open for only milliseconds, making these molecular devices effective for very fast signal transmission in the nervous system.

Because a single ion channel typically remains open for only a few milliseconds, monitoring this process is beyond the limit of most biochemical measurements. Ion fluxes must therefore be measured electrically, either as changes in V_m (in the millivolt range) or as electric current I (in the microampere or picoampere range), using microelectrodes and appropriate amplifiers. In **patch-clamping**, very small currents are measured through a tiny region of the membrane surface containing only one or a few ion-channel molecules (Fig. 11-44). The researcher can measure the size and duration of the current that flows during one opening of an ion channel and can determine how often a channel opens and how that frequency is affected by membrane potential, regulatory ligands, toxins, and other agents. Patch-clamp studies

FIGURE 11-44 Electrical measurements of ion-channel function. The “activity” of an ion channel is estimated by measuring the flow of ions through it, using the patch-clamp technique. A finely drawn-out pipette (micropipette) is pressed against the cell surface, and negative pressure in the pipette forms a pressure seal between pipette and membrane. As the pipette is pulled away from the cell, it pulls off a tiny patch of membrane (which may contain one or a few ion channels). After placing the pipette and attached patch in an aqueous solution, the researcher can measure channel activity as the electric current that flows between the contents of the pipette and the aqueous solution. In practice, a circuit is set up that “clamps” the transmembrane potential at a given value and measures the current that must flow to maintain this voltage. With highly sensitive detectors, researchers can measure the current flowing through a single ion channel, typically a few picoamperes. The trace shows the current through a single acetylcholine receptor channel as a function of time (in milliseconds), revealing how fast the channel opens and closes, how frequently it opens, and how long it stays open. Downward deflection represents channel opening. Clamping the V_m at different values permits determination of the effect of membrane potential on these parameters of channel function.

[Information from V. Witzemann et al., *Proc. Natl. Acad. Sci. USA* 93:13,286, 1996.]



have revealed that as many as 10^4 ions can move through a single ion channel in 1 ms. Such an ion flux represents a huge amplification of the initial signal, which may be just one or two signaling molecules (neurotransmitters, for example).

The Structure of a K^+ Channel Reveals the Basis for Its Specificity

The structure of a potassium channel from the bacterium *Streptomyces lividans* provides important insight into the way ion channels work. This bacterial ion channel is related in sequence to all other known K^+ channels and serves as the prototype for such channels, including the voltage-gated K^+ channel of neurons. Among the members of this protein family, the similarities in sequence are greatest in the “pore region,” which contains the ion selectivity filter that allows K^+ (radius 1.33 Å) to pass 10^4 times more readily than Na^+ (radius 0.95 Å) — at a rate (about 10^8 ions/s) approaching the theoretical limit for unrestricted diffusion.

The K^+ channel consists of four identical subunits that span the membrane and form a cone within a cone surrounding the ion channel, with the wide end of the double cone facing the extracellular space (Fig. 11-45a). Each subunit has two transmembrane α helices and a third, shorter helix that contributes to the pore region. The outer cone is formed by one of the transmembrane helices of each subunit. The inner cone, formed by the other four transmembrane helices, surrounds the ion channel and cradles the ion selectivity filter. Viewed perpendicular to the plane of the membrane, the central

channel is seen to be just wide enough to accommodate an unhydrated metal ion such as potassium (Fig. 11-45b).

Both the ion specificity and the high flux through the channel are understandable from what we know of the channel's structure (Fig. 11-45c). At the inner and outer plasma membrane surfaces, the entryways to the channel have several negatively charged amino acid residues, which presumably increase the local concentration of cations such as K^+ and Na^+ . The ion path through the membrane begins (on the inner surface) as a wide, water-filled channel in which the ion can retain its hydration sphere. Further stabilization is provided by the short helices in the pore region of each subunit, with the partial negative charges of their electric dipoles pointed at K^+ in the channel. About two-thirds of the way through the membrane, this channel narrows in the region of the selectivity filter, forcing the ion to give up its hydrating water molecules. Carbonyl oxygen atoms in the backbone of the selectivity filter replace the water molecules in the hydration sphere, forming a series of perfect coordination shells through which the K^+ moves. This favorable interaction with the filter is not possible for Na^+ , which is too small to make contact with all the potential oxygen ligands. The preferential stabilization of K^+ is the basis for the ion selectivity of the filter, and mutations that change residues in this part of the protein eliminate the channel's ion selectivity. The K^+ -binding sites of the filter are flexible enough to collapse to fit any Na^+ that enters the channel, and this conformational change closes the channel.

There are four potential K^+ -binding sites along the selectivity filter, each composed of an oxygen “cage” that provides ligands for the K^+ ions (Fig. 11-45c). In

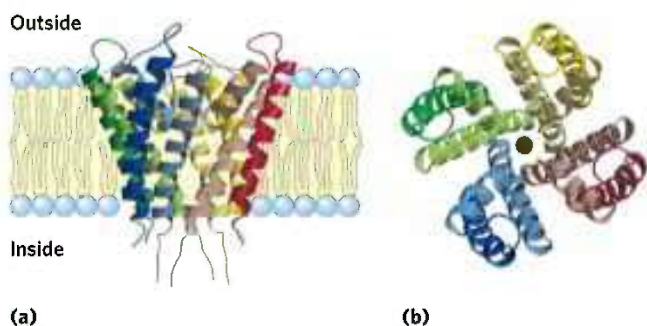
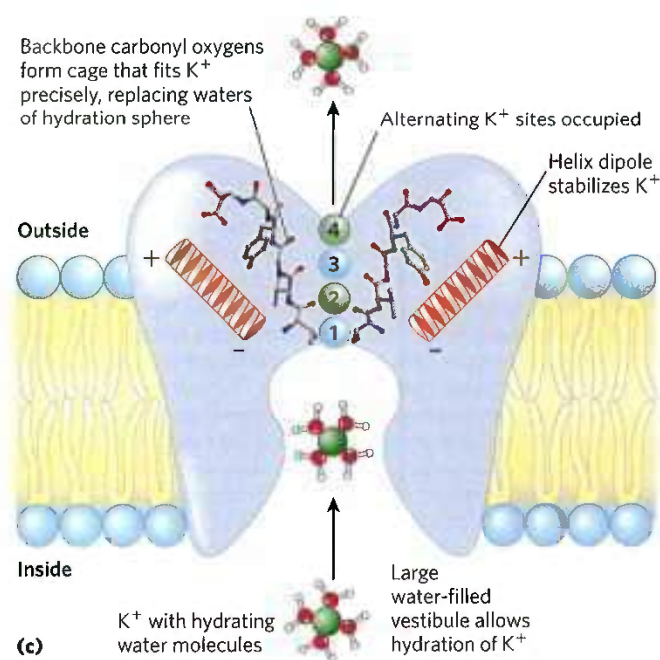


FIGURE 11-45 The K^+ channel of *Streptomyces lividans*. (a) Viewed in the plane of the membrane, the channel consists of eight transmembrane helices (two from each of four identical subunits), forming a cone with its wide end toward the extracellular space. The inner helices of the cone (lighter colored) line the transmembrane channel, and the outer helices interact with the lipid bilayer. Short segments of each subunit converge in the open end of the cone to make a selectivity filter. (b) This view, perpendicular to the plane of the membrane, shows the four subunits arranged around a central channel just wide enough for a single K^+ ion to pass. (c) Diagram of a K^+ channel in cross section, showing the structural features critical to function. K^+ ions go through the channel in pairs, first in sites 1 and 3, then in sites 2 and 4. Carbonyl oxygens (red) of the peptide backbone in the selectivity filter protrude into the channel, interacting with and stabilizing the K^+ ions that are passing through. [Data from (a, b) PDB ID 1BL8, D. A. Doyle et al., *Science* 280:69, 1998; (c) G. Yellen, *Nature* 419:35, 2002, and PDB ID 1J95, M. Zhou et al., *Nature* 411:557, 2001.]



the crystal structure, two K^+ ions are visible within the selectivity filter, about 7.5 Å apart, and two water molecules occupy the unfilled positions. K^+ ions pass through the filter in single file; their mutual electrostatic repulsion probably just balances the interaction of each ion with the selectivity filter and keeps them moving. Movement of the two K^+ ions is concerted: first they occupy positions 1 and 3, then they hop to positions 2 and 4. The energetic difference between these two configurations (1, 3 and 2, 4) is very small; energetically, the selectivity pore is not a series of hills and valleys but a flat surface, which is ideal for rapid ion movement through the channel. The structure of the channel seems to have been optimized during evolution to give maximal flow rates and high specificity.

SUMMARY 11.3 Solute Transport across Membranes

- Some transporters simply facilitate passive diffusion of a solute across the membrane, from a higher concentration to a lower concentration. Others transport solutes against an electrochemical gradient; this requires a source of metabolic energy.

- Transporters move solutes across a membrane one or a few at a time, providing a binding site on each side of the membrane. The binding sites alternate between being accessible from the outside and from the inside. Ion channels provide a path across the membrane, which is either open or closed. When open, the channel allows the movement of large numbers of solute ion across the membrane at nearly the speed of unhindered diffusion.

- A family of glucose transporters in humans includes the passive transporter GLUT1, which is saturated at normal levels of glucose in the blood. GLUT1 facilitates movement of glucose from blood into erythrocytes.

- The chloride-bicarbonate exchanger of erythrocytes exchanges one Cl^- ion for one HCO_3^- ion across the erythrocyte plasma membrane, mediating the uptake of CO_2 in the tissues and its release in the lungs.

- Active transporters use energy to pump solutes against an electrochemical gradient.

- P-type ATPases, including the Na^+K^+ ATPase of the plasma membrane and the Ca^{2+} transporters of the sarcoplasmic/endoplasmic reticulum, couple phosphorylation and dephosphorylation of the transporter to alternate exposure of solute binding sites on the inside and the outside of the membrane. In animal cells, the Na^+K^+ ATPase maintains the differences in cytosolic and extracellular concentrations of Na^+ and K^+ , and the resulting Na^+ gradient is used as the energy source for a variety of secondary active transport processes.

- V-type and F-type ATPases are active transporters that couple ATP cleavage to the uphill transport of H^+ ions. The same mechanism, working in reverse, allows

the synthesis of ATP, driven by movement of protons down their electrochemical gradient.

- ABC transporters carry a variety of substrates (including many drugs) out of cells, using ATP as the energy source. The ATP-using domain is conserved in many ABC transporters, and it is coupled with various transmembrane domains that give substrate specificity.

- Some active cotransporters use the energy in an ion gradient generated catabolically to move a solute uphill. The Na^+ -glucose cotransporter of the kidney and intestine is such a transporter.

- Water moves across membranes through aquaporins. Some aquaporins are regulated; some also transport glycerol or urea.

- Ion channels provide hydrophilic pores through which select ions can diffuse, moving down their electrical or chemical concentration gradients. Ion channels are unsaturable, have very high flux rates, and are highly specific for one ion.

- Structural studies of K^+ channels reveal the mechanism that allows great discrimination between K^+ and other ions like Na^+ . The polar transmembrane passage precisely fits the K^+ ion, but allows neither larger ions nor smaller ions to pass.

KEY TERMS

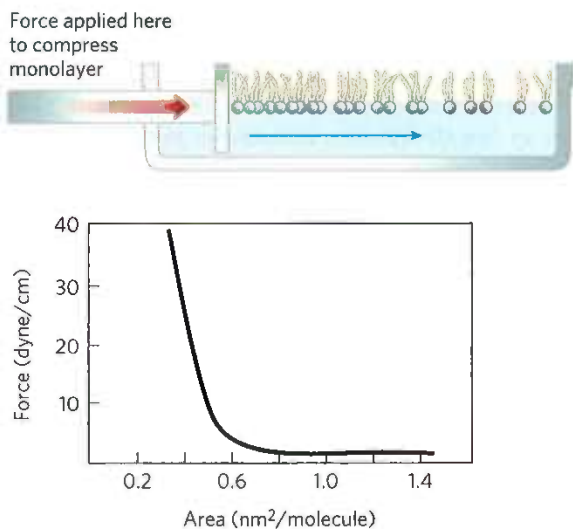
Terms in bold are defined in the glossary.

micelle 368	v-SNAREs 383
bilayer 368	t-SNAREs 383
vesicle 368	integrin 383
fluid mosaic model 369	selectin 384
lipid transfer protein 371	simple diffusion 385
integral proteins 372	membrane potential
peripheral proteins 372	(V_m) 385
amphitropic proteins 372	electrochemical
monotopic 372	gradient 385
bitopic 372	electrochemical
polytopic 372	potential 385
hydropathy index 374	transporters 385
β barrel 374	passive transport 385
porin 374	active transport 386
positive-inside rule 375	ion channels 386
GPI-anchored protein 376	K_t ($K_{transport}$) 388
flippases 378	amphipathic 388
floppases 378	electroneutral 391
scramblases 379	cotransport 391
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F-type ATPases 395	ionophore 399
ATP synthase 395	aquaporins (AQPs) 400
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multidrug transporters 396	voltage-gated ion channel 401
Na ⁺ -glucose symporter 398	patch-clamping 401

PROBLEMS

1. Determining the Cross-Sectional Area of a Lipid Molecule When phospholipids are layered gently onto the surface of water, they orient at the air-water interface with their head groups in the water and their hydrophobic tails in the air. The experimental apparatus shown can be used to progressively reduce the surface area available to a layer of lipids. By measuring the force necessary to push the lipids together, we can determine when the molecules are packed tightly in a continuous monolayer; as that area is approached, the force needed to further reduce the surface area increases sharply, as shown in the graph. How would you use this apparatus to determine the average area occupied by a single lipid molecule in the monolayer?



2. Properties of Lipids and Lipid Bilayers Lipid bilayers form when phospholipids are suspended in water. The edges of these sheets close upon each other and undergo self-sealing to form vesicles (liposomes).

- What properties of lipids are responsible for this property of bilayers? Explain.
- What are the consequences of this property for the structure of biological membranes?

3. Length of a Fatty Acid Molecule The carbon-carbon bond distance for single-bonded carbons, such as those in a saturated fatty acyl chain, is about 1.5 Å. Estimate to one significant figure the length of a single molecule of palmitate in its fully extended form. If two molecules of palmitate were placed end to end, how would their total length compare with the thickness of the lipid bilayer in a biological membrane?

4. Membrane Proteins What are the three main categories of membrane proteins, and how are they distinguished experimentally?

5. Location of a Membrane Protein Treatment of disrupted erythrocyte membranes with a concentrated salt solution released

an unknown membrane protein, X. Proteolytic enzymes cleaved X into fragments. In additional experiments, intact erythrocytes were treated with proteolytic enzymes, washed, then disrupted. Extraction of membrane components yielded intact X. What do these observations indicate about the location of X in the plasma membrane? Do the properties of X resemble those of an integral membrane protein or a peripheral membrane protein?

6. Predicting Membrane Protein Topology from Sequence

You have cloned the gene for a human erythrocyte protein, which you suspect is a membrane protein. You deduce the amino acid sequence of the protein from the nucleotide sequence of the gene. From this sequence alone, how would you evaluate the possibility that the protein is an integral protein? Suppose the protein proves to be an integral protein with one transmembrane segment. Suggest biochemical or chemical experiments that might allow you to determine whether the protein is oriented with the amino terminus on the outside of the cell or on the inside of the cell.

7. Surface Density of a Membrane Protein

E. coli can be induced to make about 10,000 copies of the lactose transporter (M_r 31,000) per cell. Assume that *E. coli* is a cylinder 1 μm in diameter and 2 μm long. The diameter of the lactose transporter is approximately 6 nm. What fraction of the plasma membrane surface is occupied by the lactose transporter molecules? Explain how you arrived at this conclusion.

8. Molecular Species in the Plasma Membrane

The plasma membrane of *E. coli* is about 75% protein and 25% phospholipid by weight. How many molecules of membrane lipid are present for each molecule of membrane protein? Assume an average protein M_r of 50,000 and an average phospholipid M_r of 750. What more would you need to know to estimate the fraction of the membrane surface that is covered by lipids?

9. Temperature Dependence of Lateral Diffusion

The experiment described in Figure 11-19 was performed at 37 °C. If the experiment were carried out at 10 °C, what effect would you expect on the rate of diffusion? Why?

10. Membrane Self-Sealing

Cellular membranes are self-sealing—if they are punctured or disrupted mechanically, they quickly and automatically reseal. What properties of membranes are responsible for this important feature?

11. Lipid Melting Temperatures

Membrane lipids in tissue samples obtained from different parts of a reindeer's leg have different fatty acid compositions. Membrane lipids from tissue near the hooves contain a larger proportion of unsaturated fatty acids than those from tissue in the upper leg. What is the significance of this observation?

12. Flip-Flop Diffusion

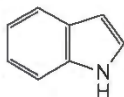
What is the physical explanation for the very slow movement of membrane phospholipids from one leaflet of a biological membrane to the other? What factors influence this rate?

13. Bilayer Asymmetry

The inner leaflet (monolayer) of the human erythrocyte membrane consists predominantly of phosphatidylethanolamine and phosphatidylserine. The outer leaflet consists predominantly of phosphatidylcholine and sphingomyelin. Although the phospholipid components of the membrane can diffuse in the fluid bilayer, this sidedness is preserved at all times. How?

14. Scramblase and Flippase Explain the difference between the scramblase enzymes and flippase enzymes based on the membranes with which they are associated, the symmetry of these membranes, and their energy requirements.

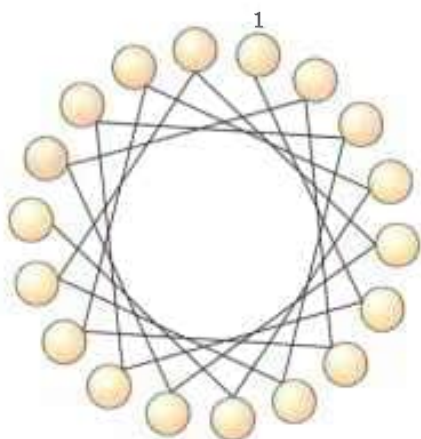
15. Membrane Permeability At pH 7, tryptophan crosses a lipid bilayer at about one-thousandth the rate of indole, a closely related compound:



Suggest an explanation for this observation.

16. Glucose Transporters A cell biologist working with cultured cells from intestinal epithelium finds that the cells take up glucose from the growth medium 10 times faster when the glucose concentration is 5 mM than when it is 0.2 mM. She also finds that glucose uptake requires Na^+ in the growth medium. What can you say about the glucose transporter in these cells?

17. Use of the Helical Wheel Diagram A helical wheel is a two-dimensional representation of a helix, a view along its central axis. Use the helical wheel diagram shown here to determine the distribution of amino acid residues in a helical segment with the sequence -Val-Asp-Arg-Val-Phe-Ser-Asn-Val-Cys-Thr-His-Leu-Lys-Thr-Leu-Gln-Asp-Lys-



What can you say about the surface properties of this helix? How would you expect the helix to be oriented in the tertiary structure of an integral membrane protein?

18. Synthesis of Gastric Juice: Energetics Parietal cells acidify gastric juice (pH 1.5) by pumping HCl from blood plasma (pH 7.4) into the stomach. How much ATP (in moles) is required to pump a mole of protons (H^+) against this concentration gradient? The free-energy change for ATP hydrolysis under cellular conditions is about -58 kJ/mol. Ignore the effects of the transmembrane electrical potential.

19. Electrogenic Transporters A single-cell organism, *Paramecium*, is large enough to allow the insertion of a microelectrode, permitting the measurement of the electrical potential between the inside of the cell and the surrounding medium (the membrane potential). The measured membrane potential is -50 mV (inside negative) in a living cell. What would happen if you added valinomycin to the surrounding medium, which contains K^+ and Na^+ ?

20. Energetics of the Na^+K^+ ATPase For a typical vertebrate cell with a membrane potential of -0.070 V (inside negative), what is the free-energy change for transporting 1 mol of Na^+ from the cell into the blood at 37 °C? Assume the Na^+ concentration is 12 mM inside the cell and 145 mM in blood plasma.

21. Action of Ouabain on Kidney Tissue Ouabain specifically inhibits the Na^+K^+ ATPase activity of animal tissues but is not known to inhibit any other enzyme. When ouabain is added to thin slices of living kidney tissue, it inhibits oxygen consumption by 66%. Why? What does this observation tell us about the use of respiratory energy by kidney tissue?

22. Digoxin to Inhibit Na^+K^+ ATPase The $\text{Na}^+\text{Ca}^{2+}$ exchanger expressed in cardiac myocytes is a bidirectional antiporter protein that removes calcium from the cytoplasm by exchanging it with sodium. Cardiac myocytes also express the Na^+K^+ ATPase. Suppose that a Na^+K^+ ATPase inhibitor (digoxin) is added to cardiac myocytes. Using your knowledge of the relative concentrations of ions (intracellular versus extracellular) and the important role of the Na^+K^+ ATPase in maintaining the electrochemical gradient, what change would you expect in the intracellular $[\text{Ca}^{2+}]$? Why?

23. Energetics of Symport Suppose you determined experimentally that a cellular transport system for glucose, driven by symport of Na^+ , could accumulate glucose to concentrations 25 times greater than in the external medium, while the external $[\text{Na}^+]$ was only 10 times greater than the intracellular $[\text{Na}^+]$. Would this violate the laws of thermodynamics? If not, how could you explain this observation?

24. Labeling the Lactose Transporter A bacterial lactose transporter, which is highly specific for lactose, contains a Cys residue that is essential to its transport activity. Covalent reaction of *N*-ethylmaleimide (NEM) with this Cys residue irreversibly inactivates the transporter. A high concentration of lactose in the medium prevents inactivation by NEM, presumably by sterically protecting the Cys residue, which is in or near the lactose-binding site. You know nothing else about the transporter protein. Suggest an experiment that might allow you to determine the M_r of this Cys-containing transporter polypeptide.

25. Transport Types You have just discovered a new L-alanine transporter in liver cells (hepatocytes). Poisoning hepatocytes with cyanide (which blocks ATP synthesis) reduces alanine transport by 90%. Tenfold reduction in extracellular $[\text{Na}^+]$ has no immediate effect on alanine transport. How would you use these observations to decide whether the alanine transporter is passive or active, primary or secondary?

26. Intestinal Uptake of Leucine You are studying the uptake of L-leucine by epithelial cells of the mouse intestine. Measurements of the rates of uptake of L-leucine and several of its analogs, with and without Na^+ in the assay buffer, yield the results given in the table. What can you conclude about the properties and mechanism of the leucine transporter? Would you expect L-leucine uptake to be inhibited by ouabain (see problem 21)?

Substrate	Uptake in presence of Na^+		Uptake in absence of Na^+	
	$V_{\max}$	$K_t(\text{mM})$	$V_{\max}$	$K_t(\text{mM})$
L-Leucine	420	0.24	23	0.2
D-Leucine	310	4.7	5	4.7
L-Valine	225	0.31	19	0.31

27. Ion Channel Selectivity Potassium channels consist of four subunits that form a channel just wide enough for K^+ ions to pass through. Although Na^+ ions are smaller (M_r 23, radius 0.95 Å) than K^+ ions (M_r 39, radius 1.33 Å), the potassium channels in the bacterium *Streptomyces lividans* transport 104 times more K^+ ions than Na^+ ions. What prevents Na^+ ions from passing through potassium channels?

28. Effect of an Ionophore on Active Transport Consider the leucine transporter described in Problem 26. Would V_{max} and/or K_t change if you added a Na^+ ionophore to the assay solution containing Na^+ ? Explain.

BIOCHEMISTRY ONLINE

29. Predicting Membrane Protein Topology I Online bioinformatics tools make hydropathy analysis easy if you know the amino acid sequence of a protein. At the Protein Data Bank (www.rcsb.org), the Protein Feature View displays additional information about a protein gleaned from other databases, such as UniProt and SCOP2. A simple graphical view of a hydropathy plot created using a window of 15 residues shows hydrophobic regions in red and hydrophilic regions in blue.

(a) Looking only at the displayed hydropathy plots in the Protein Feature View, what predictions would you make about the membrane topology of these proteins: glycophorin A (PDB ID 1AFO), myoglobin (PDB ID 1MBO), and aquaporin (PDB ID 2B6O)?

(b) Now, refine your information using the ProtScale tools at the ExPASy bioinformatics resource portal. Each of the PDB Protein Feature Views was created with a UniProt Knowledgebase ID. For glycophorin A, the UniProtKB ID is P02724; for myoglobin, P02185; and for aquaporin, Q6J8I9. Go to the ExPASy portal (<http://web.expasy.org/protscale>) and select the Kyte & Doolittle hydropathy analysis option, with a window of 7 amino acids. Enter the UniProtKB ID for aquaporin (Q6J8I9, which you can also get from the PDB's Protein Feature View page), then select the option to analyze the complete chain (residues 1 to 263). Use the default values for the other options and click Submit to get a hydropathy plot. Save a GIF image of this plot. Now repeat the analysis using a window of 15 amino acids. Compare the results for the 7-residue and 15-residue window analyses. Which window size gives you a better signal-to-noise ratio?

(c) Under what circumstances would it be important to use a narrower window?

30. Predicting Membrane Protein Topology II The epinephrine receptor in animal cells is an integral membrane protein (M_r 64,000) that is believed to have seven transmembrane α -helical regions.

(a) Show that a protein of this size is capable of spanning the membrane seven times.

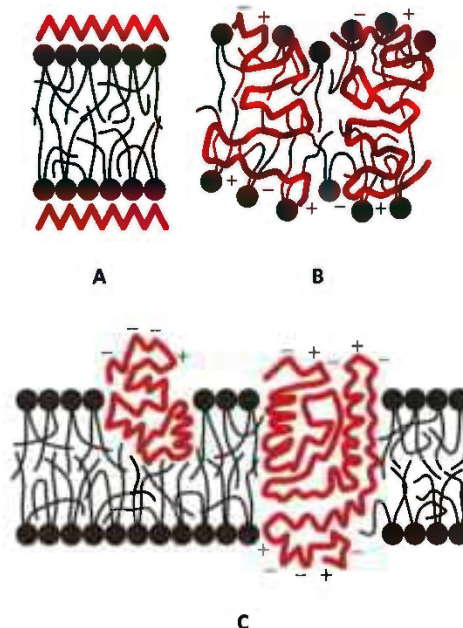
(b) Given the amino acid sequence of this protein, how would you predict which regions of the protein form the membrane-spanning helices?

(c) Go to the Protein Data Bank (www.rcsb.org). Use the PDB identifier 1DEP to retrieve the data page for a portion of the β -adrenergic receptor (one type of epinephrine receptor) isolated from turkey. Using JSmol to explore the structure, predict whether this portion of the receptor is located within the membrane or at the membrane surface. Explain your answer. Now use the Protein Feature View to see the hydrophobicity analysis of the sequence. Does this support your answer?

(d) Retrieve the data for a portion of another receptor, the acetylcholine receptor of neurons and myocytes, using the PDB identifier 1A11. As in (c), predict where this portion of the receptor is located and explain your answer.

DATA ANALYSIS PROBLEM

31. The Fluid Mosaic Model of Biological Membrane Structure Figure 11-3 shows the currently accepted fluid mosaic model of biological membrane structure. This model was presented in detail in a review article by S. J. Singer in 1971. In the article, Singer presented the three models of membrane structure that had been proposed up to that time:



A. The Davson-Danielli-Robertson Model. This was the most widely accepted model in 1971, when Singer's review was published. In this model, the phospholipids are arranged as a bilayer. Proteins are found on both surfaces of the bilayer, attached to it by ionic interactions between the charged head groups of the phospholipids and charged groups of the proteins. Crucially, there is no protein in the interior of the bilayer.

B. The Benson Lipoprotein Subunit Model. Here the proteins are globular and the membrane is a protein-lipid mixture. The hydrophobic tails of the lipids are embedded in the hydrophobic parts of the proteins. The lipid head groups are exposed to the solvent. There is no lipid bilayer.

C. The Lipid-Globular Protein Mosaic Model. This is the model shown in Figure 11-3. The lipids form a bilayer and proteins are embedded in it, some extending through the bilayer and others not. Proteins are anchored in the bilayer by interactions between the hydrophobic tails of the lipids and hydrophobic portions of the protein.

For the data given below, consider how each piece of information aligns with each of the three models of membrane structure. Which model(s) are supported, which are not supported, and what reservations do you have about the data or their interpretation? Explain your reasoning.

(a) When cells were fixed, stained with osmium tetroxide, and examined in the electron microscope, the membranes showed

a "railroad track" appearance, with two dark-staining lines separated by a light space.

(b) The thickness of membranes in cells fixed and stained in the same way was found to be 5 to 9 nm. The thickness of a "naked" phospholipid bilayer, without proteins, was 4 to 4.5 nm. The thickness of a single monolayer of proteins was about 1 nm.

(c) Singer wrote in his article: "The average amino acid composition of membrane proteins is not distinguishable from that of soluble proteins. In particular, a substantial fraction of the residues is hydrophobic" (p. 165).

(d) As described in Problems 1 and 2 of this chapter, researchers had extracted membranes from cells, extracted the lipids, and compared the area of the lipid monolayer with the area of the original cell membrane. The interpretation of the results was complicated by the issue illustrated in the graph of Problem 1: the area of the monolayer depended on how hard it was pushed. With very light pressures, the ratio of monolayer area to cell membrane area was about 2.0. At higher pressures—thought to be more like those found in cells—the ratio was substantially lower.

(e) Circular dichroism spectroscopy uses changes in polarization of UV light to make inferences about protein secondary structure (see Fig. 4-9). On average, this technique showed that membrane proteins have a large amount of α helix

and little or no β sheet. This finding was consistent with most membrane proteins having a globular structure.

(f) Phospholipase C is an enzyme that removes the polar head group (including the phosphate) from phospholipids. In several studies, treatment of intact membranes with phospholipase C removed about 70% of the head groups without disrupting the "railroad track" structure of the membrane.

(g) Singer described in his article a study in which "a glycoprotein of molecular weight about 31,000 in human red blood cell membranes is cleaved by tryptic treatment of the membranes into soluble glycopeptides of about 10,000 molecular weight, while the remaining portions are quite hydrophobic" (p. 199). Trypsin treatment did not cause gross changes in the membranes, which remained intact.

Singer's review also included many more studies in this area. In the end, though, the data available in 1971 did not conclusively prove Model C was correct. As more data have accumulated, this model of membrane structure has been accepted by the scientific community.

Reference

Singer, S.J. 1971. The molecular organization of biological membranes. In *Structure and Function of Biological Membranes* (L.I. Rothfield, ed.), pp. 145–222. New York: Academic Press.

BIOCHEMICAL SIGNALING

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The ability of cells to receive and act on signals from outside the plasma membrane is fundamental to life. Cells and organisms constantly sample the surrounding medium for nutrients, oxygen, light, and sexual partners, and for the presence of noxious chemicals, predators, or competitors for food. These signals elicit appropriate responses, such as motion toward food or away from toxic substances. In multicellular organisms, cells exchange hundreds of signals—neurotransmitters, hormones, key metabolites—which trigger appropriate responses in such cellular activities as metabolism, cell division, embryonic growth and development, movement, and defense. In all these cases, the signal represents *information* that is detected by specific receptors and converted to a cellular response, which always involves a *chemical* process. This conversion of information into a chemical change, **signal transduction**, is a universal property of living cells.

In this chapter, we present examples of specific biological signaling systems from which we have acquired our current understanding of the biochemistry

of signal transduction in animals. We emphasize the following principles:

- P1** Cells respond to external signals through receptor-mediated processes that amplify the signal, integrate it with input from other receptors, transmit the signal to the appropriate effectors, and eventually end the response.
- P2** There is a high degree of evolutionary conservation of signaling proteins and transduction mechanisms across the animal phyla. At least a billion years of evolution have passed since the plant and animal branches of the eukaryotes diverged, which is reflected in the differences in signaling mechanisms used in the two kingdoms. We focus here on the animal kingdom.
- P3** In multicellular animals, GPCRs with seven transmembrane helices are the largest group of plasma membrane receptors. These G Protein–Coupled Receptors act through G proteins, which are turned on when they bind guanosine triphosphate (GTP). Animals have hundreds of different GPCRs, each able to respond to a specific signal.
- P4** Plasma membrane receptors with an intracellular tyrosine kinase activity act through cascades of protein kinases to transduce signals about the metabolic state, including growth factors.
- P5** Phosphorylation of intrinsically disordered regions of signaling proteins acts as a switch, toggling enzyme activity, or creating binding sites for other molecules. Signal responses are integrated by multiprotein signaling complexes with modular domains that bind phosphorylated Tyr, Ser, or Thr residues.
- P3** Ion channels gated by membrane potential or ligands are central to signaling in all cells, including bacteria, plants, and animals.

- P7** Some hormone signals act inside the cell, not at the plasma membrane, forming complexes with specific receptor proteins that regulate gene expression.
- P8** Cells receive extracellular signals that determine progress through the cell division cycle, or trigger cell death—processes regulated by phosphorylation and dephosphorylation of key regulatory proteins.
- P9** Defective signaling proteins or defective regulation of their synthesis and breakdown can disrupt cell cycle regulation and lead to tumor formation (cancer).

12.1 General Features of Signal Transduction

Signal transductions are remarkably specific and exquisitely sensitive. **Specificity** is achieved by precise molecular complementarity between the signal and receptor molecules (Fig. 12-1a), mediated by the same kinds of weak (noncovalent) forces that mediate enzyme-substrate and antigen-antibody interactions. Multicellular organisms have an additional level of specificity, because the receptors for a given signal, or the intracellular targets of a given signal pathway, are present only in certain cell types. Thyrotropin-releasing hormone, for example, triggers responses in the cells of the anterior pituitary but not in hepatocytes, which lack receptors for this hormone. Epinephrine alters glycogen metabolism in hepatocytes but not in adipocytes; in this case, both cell types have receptors for the hormone, but whereas hepatocytes contain glycogen and the glycogen-metabolizing enzyme that is stimulated by epinephrine, adipocytes contain neither.

Signal-Transducing Systems Share Common Features

The extraordinary **sensitivity** of signal transduction results from the high affinity of signal receptors for their ligands (Fig. 12-1b). This affinity can be expressed as the dissociation constant, K_d , for the receptor-ligand interaction. K_d is commonly 10^{-7} M or less, meaning that the receptor detects micromolar to nanomolar concentrations of a signaling ligand. In some cases, cooperativity in receptor-ligand interactions results in relatively large changes in receptor activation with small changes in ligand concentration, further increasing the sensitivity of signal detection.

P1 **Amplification** results when an enzyme is activated by a signal receptor and, in turn, catalyzes the activation of many molecules of a second enzyme, each of which activates many molecules of a third enzyme and so on, in an **enzyme cascade** (Fig. 12-1c). Such cascades

can produce amplifications of several orders of magnitude within milliseconds. The response to a signal must also be terminated, such that the downstream effects are in proportion to the strength of the original stimulus.

Interacting signaling proteins are **modular**. Many signaling proteins have multiple domains that recognize specific features in several other proteins, or in the cytoskeleton or plasma membrane. **P5** This modularity allows a cell to mix and match a set of signaling molecules to create a wide variety of multienzyme complexes with different functions or cellular locations. One common theme in these interactions is the binding of one modular signaling protein to *phosphorylated* residues in another protein; the resulting interaction can be regulated by phosphorylation or dephosphorylation of the protein partner (Fig. 12-1d). **Nonenzymatic scaffold proteins** with affinity for several enzymes that interact in cascades bring these enzymes together, ensuring that they interact at specific cellular locations and at specific times. Many of the regions involved in protein-protein interactions are intrinsically disordered (see Fig. 4-20), capable of folding differently depending on which protein they interact with. As a result, a single protein can have multiple functions in signaling pathways.

The sensitivity of receptor systems is subject to modification. When a signal is present continuously, the receptor system becomes **desensitized** (Fig. 12-1e), so that it no longer responds to the signal. When the stimulus falls below a certain threshold, the system again becomes sensitive. Think of what happens to your visual transduction system when you walk from bright sunlight into a darkened room or from darkness into the light.

Signal integration (Fig. 12-1f) is the ability of the system to receive multiple signals and produce a unified response appropriate to the combined needs of the cell or organism. Different signaling pathways converse with each other at several levels, generating complex cross talk that maintains homeostasis in the cell and the organism.

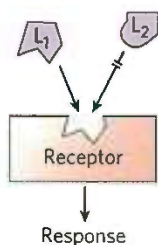
Signaling pathways are often **divergent**—branched rather than linear; one stimulus acts through one receptor, which activates two or more pathways with different downstream targets and responses (Fig. 12-1g).

A final noteworthy feature of signal-transducing systems is **response localization** within a cell (Fig. 12-1h). When the components of a signaling system are confined to a specific subcellular structure (a raft in the plasma membrane, for example), a cell can regulate a process locally, without affecting distant regions of the cell.

One of the revelations of research on signaling is the remarkable degree to which signaling mechanisms have been conserved during evolution. **P5** Although the number of different biological signals to which cells respond is probably in the thousands (Table 12-1 lists a few important types), and the kinds of responses elicited

(a) Specificity

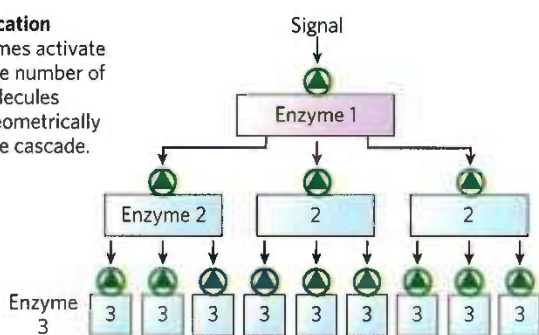
Signaling ligand fits binding site on its complementary receptor; other ligands do not fit.

**(b) Sensitivity**

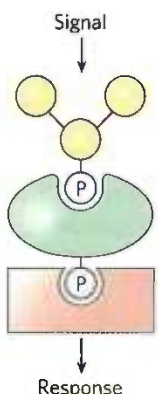
Receptor has high affinity (low K_d) for signaling ligand (L).

**(c) Amplification**

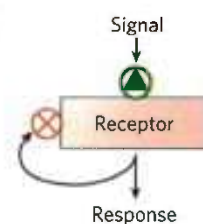
When enzymes activate enzymes, the number of affected molecules increases geometrically in an enzyme cascade.

**(d) Modularity**

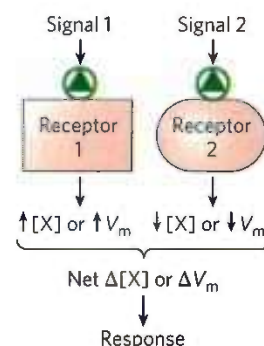
Proteins with multivalent affinities form diverse signaling complexes from interchangeable parts. Phosphorylation provides reversible points of interaction.

**(e) Desensitization/Adaptation**

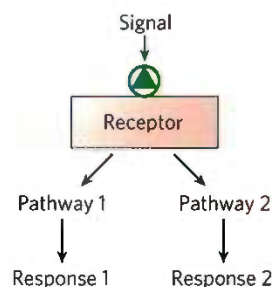
Receptor activation triggers a feedback circuit that shuts off the receptor or removes it from the cell surface.

**(f) Integration**

When two signals have opposite effects on a metabolic characteristic such as the concentration of a second messenger X, or the membrane potential V_m , the regulatory outcome results from the integrated input from both receptors.

**(g) Divergence**

When a receptor is activated by a signal, it activates two or more pathways with different end effects.

**(h) Localized response**

When the enzyme that destroys an intracellular message is clustered with the message producer, the message is degraded before it can diffuse to distant points, so the response is only local and brief.

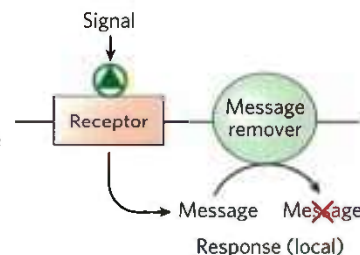


FIGURE 12-1 Eight features of signal-transducing systems.

TABLE 12-1 Some Signals to Which Cells Respond

Antigens	Mechanical touch
Cell surface glycoproteins, oligosaccharides	Microbial, insect pathogens
Developmental signals	Neurotransmitters
Extracellular matrix components	Nutrients
Growth factors	Odorants
Hormones	Pheromones
Hypoxia	Tastants
Light	

by these signals are comparably numerous, the machinery for transducing all of these signals is built from about 10 basic types of protein components (Table 12-2).

The General Process of Signal Transduction in Animals Is Universal

In this chapter we consider the molecular details of several representative signal-transduction systems, classified according to the type of receptor. The trigger for each system is different, but **P1** the general features of signal transduction are common to all: a signal (ligand) interacts with a receptor; the activated receptor

TABLE 12-2 Some Conserved Elements of Animal Signaling Systems

Plasma membrane receptors with 7 transmembrane (7tm) helices
G proteins that bind GTP or GDP and interface with membrane receptors
Membrane enzymes with cyclic nucleotides as substrates or products
Protein kinases that phosphorylate GPCR receptors
Membrane protein tyrosine kinases
Cyclic nucleotide-dependent protein kinases
Ca ²⁺ -binding proteins
Ca ²⁺ -dependent protein kinases
Protein kinases that are activated during cell division
Nonenzymic protein scaffolds that bring modules together

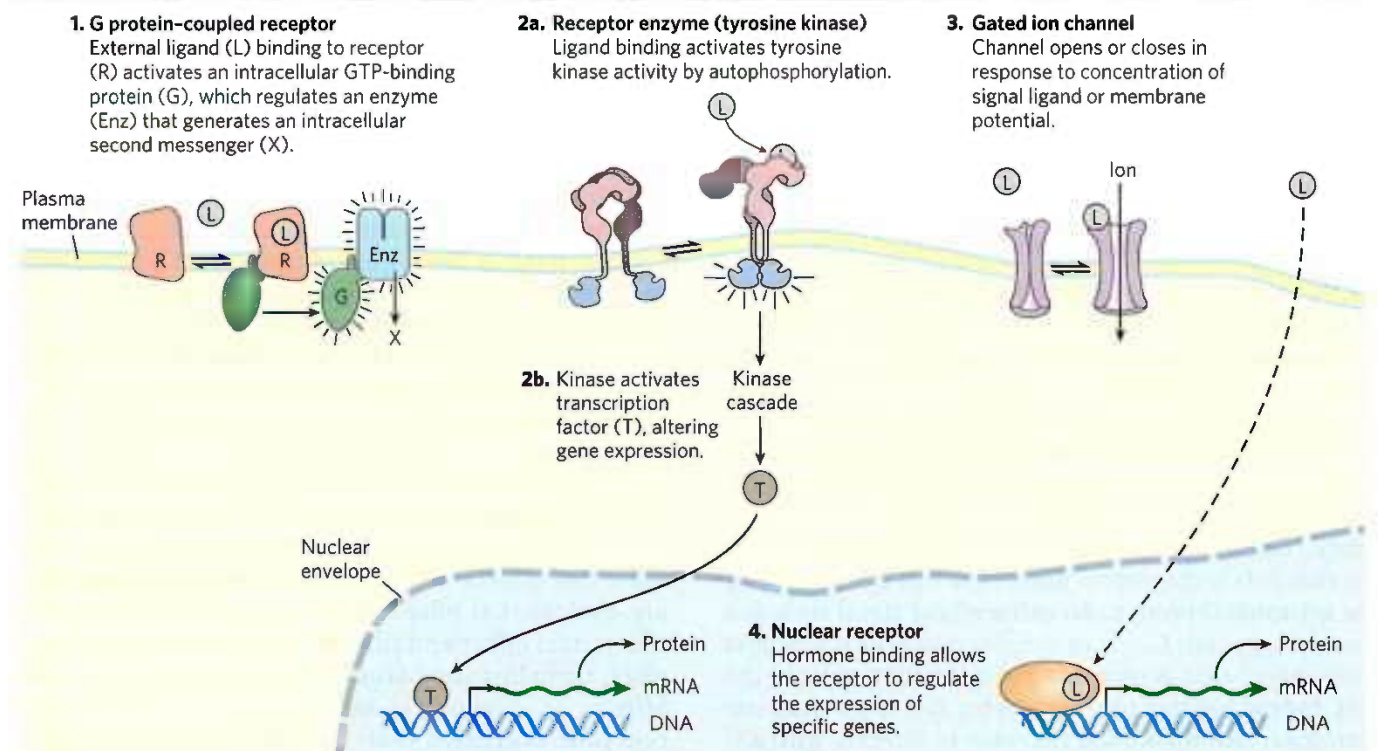
interacts with cellular machinery, producing a second signal or a change in the activity of a cellular protein; the metabolic activity of the target cell undergoes a change; and finally, the transduction event ends. To illustrate these general features of signaling systems, we will look at examples of four basic receptor types (Fig. 12-2).

1. *G protein-coupled receptors* that indirectly activate (through GTP-binding proteins, or G proteins) enzymes that generate intracellular second messengers. Examples include the β -adrenergic receptor

system that responds to epinephrine (Section 12.2) and the vision, olfaction, and gustation systems (Section 12.3).

2. *Receptor enzymes* in the plasma membrane that have an enzymatic activity on the cytoplasmic side, triggered by ligand binding on the extracellular side. Receptors with tyrosine kinase activity, for example, catalyze the phosphorylation of Tyr residues in specific intracellular target proteins. Examples include the insulin receptor (Section 12.4) and receptor guanylyl cyclases (see Box 12-2).
3. *Gated ion channels* of the plasma membrane that open and close (hence the term “gated”) in response to the binding of chemical ligands or changes in transmembrane potential. These are the simplest signal transducers (Section 12.6).
4. *Nuclear receptors* that bind specific ligands (such as the hormone estrogen) and alter the rate at which specific genes are transcribed and translated into cellular proteins. Because steroid hormones function through mechanisms intimately related to the regulation of gene expression, we consider them briefly in Section 12.7 and defer a detailed discussion of their action until Chapters 23 and 28.

As we begin this discussion of biological signaling, a word about the nomenclature of signaling proteins is in order. These proteins are typically discovered in one context and named accordingly, then prove to be involved in

**FIGURE 12-2** Four general types of signal transducers.

a broader range of biological functions for which the original name is not helpful. For example, the retinoblastoma protein, pRb, was initially identified as the product of a mutation that contributes to cancer of the retina (retinoblastoma), but it is now known to function in many pathways essential to cell division in all cells, not just those of the retina. Some genes and proteins are given noncommittal names: the tumor suppressor protein p53, for example, is a protein of 53 kDa, but its name gives no clue to its great importance in the regulation of cell division and the development of cancer. In this chapter we generally define these protein names as we encounter them, introducing the names that are commonly used by researchers in the field.

SUMMARY 12.1 General Features of Signal Transduction

- All cells have specific and highly sensitive signal-transducing mechanisms, which have been conserved during evolution.
- A wide variety of stimuli act through specific protein receptors in the plasma membrane.
- In all types of signal transduction, receptors bind the signal ligand and initiate a process that amplifies the signal, integrates it with input from other receptors, and transmits the information throughout the cell. If the signal persists, receptor desensitization reduces or ends the response.
- Multicellular organisms have four general types of signaling mechanisms: plasma membrane proteins that act through G proteins, receptors with internal enzyme activity (such as tyrosine kinase), gated ion channels, and nuclear receptors that bind steroids and alter gene expression.

12.2 G Protein–Coupled Receptors and Second Messengers

As their name implies, **G protein–coupled receptors (GPCRs)** are receptors that act through a member of the **guanosine nucleotide–binding protein**, or **G protein**, family. **P3** Three essential components define signal transduction through GPCRs: a plasma membrane receptor with seven transmembrane helical segments, a G protein that cycles between active (guanosine triphosphate (GTP)-bound) and inactive (guanosine diphosphate (GDP)-bound) forms, and an effector enzyme (or ion channel) in the plasma membrane that is regulated by the activated G protein. An extracellular signal such as a hormone, growth factor, or neurotransmitter is the “first messenger” that activates a receptor from outside the cell. Ligand binding to the receptor forces an allosteric transition that allows the receptor to interact with a G

protein, causing it to exchange its bound GDP for a GTP from the cytosol. The G protein then dissociates from the activated receptor and binds to the nearby effector enzyme, altering its activity. The effector enzyme then causes a change in the cytosolic concentration of a low molecular weight metabolite (such as 3',5'-cyclic AMP) or inorganic ion (Ca^{2+}), which acts as a **second messenger** to activate or inhibit one or more downstream targets, often protein kinases.



The human genome encodes just over 800 GPCRs, about 350 for detecting hormones, growth factors, and other endogenous ligands, and up to 500 that serve as olfactory (smell) and gustatory (taste) receptors. The largest superfamily of proteins encoded in the human genome, GPCRs have been implicated in many common medical conditions, including allergies, depression, blindness, diabetes, and various cardiovascular defects. GPCR mutations are also found in 20% of all cancers. In the United States, more than a third of *all* pharmaceuticals on the market target a GPCR. For example, the β -adrenergic receptor, which mediates the effects of epinephrine, is the target of the “beta blockers,” prescribed for such diverse conditions as hypertension, cardiac arrhythmia, glaucoma, anxiety, and migraine headache. More than 100 of the GPCRs found in the human genome are still “orphan receptors,” meaning that their natural ligands are not yet identified, and so we know little about their biology. The β -adrenergic receptor, with well-understood biology and pharmacology, is the prototype for all GPCRs, and our discussion of signal-transducing systems begins there. ■

The β -Adrenergic Receptor System Acts through the Second Messenger cAMP

Epinephrine released from the adrenal glands sounds the alarm when a threat requires an animal to mobilize its energy-generating machinery; it signals the need to fight or flee. Epinephrine action begins when the hormone binds to a protein receptor in the plasma membrane of an epinephrine-sensitive cell (a myocyte in muscle, for example). **Adrenergic receptors** (“adrenergic” reflects the alternative name for epinephrine, adrenaline) are of four general types, α_1 , α_2 , β_1 , and β_2 , defined by differences in their affinities and responses to a group of agonists and antagonists. **Agonists** are molecules (natural ligands or their structural analogs) that bind a receptor and produce the effects of the natural ligand; **antagonists** are analogs that bind the receptor without triggering the normal effect and thereby block the effects of agonists, including the natural ligand. In some cases, the affinity of a synthetic agonist or antagonist for the receptor is greater than that of the natural agonist

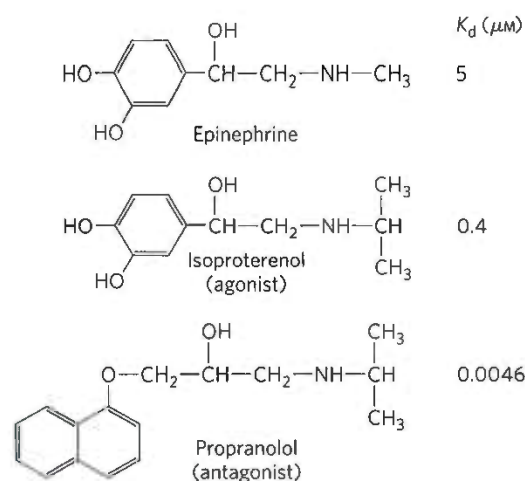


FIGURE 12-3 Epinephrine and its synthetic analogs. Epinephrine regulates energy-yielding metabolism in muscle, liver, and adipose tissue. Its affinity for its receptor is expressed as a dissociation constant for the receptor–ligand complex. Isoproterenol and propranolol are synthetic analogs, one an agonist with an affinity for the receptor that is higher than that of epinephrine, and the other an antagonist with extremely high affinity.

(Fig. 12-3). The four types of adrenergic receptors are found in different target tissues and mediate different responses to epinephrine. Here we focus on the **β -adrenergic receptors** of muscle, liver, and adipose tissue. These receptors mediate changes in fuel metabolism, as described in Chapter 23, including the increased breakdown of glycogen and fat. Adrenergic receptors of the β_1 and β_2 subtypes act through the same mechanism, so in our discussion, “ β -adrenergic” applies to both types.

P3 Like all GPCRs, the β -adrenergic receptor is an integral protein with seven hydrophobic, helical regions of 20 to 28 amino acid residues that span the plasma membrane seven times, thus the alternative names for GPCRs: **seven-transmembrane (7tm)** or **heptahelical receptors**. GPCRs effect signal transduction through interaction with **heterotrimeric G proteins**, a conserved family of signaling proteins with three subunits, α , β , and γ . The binding site for GDP or GTP is on the α subunit. When GDP is bound, the G protein is in its trimeric, inactive form.

GPCRs are allosteric proteins. When epinephrine binds to the β -adrenergic receptor (Fig. 12-4a, step 1), allosteric transitions in the receptor and its associated G protein favor the replacement of GDP with GTP. Thus the hormone-bound GPCR acts as a **guanosine nucleotide-exchange factor (GEF)**. In this active form, the G protein (step 2) can transmit the signal from the activated receptor to the downstream effector protein, adenylyl cyclase. Because this G protein stimulates its effector, it is referred to as a **stimulatory G protein**, or G_s . In the active form, the β and γ subunits of G_s dissociate from the α subunit as a $\beta\gamma$ dimer, and $G_{s\alpha}$, with its bound GTP,

moves in the plane of the membrane from the receptor to a nearby molecule of adenylyl cyclase (step 3).

Adenylyl cyclase is an integral protein of the plasma membrane, with its active site on the cytoplasmic face. The association of active $G_{s\alpha}$ with adenylyl cyclase stimulates the cyclase to catalyze the synthesis of second messenger cAMP from ATP (Fig. 12-4a, step 4; Fig. 12-4b), raising the cytosolic [cAMP]. The interaction between $G_{s\alpha}$ and adenylyl cyclase occurs only when $G_{s\alpha}$ is bound to GTP.

The stimulation by $G_{s\alpha}$ is self-limiting; $G_{s\alpha}$ has *intrinsic GTPase activity that switches $G_{s\alpha}$ to its inactive form* by converting its bound GTP to GDP (Fig. 12-5). The inactive $G_{s\alpha}$ dissociates from adenylyl cyclase, rendering the cyclase inactive. $G_{s\alpha}$ reassociates with the $\beta\gamma$ dimer ($G_{s\beta\gamma}$), and inactive G_s is again available to interact with a hormone-bound receptor.

Cyclic AMP Activates Protein Kinase A

Epinephrine exerts its downstream effects through the increase in [cAMP] that results from activation of adenylyl cyclase. Cyclic AMP, the second messenger, allosterically activates **cAMP-dependent protein kinase**, also called **protein kinase A** or **PKA** (Fig. 12-4a, step 5), which catalyzes the phosphorylation of specific Ser or Thr residues of targeted proteins, such as glycogen phosphorylase *b* kinase. The latter enzyme is active when phosphorylated and can begin the process of mobilizing glycogen stores in muscle and liver in anticipation of the need for energy, as signaled by epinephrine.

The inactive form of PKA has two identical catalytic subunits (C) and two identical regulatory subunits (R) (Fig. 12-6a). The tetrameric R_2C_2 complex is catalytically inactive, because an autoinhibitory domain of each R subunit occupies the substrate-binding cleft of each C subunit. The autoinhibitory domain is an intrinsically disordered region that can fold to fit any of several protein partners. Cyclic AMP is an allosteric activator of PKA. When cAMP binds to the R subunits, they undergo a conformational change that moves the autoinhibitory domain of R out of the catalytic domain of C, and the R_2C_2 complex dissociates to yield two free, catalytically active C subunits. **P5** This same basic mechanism—displacement of an autoinhibitory domain—mediates the allosteric activation of many types of protein kinases by their second messengers (as in Figs 12-21 and 12-29, for example). The structure of the substrate-binding cleft in PKA is the prototype for all known protein kinases (Fig. 12-6b); certain residues in this cleft region have identical counterparts in all of the 544 protein kinases encoded in the human genome. The ATP-binding site of each catalytic subunit positions ATP perfectly for the transfer of its

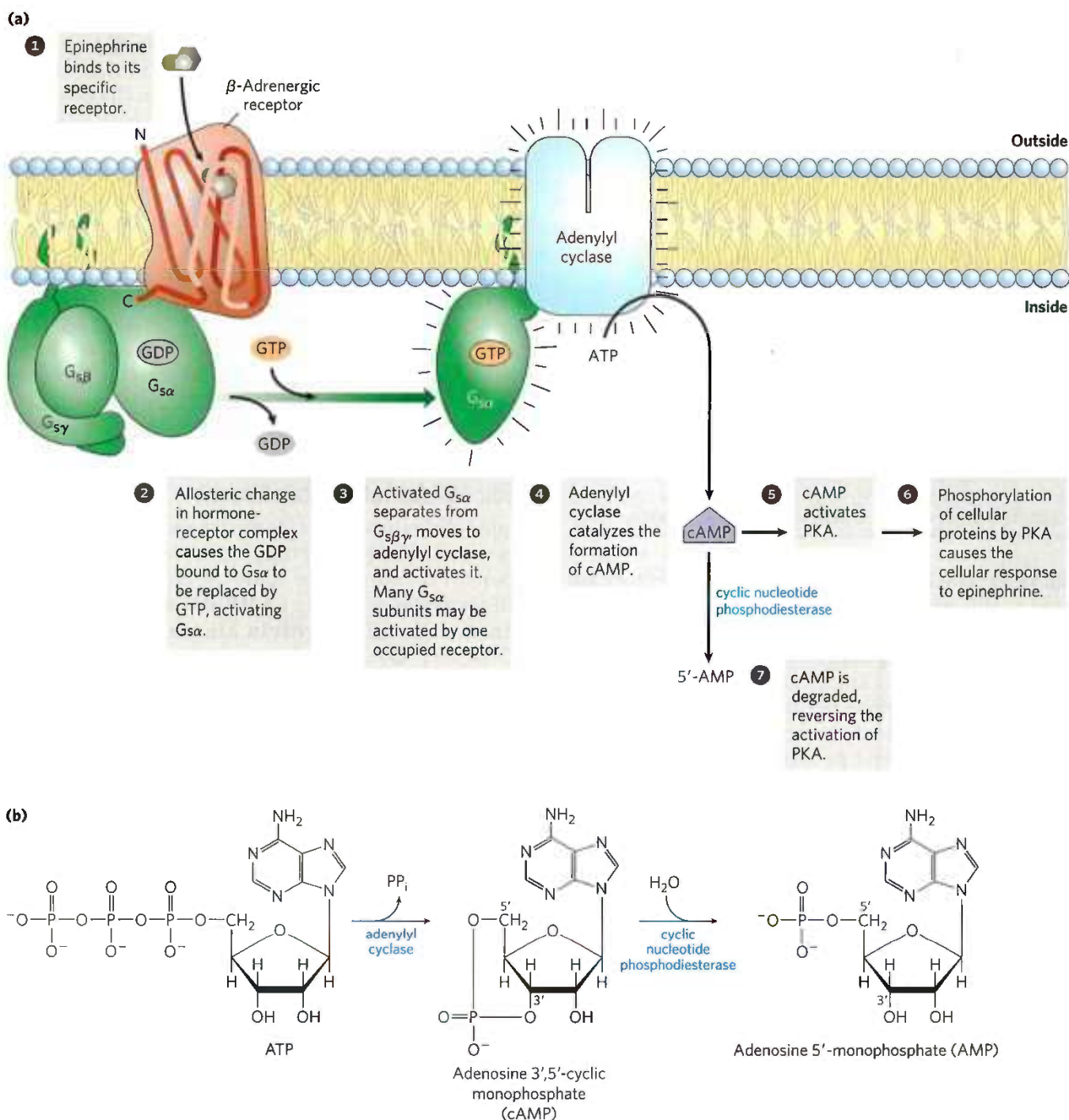


FIGURE 12-4 Transduction of the epinephrine signal: the β -adrenergic pathway. (a) The mechanism that couples binding of epinephrine to its receptor with activation of adenylyl cyclase; the seven steps are discussed in the text. The same adenylyl cyclase molecule in the plasma membrane may be regulated by a stimulatory G protein (G_s), as shown, or by

an inhibitory G protein (G_i , not shown). G_s and G_i are under the influence of different hormones. Hormones that induce GTP binding to G_i cause *inhibition* of adenylyl cyclase, resulting in lower cellular [cAMP]. (b) The combined action of the enzymes that catalyze steps 4 and 7, synthesis and hydrolysis of cAMP by adenylyl cyclase and cAMP phosphodiesterase, respectively.

terminal (γ) phosphoryl group to the —OH in the side chain of a Ser or Thr residue in the target protein.

As indicated in Figure 12-4a (step 6), PKA regulates many enzymes downstream in the signaling pathway. Although these downstream targets have diverse functions, they share a region of sequence similarity

around the Ser or Thr residue that undergoes phosphorylation, a sequence that marks them for regulation by PKA. The substrate-binding cleft of PKA recognizes these sequences and phosphorylates their Thr or Ser residue. Comparison of the sequences of various protein substrates for PKA has yielded the short **consensus**

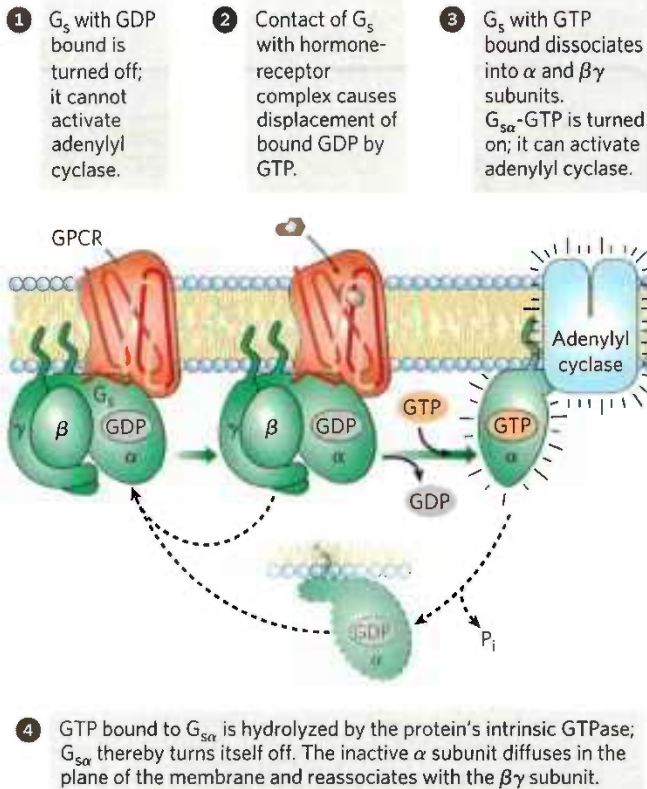


FIGURE 12-5 The GTPase switch. G proteins cycle between GDP-bound (off) and GTP-bound (on). The protein's GTPase activity, in many cases stimulated by RGS proteins (regulators of G-protein signaling; see Fig. 12-8), determines how quickly bound GTP is hydrolyzed to GDP and thus how long the G protein remains active.

sequence in which the target for phosphorylation (Ser or Thr) is most commonly embedded. For PKA, the consensus sequence is $xR[RK]x[ST]B$, where x can be any residue, R is Arg, $[RK]$ can be either Arg or Lys, $[ST]$ is either Ser or Thr (which is the residue phosphorylated), and B is any basic residue. The consensus sequences of a number of protein kinases are shown in Table 6-10; they define the targets of the hundreds of protein kinases in the eukaryotic cell.

How do we know that two proteins interact, or where in the cell they interact? Changes in the state of association of two proteins (such as the R and C subunits of PKA) can be seen by measuring the nonradiative transfer of energy between fluorescent probes attached to each protein, a technique called fluorescence resonance energy transfer (FRET), which we describe in **Box 12-1**.

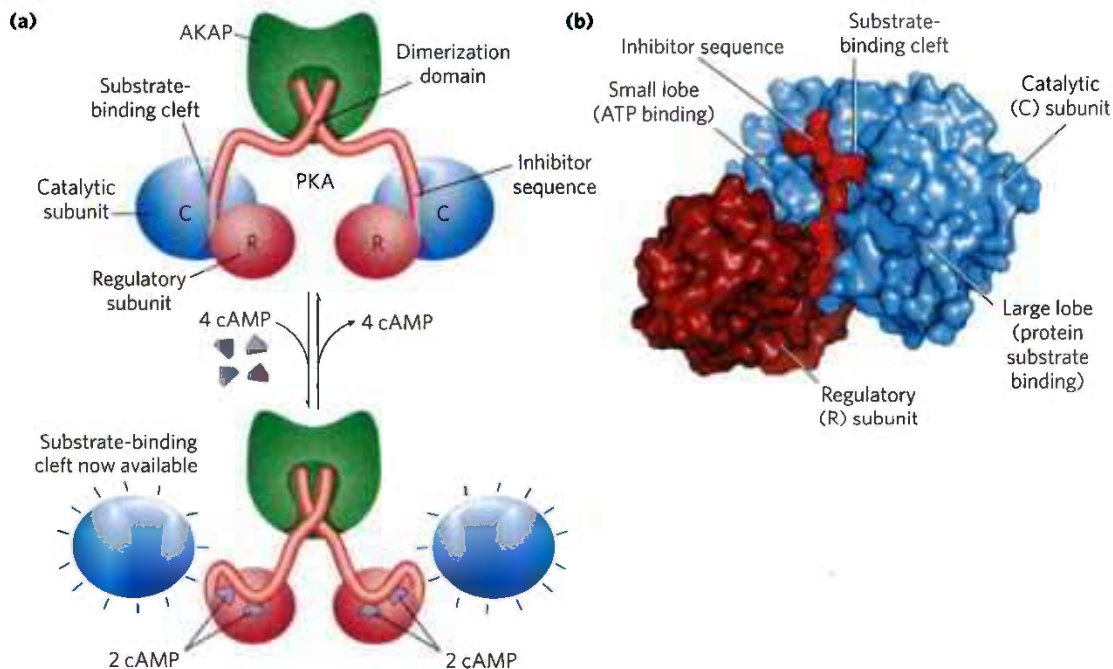


FIGURE 12-6 Activation of cAMP-dependent protein kinase (PKA). (a) When $[cAMP]$ is low, the two identical regulatory subunits (R ; red) associate with the two identical catalytic subunits (C ; blue). In this R_2C_2 complex, the inhibitor sequences of the R subunits lie in the substrate-binding cleft of the C subunits and prevent binding of protein substrates; the complex is therefore catalytically inactive. The amino-terminal sequences of the R subunits interact to form an R_2 dimer, the site of binding to an A kinase anchoring protein (AKAP; Fig. 12-11). When $[cAMP]$ rises in response to a hormonal signal, each R subunit binds two $cAMP$ molecules and undergoes a dramatic reorganization that pulls its inhibitory

sequence away from the C subunit, opening up the substrate-binding cleft and releasing each C subunit in its catalytically active form. (b) A crystal structure showing part of the R_2C_2 complex — one C subunit and part of one R subunit. The amino-terminal dimerization region of the R subunit is omitted for simplicity. The small lobe of C contains the ATP-binding site, and the large lobe surrounds and defines the cleft where the protein substrate binds and undergoes phosphorylation at a Ser or Thr residue, with a phosphoryl group transferred from ATP. In this inactive form, the inhibitor sequence of R blocks the substrate-binding cleft of C , inactivating it. [(b) Data from PDB ID 3FHI, C. Kim et al., *Science* 307:690, 2005.]

FRET: Biochemistry in a Living Cell

Fluorescent probes are commonly used to detect biochemical changes in single living cells on a nanosecond time scale. In one widely used procedure, **green fluorescent protein (GFP)** and variants with different fluorescence spectra, described in Chapter 9 (see Fig. 9-16), are genetically fused with another protein. These fluorescent hybrid proteins act as spectroscopic rulers for measuring distances between interacting proteins within a cell. They can be used indirectly as measures of local concentrations of compounds that change the distance between two proteins.

An excited fluorescent molecule such as GFP or YFP (yellow fluorescent protein) can dispose of the energy from the absorbed photon in either of two ways: (1) by fluorescence, emitting a photon of slightly longer wavelength (lower energy) than the exciting light, or (2) by nonradiative **fluorescence resonance energy transfer (FRET)**, in which the energy of the excited molecule (the donor) passes directly to a nearby molecule (the acceptor) *without emission of a photon*, exciting the acceptor (Fig. 1). The acceptor can now decay to its ground state by fluorescence; the emitted photon has a longer wavelength (lower energy) than both the original exciting light and the fluorescence emission of the donor. This second mode of decay (FRET) is possible only when donor and acceptor are close to each other (within 1 to 50 Å); the efficiency of FRET is inversely proportional to the *sixth power* of the distance between donor and acceptor. Thus, very small changes in the distance between donor and acceptor register as very large changes in FRET, measured as the fluorescence of the acceptor molecule when the donor is excited. With sufficiently sensitive light detectors, this fluorescence signal can be located to specific regions of a single, living cell.

FRET has been used to measure [cAMP] in living cells. The gene for BFP (blue fluorescent protein) is fused with that for the regulatory subunit (R) of cAMP-dependent protein kinase (PKA), and the gene for GFP is fused with that for the catalytic subunit (C) (Fig. 2). When these two hybrid proteins are expressed in a cell, BFP (donor) and GFP (acceptor) in the inac-

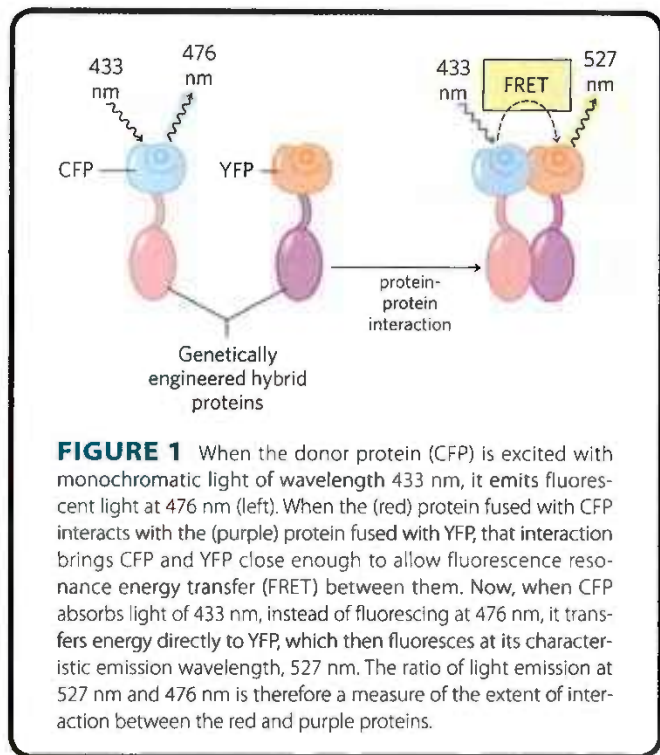


FIGURE 1 When the donor protein (CFP) is excited with monochromatic light of wavelength 433 nm, it emits fluorescent light at 476 nm (left). When the (red) protein fused with CFP interacts with the (purple) protein fused with YFP, that interaction brings CFP and YFP close enough to allow fluorescence resonance energy transfer (FRET) between them. Now, when CFP absorbs light of 433 nm, instead of fluorescing at 476 nm, it transfers energy directly to YFP, which then fluoresces at its characteristic emission wavelength, 527 nm. The ratio of light emission at 527 nm and 476 nm is therefore a measure of the extent of interaction between the red and purple proteins.

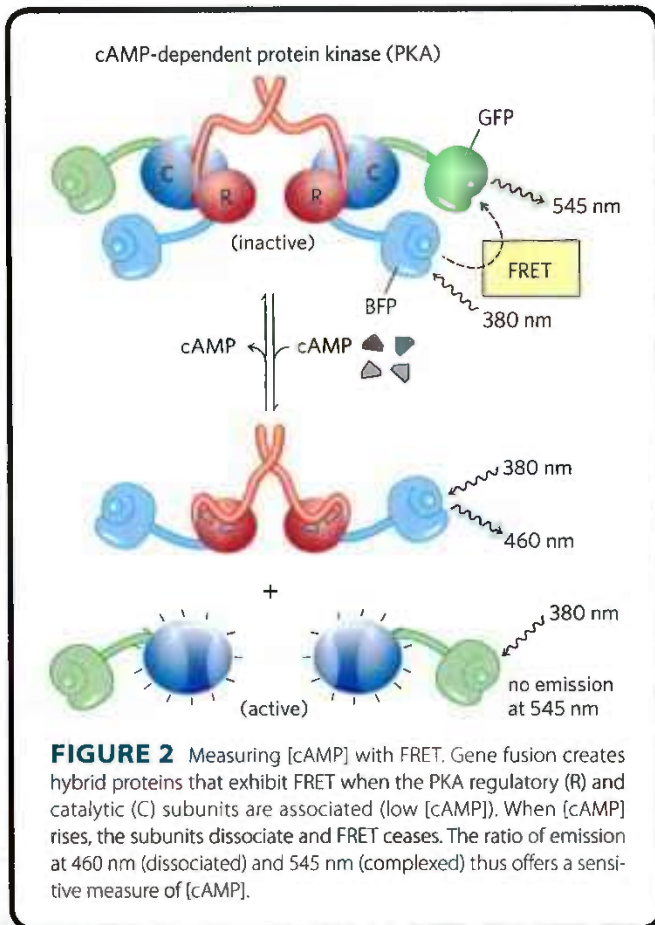
tive PKA (R_2C_2 tetramer) are close enough to undergo FRET. Wherever in the cell [cAMP] increases, the R_2C_2 complex dissociates into R_2 and 2 C and the FRET signal is lost, because donor and acceptor are now too far apart for efficient FRET. Viewed in the fluorescence microscope, the region of higher [cAMP] has a minimal GFP signal and higher BFP signal. Measuring the ratio of emission at 460 nm and 545 nm gives a sensitive measure of the change in [cAMP]. By determining this ratio for all regions of the cell, the investigator can generate a false color image of the cell in which the ratio, or relative [cAMP], is represented by

P1 As in many signaling pathways, signal transduction by the β -adrenergic receptor and adenylyl cyclase entails several steps that *amplify* the original hormone signal (Fig. 12-7). First, the binding of one hormone molecule to one receptor molecule catalytically activates many G_s molecules that associate with the activated receptor, one after the other. Next, by activating one molecule of adenylyl cyclase, each active G_{sa} molecule stimulates the catalytic synthesis of *many* molecules of cAMP. The second messenger cAMP now activates PKA, each molecule of which catalyzes the phosphorylation of *many* molecules of the target protein—phosphorylase *b* kinase in Figure 12-7. This kinase activates glycogen phosphorylase *b*, which leads to the rapid mobilization of glucose from glycogen. The

net effect of the cascade is amplification of the hormonal signal by orders of magnitude, which accounts for the very low concentration of epinephrine (or any other hormone) required for hormone activity. This signaling pathway is also rapid: the signal leads to intracellular changes within fractions of a second.

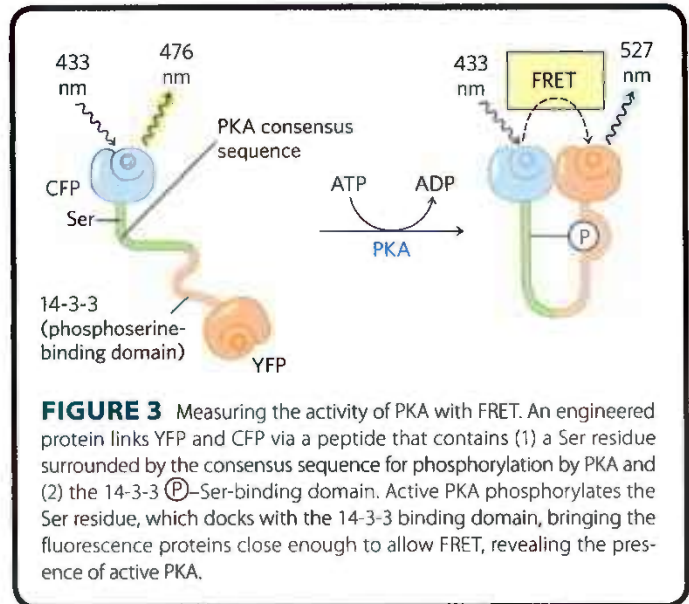
Several Mechanisms Cause Termination of the β -Adrenergic Response

P1 To be useful, a signal-transducing system has to *turn off* after the hormonal or other stimulus has ended, and mechanisms for shutting off the signal are part of all signaling systems. Most systems also adapt to the continued presence of the signal by becoming less sensitive to



the intensity of the color. Images recorded at timed intervals reveal changes in [cAMP] over time.

A variation of this technology has been used to measure the activity of PKA in a living cell (Fig. 3). Researchers create a phosphorylation target for PKA by producing a hybrid protein



containing four elements: YFP (acceptor); a short peptide with a Ser residue surrounded by the consensus sequence for PKA; P-Ser -binding domain (called 14-3-3); and CFP (donor). When the Ser residue is not phosphorylated, 14-3-3 has no affinity for the Ser residue and the hybrid protein exists in an extended form, with the donor and acceptor too far apart to generate a FRET signal. Wherever PKA is active in the cell, it phosphorylates the Ser residue of the hybrid protein, and 14-3-3 binds to the P-Ser . In doing so, it draws YFP and CFP together and a FRET signal is detected with the fluorescence microscope, revealing the presence of active PKA.

With more sophisticated microscopic equipment, the FRET signal from a single protein molecule can be detected with much higher spatial and temporal resolution than is available in FRET with an ensemble of molecules.

it, by *desensitizing*. The β -adrenergic system illustrates both. Here, our focus is on termination.

The response to β -adrenergic stimulation will end when the concentration of the ligand (epinephrine) in the blood drops below the K_d for its receptor. The epinephrine then dissociates from the receptor, and the latter reassumes its inactive conformation, in which it can no longer activate G_s .

A second means of ending the response is the hydrolysis of GTP bound to the G_α subunit, catalyzed by the GTPase activity of the G protein. Conversion of bound GTP to GDP favors the return of G_α to the conformation in which it binds the $G_{\beta\gamma}$ subunits—the conformation in which the G protein is unable to interact with or stimulate adenylyl cyclase. This ends the

production of cAMP. The rate of inactivation of G_s depends on the GTPase activity, which for G_α alone is very feeble. However, **GTPase activator proteins (GAPs)** strongly stimulate this GTPase activity, causing more-rapid inactivation of the G protein (Fig. 12-8). GAPs can themselves be regulated by other factors, providing a fine-tuning of the response to β -adrenergic stimulation. A third mechanism for terminating the response is to remove the second messenger: cAMP is hydrolyzed to 5'-AMP (which is not active as a second messenger) by **cyclic nucleotide phosphodiesterase** (Fig. 12-4a, step 7; 12-4b).

Finally, at the end of the signaling pathway, the metabolic effects that result from phosphorylation of target enzymes by PKA are reversed by the action of

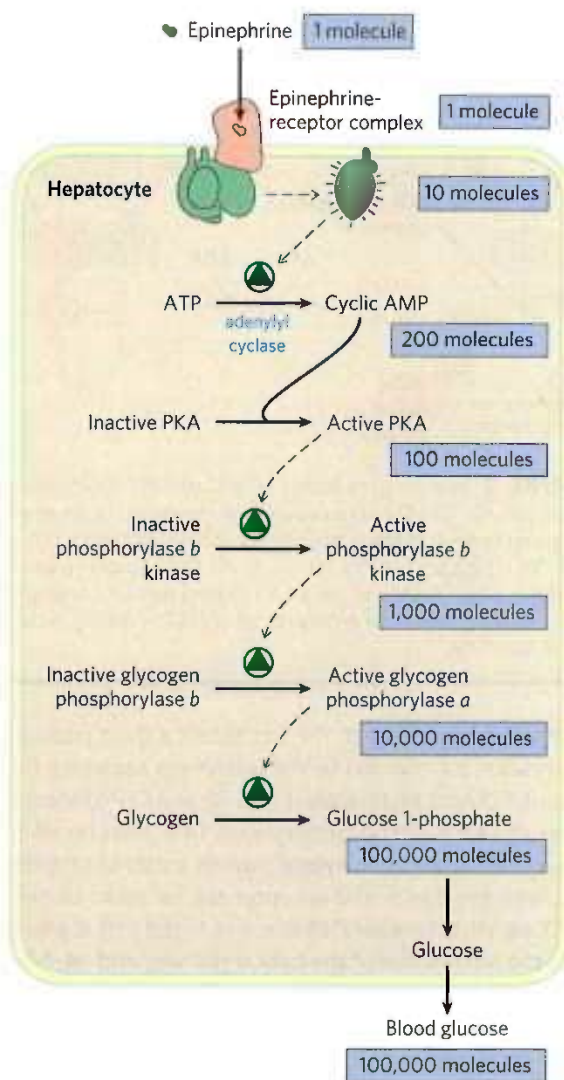


FIGURE 12-7 Epinephrine cascade. Epinephrine triggers a series of reactions in hepatocytes in which catalysts activate catalysts, resulting in great amplification of the original hormone signal. The numbers of molecules shown are simply to illustrate amplification and are almost certainly gross underestimates. Binding of one molecule of epinephrine to one β -adrenergic receptor on the cell surface activates many (possibly hundreds of) G proteins, one after another, each of which goes on to activate a molecule of the enzyme adenylyl cyclase. Adenylyl cyclase acts catalytically, producing many molecules of cAMP for each activated adenylyl cyclase. (Because two molecules of cAMP are required to activate one PKA catalytic subunit, this step does not amplify the signal.)

phosphoprotein phosphatases, which hydrolyze phosphorylated Tyr, Ser, or Thr residues, releasing inorganic phosphate (P_i). About 190 genes in the human genome encode phosphoprotein phosphatases, fewer than the number (about 540) that encode protein kinases, reflecting the relative promiscuity of the phosphoprotein phosphatases. A single **phosphoprotein phosphatase** (PP1) dephosphorylates some 200 different phosphoprotein targets, including the β -adrenergic receptor and other GPCRs. Some phosphatases are known to be regulated; others may be constantly active. When [cAMP] drops and

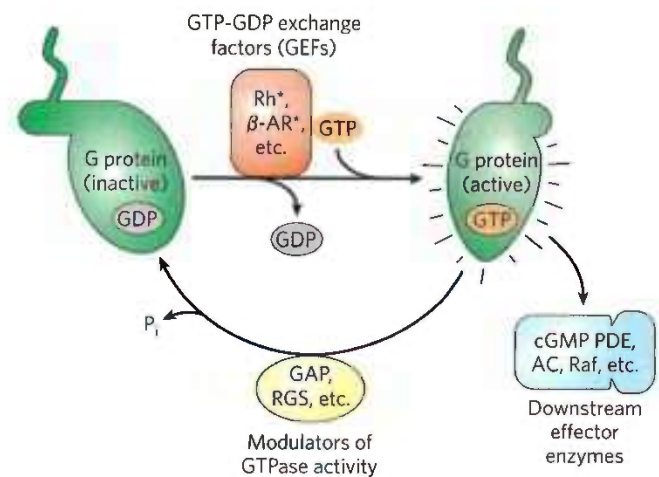


FIGURE 12-8 Factors that regulate G-protein activity. Inactive G proteins, both small G proteins such as Ras and heterotrimeric G proteins such as G_s , interact with upstream GTP-GDP exchange factors (red). Often these exchange factors are activated (*) receptors such as rhodopsin (Rh) and β -adrenergic receptors (AR). The G proteins are activated by GTP binding, and in the GTP-bound form, activate downstream effector enzymes (blue), such as cGMP phosphodiesterase (PDE), adenylyl cyclase (AC), and Raf. By modulating the GTPase activity of G proteins, GTPase activator proteins (GAPs, in the case of small G proteins) and regulators of G-protein signaling (RGSs) (yellow), determine how long the G protein will remain active.

PKA returns to its inactive form (step 7 in Fig. 12-4a), the balance between phosphorylation and dephosphorylation is tipped toward dephosphorylation by these phosphatases.

The β -Adrenergic Receptor Is Desensitized by Phosphorylation and by Association with Arrestin

The mechanisms for signal termination described above take effect when the stimulus ends. A different mechanism, desensitization, damps the response *even while the signal persists*. When the receptor remains occupied with epinephrine, **β -adrenergic receptor kinase**, or **β ARK**, phosphorylates several Ser residues near the receptor's carboxyl terminus, which is on the cytoplasmic side of the plasma membrane (Fig. 12-9). β ARK is drawn to the plasma membrane by its association with the $G_{s\beta\gamma}$ subunits and is thus positioned to phosphorylate the receptor. Receptor phosphorylation creates a binding site for the protein **β -arrestin**, or **β arr** (also called arrestin 2), and binding of β -arrestin blocks the sites in the receptor that interact with the G protein (Fig. 12-10). The binding of β -arrestin also facilitates the sequestration of receptor molecules and their removal from the plasma membrane by endocytosis into small intracellular vesicles (endosomes). The arrestin-receptor complex recruits clathrin and other proteins involved in vesicle formation, which initiate membrane invagination, leading to the formation of endosomes containing the adrenergic receptor. In this state, the receptors are inaccessible to epinephrine and

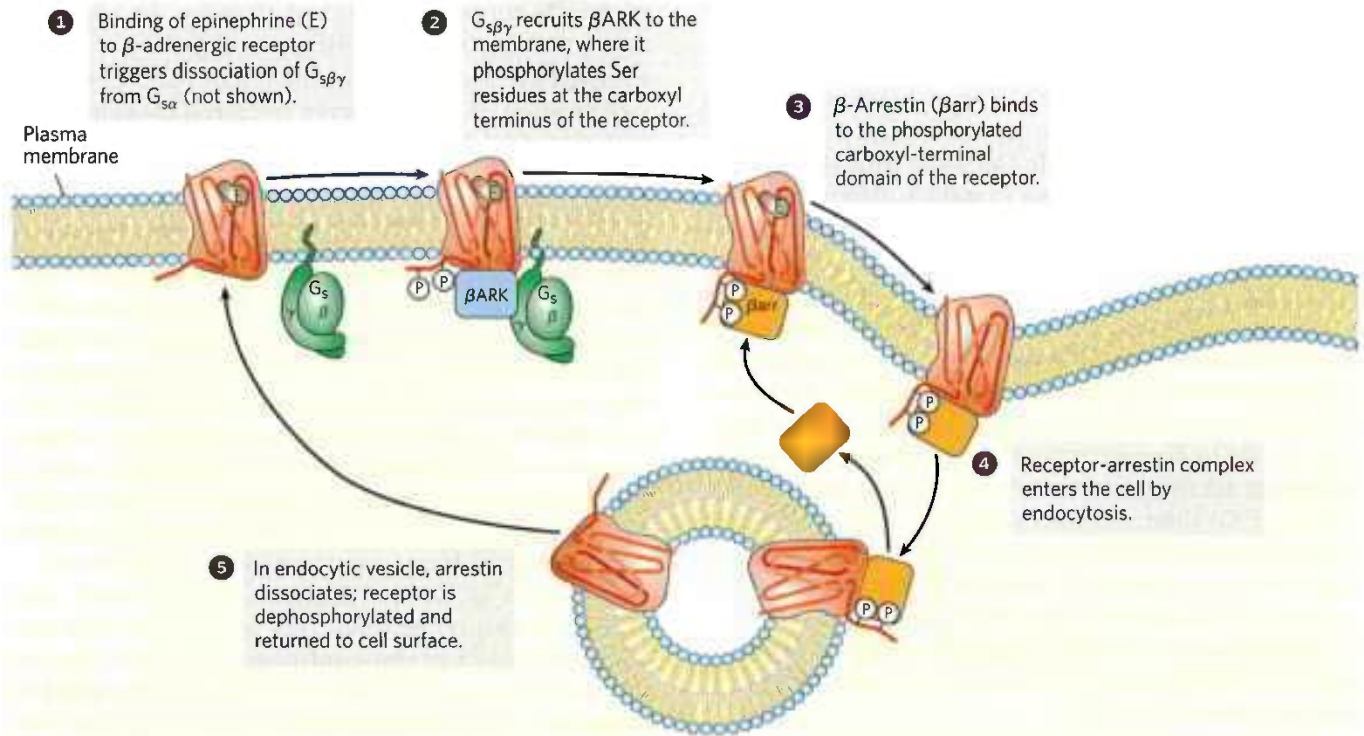


FIGURE 12-9 Desensitization of the β -adrenergic receptor in the continued presence of epinephrine. This process is mediated by two proteins: β -adrenergic protein kinase (β ARK) and β -arrestin (β arr).

Not shown here is the phosphorylation and activation of β ARK by PKA. PKA is activated by the rise in [cAMP] in response to the initial signal (epinephrine).

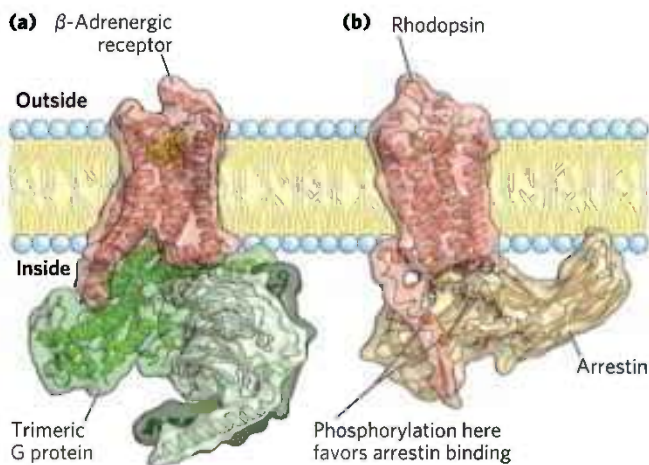


FIGURE 12-10 Mutual exclusion of trimeric G protein and arrestin in their interaction with a GPCR. (a) The β -adrenergic receptor complexed with its trimeric G protein, G_s . (b) Another GPCR, the rhodopsin receptor, phosphorylated near its carboxyl terminus, and bound to arrestin. Binding of arrestin blocks binding and further activation of the G protein and ends the response to the initial signal. [Data from (a) PDB ID 3SN6, S. G. F. Rasmussen et al., *Nature* 477:549, 2011; (b) PDB ID 4ZWI, Y. Kang et al., *Nature* 523:561, 2015.]

β -Adrenergic receptor kinase is a member of a family of **G protein–coupled receptor kinases (GRKs)**, all of which phosphorylate GPCRs on their carboxyl-terminal cytoplasmic domains and play roles similar to that of β ARK in desensitization and resensitization of their receptors. Seven GRKs and four different arrestins are encoded in the human genome; each GRK is capable of desensitizing a particular subset of GPCRs, and each arrestin can interact with many different types of phosphorylated receptors.

 The receptor-arrestin complex has another important role: it initiates signaling by a second pathway, the MAPK cascade described below. Thus, acting through a single GPCR, epinephrine triggers two divergent signaling pathways. The two pathways, one initiated by the receptor's interaction with a G protein and the other initiated by its interaction with arrestin, can be differentially affected by the agonist; in some cases, one agonist favors the G-protein pathway and another favors the arrestin pathway. This bias is an important consideration in the development of a medication that acts through a GPCR. For example, the most addictive of the opioid drugs of abuse act more strongly through G-protein signaling than through arrestin. An ideal opioid pain medication would act through the branch of the pathway that has therapeutic effects and not through the pathway that leads to addiction. ■

therefore inactive. These receptor molecules are eventually dephosphorylated and returned to the plasma membrane, completing the circuit and resensitizing the system to epinephrine.

Cyclic AMP Acts as a Second Messenger for Many Regulatory Molecules

P2 Epinephrine is just one of many hormones, growth factors, and other regulatory molecules that act by changing the intracellular [cAMP] and thus the activity of PKA. **Table 12-3** lists a few examples. Glucagon binds to its receptors in the plasma membrane of adipocytes, activating (via a G_s protein) adenylyl cyclase. PKA, stimulated by the resulting rise in [cAMP], phosphorylates and activates two proteins critical to the mobilization of the fatty acids of stored fats (see Fig. 17-2). Similarly, the peptide hormone ACTH (adrenocorticotrophic hormone, also called corticotropin), produced by the anterior pituitary, binds to specific receptors in the adrenal cortex, activating adenylyl cyclase and raising the intracellular [cAMP]. PKA then phosphorylates and activates several of the enzymes required for the synthesis of cortisol and other steroid hormones. In many cell types, the catalytic subunit of PKA can also move into the nucleus, where it phosphorylates the **cAMP response element binding protein (CREB)**, which alters the expression of specific genes regulated by cAMP.

Some hormones act by *inhibiting* adenylyl cyclase, thus *lowering* [cAMP] and *suppressing* protein phosphorylation by PKA. For example, the binding of somatostatin to its receptor in the pancreas leads to activation of an **inhibitory G protein**, or G_i , structurally homologous to G_s , that inhibits adenylyl cyclase and lowers [cAMP]. In this way, somatostatin inhibits the secretion

of several hormones, including glucagon. In adipose tissue, prostaglandin E_2 (PGE_2 ; see Fig. 10-17) inhibits adenylyl cyclase, thus lowering [cAMP] and slowing the mobilization of lipid reserves triggered by epinephrine and glucagon. In certain other tissues, PGE_2 stimulates cAMP synthesis: its receptors are coupled to adenylyl cyclase through a stimulatory G protein, G_s . In tissues with α_2 -adrenergic receptors, epinephrine lowers [cAMP]; in this case, the receptors are coupled to adenylyl cyclase through an inhibitory G protein, G_i . In short, an extracellular signal such as epinephrine or PGE_2 can have different effects on different tissues or cell types, depending on three factors: the type of receptor in the tissue, the type of G protein (G_s or G_i) with which the receptor is coupled, and the set of PKA target enzymes in the cell. By summing the influences that tend to increase and decrease [cAMP], a cell achieves the integration of signals that is a general feature of signal-transducing mechanisms (Fig. 12-1f).

Another factor that explains how so many types of signals can be mediated by a single second messenger (cAMP) is the confinement of the signaling process to a specific region of the cell by **adaptor proteins**—noncatalytic proteins that hold together other protein molecules that function in concert (further described below). **P5 AKAPs (A kinase anchoring proteins)** have multiple, distinct protein-binding domains, often intrinsically disordered regions; they are multivalent adaptor proteins. One domain binds to the R subunits of PKA and another binds to a specific structure in the cell, confining the PKA to the vicinity of that structure. For example, specific AKAPs bind PKA to microtubules, actin filaments, ion channels, mitochondria, or the nucleus. Different types of cells have different complements of AKAPs, so cAMP might stimulate phosphorylation of mitochondrial proteins in one cell and stimulate phosphorylation of actin filaments in another. In some cases, an AKAP connects PKA with the enzyme that triggers PKA activation (adenylyl cyclase) or terminates PKA action (cAMP phosphodiesterase or phosphoprotein phosphatase) (Fig. 12-11). The very close proximity of these activating and inactivating enzymes presumably achieves a highly localized, and very brief, response.

In a process analogous with the cAMP-PKA pathway, a cyclic derivative of GTP (3',5'-cyclic GMP; cGMP) is generated in response to an extracellular signal. The cGMP activates a cGMP-dependent protein kinase (PKG), which phosphorylates specific protein substrates, changing their activities in response to the initial signal (**Box 12-2**).

TABLE 12-3 Some Signals That Use cAMP as Second Messenger

Corticotropin (ACTH)
Corticotropin-releasing hormone (CRH)
Dopamine [D_1 , D_2]
Epinephrine (β -adrenergic)
Follicle-stimulating hormone (FSH)
Glucagon
Histamine [H_2]
Luteinizing hormone (LH)
Melanocyte-stimulating hormone (MSH)
Odorants (many)
Parathyroid hormone
Prostaglandins E_1 , E_2 (PGE_1 , PGE_2)
Serotonin [$5-HT_1$, $5-HT_4$]
Somatostatin
Tastants (sweet, bitter)
Thyroid-stimulating hormone (TSH)

Note: Receptor subtypes in square brackets. Subtypes may have different transduction mechanisms. For example, serotonin is detected in some tissues by receptor subtypes $5-HT_1$ and $5-HT_4$, which act through adenylyl cyclase and cAMP, and in other tissues by receptor subtype $5-HT_2$, acting through the phospholipase $C-IP_3$ mechanism (see Table 12-4).

G Proteins Act as Self-Limiting Switches in Many Processes

Proteins sensitive to the binding of either GTP or GDP play critical roles in many cellular processes, including sensory perception, signaling for cell division, growth

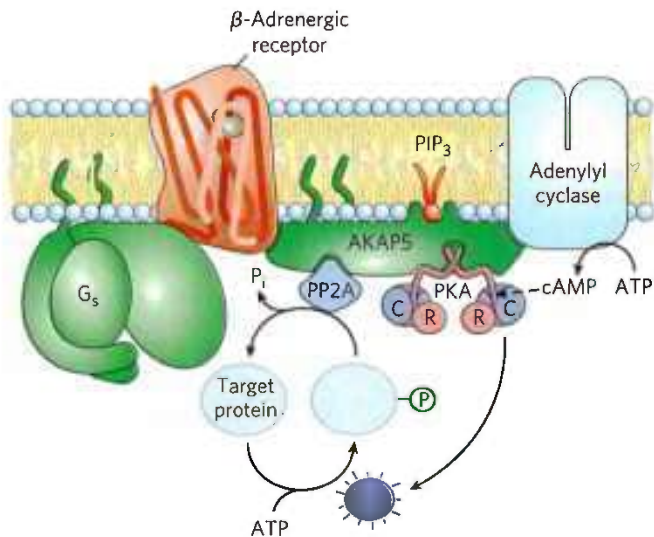


FIGURE 12-11 Nucleation of supramolecular complexes by A kinase anchoring proteins (AKAPs). AKAP5 is one of a family of proteins that act as multivalent scaffolds, holding PKA catalytic subunits — through interaction of the AKAP with the PKA regulatory subunits — in proximity to a particular region or structure in the cell. AKAP5 is targeted to rafts in the cytoplasmic face of the plasma membrane by two covalently attached palmitoyl groups and a site that binds phosphatidylinositol 3,4,5-trisphosphate (PIP₃) in the membrane. AKAP5 also has binding sites for the β -adrenergic receptor, adenylyl cyclase, PKA, and a phosphoprotein phosphatase (PP2A), bringing them all together in the plane of the membrane. When epinephrine binds to the β -adrenergic receptor, adenylyl cyclase produces cAMP, which reaches the nearby PKA quickly and with very little dilution. PKA phosphorylates its target protein, altering its activity, until the phosphoprotein phosphatase removes the phosphoryl group and returns the target protein to its prestimulus state. The AKAPs in this and other cases bring about a high local concentration of enzymes and second messengers, so that the signaling circuit remains highly localized and the duration of the signal is limited.

and differentiation, intracellular movements of proteins and membrane vesicles, and protein synthesis. The human genome encodes nearly 200 of these proteins, which differ in size and subunit structure, intracellular location, and function. But all G proteins share a common feature: they can become activated by binding GTP and then, after a brief period, can inactivate themselves with their GTPase activity, thereby serving as molecular binary switches with built-in timers. This superfamily of proteins includes the trimeric G proteins involved in β -adrenergic signaling (G_s or G_i) and vision (transducin); small, monomeric G proteins such as that involved in insulin signaling (Ras; see below) and others that function in vesicle trafficking (ARF, RAC1, and Rab), transport into and out of the nucleus (Ran), and timing of the cell cycle (Rho); and several proteins involved in protein synthesis (initiation factor IF2 and elongation factors EF-Tu and EF-G; see Chapter 27). Among the trimeric G proteins, G_α subunits have covalently linked lipids: the amino terminus is palmitoylated, and some are also myristoylated. G_γ

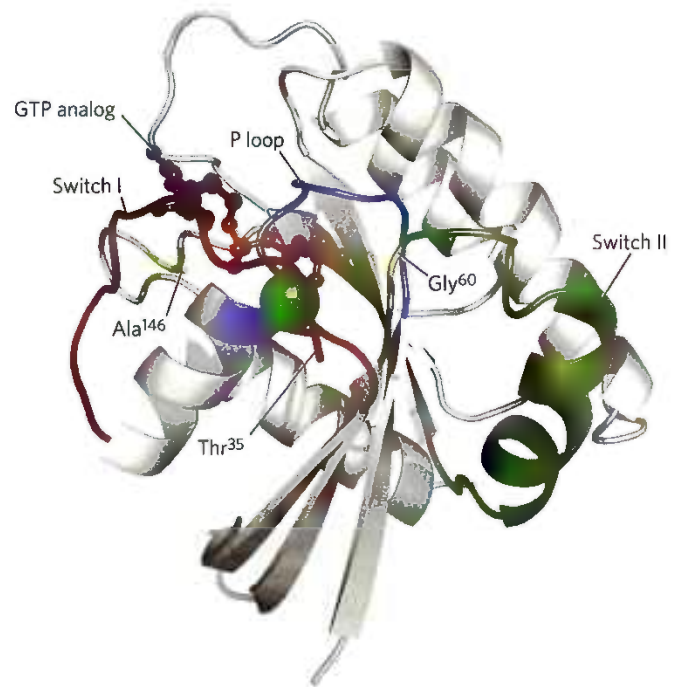


FIGURE 12-12 Ras, the G-protein prototype. Mg^{2+} -GTP is held by critical residues in the phosphate-binding P loop (blue) and by Thr³⁵ in the switch I region (red) and Gly⁶⁰ in the switch II region (green). Ala¹⁴⁶ gives specificity for GTP over ATP. In the structure shown here, the nonhydrolyzable GTP analog Gpp(NH)p is in the GTP-binding site. [Data from PDB ID 5P21, E. F. Pai et al., *EMBO J.* 9:2351, 1990.]

and the monomeric Ras protein have an isoprenyl lipid at their carboxyl termini. The attached lipids keep them anchored to the plasma membrane, restricting their action to that two-dimensional plane.

All G proteins have the same core structure and use the same mechanism for switching between an inactive conformation, favored when GDP is bound, and an active conformation, favored when GTP is bound. We can use **Ras** (~20 kDa, a minimal signaling unit) as a prototype for all members of this superfamily (**Fig. 12-12**).

In the nucleotide-binding site of Ras, Ala¹⁴⁶ hydrogen bonds to the guanine oxygen, allowing GTP, but not ATP, to bind. In the GTP-bound conformation, the G protein exposes previously buried regions (called **switch I** and **switch II**) that interact with proteins downstream in the signaling pathway until the G protein inactivates itself by hydrolyzing its bound GTP to GDP. The critical determinant of G-protein conformation is the γ phosphate of GTP, which interacts with a region called the **P loop** (phosphate-binding). In Ras, the γ phosphate of GTP binds to a Lys residue in the P loop and to two critical residues, Thr³⁵ in switch I and Gly⁶⁰ in switch II, that hydrogen-bond with the oxygens of the γ phosphate of GTP. These hydrogen bonds act like a pair of springs holding the protein in its active



Receptor Guanylyl Cyclases, cGMP, and Protein Kinase G

Guanylyl cyclases (Fig. 1) are receptor enzymes that, when activated, convert GTP to the second messenger **guanosine 3',5'-cyclic monophosphate (cGMP)** (Fig. 2). Many of the actions of cGMP in animals are mediated by **cGMP-dependent protein kinase**, also called **protein kinase G (PKG)**. On activation by cGMP, PKG phosphorylates Ser and Thr residues in target proteins. The catalytic and regulatory domains of this enzyme are in a single polypeptide ($M_r \sim 80,000$). Part of the regulatory domain fits snugly in the substrate-binding cleft. Binding of cGMP forces this activation loop out of the binding site, opening the site to PKG target proteins.

Cyclic GMP carries different messages in different tissues. In cardiac muscle (a type of smooth muscle), it signals relaxation. In the kidney and the intestine, it triggers changes in ion transport and water retention. Guanylyl cyclase in the kidney is activated by the peptide hormone **atrial natriuretic factor (ANF)** (Fig. 1a), which is released by cells in the cardiac atrium when the heart is stretched by increased blood volume. Carried in the blood to the kidney, ANF activates guanylyl cyclase in cells of the collecting ducts. The resulting rise in [cGMP] triggers increased renal excretion of Na^+ and consequently of water, driven by the change in osmotic pressure.

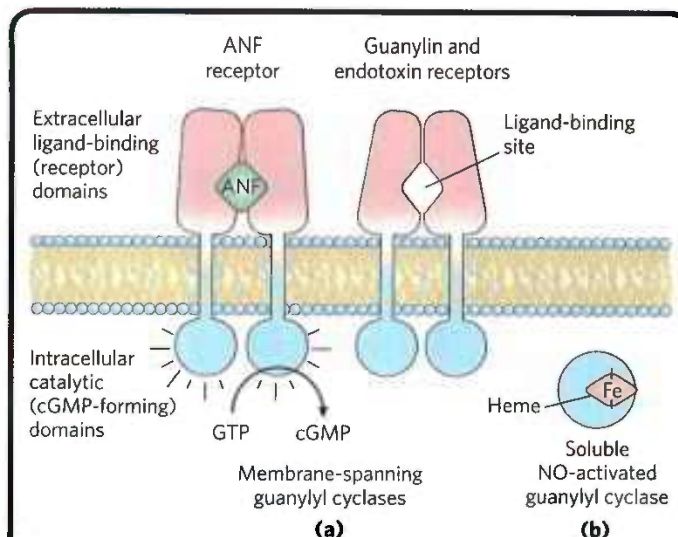


FIGURE 1 Two types of signal-transducing guanylyl cyclase. (a) Membrane-spanning guanylyl cyclases such as the ANF and guanylin receptors are homodimers with an extracellular ligand-binding domain and an intracellular guanylyl cyclase domain. (b) A soluble heme-containing guanylyl cyclase is activated by intracellular NO.

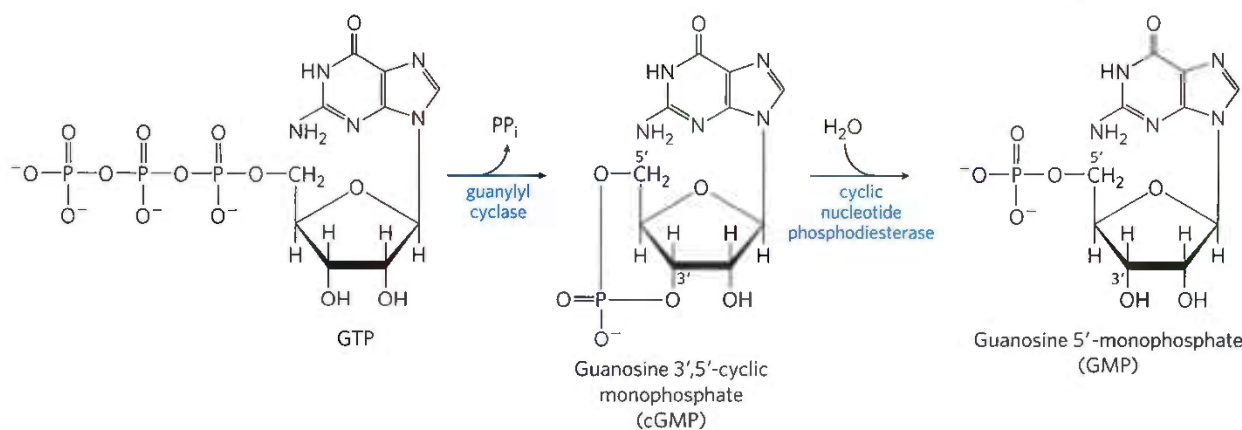


FIGURE 2 Synthesis of cGMP by guanylyl cyclase and its hydrolysis by cGMP phosphodiesterase.

conformation (Fig. 12-13). When GTP is cleaved to GDP, and P_i is released, these hydrogen bonds are lost; the protein then relaxes into its inactive conformation, burying switch I and II so they are no longer available to interact with other partners.

The GTPase activity of most G proteins is very weak, but it is increased up to 10^5 -fold by **GTPase activator proteins (GAPs)**, also called, in the case of heterotrimeric G proteins, **regulators of G-protein signaling (RGSs)** (Fig. 12-8). GAPs and RGSs thus determine how

long the G-protein switch remains on. There are about 40 different RGS proteins associated with a variety of processes and expressed in most tissues. They have a critical Arg residue that reaches into the G-protein GTPase active site and contributes to catalysis. The intrinsically slow process of replacing bound GDP with GTP, switching the protein on, is catalyzed by **guanosine nucleotide-exchange factors (GEFs)**, such as the β -adrenergic receptor associated with the G protein. The ligand-bound β -adrenergic receptor is one of many GEFs, and a broad

Water loss reduces the blood volume, countering the stimulus that initially led to ANF secretion. Vascular smooth muscle also has an ANF receptor–guanylyl cyclase; on binding to this receptor, ANF causes relaxation (vasodilation) of the blood vessels, which increases blood flow while decreasing blood pressure.

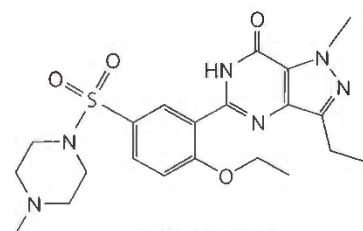
A similar receptor guanylyl cyclase in the plasma membrane of epithelial cells lining the intestine is activated by the peptide **guanylin** (Fig. 1a), which regulates Cl^- secretion in the intestine. This receptor is also the target of a heat-stable peptide endotoxin produced by *Escherichia coli* and other gram-negative bacteria. The elevation in [cGMP] caused by the endotoxin increases Cl^- secretion and consequently decreases reabsorption of water by the intestinal epithelium, producing diarrhea.

A distinctly different type of guanylyl cyclase is a soluble cytosolic protein with a tightly associated heme group (Fig. 1b), an enzyme activated by nitric oxide (NO). Nitric oxide is produced from arginine by Ca^{2+} -dependent **NO synthase**, present in many mammalian tissues, and diffuses from its cell of origin into nearby cells (see Chapter 22).

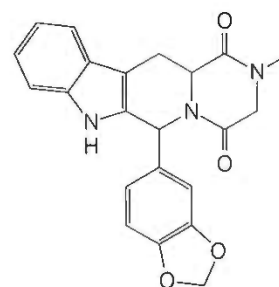
In the target cell, NO binds to the heme group of guanylyl cyclase and activates cGMP production. In the heart, cGMP-dependent protein kinase reduces the forcefulness of contractions by stimulating the ion pump(s) that remove Ca^{2+} from the cytosol. NO-induced relaxation of cardiac muscle is the same response brought about by nitroglycerin and other nitrovasodilators taken to relieve **angina pectoris**, the pain caused by contraction of a heart deprived of O_2 because of blocked coronary arteries. Nitric oxide is unstable and its action is brief; within seconds of its formation, it undergoes oxidation to nitrite or nitrate. Nitrovasodilator drugs produce long-lasting relaxation of cardiac muscle because they break down over several hours, yielding a steady stream of NO. The value of nitroglycerin as a treatment for angina was discovered serendipitously in factories producing nitroglycerin as an explosive in the 1860s. Workers with angina reported that

their condition was much improved during the workweek but worsened on weekends. The physicians treating these workers heard this story so often that they made the connection, and a drug was born.

The effects of increased cGMP synthesis diminish after the stimulus ceases, because a specific phosphodiesterase (cGMP PDE) converts cGMP to the inactive 5'-GMP (see Fig. 2). Humans have several isoforms of cGMP PDE, with different tissue distributions. The isoform in the blood vessels of the penis is inhibited by the drugs sildenafil (Viagra) and tadalafil (Cialis), which therefore cause [cGMP] to remain elevated once raised by an appropriate stimulus, accounting for the usefulness of this drug in the treatment of erectile dysfunction.



Sildenafil (Viagra)



Tadalafil (Cialis)

range of proteins act as GAPs. Their combined effects set the level of GTP-bound G proteins, and thus the strength of the response to signals that arrive at the receptors.



P9

Because G proteins play crucial roles in so many signaling processes, it is not surprising that defects in G proteins lead to a variety of diseases. In about 25% of all human cancers (and in a much higher proportion of certain types of cancer), a mutation in a Ras protein—typically in one of the critical residues around the GTP-binding site or in the P loop—virtually

eliminates its GTPase activity. Once activated by GTP binding, this Ras protein remains constantly active, promoting cell division in cells that should not divide. The tumor suppressor gene *NF1* encodes a GAP that enhances the GTPase activity of normal Ras. Mutations in *NF1* that result in a nonfunctioning GAP leave Ras with only its GTPase activity, which is very weak (that is, has a very low turnover number); once activated by GTP binding, Ras stays active for an extended period, continuing to send the signal: divide.

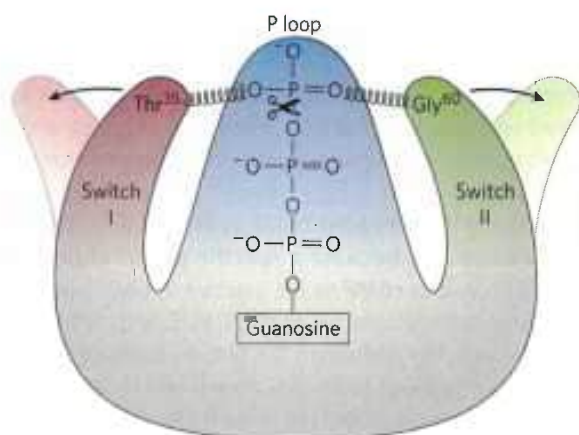


FIGURE 12-13 GTP hydrolysis flips the switches in Ras. When bound GTP is hydrolyzed by the GTPase activities of Ras and its GTPase activator protein (GAP), loss of hydrogen bonds to Thr³⁵ and Gly⁶⁰ allows the switch I and switch II regions to relax into a conformation in which they are no longer available to interact with downstream targets. [Information from I. R. Vetter and A. Wittinghofer, *Science* 294:1299, 2001, Fig. 3.]

Defective heterotrimeric G proteins can also lead to disease. Mutations in the gene that encodes the α subunit of G_s (which mediates changes in [cAMP] in response to hormonal stimuli) may result in a $G_{s\alpha}$ that is permanently active or permanently inactive. “Activating” mutations generally occur in residues crucial to GTPase activity; they lead to a continuously elevated [cAMP], with significant

downstream consequences, including undesirable cell proliferation. Such mutations are found in about 40% of pituitary tumors (adenomas). Individuals with “inactivating” mutations in G_{α} are unresponsive to hormones (such as thyroid hormone) that act through cAMP. Mutation in the gene for the transducin α subunit (T_{α}), which is involved in visual signaling, leads to a type of night blindness, apparently due to defective interaction between the activated T_{α} subunit and the phosphodiesterase of the rod outer segment (see Fig. 12-19). A sequence variation in the gene encoding the β subunit of a heterotrimeric G protein is commonly found in individuals with hypertension (high blood pressure), and this variant gene is suspected of involvement in obesity and atherosclerosis.

The pathogenic bacterium that causes cholera produces a toxin that targets a G protein, interfering with normal signaling in host cells. **Cholera toxin**, secreted by *Vibrio cholerae* in the intestine of an infected person, is a heterodimeric protein. Subunit B recognizes and binds to specific gangliosides on the surface of intestinal epithelial cells and provides a route for subunit A to enter these cells. After entry, subunit A is broken into two fragments, A1 and A2. A1 associates with the host cell’s ADP-ribosylation factor ARF6, a small G protein, through residues in its switch I and switch II regions—which are accessible only when ARF6 is in its active (GTP-bound) form. This association with ARF6 activates A1, which catalyzes the transfer of ADP-ribose from NAD⁺ to the critical Arg residue in the P loop of the α subunit of G_s (Fig. 12-14). ADP-ribosylation

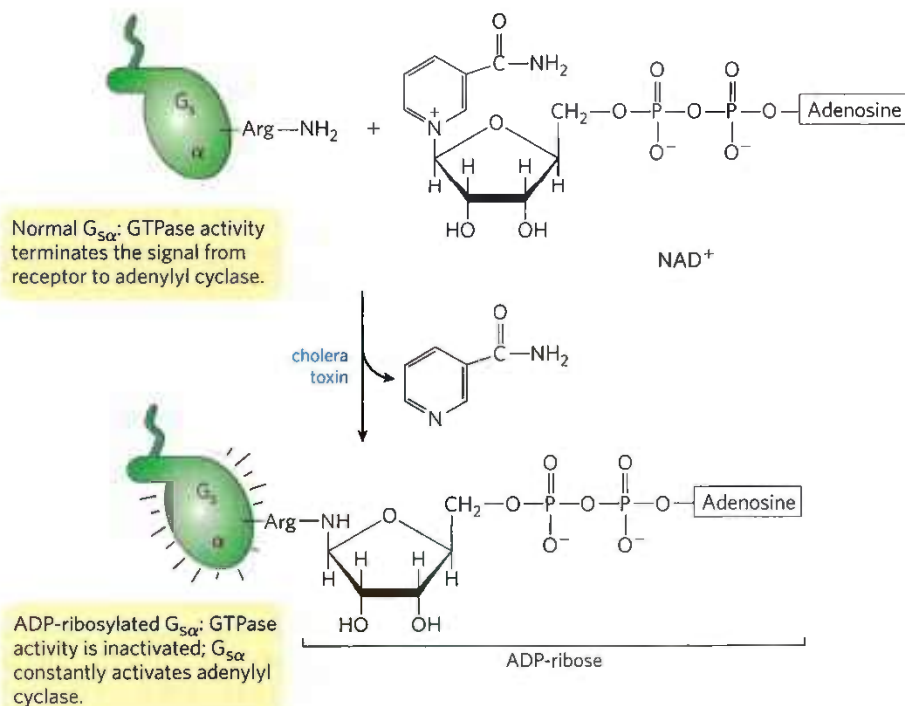


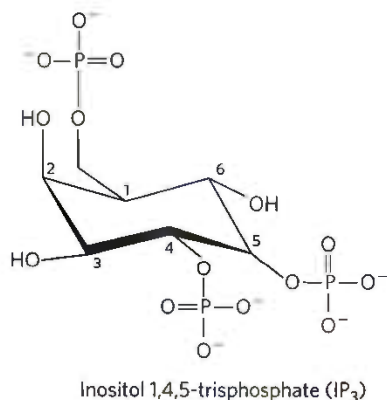
FIGURE 12-14 ADP-ribosylation locks $G_{s\alpha}$ in the active conformation. The bacterial toxin that causes cholera is an enzyme that catalyzes transfer of the ADP-ribose moiety of NAD⁺ (nicotinamide adenine

dinucleotide) to an Arg residue of $G_{s\alpha}$. The G proteins thus modified fail to respond to normal hormonal stimuli. The pathology of cholera results from defective regulation of adenylyl cyclase and overproduction of cAMP.

blocks the GTPase activity of G_s and thereby renders G_s permanently active. This results in continuous activation of the adenylyl cyclase of intestinal epithelial cells, chronically high [cAMP], and chronically active PKA. PKA phosphorylates the CFTR Cl^- channel (see Box 11-2) and a Na^+/H^+ exchanger in the intestinal epithelial cells. The resultant efflux of NaCl triggers massive water loss through the intestine as cells respond to the ensuing osmotic imbalance. Severe dehydration and electrolyte loss are the major pathologies in cholera. These can be fatal in the absence of prompt rehydration therapy. ■

Diacylglycerol, Inositol Trisphosphate, and Ca^{2+} Have Related Roles as Second Messengers

A second broad class of GPCRs is coupled through a G protein to a plasma membrane **phospholipase C (PLC)** that catalyzes cleavage of the membrane phospholipid phosphatidylinositol 4,5-bisphosphate, or PIP_2 (see Fig. 10-14). When one of the agonists (hormone, neurotransmitter, growth factor; Table 12-4) that act by this mechanism binds its specific receptor in the plasma membrane (Fig. 12-15, step 1), the receptor-hormone complex catalyzes GTP-GDP exchange on an associated trimeric G protein, G_q (step 2). This activates the G_q proteins in much the same way that the β -adrenergic receptor activates G_s (Fig. 12-4). The activated G_q activates the PIP_2 -specific PLC (Fig. 12-15, step 3), which catalyzes the production of two potent second messengers (step 4), **diacylglycerol** and **inositol 1,4,5-trisphosphate**, or **IP_3** (not to be confused with PIP_3 , p. 435 or Fig. 12-24).



Inositol trisphosphate, a water-soluble compound, diffuses from the plasma membrane to the endoplasmic reticulum (ER), where it binds to the **IP_3 -gated Ca^{2+} channel**, causing the channel to open. The action of the SERCA pump (pp. 393–394) ensures that $[\text{Ca}^{2+}]$ in the ER is orders of magnitude higher than that in the cytosol, so when these gated Ca^{2+} channels open, Ca^{2+} rushes into the cytosol (Fig. 12-15, step 5), and the cytosolic $[\text{Ca}^{2+}]$ rises sharply to about 10^{-6} M. One effect of elevated $[\text{Ca}^{2+}]$ is the activation of **protein kinase C (PKC)** (C for Ca^{2+}). Diacylglycerol cooperates with Ca^{2+} in activating PKC, thus also acting as a second messenger (step 6). Activation involves the movement of a PKC domain (the pseudosubstrate domain) away from its location in the substrate-binding region of the enzyme, allowing the enzyme to bind and phosphorylate proteins that contain a PKC consensus sequence—Ser or Thr residues embedded in an amino acid sequence recognized by PKC (step 7). Figure 12-16 shows the structure of the IP_3 receptor and a proposed mechanism of its action as a gated Ca^{2+} channel.

Several isozymes of PKC are known, each with a characteristic tissue distribution, target protein specificity, and role. $\text{PLC}\beta$ is activated by GPCRs; another isoform, $\text{PLC}\gamma$, is activated by receptor tyrosine kinases, as described below. The ultimate PKC targets include cytoskeletal proteins, enzymes, and nuclear proteins that regulate gene expression. Taken together, this family of enzymes has a wide range of cellular actions, affecting neuronal and immune function and the regulation of cell division. Compounds that lead to overexpression of PKC or that increase its activity to abnormal levels act as tumor promoters; animals exposed to these substances have increased rates of cancer.

Calcium Is a Second Messenger That Is Limited in Space and Time

There are many variations on this basic scheme for Ca^{2+} signaling. In many cell types that respond to extracellular signals, Ca^{2+} serves as a second messenger that triggers intracellular responses, such as exocytosis in neurons and endocrine cells, contraction in skeletal muscle, and cytoskeletal rearrangements during amoeboid movement. In unstimulated cells, cytosolic $[\text{Ca}^{2+}]$ is kept very low

TABLE 12-4 Some Signals That Act through Phospholipase C, IP_3 , and Ca^{2+}

Acetylcholine [muscarinic M_1]	Gastrin-releasing peptide	Platelet-derived growth factor (PDGF)
α_1 -Adrenergic agonists	Glutamate	Serotonin [5-HT ₂]
Angiogenin	Gonadotropin-releasing hormone (GRH)	Thyrotropin-releasing hormone (TRH)
Angiotensin II	Histamine [H_1]	Vasopressin
ATP [P_{2x} , P_{2y}]	Light (<i>Drosophila</i>)	
Auxin	Oxytocin	

Note: Receptor subtypes are in square brackets; see footnote to Table 12-3.

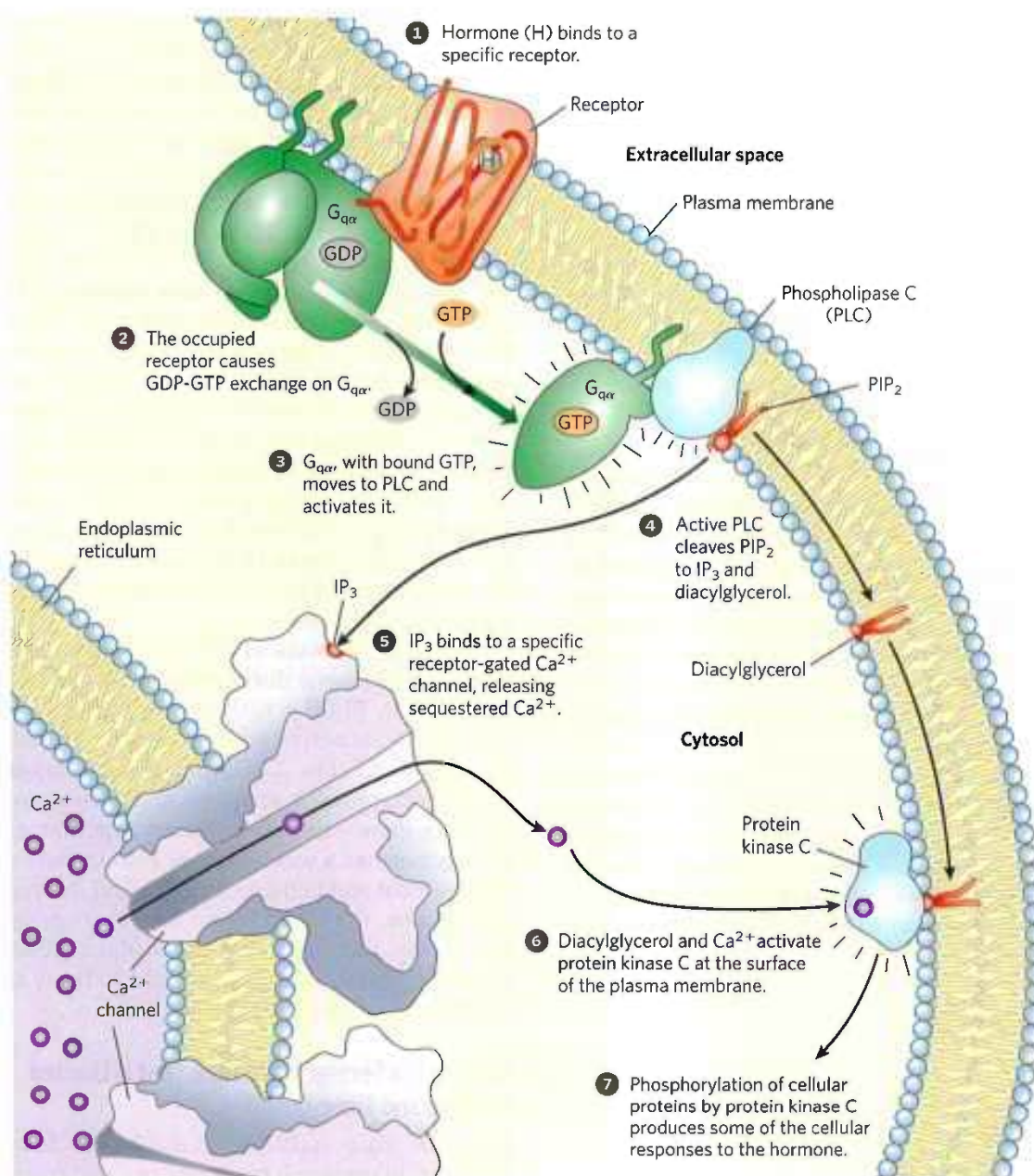


FIGURE 12-15 Hormone-activated phospholipase C and IP_3 .

Two intracellular second messengers are produced in the hormone-sensitive phosphatidylinositol system: inositol 1,4,5-trisphosphate (IP_3) and diacylglycerol

are cleaved from phosphatidylinositol 4,5-bisphosphate (PIP_2). Both contribute to the activation of protein kinase C. By raising cytosolic $[Ca^{2+}]$, IP_3 also activates other Ca^{2+} -dependent enzymes; thus Ca^{2+} also acts as a second messenger.

($<10^{-7}$ M) by the action of Ca^{2+} pumps in the ER, mitochondria, and plasma membrane (as further discussed below).

P6 Hormonal, neural, or other stimuli cause either an influx of Ca^{2+} into the cell through specific Ca^{2+} channels in the plasma membrane or the release of sequestered Ca^{2+} from the ER or mitochondria, in either case raising the cytosolic $[Ca^{2+}]$ and triggering a cellular response.

Changes in intracellular $[Ca^{2+}]$ are detected by Ca^{2+} -binding proteins that regulate a variety of Ca^{2+} -dependent enzymes. **Calmodulin (CaM; M_r 17,000)** is an acidic protein with four high-affinity Ca^{2+} -binding sites ($K_d \approx 0.1$ to $1 \mu M$) (Fig. 12-17). When intracellular $[Ca^{2+}]$ rises to about 10^{-6} M ($1 \mu M$), the binding of Ca^{2+} to

calmodulin drives a conformational change in the protein. Calmodulin associates with a variety of proteins and, in its Ca^{2+} -bound state, modulates their activities. It is a member of a family of Ca^{2+} -binding proteins that also includes troponin (see Fig. 5-30), which triggers skeletal muscle contraction in response to increased $[Ca^{2+}]$. Members of this family share a characteristic Ca^{2+} -binding structure, the EF hand.

Calmodulin is a subunit of the **Ca^{2+} /calmodulin-dependent protein kinases (CaM kinases, types I through IV)**. When intracellular $[Ca^{2+}]$ increases in response to a stimulus, calmodulin binds Ca^{2+} , undergoes a change in conformation, and activates the CaM kinase. The kinase

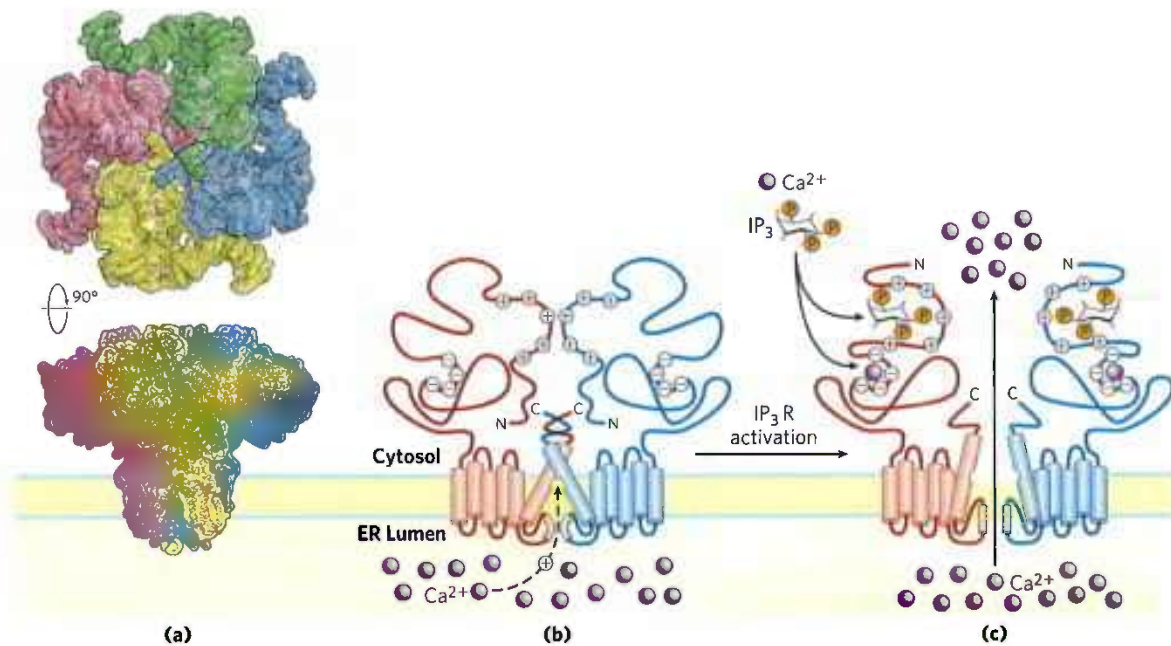


FIGURE 12-16 Proposed mechanism of action of the IP₃-gated Ca²⁺ channel. (a) The IP₃ receptor in its closed conformation, determined by cryo-EM. The 1.3 MDa tetramer has 24 transmembrane helices that surround a central pore. The bulk of the protein protrudes out of the ER into the cytosol and contains the IP₃-binding sites. The central pore is the channel for Ca²⁺ movement, which does not conduct Ca²⁺ in the absence of IP₃.

(b, c) Model for receptor activation by IP₃, showing only two of the four identical subunits (b) in the absence of IP₃, and (c) with one IP₃ bound to each subunit. According to this model, IP₃ binding near the amino-terminal end of a subunit causes a major rearrangement of the α helix at the carboxyl-terminal end, opening the Ca²⁺ channel. [(a) Data from PDB ID 6MU2, G. Fan et al., *Cell Res.* 28:1158, 2018. (b, c) Information from M. J. Berridge, *Physiol. Rev.* 96:1261, 2016, Fig. 3.]

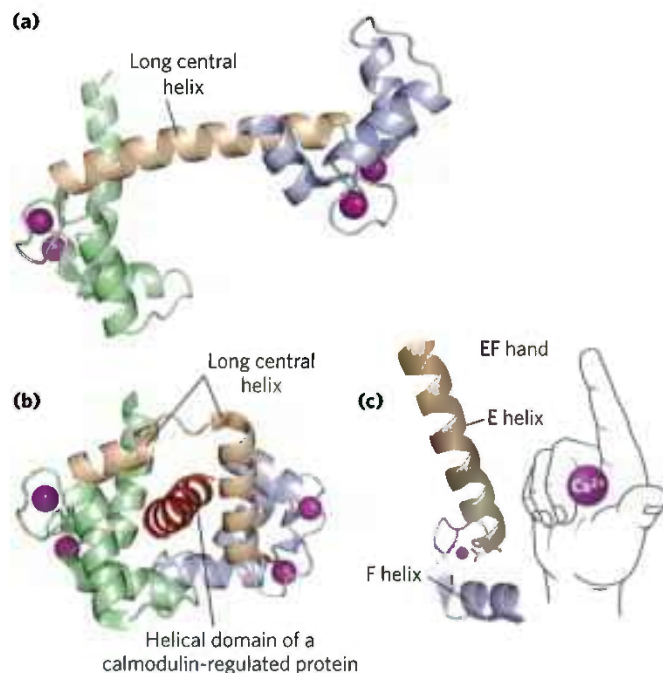


FIGURE 12-17 Calmodulin, the protein mediator of many Ca²⁺-stimulated enzymatic reactions. (a) In this ribbon model of the crystal structure of calmodulin, the four high-affinity Ca²⁺-binding sites are occupied by Ca²⁺ (purple). The amino-terminal domain is on the left; the carboxyl-terminal domain on the right. (b) Calmodulin associated with a helical domain of one of the many enzymes it regulates, calmodulin-dependent protein kinase II. Notice that the long central α helix of calmodulin visible in (a) has bent back on itself in binding to the helical substrate domain. The central helix of calmodulin is clearly more flexible in solution than in the crystal. (c) Each of the four Ca²⁺-binding sites occurs in a helix-loop-helix motif called the EF hand, found in many Ca²⁺-binding proteins. [Data from (a) PDB ID 1CLL, R. Chattopadhyaya et al., *J. Mol. Biol.* 228:1177, 1992; (b, c) PDB ID 1CDL, W. E. Meador et al., *Science* 257:1251, 1992.]

then phosphorylates target enzymes, regulating their activities. Calmodulin is also a regulatory subunit of phosphor-ylase *b* kinase of muscle, which is activated by Ca²⁺. Thus Ca²⁺ triggers ATP-requiring muscle contractions while also activating glycogen breakdown, providing fuel for ATP syn-thesis. Many other enzymes are known to be modulated by Ca²⁺ through calmodulin (Table 12-5). The activity of the second messenger Ca²⁺, like that of cAMP, can be spatially restricted; after its release triggers a local response, Ca²⁺ is generally removed before it can diffuse to distant parts of the cell.

TABLE 12-5 Some Proteins Regulated by Ca²⁺ and Calmodulin

Adenylyl cyclase (brain)
Ca ²⁺ /calmodulin-dependent protein kinases (CaM kinases I to IV)
Ca ²⁺ -dependent Na ⁺ channel (<i>Paramecium</i>)
Ca ²⁺ -release channel of sarcoplasmic reticulum
Calcineurin (phosphoprotein phosphatase 2B)
cAMP phosphodiesterase
cAMP-gated olfactory channel
cGMP-gated Na ⁺ , Ca ²⁺ channels (rod and cone cells)
Glutamate decarboxylase
Myosin light-chain kinases
NAD ⁺ kinase
Nitric oxide synthase
Phosphatidylinositol 3-kinase
Plasma membrane Ca ²⁺ ATPase (Ca ²⁺ pump)
RNA helicase (p68)

Commonly, Ca^{2+} level does not simply rise and then fall, but rather oscillates with a period of a few seconds (Fig. 12-18) — even when the extracellular concentration of the triggering hormone remains constant. The mechanism underlying $[\text{Ca}^{2+}]$ oscillations presumably entails feedback regulation by Ca^{2+} on some part of the Ca^{2+} -release process. Whatever the mechanism, the effect is that one kind of signal (hormone concentration, for example) is converted into another (frequency and amplitude of intracellular $[\text{Ca}^{2+}]$ “spikes”). The Ca^{2+} signal diminishes as Ca^{2+} diffuses away from the initial source (the Ca^{2+} channel), is sequestered in the ER, or is pumped out of the cell.

There is significant cross talk between the Ca^{2+} and cAMP signaling systems. In some tissues, both the enzyme that produces cAMP (adenylyl cyclase) and

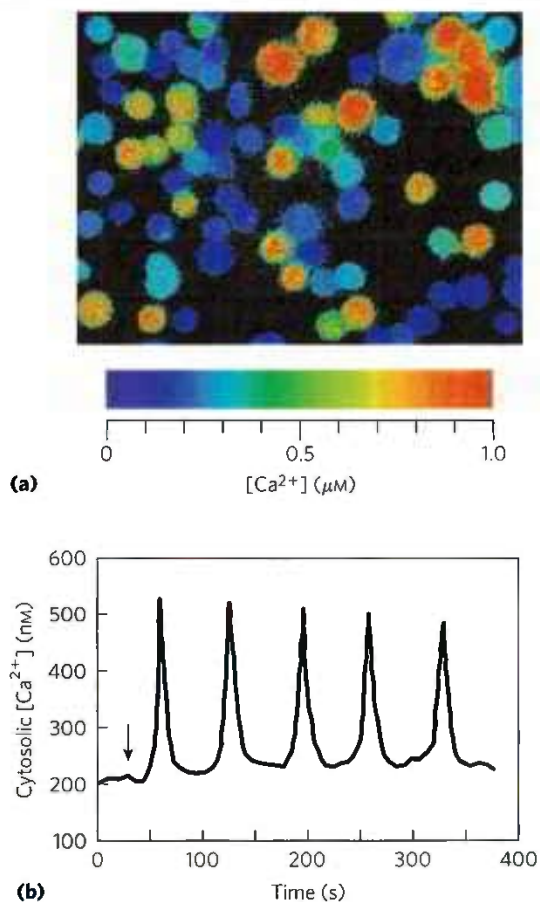


FIGURE 12-18 Triggering of oscillations in intracellular $[\text{Ca}^{2+}]$ by extracellular signals. (a) The dye fura, which undergoes fluorescence changes when it binds Ca^{2+} , can be used with fluorescence microscopy to measure the instantaneous Ca^{2+} in cells. The color scale relates fluorescence intensity to $[\text{Ca}^{2+}]$. Here, thymus cells have been stimulated with extracellular ATP, which raises their internal $[\text{Ca}^{2+}]$. The cells are heterogeneous in their responses: some have high intracellular $[\text{Ca}^{2+}]$ (red), others have much lower $[\text{Ca}^{2+}]$ (blue). (b) When such a probe is used in a single hepatocyte, the agonist norepinephrine (added at the arrow) causes oscillations of $[\text{Ca}^{2+}]$ from 200 to 500 nm. Similar oscillations are induced in other cell types by other extracellular signals. (a) Courtesy Michael D. Cahalan, University of California, Irvine, Department of Physiology and Biophysics. (b) Data from T. A. Rooney et al., *J. Biol. Chem.* 264:17,131, 1989.]

the enzyme that degrades cAMP (phosphodiesterase) are stimulated by Ca^{2+} . Temporal and spatial changes in $[\text{Ca}^{2+}]$ can therefore produce transient, localized changes in $[\text{cAMP}]$. We have noted already that PKA, the enzyme that responds to cAMP, is often part of a highly localized supramolecular complex assembled on scaffold proteins such as AKAPs. This subcellular localization of target enzymes, combined with temporal and spatial gradients in $[\text{Ca}^{2+}]$ and $[\text{cAMP}]$, allows a cell to respond to one or several signals with subtly nuanced metabolic changes.

SUMMARY 12.2 G Protein–Coupled Receptors and Second Messengers

- G protein–coupled receptors (GPCRs) have seven transmembrane helices and act through heterotrimeric G proteins. Ligand binding activates the G protein, which then stimulates or inhibits the activity of an effector enzyme, changing the local concentration of its second-messenger product cAMP.

- Epinephrine, acting through its GPCR and the G_s protein, stimulates adenylyl cyclase, which produces cAMP. Cyclic AMP activates protein kinase A (PKA), which then phosphorylates target proteins on a Ser or Thr residue, changing their biological activity.

- To end the response to epinephrine, a phosphodiesterase breaks down cAMP, and the G protein is inactivated by its own GTPase activity. Phosphoprotein phosphatases reverse the effects of PKA.

- When the epinephrine signal persists, β -adrenergic receptor–specific protein kinase phosphorylates the GPCR, creating a binding site for the protein β -arrestin, which prevents interaction between the GPCR and its G protein. Arrestin triggers desensitization by movement of the receptor into intracellular vesicles.

- GPCRs typically have their effects through the G protein–cAMP–PKA pathway; some act through G_s , raising $[\text{cAMP}]$, others through G_i , lowering $[\text{cAMP}]$. Adaptor proteins such as AKAPs tether PKA and limit its area of action and its target proteins.

- Trimeric G proteins coupled to GPCRs are activated when their bound GDP is exchanged for GTP, and they remain active until their GTPase converts bound GTP to GDP. Their GTPase activity is modulated by GTPase activator proteins (GAPs) and regulators of G-protein signaling (RGSs).

- Monomeric (small) G proteins such as Ras, Rab, and Ran also serve as self-limiting switches. Defective G-protein signaling is common in individuals with some types of cancer.

- Some GPCRs are coupled to a G protein (G_q) that acts through a plasma membrane phospholipase C that cleaves PIP_2 to diacylglycerol and IP_3 . By opening Ca^{2+} channels in the endoplasmic reticulum, IP_3 raises cytosolic

[Ca²⁺]. Diacylglycerol and Ca²⁺ act together to activate protein kinase C, which phosphorylates and changes the activity of specific cellular proteins.

■ Cellular [Ca²⁺] also regulates (often through calmodulin) many other enzymes and proteins involved in secretion, cytoskeletal rearrangements, or contraction. Many of the target enzymes are in the family of Ca²⁺-activated protein kinases (PKCs).

12.3 GPCRs in Vision, Olfaction, and Gustation

The detection of light, odors, and tastes (vision, olfaction, and gustation, respectively) in animals is accomplished by specialized sensory neurons that use signal-transduction mechanisms fundamentally similar to those that detect hormones, neurotransmitters, and growth factors. An

initial sensory signal is greatly amplified by mechanisms that include gated ion channels and intracellular second messengers; the system adapts to continued stimulation by changing its sensitivity to the stimulus (desensitization); and sensory input from several receptors is integrated before the final signal goes to the brain.

The Vertebrate Eye Uses Classic GPCR Mechanisms

Visual transduction (**Fig. 12-19**) begins when light falls on **rhodopsin**, a GPCR in the disk membranes of rod cells of the vertebrate eye. (Rod cells do not detect colors; cone cells do, as we shall see in **Box 12-3**.) The light-absorbing pigment (chromophore) 11-*cis*-retinal is covalently attached to **opsin**, the protein component of rhodopsin, which lies near the middle of the disk membrane bilayer. When a photon is absorbed by the retinal component of rhodopsin (step 1), the energy causes a photochemical change; 11-*cis*-retinal is converted

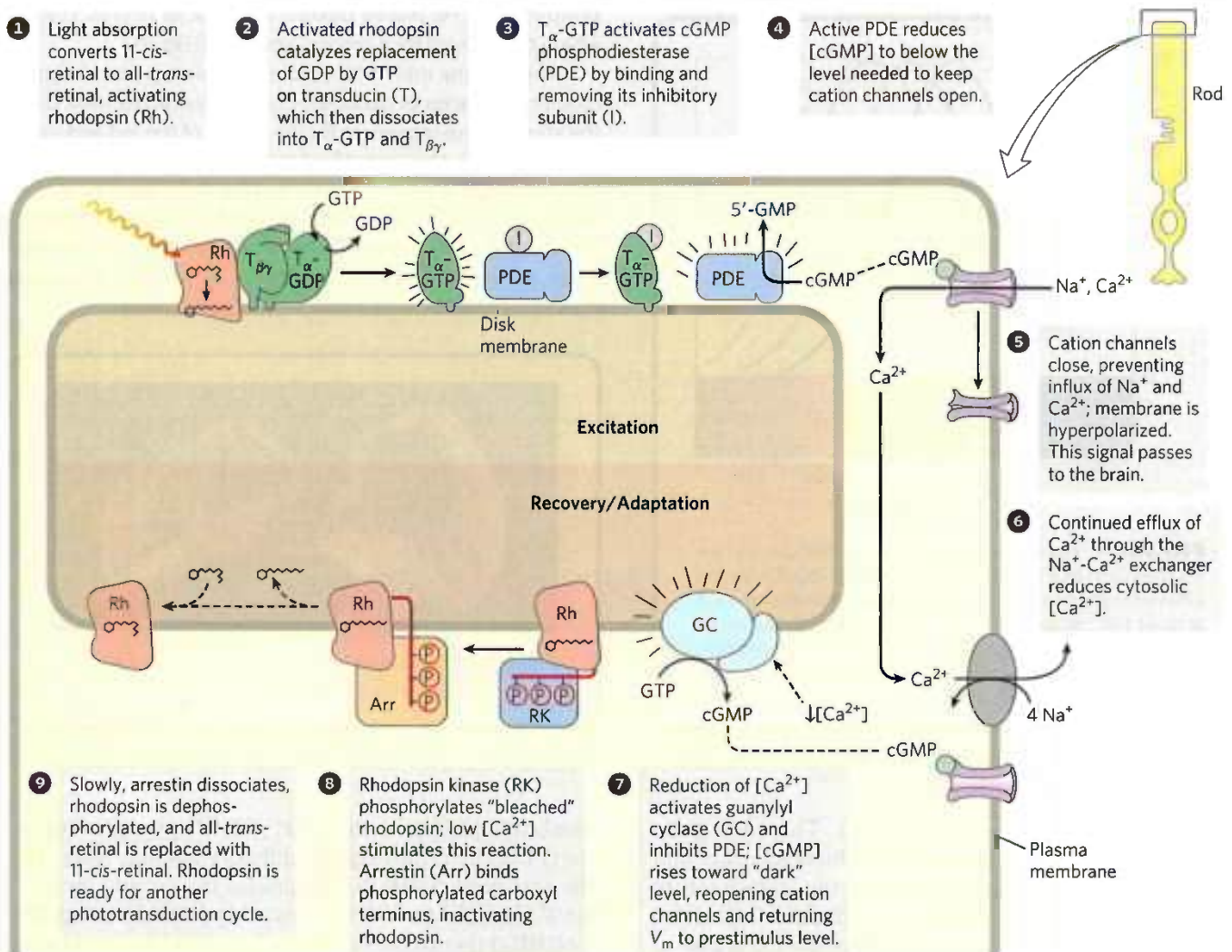


FIGURE 12-19 Molecular consequences of photon absorption by rhodopsin in the rod outer segment. The top half of the figure (steps 1 to 5) describes excitation; the bottom shows post-illumination steps: recovery (steps 6 and 7) and adaptation (steps 8 and 9).



Color Blindness: John Dalton's Experiment from the Grave

Color vision involves a path of sensory transduction in specialized cells in the retina. Three types of cone cells are specialized to detect light from different regions of the spectrum, using three related photoreceptor proteins (opsins). Each cone cell expresses only one kind of opsin, but each type is closely related to rhodopsin in size, amino acid sequence, and, presumably, three-dimensional structure. The differences among the opsins, however, are great enough to place the chromophore, 11-*cis*-retinal, in three slightly different environments, with the result that the three photoreceptors have different absorption spectra (Fig. 1). We discriminate colors and hues by integrating the output from the three types of cone cells, each containing one of the three types of photoreceptors.

Color blindness, such as the inability to distinguish red from green, is a fairly common, genetically inherited trait in humans. The various types of color blindness result from different opsin mutations. One form is due to loss of the red photoreceptor; affected individuals are **red⁻ dichromats** (they

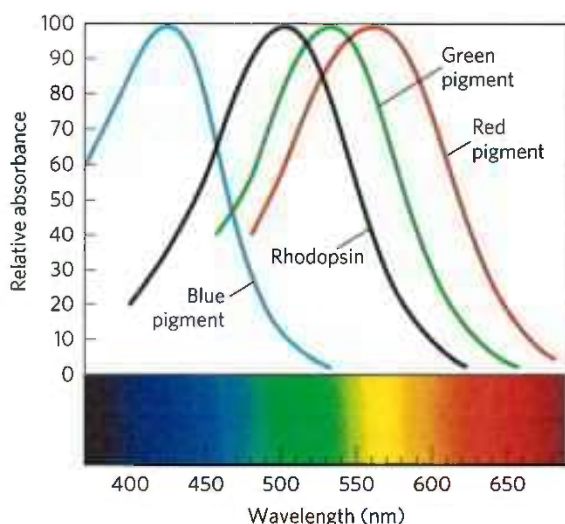


FIGURE 1 Absorption spectra of purified rhodopsin and the red, green, and blue receptors of cone cells. The receptor spectra, obtained from individual cone cells isolated from cadavers, peak at about 420, 530, and 560 nm, and the maximum absorption for rhodopsin is at about 500 nm. For reference, the visible spectrum for humans is about 380 to 750 nm. [Data from J. Nathans, *Sci. Am.* 260 (February):42, 1989.]

to all-*trans*-retinal (see Fig. 10-20). This change in the structure of the chromophore forces conformational changes in the rhodopsin molecule, allowing it to interact with and thus activate its trimeric G protein, transducin. Rhodopsin now stimulates the exchange of bound GDP on transducin for GTP from the cytosol (Fig. 12-19, step 2), and activated transducin stimulates the

see only two primary colors). Others lack the green pigment and are **green⁻ dichromats**. In some cases, the red and green photoreceptors are present but have a changed amino acid sequence that causes a change in their absorption spectra, resulting in abnormal color vision. Depending on which pigment is altered, these individuals are **red-anomalous trichromats** or **green-anomalous trichromats**. Examination of the genes for the visual receptors has allowed the diagnosis of color blindness in the chemist John Dalton more than a century after his death.

Dalton (of atomic theory fame) was color-blind. He thought it probable that the vitreous humor of his eyes (the fluid that fills the eyeball behind the lens) was tinted blue, unlike the colorless fluid of normal eyes. He proposed that after his death, his eyes should be dissected and the color of the vitreous humor determined. His wish was honored. The day after Dalton's death in July 1844, Joseph Ransome dissected his eyes and found the vitreous humor to be perfectly colorless. Ransome, like many scientists, was reluctant to throw samples away. He placed Dalton's eyes in a jar of preservative, where they stayed for a century and a half (Fig. 2).

Then, in the mid-1990s, molecular biologists in England took small samples of Dalton's retinas and extracted DNA. Using the known gene sequences for the opsins of the red and green light receptors, they amplified the relevant sequences using PCR and determined that Dalton had the opsin gene for the red photopigment but lacked the opsin gene for the green photopigment. Dalton was a green⁻ dichromat. So, 150 years after his death, the experiment Dalton started — by hypothesizing about the cause of his color blindness — was finally finished.



FIGURE 2 Dalton's eyes. [Professor J. D. Mollon, Cambridge University, Department of Experimental Psychology.]

membrane protein cyclic GMP (cGMP) phosphodiesterase (PDE) by removing an inhibitory subunit (step 3). The activated cGMP PDE degrades the second messenger 3',5'-cGMP to 5'-GMP, lowering [cGMP] (step 4). A cGMP-dependent Na⁺ or Ca²⁺ channel in the plasma membrane closes (step 5), while a Na⁺-Ca²⁺ active antiporter continues to pump Ca²⁺ outward across the

plasma membrane (step 6), making the transmembrane electrical potential more negative inside (that is, hyperpolarizing the rod cell). This electrical change passes through a series of specialized nerve cells to the visual cortex of the brain.

Several steps in the visual-transduction process result in a huge amplification of the signal. Each excited rhodopsin molecule activates at least 500 molecules of transducin, and each transducin molecule can activate a molecule of cGMP PDE. This phosphodiesterase has a remarkably high turnover number: each activated molecule hydrolyzes 4,200 molecules of cGMP per second. The binding of cGMP to cGMP-gated ion channels is cooperative, and a relatively small change in [cGMP] therefore registers as a large change in ion conductance. The result of these amplifications is exquisite sensitivity to light. Absorption of a single photon closes 1,000 or more ion channels for Na^+ and Ca^{2+} , hyperpolarizing the cell's membrane by about 1 mV.

As your eyes move across this line of type, the retinal images of the first words disappear rapidly—before you see the next series of words. In that short interval, a great deal of biochemistry has taken place. Very soon after illumination of the rod or cone cells stops, the photosensory system shuts off. The α subunit of transducin (T_α , with bound GTP) has GTPase activity. Within milliseconds after the decrease in light intensity, GTP is hydrolyzed and T_α reassociates with $T_{\beta\gamma}$. The inhibitory subunit of PDE, which had been bound to T_α -GTP, is released and reassociates with PDE, strongly inhibiting its activity and thus slowing cGMP breakdown.

At the same time, a second factor that helps to end the response to light is the reduction of intracellular $[\text{Ca}^{2+}]$ that results from continued Ca^{2+} efflux through the Na^+ - Ca^{2+} exchanger (Fig. 12-19, step 6). High $[\text{Ca}^{2+}]$ inhibits the enzyme that makes cGMP (guanylyl cyclase; step 7), so cGMP production rises when $[\text{Ca}^{2+}]$ falls, quickly reaching its prestimulus level.

In response to prolonged illumination, rhodopsin itself undergoes changes that limit the duration of its signaling activity. The conformational change induced in rhodopsin by light absorption exposes several Thr and Ser residues in its carboxyl-terminal domain, and these residues are phosphorylated by **rhodopsin kinase** (step 8), which is functionally and structurally homologous to the β -adrenergic kinase (β ARK) that desensitizes the β -adrenergic receptor. The phosphorylated carboxyl-terminal domain of rhodopsin is bound by the protein **arrestin 1**, preventing further interaction between activated rhodopsin and transducin (see Fig. 12-10b). Arrestin 1 is a close homolog of arrestin 2 (β arr) of the β -adrenergic system. On a much longer time scale (step 9), the all-*trans*-retinal bound to light-bleached rhodopsin is removed and replaced with 11-*cis*-retinal, making rhodopsin ready to detect another photon.

Vertebrate Olfaction and Gustation Use Mechanisms Similar to the Visual System

The sensory cells that detect odors and tastes have much in common with the visual receptor system. Binding of an odorant molecule to one of its specific GPCRs (humans have about 800 different GPCRs; rodents have about 1,200) triggers a change in receptor conformation, activating a G protein, G_{olf} , analogous to transducin and to G_s of the β -adrenergic system. The activated G_{olf} activates adenylyl cyclase, raising the local [cAMP]. **P5** The cAMP-gated Na^+ and Ca^{2+} channels of the plasma membrane open, and the influx of Na^+ and Ca^{2+} produces a small depolarization called the **receptor potential**. If a sufficient number of odorant molecules encounter receptors, the receptor potential is strong enough to cause the neuron to fire an action potential. This signal is relayed to the brain in several stages and registers as a specific smell. All these events occur within 100 to 200 ms. When the olfactory stimulus is no longer present, the transducing machinery shuts itself off in several ways. A cAMP phosphodiesterase returns [cAMP] to the prestimulus level. G_{olf} hydrolyzes its bound GTP to GDP, thereby inactivating itself. Phosphorylation of the receptor by a specific kinase prevents its interaction with G_{olf} , by a mechanism analogous to that used to desensitize the β -adrenergic receptor and rhodopsin. Some odorants are detected by another mechanism we have seen in other signal transductions: activation of a phospholipase and production of IP_3 , leading to a rise in intracellular $[\text{Ca}^{2+}]$.

The sense of taste in vertebrates reflects the activity of gustatory neurons clustered in taste buds on the surface of the tongue. In taste sensory neurons, GPCRs are coupled to the heterotrimeric G protein **gustducin**. When the tastant molecule binds its receptor, gustducin is activated and stimulates cAMP production by adenylyl cyclase. The resulting elevation of [cAMP] activates PKA, which phosphorylates K^+ channels in the plasma membrane, causing them to close and sending an electrical signal to the brain. Other taste buds specialize in detecting bitter, sour, salty, or umami (savory) tastants, using various combinations of second messengers and ion channels in the transduction mechanisms.

All GPCR Systems Share Universal Features

We have now looked at several types of signaling systems (hormone signaling, vision, olfaction, and gustation) in which membrane receptors are coupled to second messenger-generating enzymes through G proteins. As we have intimated, signaling mechanisms must have arisen early in evolution; genomic studies have revealed hundreds of genes encoding GPCRs in vertebrates, arthropods (*Drosophila* and mosquito), and the roundworm *Caenorhabditis elegans*. Even the common budding yeast *Saccharomyces* uses GPCRs and G proteins to detect the opposite mating type. Overall patterns have been conserved, and the introduction of variety has given

modern organisms the ability to respond to a wide range of stimuli (**Table 12-6**). Of the approximately 20,000 genes in the human genome, as many as 800 encode GPCRs, including hundreds for olfactory stimuli and many orphan receptors, for which the natural ligand is not yet known.

P2 All well-studied signal-transducing systems that act through heterotrimeric G proteins share some common

TABLE 12-6 Some Signals That Act through GPCRs

Amines	Tachykinin
Acetylcholine (muscarinic)	Thyrotropin-releasing hormone
Dopamine	Vasopressin
Epinephrine	Protein hormones
Histamine	Follicle-stimulating hormone
Serotonin	Gonadotropin
Peptides	Lutropin-choriogonadotropic hormone
Angiotensin	Thyrotropin
Bombesin	Prostanoids
Bradykinin	Prostacyclin
Chemokine	Prostaglandin
Colecystokinin (CCK)	Thromboxane
Endothelin	Others
Gonadotropin-releasing hormone	Cannabinoids
Interleukin-8	Lysosphingolipids
Melanocortin	Melatonin
Neuropeptide Y	Olfactory stimuli
Neurotensin	Opioids
Orexin	Rhodopsin
Somatostatin	

features that reflect their evolutionary relatedness (**Fig. 12-20**). The receptors have seven transmembrane segments, a domain (generally the loop between transmembrane helices 6 and 7) that interacts with a G protein, and a carboxyl-terminal cytoplasmic domain that undergoes reversible phosphorylation on several Ser or Thr residues. The ligand-binding site (or, in the case of light reception, the light receptor) is buried deep in the membrane and includes residues from several of the transmembrane segments. Ligand binding (or light) induces a conformational change in the receptor, exposing a domain that can interact with a G protein. Heterotrimeric G proteins activate or inhibit effector enzymes (adenylyl cyclase, PDE, or PLC), which change the concentration of a second messenger (cAMP, cGMP, IP₃, or Ca²⁺). In the hormone-detecting systems, the final output is an activated protein kinase that regulates some cellular process by phosphorylating a protein critical to that process. In sensory neurons, the output is a change in membrane potential and a consequent electrical signal that passes to another neuron in the pathway connecting the sensory cell to the brain.

All these systems self-inactivate. Bound GTP is converted to GDP by the GTPase activity of G proteins, often augmented by GTPase-activating proteins (GAPs) or regulators of G-protein signaling (RGS). In some cases, the effector enzymes that are the targets of modulation by G proteins also serve as GAPs. The desensitization mechanism involving phosphorylation of the carboxyl-terminal region followed by arrestin binding is widespread and may be universal.

SUMMARY 12.3 GPCRs in Vision, Olfaction, and Gustation

■ Vision, olfaction, and gustation in vertebrates employ GPCRs, which act through heterotrimeric G proteins to change the membrane potential (V_m) of a sensory neuron.

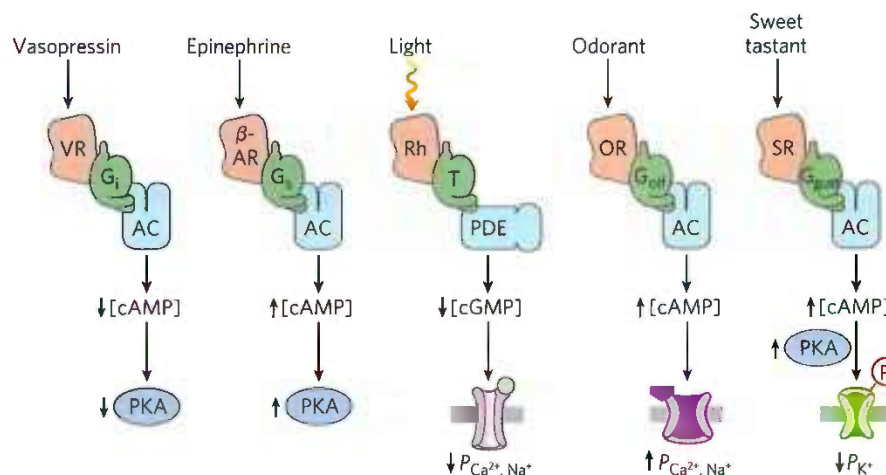


FIGURE 12-20 Common features of signaling systems that detect hormones, light, smells, and tastes. GPCRs provide signal specificity, and their interaction with G proteins provides signal amplification. Heterotrimeric G proteins activate or inhibit effector enzymes: adenylyl cyclase (AC) and phosphodiesterases (PDEs) that degrade cAMP or cGMP. Changes in concentration of the second messengers (cAMP, cGMP) result in alterations in enzymatic activities via phosphorylation or alterations in the permeability (P) of surface membranes to Ca²⁺, Na⁺, and K⁺. The resulting

depolarization or hyperpolarization of the sensory cell (the signal) passes through relay neurons to sensory centers in the brain. In the best-studied cases, desensitization includes phosphorylation of the receptor and binding of a protein (arrestin) that interrupts receptor–G protein interactions. (The path of odorant detection by production of IP₃ and increase in intracellular [Ca²⁺], mentioned in the text, is not shown here.) VR is the vasopressin receptor; β-AR, the β-adrenergic receptor; Rh, rhodopsin; OR, olfactory receptor; SR, sweet-taste receptor.

- Light activates the GPCR rhodopsin, which allosterically activates the trimeric G protein transducin. T_α activates a cGMP phosphodiesterase, lowering [cGMP] and closing cGMP-dependent ion channels. The resulting electrical impulse carries the signal to the brain.
- In olfactory neurons, olfactory stimuli, acting through GPCRs and G proteins, trigger an increase in [cAMP] (by activating adenylyl cyclase) or [Ca^{2+}] (by activating PLC). These second messengers affect ion channels and thus the V_m . Gustatory neurons have GPCRs that respond to tastants by altering levels of cAMP, which changes V_m by gating ion channels.
- There is a high degree of conservation of signaling proteins and transduction mechanisms across signaling

systems and across species. GPCRs with seven transmembrane helices, G proteins with intrinsic GTPase activities, cyclic nucleotides, and protein kinases are central to signaling.

12.4 Receptor Tyrosine Kinases

P4 The **receptor tyrosine kinases (RTKs)**, a family of plasma membrane receptors with protein kinase activity, transduce extracellular signals by a mechanism fundamentally different from that of GPCRs. RTKs have a ligand-binding domain on the extracellular face of the plasma membrane and an enzyme active site on the cytoplasmic face, connected by a single transmembrane segment, as seen for insulin in **Figure 12-21a**. The cytoplasmic domain

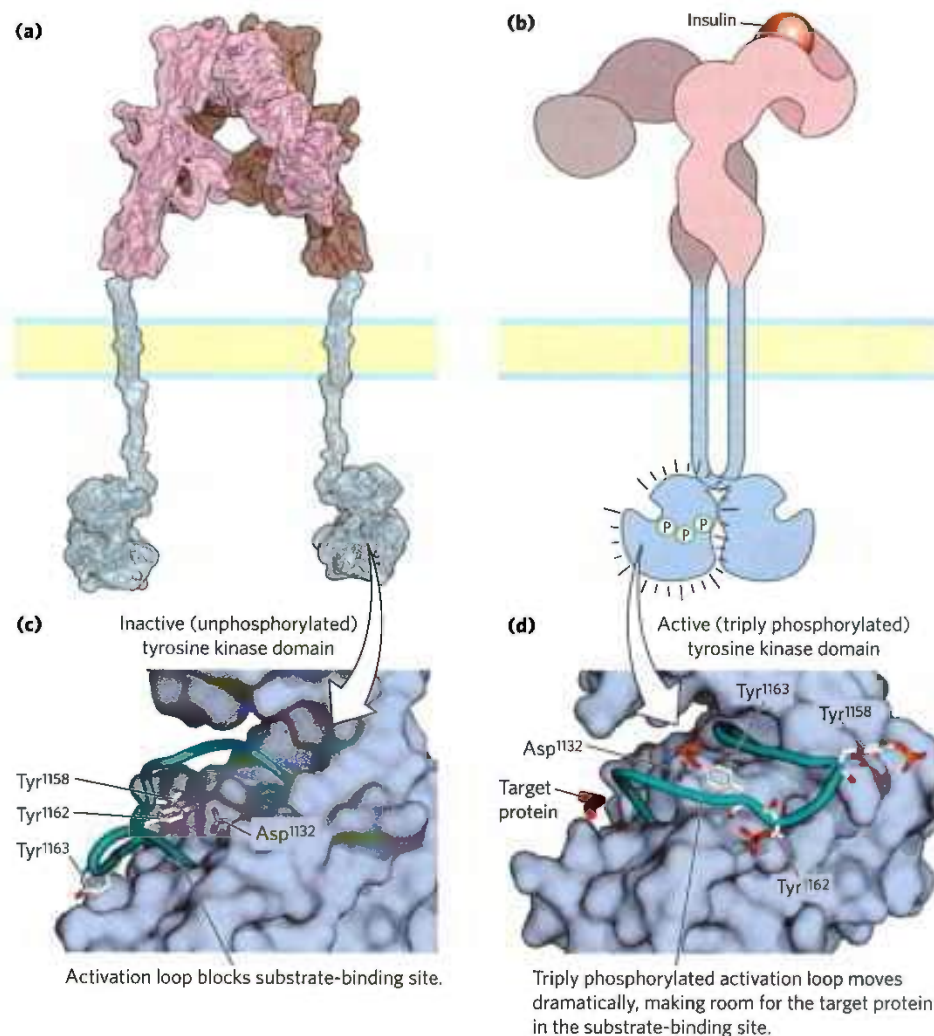


FIGURE 12-21 Activation of the insulin-receptor tyrosine kinase by autophosphorylation. (a) A structural model assembled from crystal structures of the extracellular and tyrosine kinase domains and NMR solution structure of the transmembrane domain. The insulin-binding region of the insulin receptor lies outside the cell and comprises two intertwined α subunits. (b) The binding of a single molecule of insulin is communicated through the single transmembrane helix of each β subunit to the paired Tyr kinase domains inside the cell, moving them toward each other and activating them to phosphorylate each other on three Tyr residues. (c) In the inactive form of the Tyr kinase domain, the activation loop (backbone

shown in teal) sits in the active site, and none of the critical Tyr residues (stick structures) are phosphorylated. This conformation is stabilized by hydrogen bonding between Tyr¹¹⁶² and Asp¹¹³². (d) Activation of the Tyr kinase allows each β subunit to phosphorylate three Tyr residues (Tyr¹¹⁵⁸, Tyr¹¹⁶², Tyr¹¹⁶³) on the other β subunit. (Phosphoryl groups are depicted in red and orange.) The introduction of three highly charged (P)-Tyr residues forces a 30 Å change in the position of the activation loop, away from the substrate-binding site, which thus becomes available to bind and phosphorylate a target protein. [(a) Data and information from T. Gutmann et al., *J. Cell Biol.* 21:1643, 2018, Fig. 1. Data from (c) PDB ID 1IRK, S. R. Hubbard et al., *Nature* 372:746, 1994; (d) PDB ID 1IR3, S. R. Hubbard, *EMBO J.* 16:5572, 1997.]

is a Tyr kinase, a protein kinase that phosphorylates specific Tyr residues in target proteins. The receptors for insulin and epidermal growth factor are prototypes for the 58 RTKs in humans (see Fig. 12-26).

Stimulation of the Insulin Receptor Initiates a Cascade of Protein Phosphorylation Reactions

Insulin regulates both metabolic enzymes and gene expression. It initiates a signal that travels divergent pathways from the plasma membrane receptor to insulin-sensitive enzymes in the cytosol, and to enzymes that act in the nucleus by stimulating the transcription of specific genes. The active insulin receptor protein (INSR) consists of two identical α subunits protruding from the outer face of the plasma membrane and two transmembrane β subunits with their carboxyl termini protruding into the cytosol—a dimer of $\alpha\beta$ monomers (Fig. 12-21). The α subunits contain the insulin-binding domain, and the intracellular domains of the β subunits contain the

protein kinase activity that transfers a phosphoryl group from ATP to the hydroxyl group of Tyr residues in specific target proteins.

Signaling through INSR begins with the binding of one insulin molecule between the two subunits of the receptor on the extracellular side, causing movement of the Tyr kinase domains together and stimulating their activity. Each β subunit phosphorylates three essential Tyr residues near the carboxyl terminus of the other β subunit. This **autophosphorylation** opens the active site so that the enzyme can phosphorylate Tyr residues of other target proteins. The mechanism of activation of the INSR protein kinase is similar to that described for PKA and PKC: a region of the cytoplasmic domain called the activation loop that usually occludes the active site moves out of the active site after being phosphorylated, opening the site for the binding of target proteins (Fig. 12-21c, d).

When INSR is autophosphorylated (Fig. 12-22, step 1) and becomes an active Tyr kinase, one of its targets is the protein **insulin receptor substrate 1 (IRS1)**;

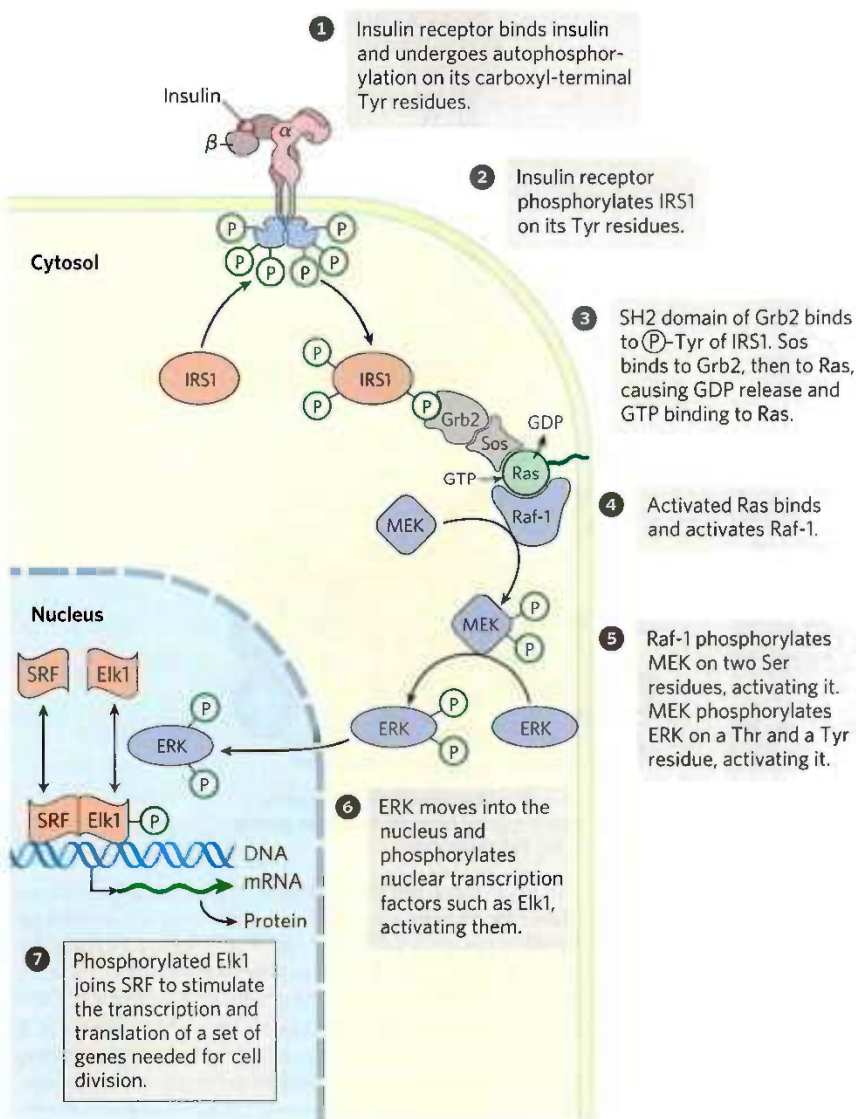


FIGURE 12-22 Regulation of gene expression by insulin through a MAP kinase cascade. The signaling pathway by which insulin regulates the expression of specific genes consists of a cascade of protein kinases, each of which activates the next. The insulin receptor is a Tyr-specific kinase; the other kinases (all shown in blue) phosphorylate Ser or Thr residues. MEK is a dual-specificity kinase that phosphorylates both a Thr residue and a Tyr residue in ERK (extracellular regulated kinase). MEK is mitogen-activated, ERK-activating kinase; SRF is serum response factor.

step ②). Once phosphorylated on several of its Tyr residues, IRS1 becomes the point of nucleation for a complex of proteins (step ③) that carry the message from the insulin receptor to end targets in the cytosol and nucleus, through a long series of intermediate proteins. First, a P-Tyr residue of IRS1 binds to the **SH2 domain** of the protein Grb2. Many signaling proteins contain SH2 (Src homology 2) domains, all of which bind P-Tyr residues in a protein partner. Grb2 (growth factor receptor-bound protein 2) is an adaptor protein, with no intrinsic enzymatic activity. Its function is to bring together two proteins (in this case, IRS1 and the protein Sos) that must interact to enable signal transduction. In addition to its SH2 (P-Tyr -binding) domain, Grb2 contains a second protein-binding domain, SH3, that binds to a proline-rich region of Sos (son of sevenless), recruiting Sos to the growing receptor complex. When bound to Grb2, Sos acts as a guanosine nucleotide-exchange factor (GEF), catalyzing the replacement of bound GDP with GTP on the small G protein Ras.

In its GTP-bound active form, Ras can activate a protein kinase, Raf-1 (Fig. 12-22, step ④), the first of three protein kinases—Raf-1, MEK, and ERK—that form a cascade in which each kinase activates the next by phosphorylation (step ⑤). The protein kinases MEK and ERK are activated by phosphorylation of both a Thr residue and a Tyr residue. When activated, ERK mediates some of the biological effects of insulin by entering the nucleus and phosphorylating transcription factors, including Elk1 (step ⑥), that modulate the transcription of about 100 insulin-regulated genes (step ⑦), some of which encode proteins essential for cell division. Thus, insulin acts as a growth factor.

The proteins Raf-1, MEK, and ERK are members of three larger families, for which several nomenclatures are used. ERK is in the **MAPK** family (mitogen-activated protein kinases; mitogens are extracellular signals that induce mitosis and cell division). Soon after discovery of the first MAPK enzyme, that enzyme was found to be activated by another protein kinase, which was named MAP kinase kinase (MEK belongs to this family), and when a third kinase that activated MAP kinase kinase was discovered, it was given the slightly ludicrous family name MAP kinase kinase kinase (Raf-1 is in this family). Somewhat less cumbersome are the abbreviations for these three families: MAPK, MAPKK, and MAPKKK. Kinases in the MAPK and MAPKKK families are specific for Ser residues or Thr residues, and MAPKKs (here, MEK) phosphorylate both a Ser residue and a Tyr residue in their substrate, a MAPK (here, ERK).

P4 This insulin pathway is but one instance of a more general scheme in which hormone signals, via pathways similar to that shown in Figure 12-22, result in a change in the phosphorylation of target enzymes by protein kinases or phosphoprotein phosphatases. The target of phosphorylation is often another protein kinase, which then phosphorylates a third protein kinase, and

so on. The result is a cascade of reactions that amplifies the initial signal by many orders of magnitude (see Fig. 12-7). **MAPK cascades** (such as the Raf-MEK-ERK sequence in Fig. 12-22) mediate signaling initiated by a variety of growth factors, such as platelet-derived growth factor (PDGF) and epidermal growth factor (EGF), both of which are receptor tyrosine kinases like the insulin receptor. Another general scheme exemplified by the insulin receptor pathway is the use of nonenzymatic adaptor proteins, often with intrinsically disordered regions, to bring together the components of divergent signaling pathways, to which we now turn.

The Membrane Phospholipid PIP_3 Functions at a Branch in Insulin Signaling

The signaling pathway from insulin branches at IRS1 (step ② in Fig. 12-22 and Fig. 12-23). Grb2 is not the only protein that associates with phosphorylated IRS1. The enzyme phosphoinositide 3-kinase (PI3K) binds IRS1 through PI3K's SH2 domain. Thus activated, PI3K converts the membrane lipid phosphatidylinositol 4,5-bisphosphate (PIP_2) to phosphatidylinositol 3,4,5-trisphosphate (PIP_3) by the transfer of a phosphoryl group from ATP (Fig. 12-24). The multiply (negatively) charged head group of PIP_3 , protruding on the cytoplasmic side of the plasma membrane, is the starting point for a second signaling branch involving another cascade of protein kinases. When bound to PIP_3 , protein kinase B (PKB; also called Akt) is phosphorylated and activated by yet another protein kinase, PDK1 (not shown in Fig. 12-23). The activated PKB then phosphorylates Ser or Thr residues in its target proteins, one of which is glycogen synthase kinase 3 (GSK3). In its active, nonphosphorylated form, GSK3 phosphorylates glycogen synthase, inactivating it and thereby contributing to the slowing of glycogen synthesis. (This mechanism is only part of the explanation for the effects of insulin on glycogen metabolism; see Fig. 15-16.) When phosphorylated by PKB, GSK3 is inactivated. By thus preventing inactivation of glycogen synthase in liver and muscle, the cascade of protein phosphorylations initiated by insulin ultimately stimulates glycogen synthesis (Fig. 12-23).

In a third signaling branch important in muscle and fat tissue, PKB, acting through two small G proteins (RAC1 and Rab) triggers the clathrin-aided movement of glucose transporters (GLUT4) from intracellular vesicles to the plasma membrane, stimulating glucose uptake from the blood (Fig. 12-23, step ⑤; see also Box 11-1). This increase in glucose uptake triggered by insulin has profound metabolic and medical consequences, as we shall see in Chapter 23.

How fast is the response to insulin? Phosphoproteomics uses high-resolution, high-throughput mass spectrometry to determine, for thousands of proteins in a cell type, which residues in which proteins are

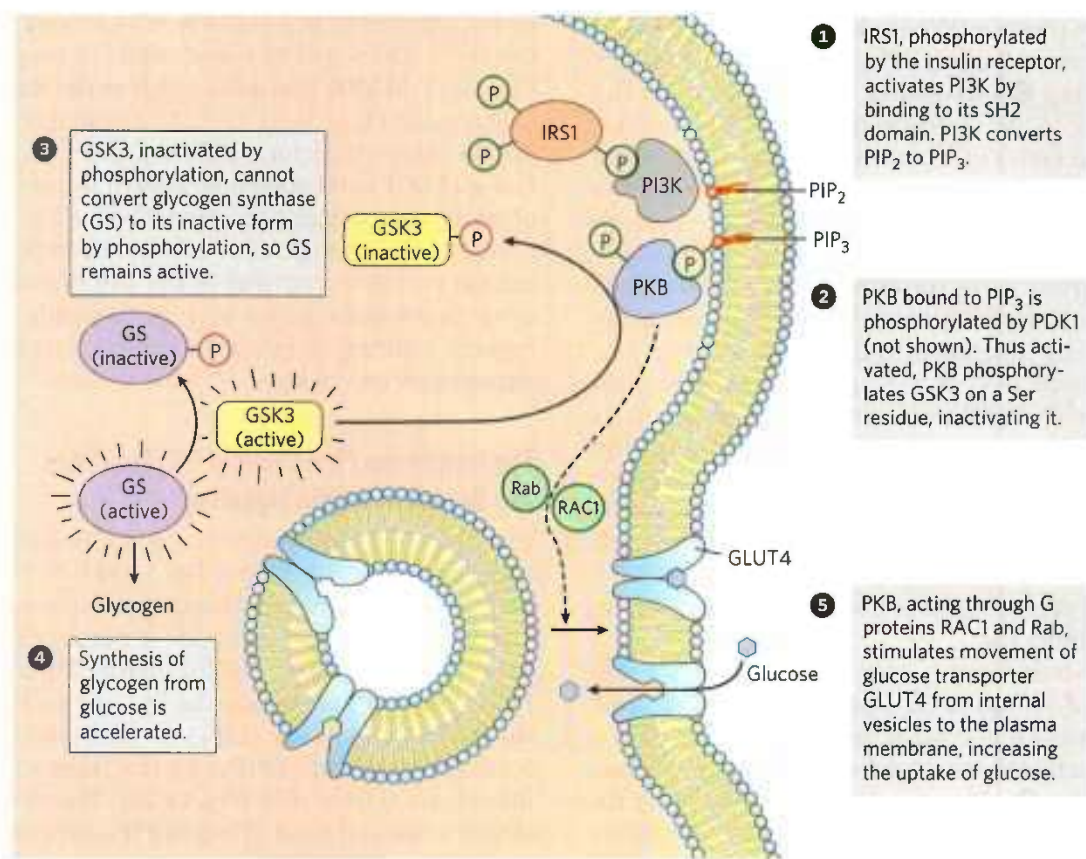


FIGURE 12-23 Insulin action on glycogen synthesis and GLUT4 movement to the plasma membrane. The activation of PI3 kinase (PI3K) by phosphorylated IRS1 initiates (through protein kinase B,

PKB) movement of the glucose transporter GLUT4 to the plasma membrane, and the activation of glycogen synthase.

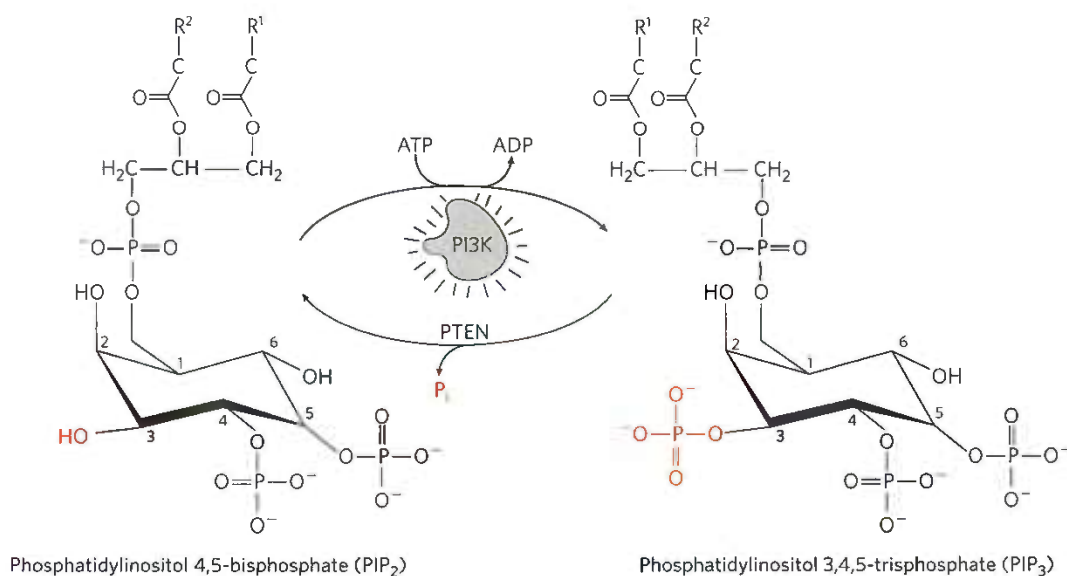


FIGURE 12-24 Regulation of PIP₃ formation and breakdown. Phosphoinositide 3-kinase (PI3K) responds to the insulin signal by catalyzing the transfer of a phosphoryl group from ATP to C-3 of the inositol ring of phosphatidylinositol 4,5-bisphosphate (PIP₂) to form phosphatidylinositol

3,4,5-trisphosphate (PIP₃), which serves as a nucleation point for proteins involved in the MAPK cascade. The enzyme PTEN (phosphatase and tensin homolog) ends the response by catalyzing removal of the same phosphoryl group, which is released as P_i.

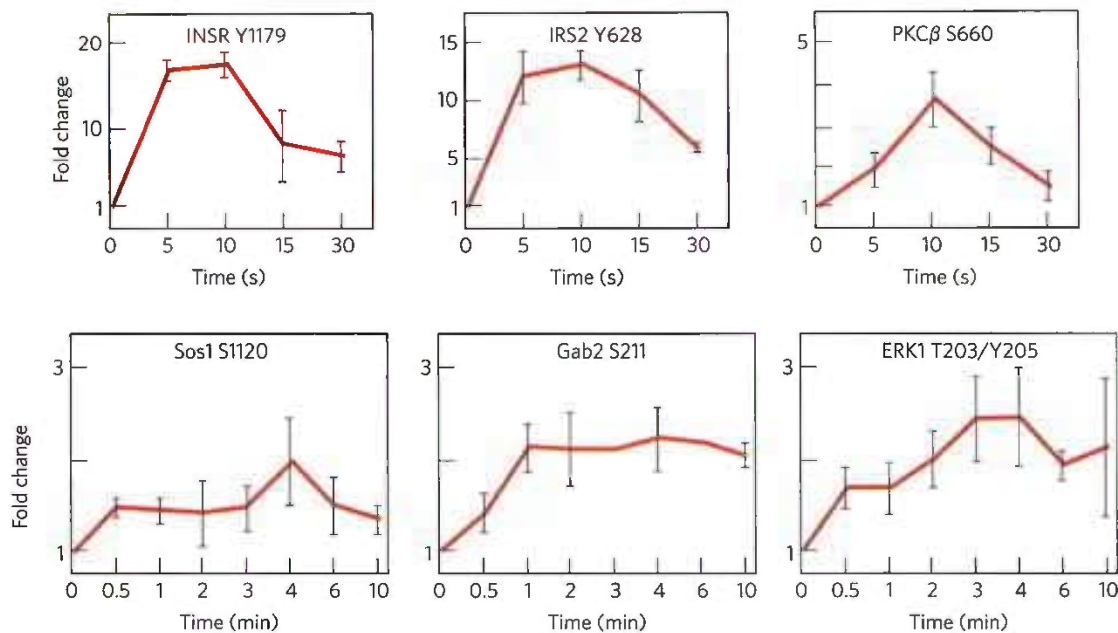


FIGURE 12-25 Time course of phosphorylations triggered by insulin. Mouse liver proteins were examined by mass spectrometry at intervals after injection of insulin to determine quantitatively for each protein which amino acid residues became phosphorylated and when phosphorylation occurred. Each graph represents the phosphorylation of a single residue

in the protein, designated by its one-letter abbreviation and position in the primary sequence. Proteins are INSR, insulin receptor; IRS2, insulin receptor substrate 2; PKC β , protein kinase C β ; Sos and ERK, as in Fig. 12-22. Gab2 is an adaptor protein involved in the MAPK and PI3K signaling pathways. (Data from S. J. Humphrey et al., *Nature Biotechnol.* 33:990, 2015, Fig. 4.)

phosphorylated in response to a stimulus such as insulin in the living animal (Fig. 12-25). Autophosphorylation of the insulin receptor and IRS1 occurs within a few seconds of insulin addition; protein kinase C β is phosphorylated slightly later (within 15 s), and Sos and Gab are phosphorylated within 0.5 and 1 min. ERK1, the end target in this signaling pathway (Fig. 12-22), is maximally phosphorylated within 3 min. The movement of glucose transporter GLUT4 into the plasma membrane takes about 15 min; and the many changes in gene expression in response to insulin occur over several hours. A recently discovered action of insulin has a different mechanism and a slower time course. In some cases, insulin enters the cell and the nucleus, where, with the help of several other nuclear proteins, it regulates gene expression by binding to promoter regions on the DNA.

 As in all signaling pathways, there is a mechanism for terminating the activity of the PI3K-PKB pathway. A PIP₃-specific phosphatase (in humans, PTEN) removes the phosphoryl group at the 3 position of PIP₃ to produce PIP₂ (see Fig. 12-24), which no longer serves as a binding site for PKB, and the signaling chain is broken. In several types of cancer, the *PTEN* gene has often undergone mutation, resulting in a defective regulatory circuit and abnormally high levels of PIP₃ activity. The result is a continuing signal for cell division and thus tumor growth. The unmutated *PTEN* gene is a tumor suppressor, the subject of Section 12.9. ■

P4 The insulin receptor is the prototype for several receptor enzymes with a similar structure and RTK activity (Fig. 12-26). The receptors for EGF and PDGF, for example, have structural and sequence similarities to INSR, and both have a protein Tyr kinase activity that phosphorylates IRS1. Many of these receptors dimerize after binding ligand and before autophosphorylation. INSR is the exception, as it is already an ($\alpha\beta$)₂ dimer before insulin binds. (The protomer of the insulin receptor is one $\alpha\beta$ unit.) The binding of adaptor proteins such as Grb2 to P-Tyr residues is a common mechanism for promoting protein-protein interactions initiated by RTKs, a subject to which we return in Section 12.5.

P2 What spurred the evolution of such complicated regulatory machinery? This system allows one activated receptor to activate several IRS1 molecules, amplifying the insulin signal, and it provides for the integration of signals from different receptors such as EGFR and PDGFR, each of which can phosphorylate IRS1. Furthermore, because IRS1 can activate any of several proteins that contain SH2 domains, a single receptor acting through IRS1 can trigger two or more signaling pathways; insulin affects gene expression through the mitogenic Grb2-Sos-Ras-MAPK pathway and affects glycogen metabolism and glucose transport through the metabolic PI3K-PKB pathway. Finally, there are several closely related IRS proteins (IRS2, IRS3), each with its own characteristic tissue distribution and function, further enriching the signaling possibilities in pathways initiated by RTKs.

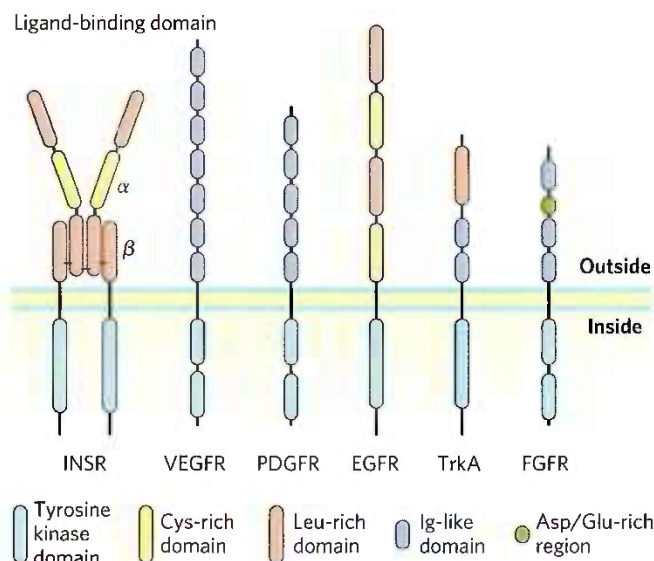


FIGURE 12-26 Receptor tyrosine kinases. Growth factor receptors that initiate signals through Tyr kinase activity include those for insulin (INSR), vascular endothelial growth factor (VEGFR), platelet-derived growth factor (PDGFR), epidermal growth factor (EGFR), high-affinity nerve growth factor (TrkA), and fibroblast growth factor (FGFR). All these receptors have a Tyr kinase domain (blue) on the cytoplasmic side of the plasma membrane. The extracellular domain is unique to each type of receptor, reflecting the different growth factor specificities. These extracellular domains are typically combinations of structural motifs such as Cys- or Leu-rich segments and segments containing one of several motifs common to immunoglobulins (Ig). Many other Tyr kinase receptors are encoded in the human genome, each with a different extracellular domain and ligand specificity. All of the receptors except INSR are monomeric and their Tyr kinase activity is silent until ligand binding triggers receptor dimerization and Tyr kinase activation. Only INSR is always dimeric, but its Tyr kinase is active only when insulin is bound.

Cross Talk among Signaling Systems Is Common and Complex

For simplicity, we have treated individual signaling pathways as separate sequences of events leading to separate metabolic consequences, but there is, in fact, extensive

cross talk among signaling systems. The regulatory circuitry that governs metabolism is richly interwoven and multilayered. We have discussed the signaling pathways for insulin and epinephrine separately, but they do not operate independently. Insulin opposes the metabolic effects of epinephrine in most tissues, and activation of the insulin signaling pathway directly reduces signaling through the β -adrenergic signaling system. For example, the INSR kinase directly phosphorylates two Tyr residues in the cytoplasmic tail of a β_2 -adrenergic receptor, and PKB, activated by insulin, phosphorylates two Ser residues in the same region (Fig. 12-27). Phosphorylation of these four residues triggers clathrin-aided internalization of the β_2 -adrenergic receptor, lowering the cell's sensitivity to epinephrine.

A second type of cross talk between these receptors occurs when P-Tyr residues on the β_2 -adrenergic receptor, phosphorylated by INSR, serve as nucleation points for SH2 domain-containing proteins such as Grb2 (Fig. 12-27, left side). Activation of the MAPK ERK by insulin is 5- to 10-fold greater in the presence of the β_2 -adrenergic receptor, presumably because of this cross talk. Signaling systems that use cAMP and Ca^{2+} also show extensive interaction; each second messenger affects the generation and concentration of the other. Yet another factor that further complicates the signaling picture is that certain metabolic intermediates, such as fatty acids and ceramides, amino acids, and bile acids, can influence insulin signaling. One of the major challenges of systems biology is to sort out the effects of such interactions on the overall metabolic patterns in each tissue—a daunting task.

SUMMARY 12.4 Receptor Tyrosine Kinases

- The insulin receptor, INSR, is the prototype of receptor enzymes with Tyr kinase activity. When insulin binds, the receptor Tyr kinase is activated, and phosphorylates Tyr residues on other proteins, such as IRS. Phosphotyrosine

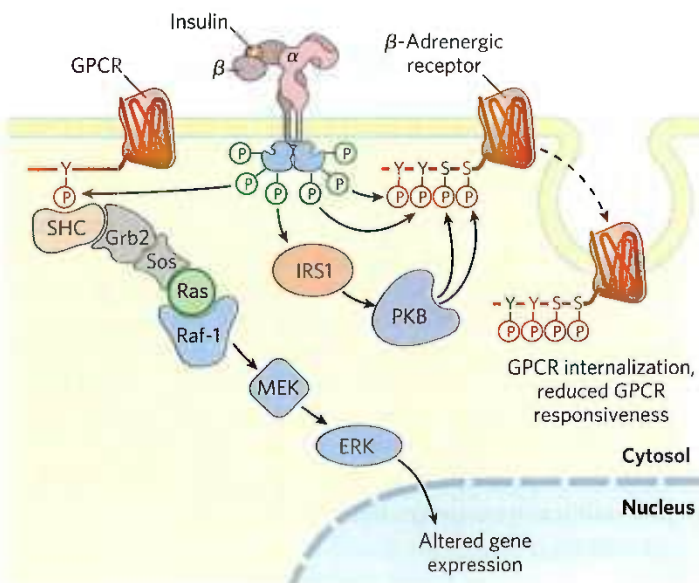


FIGURE 12-27 Cross talk between the insulin receptor and the β -adrenergic receptor (or other GPCR). When INSR is activated by insulin binding, its Tyr kinase directly phosphorylates the β -adrenergic receptor (right side) on two Tyr residues (Tyr³⁵⁰ and Tyr³⁶⁴) near its carboxyl terminus, and indirectly causes phosphorylation of two Ser residues in the same region (through activation of protein kinase B (PKB)). The effect of these phosphorylations is internalization of the adrenergic receptor, reducing the response to the adrenergic stimulus. Alternatively (left side), INSR-catalyzed phosphorylation of a GPCR (an adrenergic or other receptor) on a carboxyl-terminal Tyr creates the point of nucleation for activating the MAPK cascade (see Fig. 12-22), with Grb2 serving as the adaptor protein. In this case, INSR has used the GPCR to enhance its own signaling.

residues of IRS1 serve as binding sites for proteins with SH2 domains. These multivalent proteins can serve as adaptors that bring two proteins into proximity. Sos bound to Grb2 activates Ras, which in turn activates a MAPK cascade that ends with the phosphorylation of target proteins in the cytosol and nucleus. The result is specific metabolic changes and altered gene expression.

- The enzyme PI3K, activated by interaction with IRS1, phosphorylates the membrane lipid PIP₂ to PIP₃, which becomes the point of nucleation for proteins in a second and third branch of insulin signaling.

- Extensive interconnections among signaling pathways allow integration and fine-tuning of multiple hormonal effects.

12.5 Multivalent Adaptor Proteins and Membrane Rafts

Two generalizations have emerged from studies of signaling systems such as those we have discussed so far. First, protein kinases that phosphorylate Tyr, Ser, and Thr residues—and phosphatases that dephosphorylate them—are central to signaling, *directly* affecting the activities of a large number of protein substrates.

P4 Second, protein-protein interactions brought about by the reversible phosphorylation of Tyr, Ser, and Thr residues in signaling proteins create *docking sites* for other proteins that bring about *indirect* effects on proteins downstream in the signaling pathway. In fact, many signaling proteins are *multivalent*: they can interact with several different proteins simultaneously to form multiprotein signaling complexes. In this section we present a few examples to illustrate the general principles of phosphorylation-dependent protein interactions in signaling pathways.

Many of the proteins involved in signaling have intrinsically disordered regions (IDRs) that are flexible enough to allow specific interaction with more than one protein—and perhaps with many other proteins. Protein kinases, for example, have a well-conserved structure containing the substrate-binding and catalytic sites, but through evolution they have acquired additional sequences, typically 20–30 residues long, that are partially disordered and can fold to fit into various multienzyme regulatory networks. The activation loop found in most protein kinases is an IDR and is the universal regulator of kinase activity; its position changes when one or several residues become phosphorylated, switching on the kinase activity. Protein kinases of the PKA, PKB, and PKC families have a disordered carboxyl-terminal tail containing critical residues whose phosphorylation state switches between active and inactive kinase structures. In the MAPK cascade, an amino-terminal IDR plays a similar role as a docking site in the multienzyme complex. Given the ~1,000 protein kinase genes in humans, the many noncatalytic

scaffold proteins, and the multiple interactions possible for IDRs in many of these proteins, the number of possible permutations and combinations available for switching and regulating metabolic processes by protein phosphorylation is impressively large.

Protein Modules Bind Phosphorylated Tyr, Ser, or Thr Residues in Partner Proteins

The protein Grb2 in the insulin signaling pathway (Figs. 12-22 and 12-27) binds through its SH2 domain to other proteins that have exposed P-Tyr residues. The human genome encodes at least 87 SH2-containing proteins, many known to participate in signaling. The P-Tyr residue is bound in a deep pocket in an SH2 domain, with each of its phosphate oxygens participating in hydrogen bonding or electrostatic interactions; the positive charges on two Arg residues figure prominently in the binding. Subtle differences in the structure of SH2 domains account for the specificities of the interactions of SH2-containing proteins with various P-Tyr -containing proteins. The SH2 domain typically interacts with a P-Tyr (which is assigned the index position 0) and the next three residues toward the carboxyl terminus (designated +1, +2, +3). Some proteins with SH2 domains (Src, Fyn, Hck, Nck) favor negatively charged residues in the +1 and +2 positions; others (PLC γ 1, SHP2) have a long hydrophobic groove that binds to aliphatic residues in positions +1 to +5. These differences define subclasses of SH2 domains that have different partner specificities.

Phosphotyrosine-binding domains, or **PTB domains**, are another binding partner for P-Tyr proteins (**Fig. 12-28**), but their critical sequences and three-dimensional

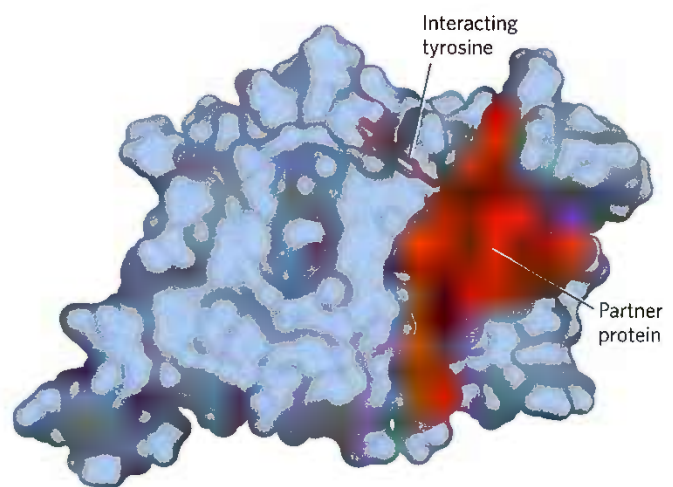


FIGURE 12-28 Interaction of a PTB domain with P-Tyr residue in a partner protein. The PTB domain is represented as a blue surface contour. The partner protein is held to the kinase by multiple noncovalent interactions, which confer specificity on the interaction and position the P-Tyr residue in a binding pocket at the enzyme's active site. [Data from PDB ID 1SHC, M. M. Zhou et al., *Nature* 378:584, 1995.]

structure distinguish them from SH2 domains. The human genome encodes at least 24 proteins that contain PTB domains, including IRS1, which we have already encountered in its role as an adaptor protein in insulin-signal transduction (Fig. 12-23). The P-Tyr -binding sites for SH2 and PTB domains on partner proteins are created by Tyr kinases and eliminated by protein tyrosine phosphatases (PTPs).

As we have seen, other signaling protein kinases, including PKA, PKC, PKG, and members of the MAPK cascade, phosphorylate Ser or Thr residues in their target proteins. In some cases, these proteins acquire the ability to interact with partner proteins through the phosphorylated residue, triggering a downstream process. An alphabet soup of domains that bind P-Ser or P-Thr residues has been identified, and more are sure to be found. Each domain favors a certain sequence around the phosphorylated residue, so proteins with that domain bind to and interact with a specific subset of phosphorylated proteins.

In some cases, the region on a protein that binds P-Tyr of a substrate protein is masked by the region's interaction with a P-Tyr in the same protein. For example, the soluble protein Tyr kinase Src, when phosphorylated on a specific Tyr residue, is rendered inactive; an SH2 domain needed to bind the substrate protein instead binds the internal P-Tyr . Removal of this P-Tyr residue by a phosphoprotein phosphatase turns on the Tyr kinase activity of Src (Fig. 12-29a). Similarly, glycogen synthase kinase 3 (GSK3) is inactive when phosphorylated on a Ser residue in its autoinhibitory domain (Fig. 12-29b). Dephosphorylation of that domain frees the enzyme to bind (and then phosphorylate) its target proteins.

In addition to the three commonly phosphorylated residues in proteins (Tyr, Ser, Thr), there is a fourth phosphorylated structure that nucleates the formation of supramolecular complexes of signaling proteins: the phosphorylated head group of the membrane phosphatidylinositols. Many signaling proteins contain domains such as SH3 and PH (pleckstrin homology) that bind tightly to PIP_3 protruding on the cytoplasmic side of the plasma membrane. Wherever the enzyme PI3K creates this head group (as it does in response to the insulin signal), proteins that bind PIP_3 will cluster at the membrane surface.

Most of the proteins involved in signaling at the plasma membrane have one or more phosphoprotein- or phospholipid-binding domain; many have three or more, and thus are multivalent in their interactions with other signaling proteins. Figure 12-30 shows just a few of the multivalent proteins known to participate in signaling. Many of the complexes include components with membrane-binding domains. Given the location of so many signaling processes at the inner surface of the plasma membrane, the molecules that must collide to produce the signaling response are effectively confined to two-dimensional space—the membrane surface; collisions are far more likely here than in the three-dimensional space of the cytosol.

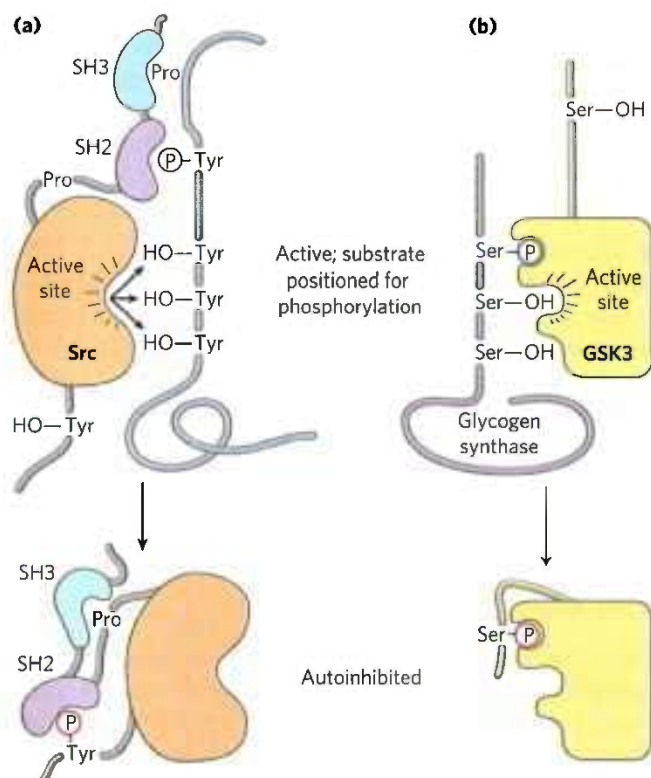


FIGURE 12-29 Mechanism of autoinhibition of Src and GSK3. (a) In the active form of the Tyr kinase Src, an SH2 domain binds a P-Tyr in the protein substrate, and an SH3 domain binds a proline-rich region of the substrate, thus lining up the active site of the kinase with several target Tyr residues in the substrate (top). When Src is phosphorylated on a specific Tyr residue (bottom), the SH2 domain binds the internal P-Tyr instead of the P-Tyr of the substrate, and the SH3 domain binds an internal proline-rich region, preventing productive enzyme-substrate binding; the enzyme is thus autoinhibited. (b) In the active form of glycogen synthase kinase 3 (GSK3), an internal P-Ser -binding domain is available to bind P-Ser in its substrate (glycogen synthase) and to position the kinase to phosphorylate neighboring Ser residues (top). Phosphorylation of an internal Ser residue allows this internal kinase segment to occupy the P-Ser -binding site, blocking substrate binding (bottom).

In summary, a remarkable picture of signaling pathways has emerged from studies of many signaling proteins and their multiple binding domains. **P5** An initial signal results in phosphorylation of the receptor or a target protein, triggering the assembly of large multiprotein complexes, held together on scaffolds with multivalent binding capacities. Some of these complexes contain several protein kinases that activate each other in turn, producing a cascade of phosphorylation and a great amplification of the initial signal. The interactions between cascade kinases are not left to the vagaries of random collisions in three-dimensional space. In the MAPK cascade, for example, a scaffold protein, KSR, binds all three kinases (MAPK, MAPKK, and MAPKKK), ensuring their proximity and correct orientation and even conferring allosteric properties on the interactions among the kinases, which makes their serial phosphorylation sensitive to very small stimuli (Fig. 12-31).

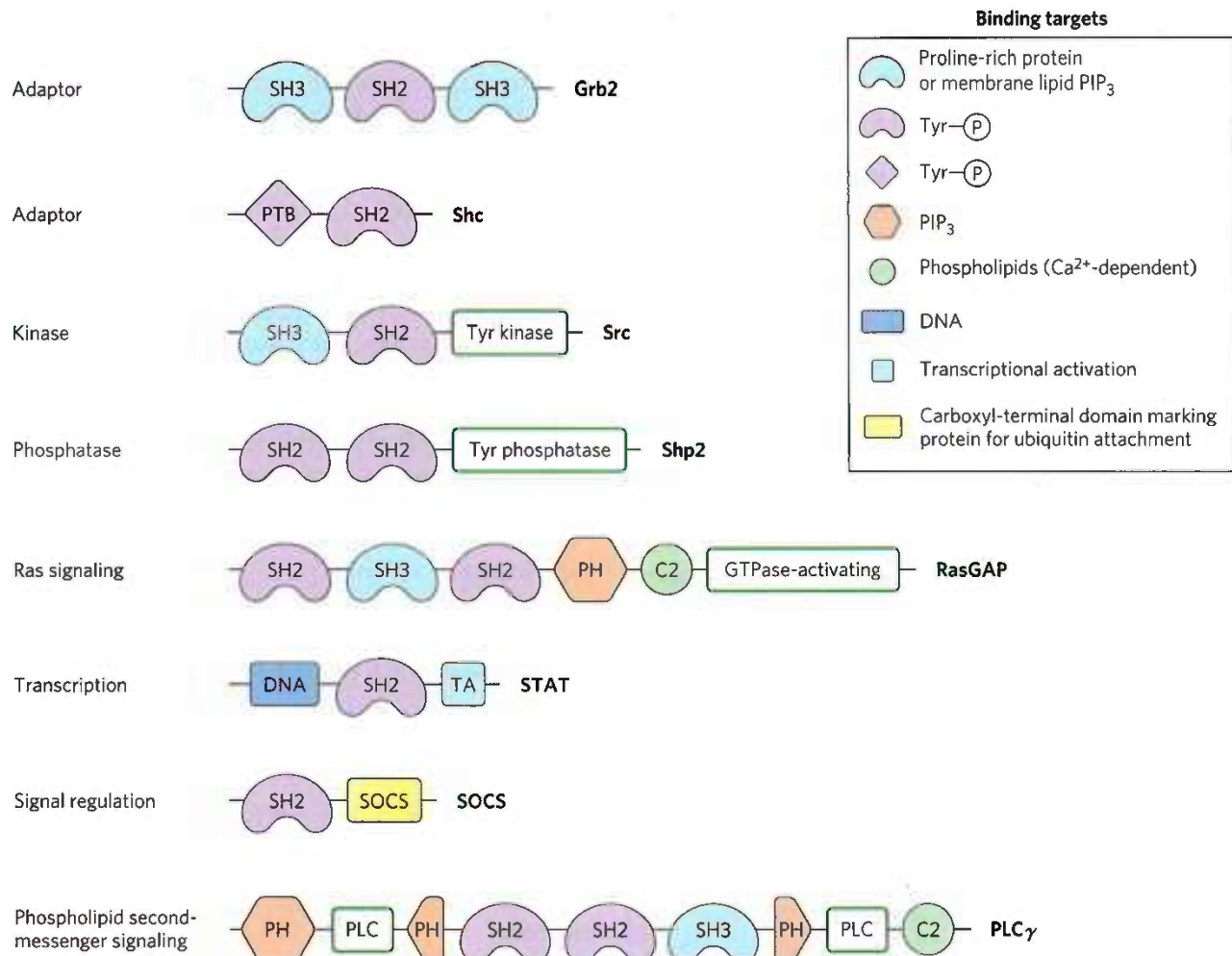


FIGURE 12-30 Some binding modules of signaling proteins.

These signaling proteins interact with phosphorylated proteins or phospholipids in many permutations and combinations to form integrated signaling complexes. Each protein is represented by a line (with the amino terminus to

the left); symbols indicate the location of conserved binding domains (with specificities as listed in the key; abbreviations are explained in the text); green boxes indicate catalytic activities. The name of each protein is given at its carboxyl-terminal end. [Information from T. Pawson et al., *Trends Cell Biol.* 11:504, 2001, Fig. 5.]

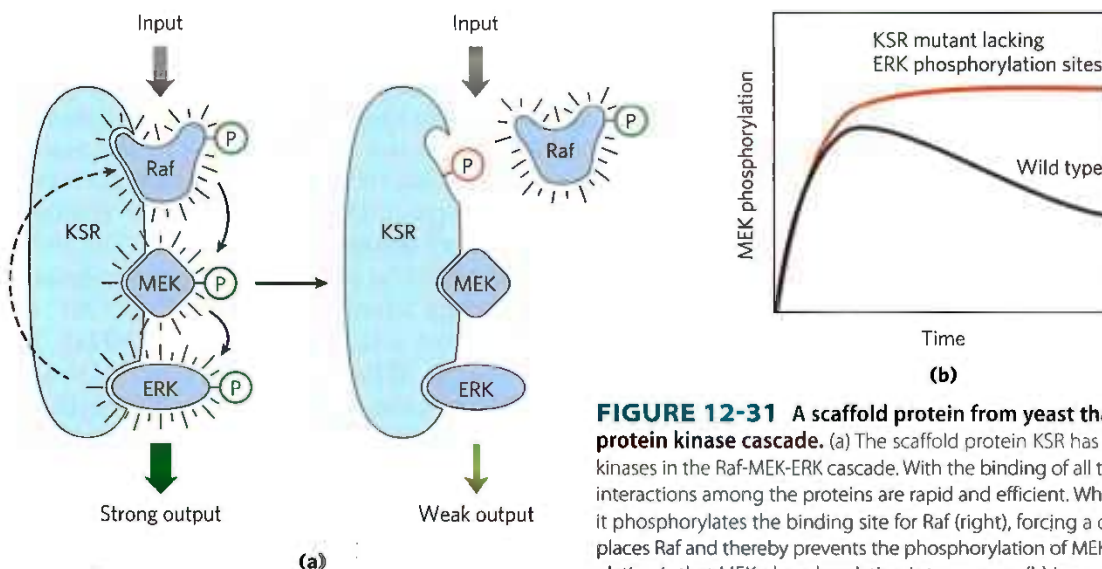


FIGURE 12-31 A scaffold protein from yeast that organizes and regulates a protein kinase cascade.

(a) The scaffold protein KSR has binding sites for all three of the kinases in the Raf-MEK-ERK cascade. With the binding of all three in appropriate orientations, interactions among the proteins are rapid and efficient. When ERK has been activated (left), it phosphorylates the binding site for Raf (right), forcing a conformational change that displaces Raf and thereby prevents the phosphorylation of MEK. The result of this feedback regulation is that MEK phosphorylation is temporary. (b) In yeast cells with mutant KSR lacking the phosphorylation sites (red curve), no feedback occurs, producing a different time course of signaling. [Information from M. C. Good et al., *Science* 332:680, 2011, Fig. 2E.]

Phosphotyrosine phosphatases remove the phosphate from P-Tyr residues, reversing the effect of phosphorylation. At least 37 genes in the human genome encode protein tyrosine phosphatases (PTPs). About half of these are receptorlike integral proteins with a single transmembrane domain; they are presumably controlled by extracellular factors not yet identified. Other PTPs are soluble and contain SH2 domains that determine their molecular partners and intracellular location. In addition, animal cells have protein P-Ser and P-Thr phosphatases such as PP1 that reverse the effects of Ser- and Thr-specific protein kinases.

We can see, then, that signaling occurs in protein circuits, which are effectively hardwired from signal receptor to response effector and can be switched off instantly by the hydrolysis of a single upstream phosphate ester bond. In these circuits, protein kinases are the writers, domains such as SH2 are the readers, and PTPs and other phosphatases are the erasers. The multivalency of signaling proteins allows the assembly of many different combinations of Lego-like signaling modules, each combination suited to particular signals, cell types, and metabolic circumstances, yielding diverse signaling circuits of extraordinary complexity.

Membrane Rafts and Caveolae Segregate Signaling Proteins

Membrane rafts (Chapter 11) are regions of the membrane bilayer enriched in sphingolipids, sterols, and certain proteins, including many proteins attached to the bilayer by GPI (glycosylated derivatives of phosphatidylinositol) anchors. The β -adrenergic receptor is segregated in rafts that also contain G proteins, adenylyl cyclase, PKA, and the protein phosphatase PP2, which together provide a highly integrated signaling unit. By segregating in a small region of the plasma membrane all of the elements required for responding to and ending the signal, the cell is able to produce a highly localized and brief “puff” of second messenger.

Some RTKs (EGFR and PDGFR) are also localized in rafts, and this sequestration very likely has functional significance. In isolated fibroblasts, EGFR is usually concentrated in specialized rafts called caveolae (see Fig. 11-23). When the cells are treated with EGF, the receptor leaves the raft, separating it from the other components of the EGF signaling pathway. This migration depends on the receptor's protein kinase activity; mutant receptors lacking this activity remain in the raft during treatment with EGF. Such experiments suggest that spatial segregation of signaling proteins in rafts is yet another dimension of the already complex processes initiated by extracellular signals.

SUMMARY 12.5 Multivalent Adaptor Proteins and Membrane Rafts

- Many signaling proteins have domains that bind phosphorylated Tyr, Ser, or Thr residues in other

proteins; the binding specificity for each domain is determined by sequences that adjoin the phosphorylated residue in the substrate. SH2 and PTB domains bind to proteins containing P-Tyr residues; other domains bind P-Ser and P-Thr residues in various contexts. SH3 and PH domains bind the membrane phospholipid PIP_3 .

- Many signaling proteins are multivalent, with several different binding modules. By combining the substrate specificities of various protein kinases with the specificities of domains that bind phosphorylated Tyr, Ser, or Thr residues, and with phosphatases that can rapidly inactivate a signaling pathway, cells create a large number of multiprotein signaling complexes.

- Membrane rafts and caveolae sequester groups of signaling proteins in small regions of the plasma membrane, effectively raising their local concentrations and making signaling more efficient.

12.6 Gated Ion Channels

Certain cells in multicellular organisms are “excitable”: they can detect an external signal, convert it into an electrical signal (specifically, a change in plasma membrane potential), and pass it on. Changes in membrane potential are effected by gated ion channels. Excitable cells play central roles in nerve conduction, muscle contraction, hormone secretion, sensory processes, and learning and memory.

Ion Channels Underlie Rapid Electrical Signaling in Excitable Cells

P6 The excitability of sensory cells, neurons, and myocytes depends on ion channels, signal transducers that provide a regulated path for the movement of inorganic ions such as Na^+ , K^+ , Ca^{2+} , and Cl^- across the plasma membrane in response to various stimuli. Recall from Chapter 11 that these ion channels are gated: they may be open or closed, depending on whether the associated receptor has been activated by the binding of its specific ligand (a neurotransmitter, for example) or by a change in the transmembrane electrical potential (**voltage-gated ion channels**). The Na^+K^+ ATPase is electrogenic; it creates a charge imbalance across the plasma membrane by carrying 3 Na^+ out of the cell for every 2 K^+ carried in (Fig. 12-32a). The action of the Na^+K^+ ATPase makes the inside of the cell negative relative to the outside. Inside the cell, $[\text{K}^+]$ is much higher and $[\text{Na}^+]$ is much lower than outside the cell (Fig. 12-32b). The direction of spontaneous ion flow across a polarized membrane is dictated by the electrochemical potential of that ion across the membrane, which has two components: the difference in concentration of the ion on the two sides of the membrane, and the difference in electrical potential (V_m),

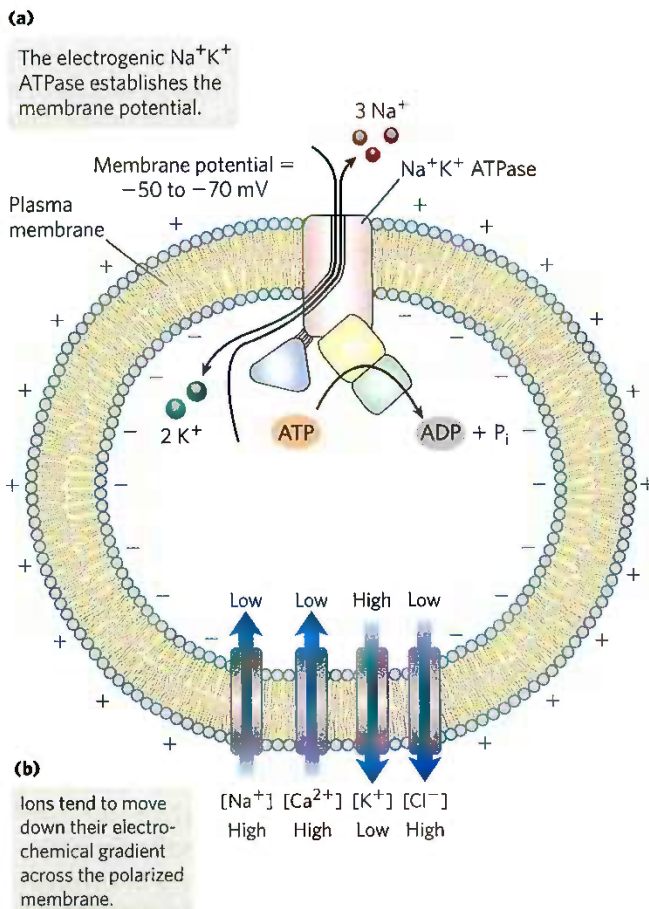


FIGURE 12-32 Transmembrane electrical potential. (a) The electrogenic Na^+K^+ ATPase produces a transmembrane electrical potential of about -60 mV (inside negative). (b) Blue arrows show the direction in which ions tend to move spontaneously across the plasma membrane in an animal cell, driven by the combination of chemical and electrical gradients. The chemical gradient drives Na^+ and Ca^{2+} inward (producing depolarization) and K^+ outward, against its electrical gradient (producing hyperpolarization). The electrical gradient drives Cl^- outward, against its concentration gradient (producing depolarization).

typically expressed in millivolts (see Eqn 11-4, p. 392). Given the ion concentration differences and a V_m of about -60 mV (inside negative), opening of a Na^+ or Ca^{2+} channel will result in a spontaneous inward flow of Na^+ or Ca^{2+} (and depolarization), whereas opening of a K^+ channel will result in a spontaneous outward flow of K^+ (and hyperpolarization) (Fig. 12-32b). In this case, K^+ moves out of the cell against the electrical gradient, because the large concentration difference exerts a stronger effect than the V_m . For Cl^- , the membrane potential predominates, so when a Cl^- channel opens, Cl^- flows outward.

The number of ions that must flow to produce a physiologically significant change in the membrane potential is negligible relative to the concentrations of Na^+ , K^+ , and Cl^- in cells and extracellular fluid, so the ion fluxes that occur during signaling in excitable cells have essentially no effect on the concentrations of these ions. With Ca^{2+} ,

the situation is different; because the intracellular $[\text{Ca}^{2+}]$ is generally very low ($\sim 10^{-7}$ M), inward flow of Ca^{2+} can significantly alter the cytosolic $[\text{Ca}^{2+}]$, allowing it to serve as a second messenger.

The membrane potential of a cell at a given time is the result of the types and numbers of ion channels open at that instant. **P6** The precisely timed opening and closing of ion channels and the resulting transient changes in membrane potential underlie the electrical signaling by which the nervous system stimulates the skeletal muscles to contract, the heart to beat, or secretory cells to release their contents. Moreover, many hormones exert their effects by altering the membrane potential of their target cells. These mechanisms are not limited to animals; ion channels play important roles in the responses of bacteria, protists, and plants to environmental signals.

To illustrate the action of ion channels in cell-to-cell signaling, we describe the mechanisms by which a neuron passes a signal along its length and across a synapse to the next neuron (or to a myocyte) in a cellular circuit, using acetylcholine as the neurotransmitter.

Voltage-Gated Ion Channels Produce Neuronal Action Potentials

Signaling in the nervous system is accomplished by networks of neurons, specialized cells that carry an electrical impulse (action potential) from one end of the cell (the cell body) through an elongated cytoplasmic extension (the axon), to the synapse with the next neuron. The electrical signal triggers release of the neurotransmitter acetylcholine into the synaptic cleft, carrying the signal to the next cell (neuron) in the circuit. Initially, the plasma membrane of the presynaptic neuron is polarized (inside negative) through the action of the electrogenic Na^+K^+ ATPase, which pumps out three Na^+ for every two K^+ pumped in (Fig. 12-32). A rapid sequence of opening and closing of several types of ion channels (Fig. 12-33) produces a wave of depolarization (an **action potential**) that sweeps the neuron from the cell body to the end of an axon. First, opening of a voltage-gated Na^+ channel allows Na^+ entry, and the resulting local depolarization causes the adjacent Na^+ channel to open, and so on (Fig. 12-33, step 1). The directionality of movement of the action potential is ensured by the brief refractory period that follows the opening of each voltage-gated Na^+ channel. A split second after the action potential passes a point in the axon, voltage-gated K^+ channels open (step 2), allowing K^+ exit, which brings about repolarization of the membrane to make it ready for the next action potential (step 3).

When the wave of depolarization reaches the axon tip, voltage-gated Ca^{2+} channels open, allowing Ca^{2+} entry (step 4). The resulting increase in internal $[\text{Ca}^{2+}]$ triggers exocytotic release of the neurotransmitter acetylcholine into the synaptic cleft (step 5). Acetylcholine binds

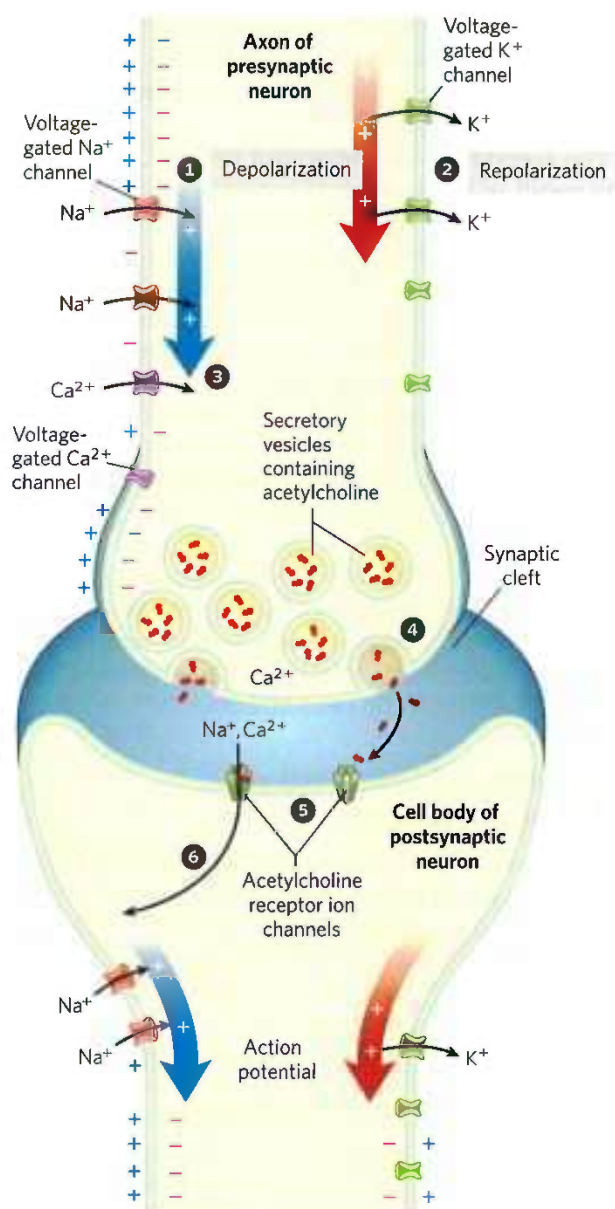


FIGURE 12-33 Role of voltage-gated and ligand-gated ion channels in neural transmission. The steps are described in the text. Stimulation of the presynaptic neuron (top) triggers a wave of depolarization (blue arrow) followed by repolarization (red arrow), producing an action potential that sweeps along the axon. At the synapse, depolarization opens Ca^{2+} channels, and Ca^{2+} entry triggers acetylcholine release into the synaptic cleft. Acetylcholine diffuses across the cleft, opening receptor channels, depolarizing the postsynaptic cell, and starting an action potential in the postsynaptic cell. Note that, for clarity, Na^+ channels and K^+ channels are drawn on opposite sides of the axon, but both types are uniformly distributed in the axonal membrane. Also, positive and negative charges are shown only on the left, but as the wave of potential sweeps the axon, the membrane potential is the same at any given point along the axon.

to a receptor on the postsynaptic neuron or myocyte, causing its ligand-gated ion channel to open (step 6). Extracellular Na^+ and Ca^{2+} enter through this channel, depolarizing the postsynaptic cell. The electrical signal has thus passed to the cell body of the postsynaptic neuron and will move along its axon to a third neuron (or a

myocyte) by this same sequence of events. We see, then, that gated ion channels convey signals in either of two ways: by changing the cytoplasmic concentration of an ion (such as Ca^{2+}), which then serves as an intracellular second messenger, or by changing V_m and affecting other membrane proteins that are sensitive to V_m . The passage of an electrical signal through one neuron and on to the next illustrates both types of mechanism.

Neurons Have Receptor Channels That Respond to Different Neurotransmitters

Animal cells, especially those of the nervous system, contain a variety of ion channels gated by ligands, voltage, or both. Receptors that are themselves ion channels are classified as **ionotropic**, to distinguish them from receptors that generate a second messenger (**metabotropic** receptors). Acetylcholine acts on an ionotropic receptor in the postsynaptic cell. The acetylcholine receptor is a cation channel. When occupied by acetylcholine, the receptor opens to the passage of cations (Na^+ , K^+ , and Ca^{2+}), triggering depolarization of the cell. The neurotransmitters serotonin, glutamate, and glycine all can act through ionotropic receptors that are structurally related to the acetylcholine receptor. Serotonin and glutamate trigger the opening of cation (Na^+ , K^+ , Ca^{2+}) channels, whereas glycine opens Cl^- -specific channels.

P6 Depending on which ion passes through a channel, binding of the ligand (neurotransmitter) for that channel results in either depolarization or hyperpolarization of the target cell. A single neuron normally receives input from many other neurons, each releasing its own characteristic neurotransmitter with its characteristic depolarizing or hyperpolarizing effect. The target cell's V_m therefore reflects the *integrated* input (see Fig. 12-1f) from multiple neurons. The cell responds with an action potential only if the integrated input adds up to a net depolarization of sufficient magnitude.

The receptor channels for acetylcholine, glycine, glutamate, and γ -aminobutyric acid (GABA) are gated by *extracellular* ligands. *Intracellular* second messengers—such as cAMP, cGMP, IP_3 , Ca^{2+} , and ATP—regulate ion channels of the type we saw in the sensory transductions of vision, olfaction, and gustation.

Toxins Target Ion Channels

Many of the most potent toxins found in nature act on ion channels. For example, dendrotoxin (from the black mamba snake) blocks the action of voltage-gated K^+ channels, tetrodotoxin (produced by puffer fish) acts on voltage-gated Na^+ channels, and cobrotoxin (from cobras) disables acetylcholine receptor ion channels. Why, in the course of evolution, have ion channels become the preferred target of toxins, rather than some critical metabolic target such as an enzyme essential in energy metabolism?

Ion channels are extraordinary amplifiers; opening of a single channel can allow the flow of 10 million ions

per second. Consequently, relatively few molecules of an ion-channel protein are needed per neuron for signaling functions. This means that a relatively small number of toxin molecules with high affinity for ion channels, acting from outside the cell, can have a pronounced effect on neurosignaling throughout the body. A comparable effect by way of a metabolic enzyme, typically present in cells at much higher concentrations than ion channels, would require far greater numbers of the toxin molecule.

SUMMARY 12.6 Gated Ion Channels

- Ion channels gated by membrane potential or ligands are central to signaling in neurons and other cells.
- The voltage-gated Na^+ and K^+ channels of neuronal membranes carry the action potential along the axon as a wave of depolarization (Na^+ influx) followed by repolarization (K^+ efflux). Arrival of an action potential at the distal end of a presynaptic neuron triggers neurotransmitter release.
- The neurotransmitter (acetylcholine, for example) diffuses to the postsynaptic neuron (or the myocyte, at a neuromuscular junction), binds to specific receptors in the plasma membrane, and triggers a change in V_m . The cell body of the neuron has receptors for a variety of neurotransmitters or extracellular signals. The neuron's V_m is the sum of the effects of all ion-channel contributions.
- Neurotoxins, produced by many organisms, attack neuronal ion channels and are therefore fast-acting and deadly.

12.7 Regulation of Transcription by Nuclear Hormone Receptors

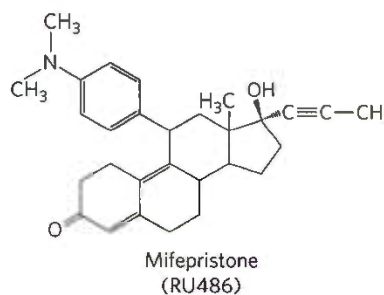
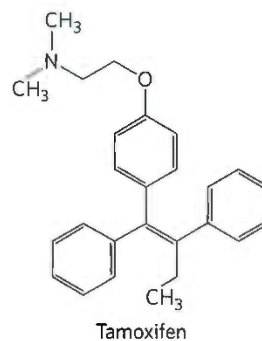
P7 The steroid, retinoic acid (retinoid), and thyroid hormones form a large group of receptor ligands that exert at least part of their effects by a mechanism fundamentally different from that of other hormones: they act directly in the nucleus to alter gene expression. We discuss their mode of action in detail in Chapter 28, along with other mechanisms for regulating gene expression. Here we give a brief overview.

Steroid hormones (estrogen, progesterone, vitamin D, and cortisol, for example), too hydrophobic to dissolve readily in the blood, are transported on specific carrier proteins from their point of release to their target tissues. In target cells, these hormones pass through the plasma membrane and nuclear membrane by simple diffusion and bind to specific receptor proteins in the nucleus (Fig. 12-34). Hormone binding triggers changes in the conformation of a receptor protein so that it becomes capable of interacting with specific regulatory sequences in DNA called **hormone response elements (HREs)**,

thus altering gene expression (see Fig. 28-34). The bound receptor-hormone complex enhances the expression of specific genes adjacent to HREs, with the help of several other proteins essential for transcription. Hours or days are required for these regulators to have their full effect—the time required for the changes in RNA synthesis and subsequent protein synthesis to become evident in altered metabolism.



The specificity of the steroid-receptor interaction is exploited in the use of the drug **tamoxifen** to treat breast cancer. In some types of breast cancer, division of the cancerous cells depends on the continued presence of estrogen. Tamoxifen is an estrogen antagonist; it competes with estrogen for binding to the estrogen receptor, but the tamoxifen-receptor complex has little or no effect on gene expression. Consequently, tamoxifen administered after surgery or during chemotherapy for hormone-dependent breast cancer slows or stops the growth of remaining cancerous cells. Another steroid analog, the drug **mifepristone (RU486)**, binds to the progesterone receptor and blocks hormone actions essential to implantation of the fertilized ovum in the uterus, and thus functions as a contraceptive.



SUMMARY 12.7 Regulation of Transcription by Nuclear Hormone Receptors

- Steroid hormones enter cells by simple diffusion and bind to specific receptor proteins, inducing a structural change that exposes a specific binding site on the DNA.
- The hormone-receptor complex binds specific regions of DNA at the hormone response elements, and interacts with other proteins to regulate the expression of nearby genes.

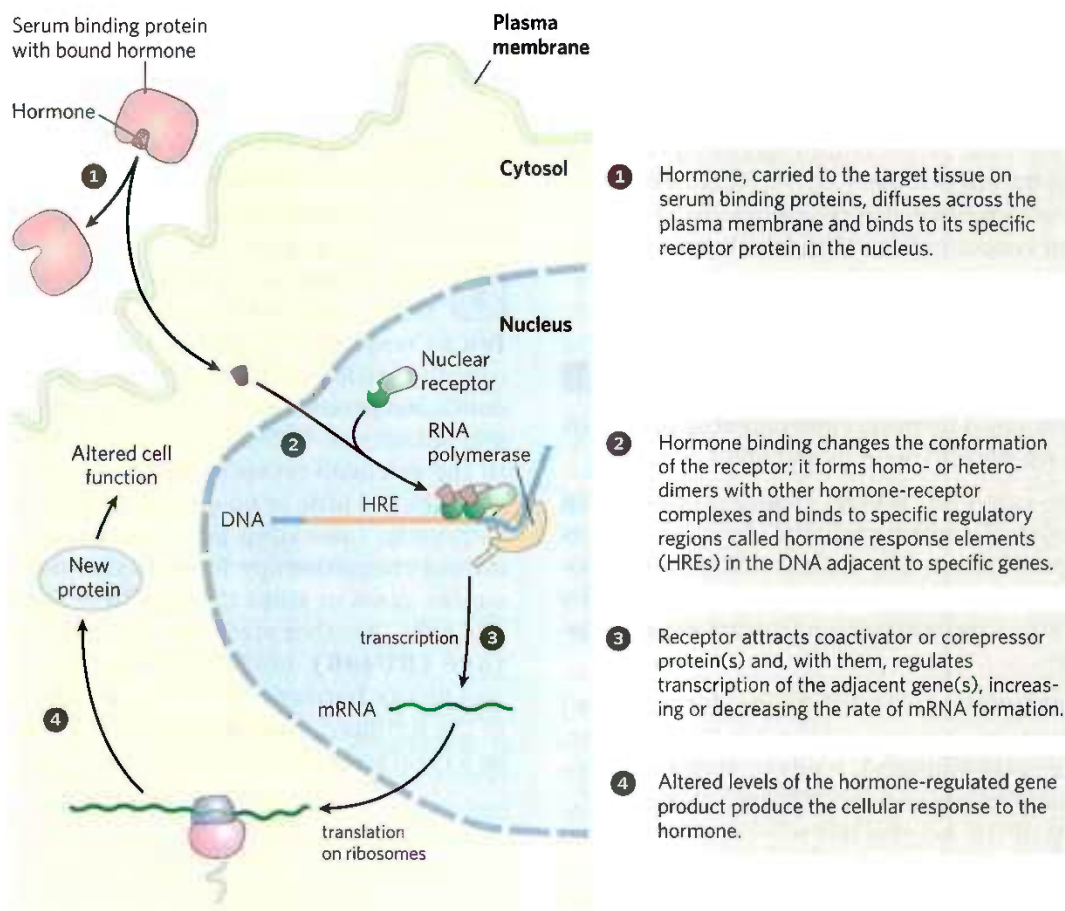


FIGURE 12-34 General mechanism by which steroid and thyroid hormones, retinoids, and vitamin D regulate gene expression. The details of transcription and protein synthesis are discussed in

Chapters 26 and 27. Some steroids also act through plasma membrane receptors by a completely different mechanism.

12.8 Regulation of the Cell Cycle by Protein Kinases

One of the most dramatic manifestations of signaling pathways is the regulation of the eukaryotic cell cycle. During embryonic growth and later development, cell division occurs in virtually every tissue. In the adult organism, cells of most tissues stop dividing, becoming quiescent. A cell's "decision" to divide or not is of crucial importance to the organism. **P9** When the regulatory mechanisms that limit cell division are defective and cells undergo unregulated division, the result is catastrophic—cancer. Proper cell division requires a precisely ordered sequence of biochemical events that assures every daughter cell a full complement of the molecules required for life. Investigations into the control of cell division in diverse eukaryotic cells have revealed universal regulatory mechanisms. Signaling mechanisms much like those discussed above are central in determining whether and when a cell undergoes cell division, and they also ensure orderly passage through the stages of the cell cycle.

The Cell Cycle Has Four Stages

Cell division accompanying mitosis in eukaryotes occurs in four well-defined stages (**Fig. 12-35**). In the S (synthesis) phase, the DNA is replicated to produce copies for both daughter cells. In the G₂ phase (G indicates the gap between divisions), new proteins are synthesized and the cell approximately doubles in size. In the M phase (mitosis), the maternal nuclear envelope breaks down, paired chromosomes are pulled to opposite poles of the cell, each set of daughter chromosomes is surrounded by a newly formed nuclear envelope, and cytokinesis pinches the cell in half, producing two daughter cells (see **Fig. 24-22**). In embryonic or rapidly proliferating tissue, each daughter cell divides again, but only after a waiting period (G₁). In animal cells in the laboratory, the entire process takes about 24 hours.

After passing through mitosis and into G₁, a cell either continues through another division or ceases to divide, entering a quiescent phase (G₀) that may last hours, days, or the lifetime of the cell. When a cell in G₀ begins to divide again, it reenters the division cycle

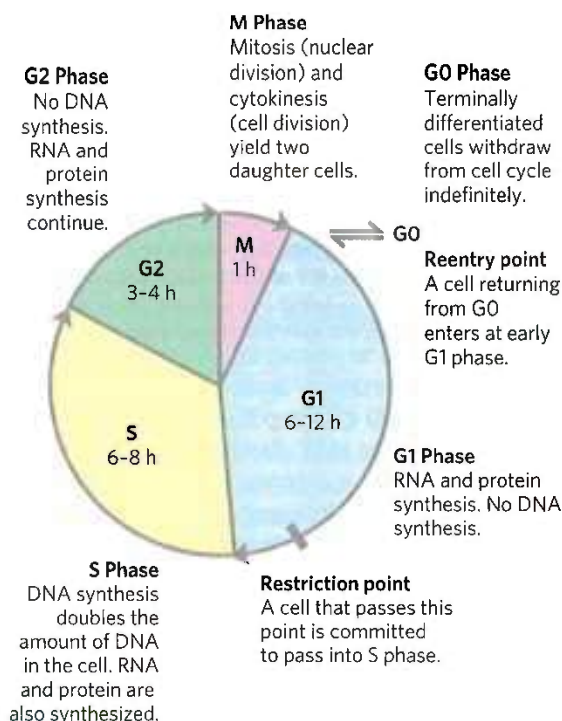


FIGURE 12-35 The eukaryotic cell cycle. The durations (in hours) of the four stages vary, but those shown are typical.

through the G1 phase. Differentiated cells such as hepatocytes or adipocytes have acquired their specialized function and form; they remain in the G0 phase. Stem cells retain their potential to divide and to differentiate into any of a number of cell types.

Levels of Cyclin-Dependent Protein Kinases Oscillate

P8 The timing of the cell cycle is controlled by a family of protein kinases with activities that change in response to cellular signals. By phosphorylating specific proteins at precisely timed intervals, these protein kinases orchestrate the metabolic activities of the cell to produce orderly cell division. The kinases are heterodimers with a regulatory subunit, a **cyclin**, and a catalytic subunit, a **cyclin-dependent protein kinase (CDK)**. In the absence of the cyclin, the catalytic subunit is virtually inactive. When the cyclin binds, the catalytic site opens up, a residue essential to catalysis becomes accessible, and the protein kinase activity of the catalytic subunit increases 10,000-fold. Animal cells have at least 10 different cyclins (designated A, B, and so forth) and at least 8 CDKs (CDK1 through CDK8), which act in various combinations at specific points in the cell cycle.

In a population of animal cells undergoing synchronous division, some CDK activities show striking oscillations (**Fig. 12-36**). These oscillations are the result of four mechanisms for regulating CDK activity: phosphorylation or dephosphorylation of the CDK, controlled degradation of the cyclin subunit, periodic synthesis of CDKs and cyclins, and the action of specific

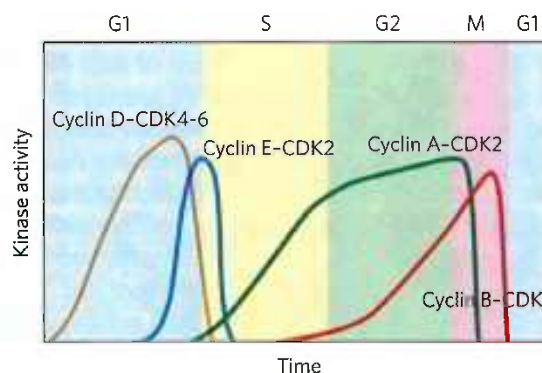


FIGURE 12-36 Variations in the activities of specific CDKs during the cell cycle in animals. Early in G1, the activity of cyclin D-CDK4-6 rises slowly, then drops sharply as G1 ends. Near the end of G1, cyclin E-CDK2 activity rises and peaks near the G1 phase-S phase boundary, when the active enzyme triggers synthesis of enzymes required for DNA synthesis (see Fig. 12-40). Cyclin A-CDK2 activity rises during the S and G2 phases, then drops sharply in the M phase, as cyclin B-CDK1 peaks. [Data from P. Icard et al., *Trends Biochem. Sci.* 44:490, 2019, Fig. 3.]

CDK-inhibiting proteins. The precisely timed activation and inactivation of a series of CDKs produces signals serving as a master clock that orchestrates the events in normal cell division and ensures that one stage is completed before the next begins.

CDKs Are Regulated by Phosphorylation, Cyclin Degradation, Growth Factors, and Specific Inhibitors

The activity of a CDK is strikingly affected by phosphorylation and dephosphorylation of two specific residues in the protein (**Fig. 12-37**). Phosphorylation of Thr¹⁶⁰ of CDK2 stabilizes a conformation in which an autoinhibitory “T loop” is moved away from the substrate-binding cleft in the kinase, opening it to bind protein substrates. Dephosphorylation of P-Tyr^{15} of CDK2 removes a negative charge that blocks ATP from approaching its binding site. This mechanism for activating a CDK is self-reinforcing; the enzyme (PTP) that dephosphorylates P-Tyr^{15} is itself a substrate for the CDK and is activated by phosphorylation. The combination of these factors activates the CDK manyfold, allowing it to phosphorylate downstream protein targets required for progression of the cell cycle (**Fig. 12-38a**).

The presence of a single-strand break in DNA signals arrest of the cell cycle in G2 by activating two proteins (ATM and ATR; see Fig. 12-40). These proteins trigger a cascade of responses that include inactivation of the PTP that dephosphorylates Tyr¹⁵ of the CDK. With the CDK inactivated, the cell is arrested in G2, unable to divide until the DNA is repaired and the effects of the cascade are reversed.

Highly specific and precisely timed proteolytic breakdown of mitotic cyclins regulates CDK activity throughout the cell cycle (**Fig. 12-38b**). How is the timing of cyclin breakdown controlled? A feedback loop occurs in

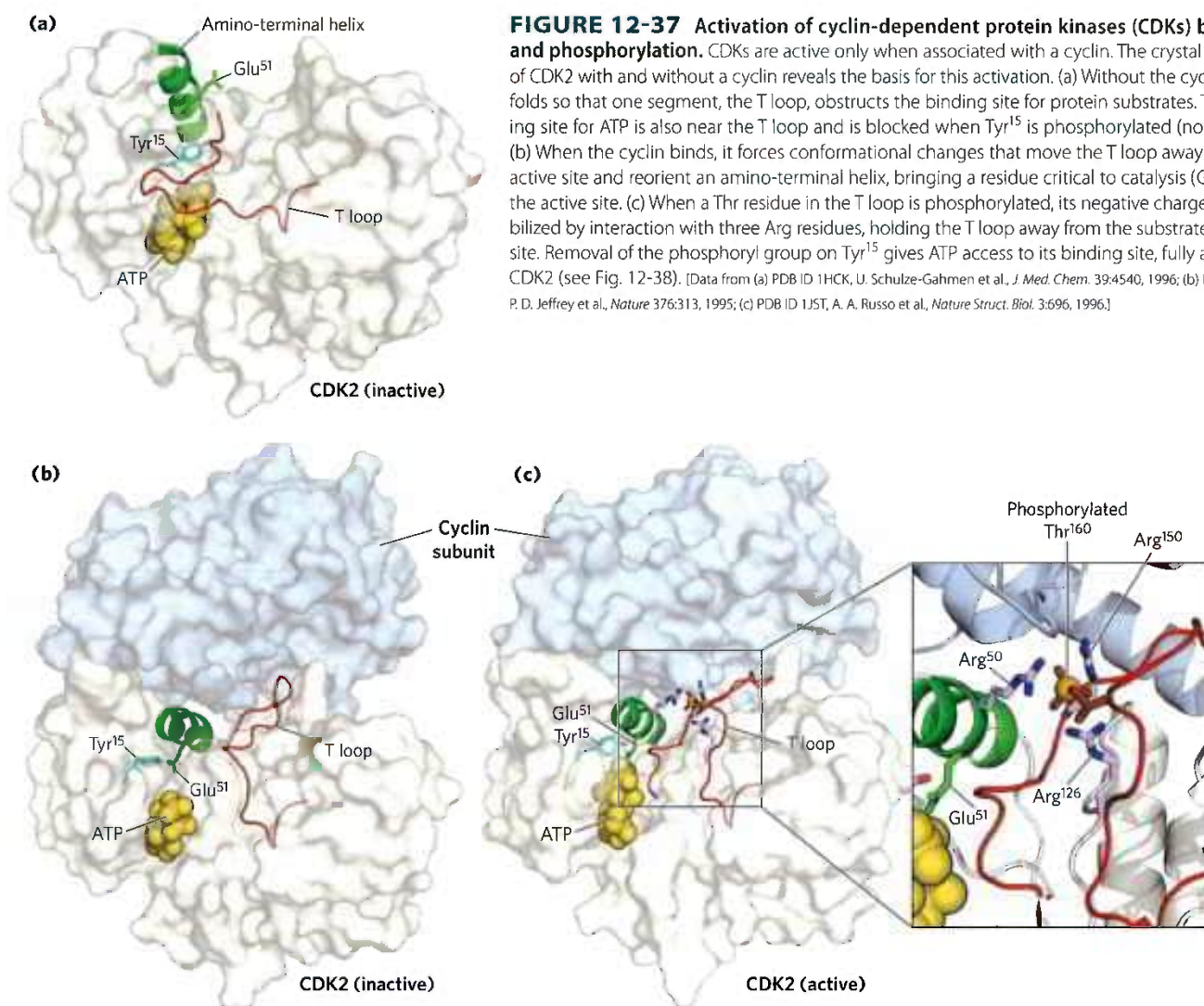


FIGURE 12-37 Activation of cyclin-dependent protein kinases (CDKs) by cyclin and phosphorylation. CDKs are active only when associated with a cyclin. The crystal structure of CDK2 with and without a cyclin reveals the basis for this activation. (a) Without the cyclin, CDK2 folds so that one segment, the T loop, obstructs the binding site for protein substrates. The binding site for ATP is also near the T loop and is blocked when Tyr¹⁵ is phosphorylated (not shown). (b) When the cyclin binds, it forces conformational changes that move the T loop away from the active site and reorient an amino-terminal helix, bringing a residue critical to catalysis (Glu⁵¹) into the active site. (c) When a Thr residue in the T loop is phosphorylated, its negative charges are stabilized by interaction with three Arg residues, holding the T loop away from the substrate-binding site. Removal of the phosphoryl group on Tyr¹⁵ gives ATP access to its binding site, fully activating CDK2 (see Fig. 12-38). [Data from (a) PDB ID 1HCK, U. Schulze-Gahmen et al., *J. Med. Chem.* 39:4540, 1996; (b) PDB ID 1FIN, P. D. Jeffrey et al., *Nature* 376:313, 1995; (c) PDB ID 1JST, A. A. Russo et al., *Nature Struct. Biol.* 3:696, 1996.]

the overall process shown in Figure 12-38. As a cell enters mitosis, the M-phase CDK is inactive (step 1). As cyclin is synthesized (step 2), the cyclin-CDK complex forms (step 3). The T loop lies in the substrate-binding site of CDK, and P-Tyr^{15} blocks its ATP-binding site, keeping the complex inactive. When Thr¹⁶⁰ in the T loop is phosphorylated, the loop moves out of the substrate-binding site, and when Tyr¹⁵ is dephosphorylated, ATP can bind. These two changes make the cyclin-CDK complex many times more active (step 4). Further activation is achieved as CDK also phosphorylates and activates the enzyme that dephosphorylates P-Tyr^{15} (step 5). The active cyclin-CDK complex triggers its own inactivation by phosphorylation of DBRP (destruction box recognizing protein; step 6). DBRP and ubiquitin ligase then attach several molecules of **ubiquitin** (U) to the cyclin (step 7), targeting it for destruction by proteolytic enzyme complexes called **proteasomes** (step 8). The role of ubiquitin and proteasomes is not limited to the regulation of cyclins; as we shall see in Chapter 27,

both also take part in the turnover of cellular proteins, a process fundamental to cellular housekeeping.

The third mechanism for changing CDK activity is regulation of the rate of synthesis of the cyclin or CDK or both. Extracellular signals such as **growth factors** and cytokines (developmental signals that trigger cell division) activate, by phosphorylation, the nuclear transcription factors Jun and Fos, which promote the synthesis of many gene products, including cyclins, CDKs, and the transcription factor E2F. In turn, E2F stimulates production of several enzymes essential for the synthesis of deoxynucleotides and DNA, and the CDK and cyclin allow the cell to enter the S phase (Fig. 12-39).

Finally, specific protein inhibitors bind to and inactivate specific CDKs. One such protein is p21, which we discuss below.

These four control mechanisms modulate the activity of specific CDKs that, in turn, control whether a cell will divide, differentiate, become permanently quiescent, or begin a new cycle of division after a period of quiescence.

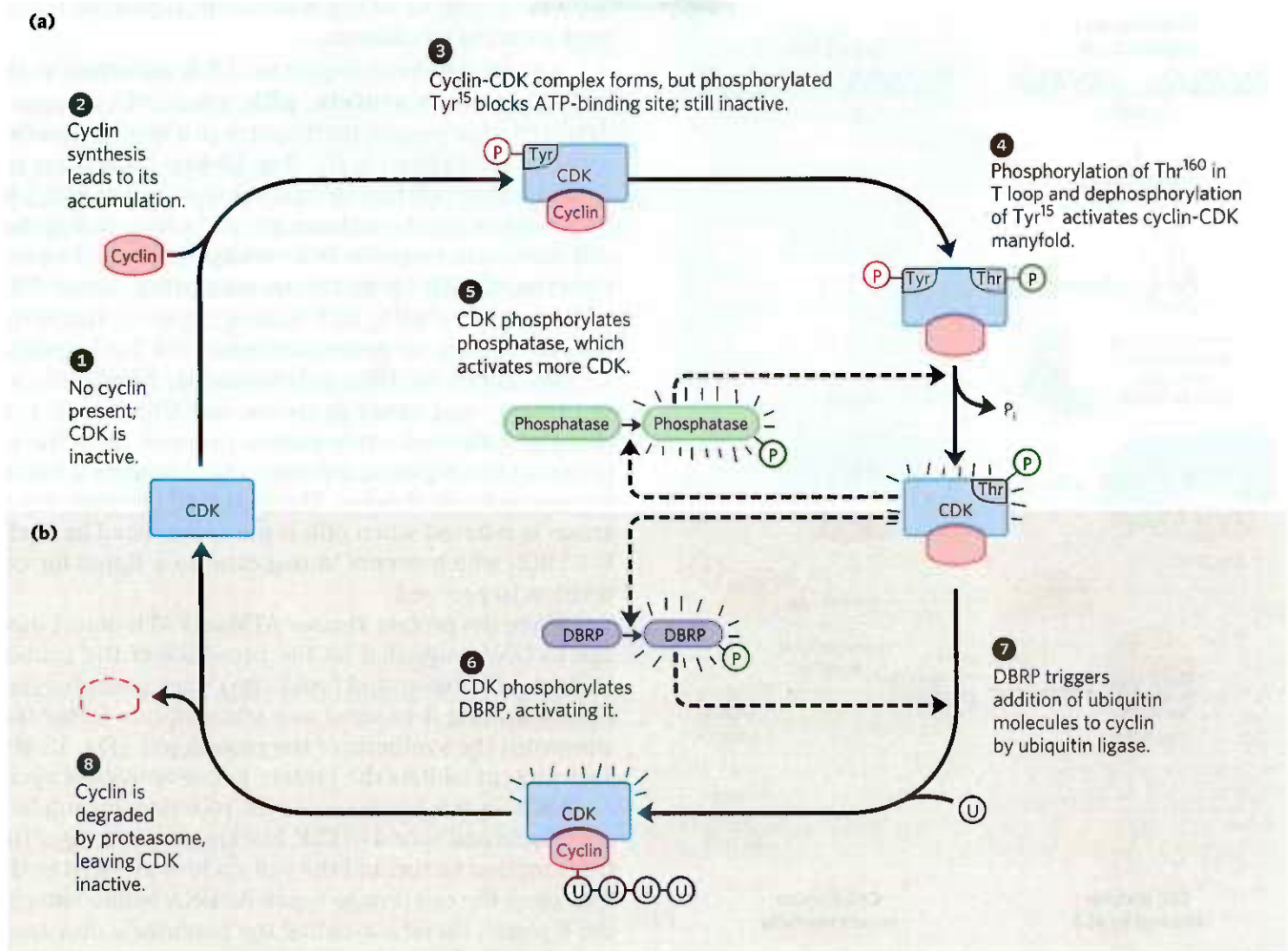


FIGURE 12-38 Regulation of CDK by phosphorylation and proteolysis. (a) The series of events that leads to activation of a cyclin-dependent protein kinase. (b) The periodic proteolytic degradation of cyclin, inactivating the cyclin-dependent protein kinase. The steps are described in the text.

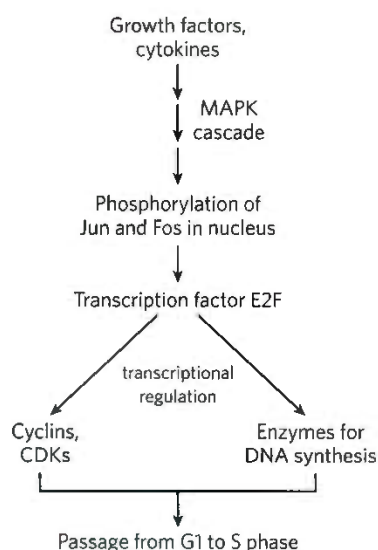


FIGURE 12-39 Regulation of cell division by growth factors.

The details of cell cycle regulation, such as the number of different cyclins and kinases and the combinations in which they act, differ from species to species, but the basic mechanism has been conserved in the evolution of all eukaryotic cells.

CDKs Regulate Cell Division by Phosphorylating Critical Proteins

We have examined how cells maintain close control of CDK activity, but how does the activity of CDKs control the cell cycle? There are scores of known CDK targets, and much remains to be learned. But we can see a general pattern behind CDK regulation by inspecting the effect of CDKs on the structures of lamin and myosin and on the activity of retinoblastoma protein.

The structure of the nuclear envelope is maintained in part by highly organized meshworks of intermediate filaments composed of the protein **lamin**. Breakdown

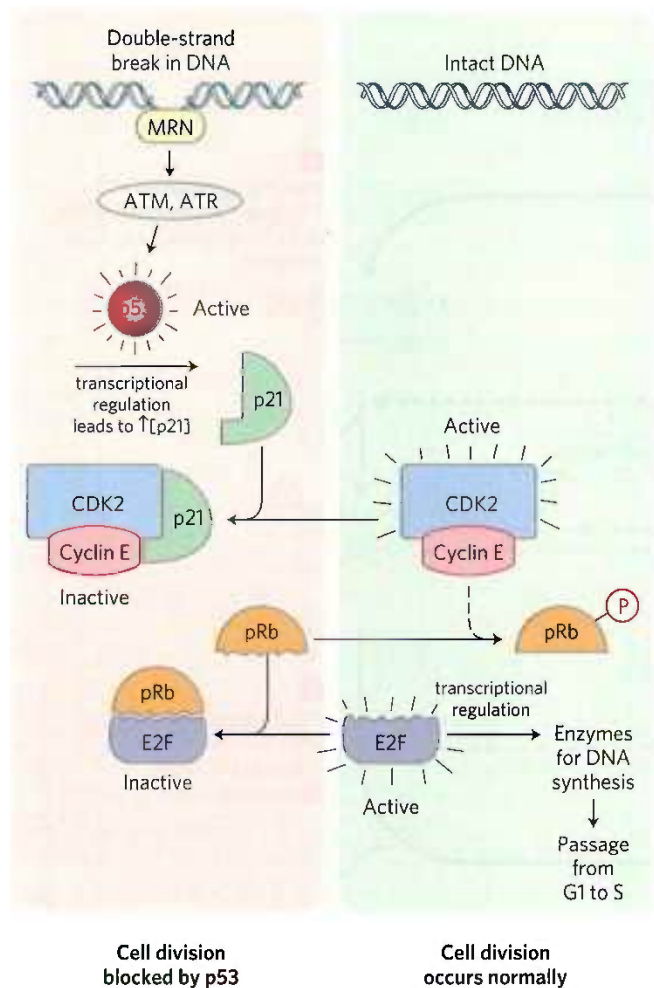


FIGURE 12-40 Regulation of passage from G1 to S by phosphorylation of pRb. Transcription factor E2F promotes transcription of genes for certain enzymes essential to DNA synthesis. The retinoblastoma protein, pRb, can bind E2F (lower left), inactivating it and preventing transcription of these genes. Phosphorylation of pRb by CDK2 prevents it from binding and inactivating E2F, and the genes are transcribed, allowing cell division. Damage to the cell's DNA (upper left) triggers a series of events that inactivate CDK2, blocking cell division. When the protein MRN detects damage to the DNA, it activates two protein kinases, ATM and ATR, and they phosphorylate and activate the transcription factor p53. Active p53 promotes the synthesis of another protein, p21, an inhibitor of CDK2. Inhibition of CDK2 stops the phosphorylation of pRb, which therefore continues to bind and inhibit E2F. With E2F inactivated, genes essential to cell division are not transcribed and cell division is blocked. When DNA has been repaired, this inhibition is released, and the cell divides.

of the nuclear envelope before segregation of the sister chromatids in mitosis is partly due to the phosphorylation of lamin by a CDK, which causes lamin filaments to depolymerize.

A second kinase target is the ATP-driven contractile machinery (actin and myosin) that pinches a dividing cell into two equal parts during cytokinesis. After the division, a CDK phosphorylates a small regulatory subunit of myosin, causing dissociation of myosin from actin filaments and inactivating the contractile machinery. Subsequent dephosphorylation

allows reassembly of the contractile apparatus for the next round of cytokinesis.

A third and very important CDK substrate is the **retinoblastoma protein, pRb**; when DNA damage is detected, this protein participates in a mechanism that arrests cell division in G1 (Fig. 12-40). Named for the retinal tumor cell line in which it was discovered, pRb functions in most, perhaps all, cell types to regulate cell division in response to a variety of stimuli. Unphosphorylated pRb binds the transcription factor E2F; while bound to pRb, E2F cannot promote transcription of a group of genes necessary for DNA synthesis (the genes for DNA polymerase α , ribonucleotide reductase, and other proteins; see Chapter 25). In this state, the cell cycle cannot proceed from the G1 phase to the S phase, the step that commits a cell to mitosis and cell division. The pRb-E2F blocking mechanism is relieved when pRb is phosphorylated by cyclin E-CDK2, which occurs in response to a signal for cell division to proceed.

When the protein kinases ATM and ATR detect damage to DNA (signaled by the presence of the protein MRN at a double-strand break site), they phosphorylate p53, activating it to serve as a transcription factor that stimulates the synthesis of the protein p21 (Fig. 12-40). This protein inhibits the protein kinase activity of cyclin E-CDK2. In the presence of p21, pRb remains unphosphorylated and bound to E2F, blocking the activity of this transcription factor, and the cell cycle is arrested in G1. This gives the cell time to repair its DNA before entering the S phase, thereby avoiding the potentially disastrous transfer of a defective genome to one or both daughter cells. When the damage is too severe to allow effective repair, this same machinery triggers apoptosis (described below), a process that leads to the death of the cell, preventing the possible development of a cancer.

SUMMARY 12.8 Regulation of the Cell Cycle by Protein Kinases

- Progression through the cell cycle is regulated by the cyclin-dependent protein kinases (CDKs), which act at specific points in the cycle, phosphorylating key proteins and modulating their activities. The catalytic subunit of CDKs is inactive unless associated with the regulatory cyclin subunit.
- The activity of a cyclin-CDK complex changes during the cell cycle through differential synthesis of CDKs, specific degradation of the cyclin, phosphorylation and dephosphorylation of critical residues in CDKs, and binding of inhibitory proteins to specific cyclin-CDKs.
- A cyclin sequence (the destruction box) marks cyclin for tagging with ubiquitin and degradation in proteasomes. The rise of cyclin concentration by its synthesis ultimately triggers its degradation, yielding oscillations in cyclin level keyed to the cell cycle.

■ Cells receive extracellular signals that determine the timing of their division. Scores of proteins are known targets of CDKs, many with unknown functions. Among the targets phosphorylated by cyclin-CDKs are proteins of the nuclear envelope and proteins required for cytokinesis and DNA repair.

12.9 Oncogenes, Tumor Suppressor Genes, and Programmed Cell Death

Tumors and cancer are the result of uncontrolled cell division. Normally, cell division is regulated by a family of extracellular growth factors, proteins that cause resting cells to divide and, in some cases, differentiate. The result is a precise balance between the formation of new cells and cell destruction. Regulation of cell division ensures that skin cells are replaced every few weeks and white blood cells are replaced every few days. This is homeostasis at the organismal level. **P9** When this balance is disturbed by defects in regulatory proteins, the result is sometimes the formation of a clone of cells that divide repeatedly and without regulation (a tumor) until their presence interferes with the function of normal tissues—cancer. The direct cause is almost always a genetic defect in one or more of the proteins that regulate cell division. In some cases, a defective gene is inherited from one parent; in other cases, the mutation occurs when a toxic compound from the environment (a mutagen or a carcinogen) or high-energy radiation interacts with the DNA of a single cell to damage it and introduce a mutation. In most cases there is both an inherited contribution and an environmental contribution, and in most cases, more than one mutation is required in order to cause completely unregulated division and full-blown cancer.

Oncogenes Are Mutant Forms of the Genes for Proteins That Regulate the Cell Cycle

P9 **Oncogenes** are mutated versions of genes encoding signaling proteins involved in cell cycle regulation. Oncogenes were originally discovered in tumor-causing viruses, then later found to be derived from genes in animal host cells, **proto-oncogenes**, which encode growth-regulating proteins. During a viral infection, the host DNA sequence of a proto-oncogene is sometimes copied into the viral genome, where it proliferates with the virus. In subsequent viral infection cycles, the proto-oncogenes can become defective by truncation or mutation. Viruses, unlike animal cells, do not have effective mechanisms for correcting mistakes during DNA replication, so they accumulate mutations rapidly. When a virus carrying an oncogene infects a new host cell, the viral DNA (and oncogene) can be incorporated into the host cell's DNA, where it can now interfere with the regulation of cell division in the host cell. In an alternative, nonviral mechanism, a single cell in a tissue

exposed to carcinogens may suffer DNA damage that renders one of its regulatory proteins defective, with the same effect as the viral oncogenic mechanism: failed regulation of cell division.

The mutations that produce oncogenes are genetically dominant; if either of a pair of chromosomes contains a defective gene, that gene product sends the signal “divide,” and a tumor may result. The oncogenic defect can be in any of the proteins involved in communicating the “divide” signal. Oncogenes discovered thus far include those that encode secreted proteins that act as signaling molecules, growth factors, transmembrane proteins (receptors), cytoplasmic proteins (G proteins and protein kinases), and the nuclear transcription factors that control the expression of genes essential for cell division (Jun, Fos).

Some oncogenes encode growth factor receptors with unregulated Tyr kinase activity; they signal continued cell division even when the growth factor is absent, leading to tumor formation. Tumor-producing mutations have been found in many of the signaling protein kinases we have discussed here, all of which use ATP as their substrate for phosphoryl transfer to another element in the signaling cascade. The development of drugs that inhibit the protein kinase activity is an obvious approach to treating cancers that result from unregulated kinase activity. However, most known protein kinase inhibitors act by blocking the binding site for ATP, which is similar in all protein kinases. Inhibitors that are effective against one protein kinase are likely to have intolerable side effects due to their inhibition of other, essential kinases. Nonetheless, the prominent role played by protein kinases in signaling processes related to normal and abnormal cell division has made these enzymes a prime target in the development of drugs for the treatment of cancer (**Box 12-4**). ■

Defects in Certain Genes Remove Normal Restraints on Cell Division

Tumor suppressor genes encode proteins that normally restrain cell division. Mutation in one or more of these genes can lead to tumor formation. Unregulated growth due to defective tumor suppressor genes, unlike that due to oncogenes, is genetically recessive; tumors form only if *both* chromosomes contain a defective gene. This is because the function of these genes is to prevent cell division, and if either copy of the gene is normal, it will produce a normal protein and normal inhibition of division. In a person who inherits one correct copy and one defective copy, every cell begins with one defective copy of the gene. If any one of the individual's 10^{12} somatic cells undergoes mutation in the one good copy, a tumor may grow from that doubly mutant cell. Mutations in both copies of the genes for pRb, p53, or p21 yield cells in which the normal restraint on cell division is lost and a tumor forms.



Development of Protein Kinase Inhibitors for Cancer Treatment

When a single cell divides without any regulatory limitation, it eventually gives rise to a clone of cells so large that it interferes with normal physiological functions (Fig. 1). This is cancer, a leading cause of death in the developed world, and increasingly so in the developing world. In all types of cancer, the normal regulation of cell division has become dysfunctional due to defects in one or more genes. For example, genes encoding proteins that normally send intermittent signals for cell division become oncogenes, producing constitutively active signaling proteins, or genes encoding proteins that normally restrain cell division (tumor suppressor genes) mutate to produce proteins that lack this braking function. In many tumors, both kinds of mutation have occurred.

Many oncogenes and tumor suppressor genes encode protein kinases or proteins that act in pathways upstream from protein kinases. It is therefore reasonable to hope that specific inhibitors of protein kinases could prove valuable in the treatment of cancer. For example, a mutant form of the EGF (epidermal growth factor) receptor is a constantly active receptor Tyr kinase (RTK), signaling cell division whether EGF is present or not. In about 30% of all women with invasive breast cancer, the gene for the receptor ErbB2 (also called HER2/neu) is over-expressed, sometimes by as much as 100-fold. Another RTK, **vascular endothelial growth factor receptor (VEGFR)**, must be activated for the formation of new blood vessels (angiogenesis) to provide a solid tumor with its own blood supply, and inhibition of VEGFR might starve a tumor of essential nutrients. Nonreceptor Tyr kinases can also mutate, resulting in constant signaling and unregulated cell division. For example, the oncogene *Abi* (from the *Abelson leukemia virus*) is associated with acute myeloid leukemia, a relatively rare blood disease (~5,000 cases a year in the United States). Another group of oncogenes encode unregulated cyclin-dependent protein kinases. In each of these cases, specific protein kinase inhibitors might be valuable chemotherapeutic agents in the treatment of disease. Not surprisingly, huge efforts are under way to develop such inhibitors. How should one approach this challenge?

Protein kinases of all types show striking conservation of structure at the active site. All share with the prototypical PKA structure the features shown in Figure 2: two lobes that enclose the active site, with a P loop that helps to align and bind the phosphoryl groups of ATP, an activation loop that moves to open the active site to the protein substrate, and a C helix that changes position as the enzyme is activated, bringing the residues in the substrate-binding cleft into their binding positions. Detailed knowledge of the structure around the ATP-binding site makes it possible to design drugs that inhibit a *specific* protein kinase by (1) blocking the critical ATP-binding site, while (2) interacting with residues around that site that are *unique* to that particular protein kinase.

The simplest protein kinase inhibitors are ATP analogs that occupy the ATP-binding site but cannot serve as phosphoryl group donors. Many such compounds are known, but their clinical usefulness is limited by their lack of selectivity—they inhibit

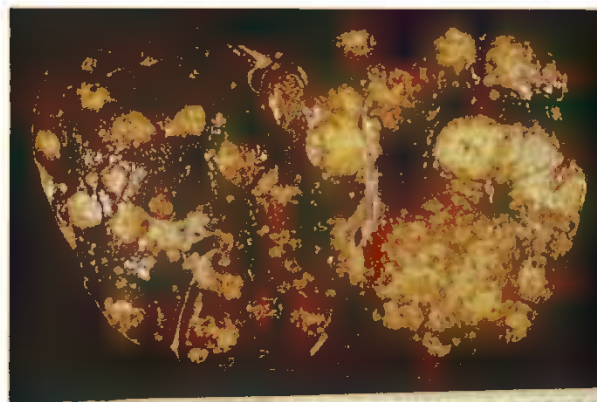


FIGURE 1 Unregulated division of a single cell in the colon led to a primary cancer that metastasized to the liver. Secondary cancers are seen as white patches in this liver obtained at autopsy. [CNRI/Science Source.]

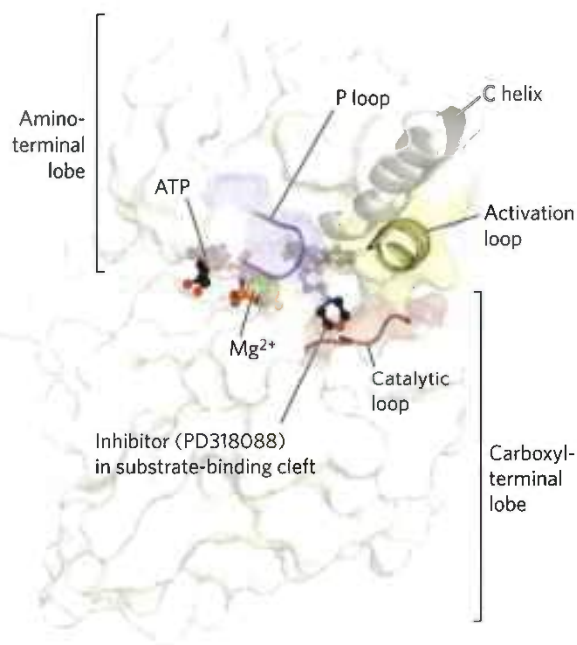
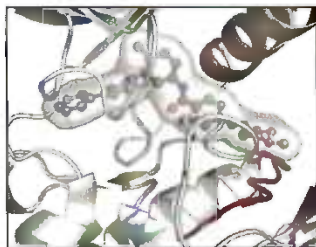
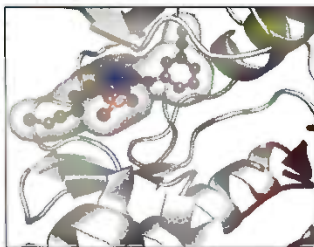


FIGURE 2 Conserved features of the active site of protein kinases. The amino-terminal and carboxyl-terminal lobes surround the active site of the enzyme, near the catalytic loop and the site where ATP binds. The activation loop of this and many other kinases undergoes phosphorylation, then moves away from the active site to expose the substrate-binding cleft, which in this image is occupied by a specific inhibitor of this enzyme, PD318088. The P loop is essential in the binding of ATP, and the C helix must also be correctly aligned for ATP binding and kinase activity. [Data from PDB ID 1S9I, J. F. Ohren et al., *Nature Struct. Mol. Biol.* 11:1192, 2004.]

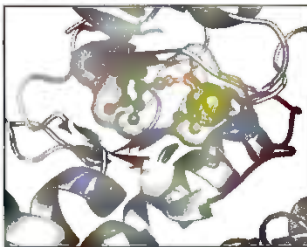
(a) Imatinib (Gleevec)
bound to Abl



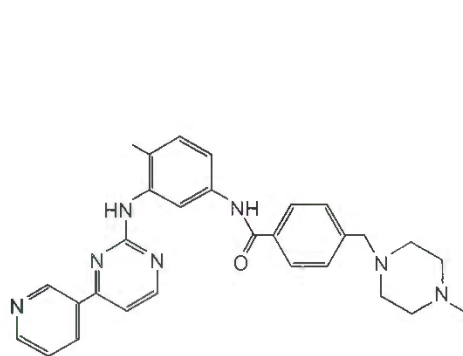
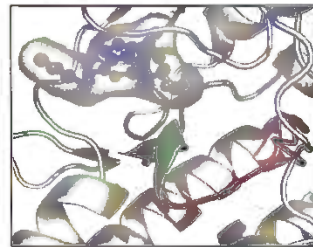
(b) Erlotinib (Tarceva)
bound to EGFR



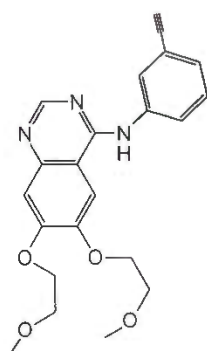
(c) ATP
bound to CDK2



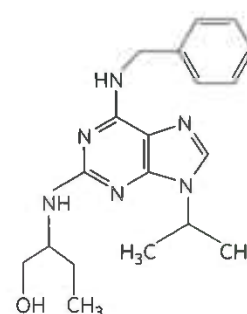
(d) Roscovitine
bound to CDK2



Imatinib (Gleevec)



Erlotinib (Tarceva)



Roscovitine

FIGURE 3 Some protein kinase inhibitors now in clinical trials or clinical use, showing their binding to the target protein. (a) Imatinib binds to the Abl kinase (an oncogene product) active site; it occupies both the ATP-binding site and a region adjacent to that site. (b) Erlotinib binds to the active site of EGFR. (c), (d) Roscovitine is an inhibitor of the cyclin-dependent kinases

CDK2, CDK7, and CDK9; shown here are normal Mg^{2+} -ATP binding at the active site (c) and roscovitine binding (d), which prevents the binding of ATP. [Data from (a) PDB ID 1IEP, B. Nagar et al., *Cancer Res.* 62:4236, 2002; (b) PDB ID 1M17, J. Stamos et al., *J. Biol. Chem.* 277:46,265, 2002; (c) PDB ID 1S9I, J. F. Ohren et al., *Nature Struct. Mol. Biol.* 11:1192, 2004; (d) PDB ID 2A4L, W. F. De Azevedo et al., *Eur. J. Biochem.* 243:518, 1997.]

virtually all protein kinases and would produce unacceptable side effects. More selectivity is seen with compounds that fill part of the ATP-binding site but also interact outside this site with parts of the protein unique to the target protein kinase. A third possible strategy is based on the fact that although the active conformations of all protein kinases are similar, their inactive conformations are not. Drugs that target the inactive conformation of a specific protein kinase and prevent its conversion to the active form may have a higher specificity of action. A fourth approach employs the great specificity of antibodies. For example, monoclonal antibodies (p. 167) that bind the extracellular portions of specific RTKs could eliminate the receptors' kinase activity by preventing dimerization or by causing their removal from the cell surface. In some cases, an antibody selectively binding to the surface of cancer cells could cause the immune system to attack those cells.

The search for drugs active against specific protein kinases has yielded encouraging results. For example, imatinib mesylate (Gleevec; Fig. 3a), a small-molecule inhibitor, has proved nearly 100% effective in bringing about remission in patients with early-stage chronic myeloid leukemia. Erlotinib (Tarceva; Fig. 3b), which targets EGFR, is effective against advanced

non-small-cell lung cancer (NSCLC). Because many cell-division signaling systems involve more than one protein kinase, inhibitors that act on several protein kinases may be useful in the treatment of cancer. Sunitinib (Sutent) and sorafenib (Nexavar) target several protein kinases, including VEGFR and PDGFR. These two drugs are in clinical use for patients with gastrointestinal stromal tumors and advanced renal cell carcinoma, respectively. Trastuzumab (Herceptin), cetuximab (Erbix), and bevacizumab (Avastin) are monoclonal antibodies that target ErbB2/HER2/neu, EGFR, and VEGFR, respectively; all three drugs are in clinical use for certain types of cancer.

At least a hundred more compounds are in preclinical trials. Among the drugs being evaluated are some obtained from natural sources and some produced by synthetic chemistry. Indirubin is a component of a Chinese herbal preparation traditionally used to treat certain leukemias; it inhibits CDK2 and CDK5. Roscovitine (Fig. 3d), a substituted adenine, has a benzyl ring that makes it highly specific as an inhibitor of CDK2. With several hundred potential anticancer drugs heading toward clinical testing, it is realistic to hope that some will prove more effective or more target-specific than those now in use.

Retinoblastoma occurs in children and causes blindness if not surgically treated. The cells of a retinoblastoma have two defective versions of the *Rb* gene (two defective alleles). Very young children who develop retinoblastoma commonly have multiple tumors in both eyes. These children have inherited one defective copy of the *Rb* gene, which is present in every cell; each tumor is derived from a single retinal cell that has undergone a mutation in its remaining good copy of the *Rb* gene. (A fetus with two mutant alleles in every cell is nonviable.) People with retinoblastoma who survive childhood also have a high incidence of cancers of the lung, prostate, and breast later in life.

A far less likely event is that a person born with two good copies of the *Rb* gene will have independent mutations in both copies in the *same* cell. Some individuals do develop retinoblastomas later in childhood, usually with only one tumor in one eye. These individuals, presumably, were born with two good copies (alleles) of *Rb* in every cell, but both *Rb* alleles in a single retinal cell have undergone mutation, leading to a tumor. After the child reaches about age 3, retinal cells stop dividing, and retinoblastomas at later ages are quite rare.

Stability genes (also called caretaker genes) encode proteins that function in the repair of major genetic defects that result from aberrant DNA replication, ionizing radiation, or environmental carcinogens. Mutations in these genes lead to a high frequency of unrepaired damage (mutations) in other genes, including proto-oncogenes and tumor suppressor genes, and thus to cancer. Among the stability genes are *ATM* (see Fig. 12-40); the *XP* gene family, in which mutations lead to xeroderma

pigmentosum; and the *BRCA1* genes associated with some types of breast cancer (see Box 25-1). Mutations in the gene for *p53* also cause tumors; in more than 90% of human cutaneous squamous cell carcinomas (skin cancers) and in about 50% of all other human cancers, *p53* is defective. Those very rare individuals who *inherit* one defective copy of *p53* commonly have the Li-Fraumeni cancer syndrome, with multiple cancers (of the breast, brain, bone, blood, lung, and skin) occurring at high frequency and at an early age. The explanation for multiple tumors in this case is the same as that for *Rb* mutations: an individual born with one defective copy of *p53* in every somatic cell is likely to suffer a second *p53* mutation in more than one cell during his or her lifetime.

P9 In summary, then, three classes of defects can contribute to the development of cancer: (1) oncogenes, in which the defect is the equivalent of a car's accelerator pedal being stuck down, with the engine racing; (2) mutated tumor suppressor genes, in which the defect leads to the equivalent of brake failure; and (3) mutated stability genes, with the defect leading to unrepaired damage to the cell's replication machinery—the equivalent of an unskilled car mechanic.

Mutations in oncogenes and tumor suppressor genes do not have an all-or-none effect. In some cancers, perhaps in all, the progression from a normal cell to a malignant tumor requires an accumulation of mutations (sometimes over several decades), none of which, alone, is responsible for the end effect. For example, the development of colorectal cancer has several recognizable stages, each associated with a mutation (Fig. 12-41). If an epithelial cell in the colon undergoes mutation of both copies of the tumor

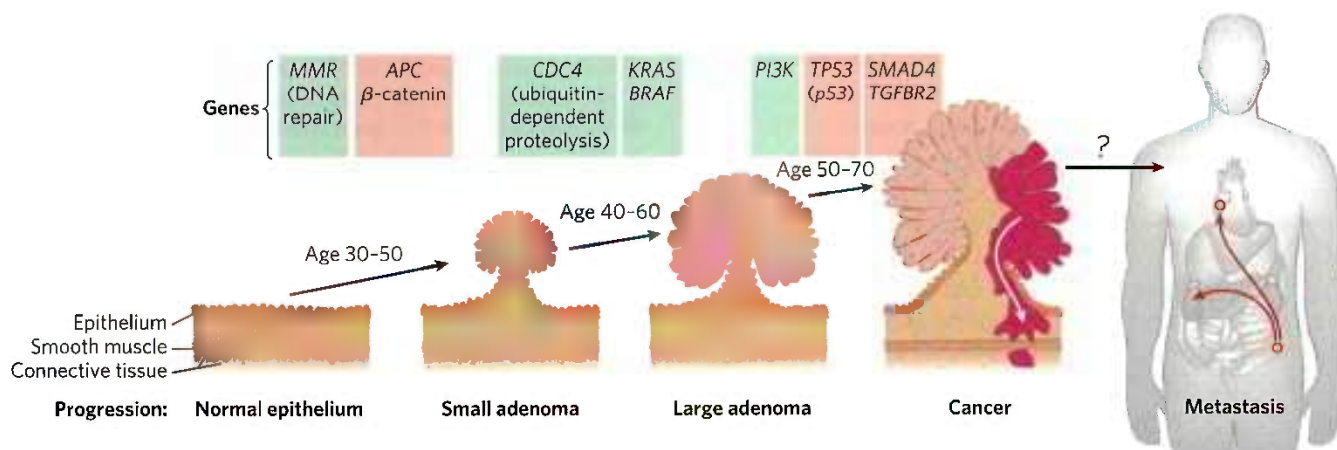


FIGURE 12-41 Multistep transition from normal epithelial cell to colorectal cancer. Serial mutations in oncogenes (green) or tumor suppressor genes (red) lead to progressively less control of cell division, until finally an active tumor forms, which can sometimes metastasize (spread from the initial site to other regions of the body). Mutation of the *MMR* gene leads to defective DNA repair and consequently to a higher rate of mutation. Mutations in both copies of the tumor suppressor gene *APC* lead to benign clusters of epithelial cells that multiply too rapidly (early adenoma). The *CDC4* oncogene results in defective ubiquitination, which is essential to the regulation of cyclin-dependent kinases (see Fig. 12-38). The oncogenes *KRAS* and *BRAF* encode Ras and Raf

proteins (see Fig. 12-22), and this further disruption of signaling leads to the formation of a large adenoma, which may be detected by colonoscopy as a benign polyp. Oncogenic mutations in the *PI3K* gene, which encodes the enzyme phosphoinositide-3 kinase, or in *PTEN*, which regulates the synthesis of this enzyme, lead to a further strengthening of the signal: divide now. When a cell in one of the polyps undergoes further mutations, such as in the tumor suppressor genes *DCC* and *p53* (see Fig. 12-40), increasingly aggressive tumors form. Finally, mutations in other tumor suppressor genes such as *SMAD4* lead to a malignant tumor and sometimes to a metastatic tumor that can spread to other tissues. [Information from S. D. Markowitz and M. M. Bertagnoli, *N. Engl. J. Med.* 361:2449, 2009, Fig. 2.]

suppressor gene *APC* (adenomatous polyposis coli), it begins to divide faster than normal and produces a clone of itself, a benign polyp (early adenoma). For reasons not yet known, the *APC* mutation results in chromosomal instability, and whole regions of a chromosome are lost or rearranged during cell division. This instability can lead to another mutation, commonly in *ras*, that converts the clone into an intermediate (precancerous) adenoma.

A third mutation (often in the tumor suppressor gene *DCC*) leads to a late adenoma. Only when both copies of *p53* become defective does this cell mass become a carcinoma—a malignant, life-threatening tumor. The full sequence therefore requires at least seven genetic “hits”: two on each of three tumor suppressor genes (*APC*, *DCC*, and *p53*) and one on the proto-oncogene *ras*. There are probably several other routes to colorectal cancer as well, but the principle that full malignancy results only from multiple mutations is likely to hold true for all of them. Because mutations accumulate over time, the chances of developing full-blown metastatic cancer rise with age.

When a polyp is detected in the early adenoma stage and the cells containing the first mutations are removed surgically, late adenomas and carcinomas will not develop; hence the importance of early detection. Cells and organisms, too, have their early detection systems. For example, the ATM and ATR proteins can detect DNA damage too extensive to be repaired effectively. They then trigger, through a pathway that includes *p53*, the process of apoptosis, in which a cell that has become dangerous to the organism kills itself.

The development of fast and inexpensive sequencing methods has opened a new window on the process by which cancer develops. In a typical study of cancers in humans, the sequences of all 20,000 genes were determined in about 3,300 different tumors, and then compared with the gene sequences in noncancerous tissue from the same patient. Almost 300,000 mutations were detected in all. Only a small fraction of these mutations, the **driver mutations**, were the *cause* of unregulated cell division; the vast majority (>99.9%) were “passenger mutations,” which occurred randomly and did not confer a selective growth advantage on the tissue in which they occurred. Among the driver mutations were those in about 75 tumor suppressor genes and about 65 oncogenes. These 140 driver mutations fell in three general categories: those that affect cell survival (in genes encoding Ras, PI3K, MAPK, for example), those that affect cells' ability to maintain an intact genome (ATM, ATR), and those that affect cell fate, causing cells to divide, differentiate, or become quiescent (*APC* is one example). A relatively small number of mutations were very common in multiple types of cancer, in the genes for Ras, *p53*, and *pRb*, for example. ■

Apoptosis Is Programmed Cell Suicide

Many cells can precisely control the time of their own death by the process of **programmed cell death**, or

apoptosis (pronounced app'-a-toe'-sis; from the Greek for “dropping off,” as in leaves dropping in the fall). One trigger for apoptosis is irreparable damage to DNA. Programmed cell death also occurs during the normal development of an embryo, when some cells must die to give a tissue or an organ its final shape. Carving fingers from stubby limb buds requires the precisely timed death of cells between developing finger bones. During development of the nematode *C. elegans* from a fertilized egg, exactly 131 cells (of a total of 1,090 somatic cells in the embryo) must undergo programmed death in order to construct the adult body.

Apoptosis also has roles in processes other than development. If a developing antibody-producing cell generates antibodies against a protein or glycoprotein that is normally present in the body, that cell undergoes programmed death in the thymus gland—an essential mechanism for eliminating anti-self antibodies (the cause of many autoimmune diseases). The monthly sloughing of cells of the uterine wall (menstruation) is another case of apoptosis mediating normal cell death. The dropping of leaves in the fall is the result of apoptosis in specific cells of the stem of a plant. Sometimes cell suicide is not programmed but occurs in response to biological circumstances that threaten the rest of the organism. For example, a virus-infected cell that dies before completion of the infection cycle prevents spread of the virus to nearby cells. Severe stresses such as heat, hyperosmolarity, UV light, and gamma irradiation also trigger cell suicide; presumably the organism is better off with any aberrant, potentially mutated cells dead.

P8 The regulatory mechanisms that trigger apoptosis involve some of the same proteins that regulate the cell cycle. The signal for suicide often comes from outside, through a surface receptor. Tumor necrosis factor (TNF), produced by cells of the immune system, interacts with cells through specific TNF receptors. These receptors have TNF-binding sites on the outer face of the plasma membrane and a “death domain” (~80 amino acid residues) that carries the self-destruct signal through the membrane to cytosolic proteins such as TRADD (TNF receptor-associated death domain) (Fig. 12-42).

When caspase-8, an “initiator” caspase, is activated by an apoptotic signal carried through TRADD, it further self-activates by cleaving its own proenzyme form. Mitochondria are one target of active caspase-8. The protease causes the release of certain proteins contained between the inner and outer mitochondrial membranes: cytochrome *c* and several “effector” caspases (see Fig. 19-39). Cytochrome *c* binds to the proenzyme form of the effector enzyme caspase-9 and stimulates its proteolytic activation. The activated caspase-9, in turn, catalyzes wholesale destruction of cellular proteins—a major cause of apoptotic cell death. One specific target of caspase action is a caspase-activated deoxyribonuclease.

In apoptosis, the monomeric products of protein and DNA degradation (amino acids and nucleotides) are

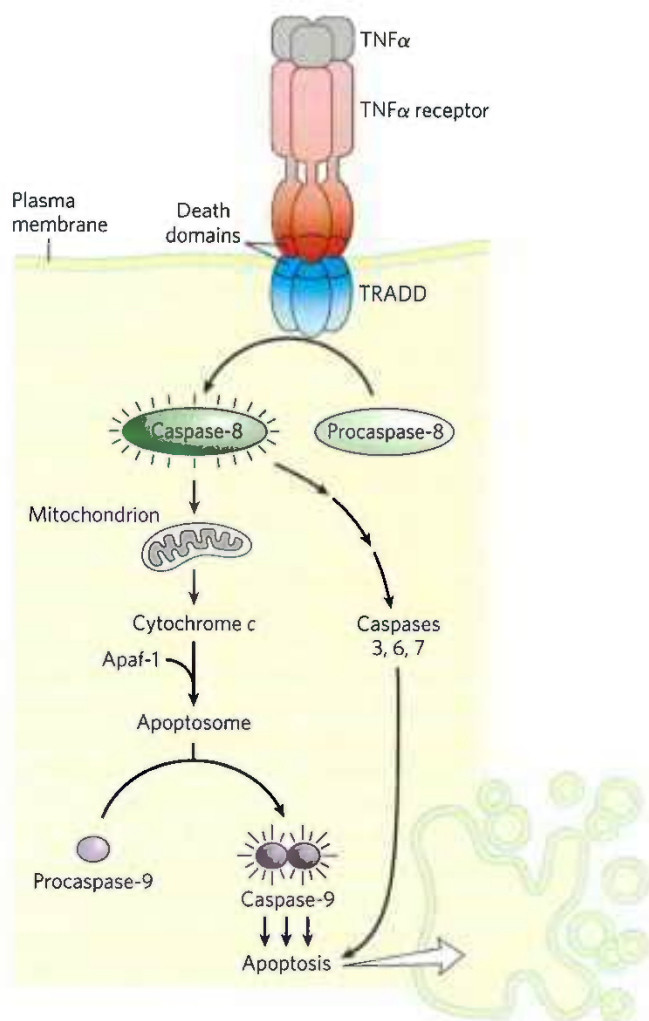


FIGURE 12-42 Initial events of apoptosis. An apoptosis-triggering signal from outside the cell (TNF α) binds to its specific receptor in the plasma membrane. The occupied receptor interacts with the cytosolic protein TRADD through "death domains" (80-residue domains on both TNF α receptor and TRADD), activating TRADD. Activated TRADD initiates a proteolytic cascade that leads to apoptosis: TRADD activates caspase-8, which acts to release cytochrome c from mitochondria, which, in concert with protein Apaf-1, activates caspase-9, triggering apoptosis.

released in a controlled process that allows them to be taken up and reused by neighboring cells. Apoptosis thus allows the organism to eliminate a cell that is unneeded or potentially dangerous without wasting its components.

SUMMARY 12.9 Oncogenes, Tumor Suppressor Genes, and Programmed Cell Death

■ **Oncogenes** encode defective signaling proteins. By continually giving the signal for cell division, they lead to tumor formation. Oncogenes are genetically dominant and may encode defective growth factors, receptors, G proteins, protein kinases, or nuclear regulators of transcription.

■ **Tumor suppressor genes** encode regulatory proteins that normally inhibit cell division; mutations in these genes are genetically recessive but can lead to tumor formation. Cancer is generally the result of an accumulation of mutations in oncogenes and tumor suppressor genes.

■ When **stability genes**, which encode proteins necessary for the repair of genetic damage, are mutated, other mutations go unrepaired, including mutations in proto-oncogenes and tumor suppressor genes that can lead to cancer.

■ **Apoptosis** is programmed and controlled cell death that functions during normal development and adulthood to destroy and recycle unnecessary, damaged, or infected cells. Apoptosis can be triggered by extracellular signals such as TNF, acting through plasma membrane receptors.

KEY TERMS

Terms in bold are defined in the glossary.

signal transduction 408	fluorescence resonance energy transfer (FRET) 416
specificity 409	β -adrenergic receptor kinase (β ARK) 418
sensitivity 409	β -arrestin (β arr) 418
amplification 409	G protein-coupled receptor kinases (GRKs) 419
enzyme cascade 409	cAMP response element binding protein (CREB) 420
modularity 409	inhibitory G protein (G_i) 420
scaffold proteins 409	adaptor proteins 420
desensitization 409	AKAPs (A kinase anchoring proteins) 420
integration 409	Ras 421
divergence 409	guanosine 3',5'-cyclic monophosphate (cyclic GMP; cGMP) 422
response localization 409	cGMP-dependent protein kinase (protein kinase G; PKG) 422
G protein-coupled receptors (GPCRs) 412	GTPase activator protein (GAP) 422
guanosine nucleotide-binding proteins 412	regulator of G-protein signaling (RGS) 422
G proteins 412	NO synthase 423
second messenger 412	phospholipase C (PLC) 425
agonist 412	inositol 1,4,5-trisphosphate (IP ₃) 425
antagonist 412	IP ₃ -gated Ca ²⁺ channel 425
β -adrenergic receptors 413	
seven-transmembrane (7tm) receptors 413	
guanosine nucleotide-exchange factor (GEF) 413	
stimulatory G protein (G_s) 413	
adenyl cyclase 413	
cAMP-dependent protein kinase (protein kinase A; PKA) 413	
consensus sequence 414	
green fluorescent protein (GFP) 416	

protein kinase		hormone response element	
C (PKC) 425		(HRE) 445	
calmodulin (CaM) 426		cyclin 447	
Ca ²⁺ /calmodulin-		cyclin-dependent protein	
dependent protein kinases		kinase (CDK) 447	
(CaM kinases) 426		ubiquitin 448	
rhodopsin 429		proteasome 448	
rhodopsin kinase 431		growth factors 448	
receptor potential 431		retinoblastoma protein	
receptor Tyr kinase		(pRb) 450	
(RTK) 433		oncogene 451	
autophosphorylation 434		proto-oncogene 451	
SH2 domain 435		tumor suppressor	
MAPKs 435		gene 451	
voltage-gated ion		programmed cell	
channels 442		death 455	
ionotropic 444		apoptosis 455	
metabotropic 444			

PROBLEMS

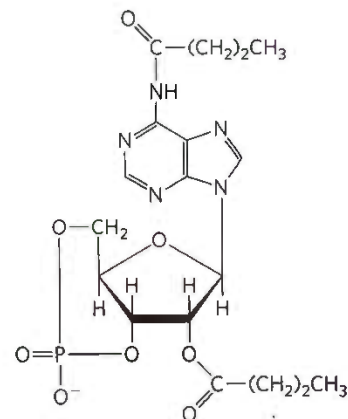
1. Hormone Experiments in Cell-Free Systems In the 1950s, Earl W. Sutherland, Jr., and his colleagues carried out pioneering experiments to elucidate the mechanism of action of epinephrine and glucagon. Given what you have learned in this chapter about hormone action, interpret each of the experiments described below. Identify substance X and indicate the significance of the results.

(a) Addition of epinephrine to a homogenate of normal liver resulted in an increase in the activity of glycogen phosphorylase. However, when the homogenate was first centrifuged at a high speed and epinephrine or glucagon was added to the clear supernatant fraction that contains phosphorylase, no increase in the phosphorylase activity occurred.

(b) When the particulate fraction from the centrifugation in (a) was treated with epinephrine, substance X was produced. The substance was isolated and purified. Unlike epinephrine, substance X activated glycogen phosphorylase when added to the clear supernatant fraction of the centrifuged homogenate.

(c) Substance X was heat-stable; that is, heat treatment did not affect its capacity to activate phosphorylase. (Hint: Would this be the case if substance X were a protein?) Substance X was nearly identical to a compound obtained when pure ATP was treated with barium hydroxide. (Fig. 8-6 will be helpful.)

2. Effect of Dibutyryl cAMP versus cAMP on Intact Cells In principle, the physiological effects of epinephrine should be mimicked by addition of cAMP to the target cells. In practice, addition of cAMP to intact target cells elicits only a minimal physiological response. Why? When the structurally related derivative dibutyryl cAMP (shown) is added to intact cells, the expected physiological response is readily apparent. Explain the basis for the difference in cellular response to these two substances. Dibutyryl cAMP is widely used in studies of cAMP function.



Dibutyryl cAMP
(N⁶,O^{2'}-Dibutyryl adenosine 3',5'-cyclic monophosphate)



3. Effect of Cholera Toxin on Adenylyl Cyclase

The gram-negative bacterium *Vibrio cholerae* produces a protein, cholera toxin (M_r 90,000), that is responsible for the characteristic symptoms of cholera: extensive loss of body water and Na⁺ through continuous, debilitating diarrhea. If body fluids and Na⁺ are not replaced, severe dehydration results; untreated, the disease is often fatal. When the cholera toxin gains access to the human intestinal tract, it binds tightly to specific sites in the plasma membrane of the epithelial cells lining the small intestine, causing adenylyl cyclase to undergo prolonged activation (hours or days).

(a) What is the expected effect of cholera toxin on [cAMP] in the intestinal cells?

(b) Based on the information above, suggest how cAMP normally functions in intestinal epithelial cells.

(c) Suggest a possible treatment for cholera.

4. Mutations in PKA Explain how mutations in the R or C subunit of cAMP-dependent protein kinase (PKA) might lead to

(a) a constantly active PKA or (b) a constantly inactive PKA.



5. Therapeutic Effects of Albuterol

The respiratory symptoms of asthma result from constriction of the bronchi and bronchioles of the lungs, caused by contraction of the smooth muscle of their walls. Raising [cAMP] in the smooth muscle reverses the constriction of the bronchi and bronchioles. Explain the therapeutic effects of albuterol, an inhaled β -adrenergic agonist, in treating asthma. Would you expect this drug to have any side effects? If so, what design change could you make to the drug to minimize side effects?

6. Termination of Hormonal Signals Signals carried by hormones must eventually be terminated. Describe several mechanisms for signal termination.

7. Using FRET to Explore Protein-Protein Interactions In Vivo Figure 12-9 shows the interaction between β -arrestin and the β -adrenergic receptor. How would you use FRET (see Box 12-1) to demonstrate this interaction in living cells? Which proteins would you fuse? Which wavelengths would you use to illuminate the cells, and which wavelengths would you monitor? What would you expect to observe if the interaction occurred? If it did not occur? How might you explain the failure of this approach to demonstrate this interaction?

8. EGTA Injection EGTA (ethylene glycol-bis(β -amino ethyl ether)- N,N,N',N' -tetraacetic acid) is a chelating agent with high affinity and specificity for Ca^{2+} . By microinjecting a cell with an appropriate Ca^{2+} -EGTA solution, an experimenter can prevent cytosolic $[\text{Ca}^{2+}]$ from rising above 10^{-7} M. How would EGTA microinjection affect a cell's response to vasopressin (see Table 12-4)? To glucagon?

9. Amplification and Termination of Hormonal Signals In the β -adrenergic system, which of these contributes to the amplification of the signal (epinephrine) and which to the termination of the signal? Do any contribute to both amplification and termination of the signal?

- (a) One G_α activates many adenylyl cyclase molecules.
- (b) One protein kinase A (PKA) phosphorylates many target proteins.
- (c) The intrinsic GTPase of G protein converts bound GTP to GDP.
- (d) A phosphodiesterase acts on many molecules of cAMP.
- (e) One epinephrine molecule activates many adrenergic receptors.
- (f) One protein kinase phosphorylates many molecules of another protein kinase.

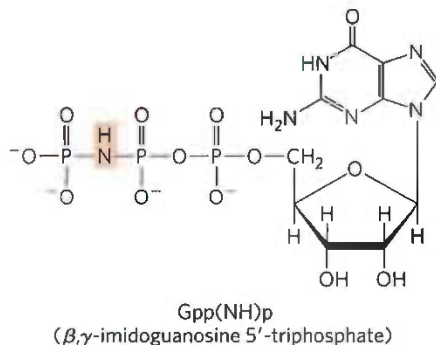
10. The Insulin Signaling System Place these components of the insulin receptor system in the order in which they occur in the sequence of events triggered by insulin: MEK, Ras, ERK, GRK, Raf, Sos, IRS1, PKA, Grb2. Some of these may not participate in that path.

11. Mutations in *ras* How would a mutation in *ras* that leads to formation of a Ras protein with no GTPase activity affect a cell's response to insulin?

12. Differences among G Proteins Compare the G protein G_s , which acts in transducing the signal from β -adrenergic receptors, and the G protein Ras. What properties do they share? How do they differ? What is the functional difference between G_s and G_i ?

13. Mechanisms for Regulating Protein Kinases Identify eight general types of protein kinases found in eukaryotic cells, and explain what factor is *directly* responsible for activating each type.

14. Nonhydrolyzable GTP Analogs Many enzymes can hydrolyze GTP between the β and γ phosphates. The GTP analog β,γ -imidoguanosine 5'-triphosphate (Gpp(NH)p), shown here, cannot be hydrolyzed between the β and γ phosphates.



Predict the effect of microinjection of Gpp(NH)p into a myocyte on the cell's response to β -adrenergic stimulation.



15. Visual Desensitization Oguchi disease is an inherited form of night blindness. Affected individuals are slow to recover vision after a flash of bright light against a dark background, such as the headlights of a car on the freeway. Suggest what the molecular defect(s) might be in Oguchi disease. Explain in molecular terms how this defect would account for night blindness.

16. Effect of a Permeant cGMP Analog on Rod Cells An analog of cGMP, 8-Br-cGMP, will permeate cellular membranes, is only slowly degraded by a rod cell's PDE activity, and is as effective as cGMP in opening the gated channel in the cell's outer segment. If you suspended rod cells in a buffer containing a relatively high [8-Br-cGMP], then illuminated the cells while measuring their membrane potential, what would you expect to see?

17. Effect of Insulin on Glycogen Synthesis Protein kinase B (PKB) inactivates glycogen synthase kinase (GSK3), and GSK3 inactivates glycogen synthase. Predict the effect of insulin on glycogen synthesis.

18. Role of Intrinsically Disordered Regions of Signaling Proteins Signaling proteins, including protein kinases, often have intrinsically disordered regions (IDRs) that are important in signaling. Describe a case in which IDRs and their interactions with other proteins are important in signaling.

19. The Action Potential Place these events in the order in which they occur after a presynaptic neuron releases acetylcholine into the synaptic cleft.

- (a) Vesicles containing a neurotransmitter fuse with the cell membrane.
- (b) Ligand-gated Na^+ channels open, causing an influx of Na^+ ions.
- (c) Voltage-gated Na^+ channels open in the axon.
- (d) Membrane depolarization triggers voltage-gated Ca^{2+} channels to open.
- (e) Local membrane depolarization in the axon triggers an efflux of K^+ .

20. Hot and Cool Taste Sensations The sensations of heat and cold are transduced by a group of temperature-gated cation channels. For example, TRPV1, TRPV3, and TRPM8 are usually closed, but they open at different temperatures. TRPV1 opens at $\geq 43^\circ\text{C}$, TRPV3 opens at $\geq 33^\circ\text{C}$, and TRPM8 opens at $< 25^\circ\text{C}$. These channel proteins are expressed in sensory neurons known to be responsible for temperature sensation.

(a) Propose a reasonable model to explain how exposing a sensory neuron containing TRPV1 to high temperature leads to a sensation of heat.

(b) Capsaicin, one of the active ingredients in "hot" peppers, is an agonist of TRPV1. Capsaicin shows 50% activation of the TRPV1 response at a concentration of 32 nM—a property known as EC_{50} . Explain why even a very few drops of hot pepper sauce can taste very "hot" without actually burning you.

(c) Menthol, one of the active ingredients in mint, is an agonist of TRPM8 ($\text{EC}_{50} = 30 \mu\text{M}$) and TRPV3 ($\text{EC}_{50} = 20 \text{ mM}$). What sensation would you expect from contact with low levels of menthol? With high levels?



21. Oncogenes, Tumor Suppressor Genes, and Tumors For each of the situations listed, provide a plausible explanation for how it could lead to unrestricted cell division.

(a) Colon cancer cells often contain mutations in the gene encoding the prostaglandin E_2 receptor. PGE_2 is a growth factor required for the division of cells in the gastrointestinal tract.

(b) Kaposi sarcoma, a common tumor in people with untreated AIDS, is caused by a virus carrying a gene for a protein similar to the chemokine receptors CXCR1 and CXCR2. Chemokines are cell-specific growth factors.

(c) Adenovirus, a tumor virus, carries a gene for the protein E1A, which binds to the retinoblastoma protein, pRb. (Hint: See Fig. 12-40.)

(d) An important feature of many oncogenes and tumor suppressor genes is their cell-type specificity. For example, mutations in the PGE_2 receptor are not typically found in lung tumors. Explain this observation. (Note that PGE_2 acts through a GPCR in the plasma membrane.)

22. Mutations in Tumor Suppressor Genes and Oncogenes Explain why mutations in tumor suppressor genes are recessive (both copies of the gene must be defective for the regulation of cell division to be defective), whereas mutations in oncogenes are dominant.

23. Retinoblastoma in Children Explain why some children with retinoblastoma develop multiple tumors of the retina in both eyes, whereas others have a single tumor in only one eye.

24. Specificity of a Signal for a Single Cell Type Discuss the validity of the proposition that a signaling molecule (hormone, growth factor, or neurotransmitter) elicits identical responses in different types of target cells if those cells contain identical receptors.

DATA ANALYSIS PROBLEM

25. Exploring Taste Sensation in Mice Pleasing tastes are an evolutionary adaptation to encourage animals to consume nutritious foods. Zhao and coauthors (2003) examined the two major pleasurable taste sensations: sweet and umami. Umami is a “distinct savory taste” triggered by amino acids, especially aspartate and glutamate, and it probably encourages animals to consume protein-rich foods. Monosodium glutamate (MSG) is a flavor-enhancer that exploits this sensitivity.

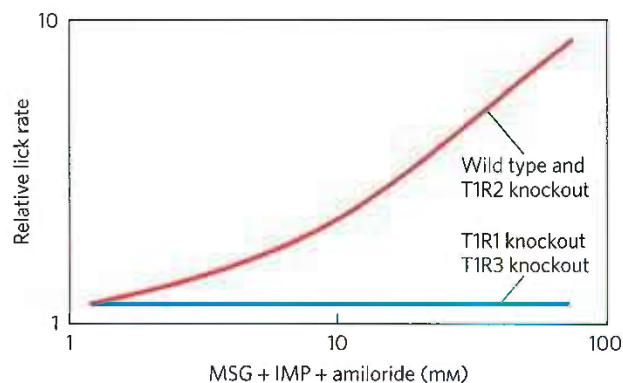
At the time the article was published, three taste receptor proteins for sweet and umami had been tentatively characterized: T1R1, T1R2, and T1R3. These proteins function as heterodimeric receptor complexes: T1R1-T1R3 was tentatively identified as the umami receptor, and T1R2-T1R3 as the sweet receptor. It was not clear how taste sensation was encoded and sent to the brain, and two possible models had been suggested. In the cell-based model, individual taste-sensing cells express only one kind of receptor; that is, there are “sweet cells,” “bitter cells,” “umami cells,” and so on, and each type of cell sends its information to the brain via a different nerve. The brain “knows” which taste is detected by the identity of the nerve fiber that transmits the message. In the receptor-based model, individual taste-sensing cells have several kinds of receptors and send different messages along the same nerve fiber to the brain, the message depending on which receptor is activated. Also unclear at the time was whether there was any interaction between the different taste sensations, or whether parts of one taste-sensing system were required for other taste sensations.

(a) Previous work had shown that different taste receptor proteins are expressed in nonoverlapping sets of taste receptor cells. Which model does this support? Explain your reasoning.

Zhao and colleagues constructed a set of “knockout mice”—mice homozygous for loss-of-function alleles for one of the three receptor proteins, T1R1, T1R2, or T1R3—and double-knockout mice with nonfunctioning T1R2 and T1R3. The researchers measured the taste perception of these mice by measuring their “lick rate” of solutions containing different taste molecules. Mice will lick the spout of a feeding bottle with a pleasant-tasting solution more often than one with an unpleasant-tasting solution. The researchers measured relative lick rates: how often the mice licked a sample solution compared with water. A relative lick rate of 1 indicated no preference; <1 , an aversion; and >1 , a preference.

(b) All four types of knockout strains had the same responses to salt and bitter tastes as did wild-type mice. Which of the above issues did this experiment address? What do you conclude from these results?

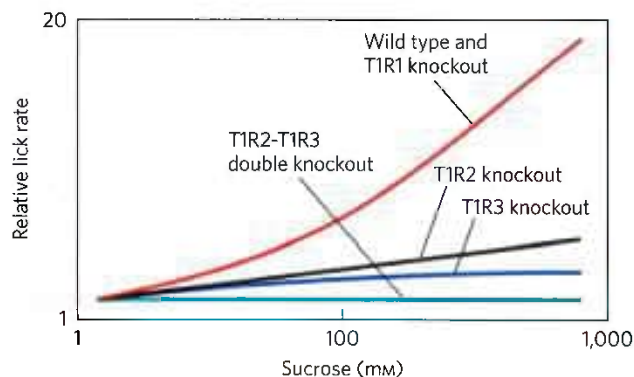
The researchers then studied umami taste reception by measuring the relative lick rates of the different mouse strains with different quantities of MSG in the feeding solution. Note that the solutions also contained inosine monophosphate (IMP), a strong potentiator of umami taste reception (and a common ingredient in ramen soups, along with MSG), and amiloride, which suppresses the pleasant salty taste imparted by the sodium of MSG. The results are shown in the graph.



(c) Are these data consistent with the umami taste receptor consisting of a heterodimer of T1R1 and T1R3? Why or why not?

(d) Which model(s) of taste encoding does this result support? Explain your reasoning.

Zhao and coworkers then performed a series of similar experiments using sucrose as a sweet taste. These results are shown below.



(e) Are these data consistent with the sweet taste receptor consisting of a heterodimer of T1R2 and T1R3? Why or why not?

(f) There were some unexpected responses at very high sucrose concentrations. How do these complicate the idea of a heterodimeric system as presented above?

In addition to sugars, humans also taste other compounds (e.g., saccharin and the peptides monellin and aspartame) as sweet; mice do not taste these as sweet. Zhao and coworkers inserted into T1R2-knockout mice a copy of the human *T1R2* gene under the control of the mouse T1R2 promoter. These modified mice now tasted monellin and saccharin as sweet. The researchers then went further, adding to T1R1-knockout mice the RASSL protein—a G protein-linked receptor for the

synthetic opiate spiradoline; the *RASSL* gene was under the control of a promoter that could be induced by feeding the mice tetracycline. These mice did not prefer spiradoline in the absence of tetracycline; in the presence of tetracycline, they showed a strong preference for nanomolar concentrations of spiradoline.

(g) Do these results strengthen your conclusions about the mechanism of taste sensation?

Reference

Zhao, G.Q., Y. Zhang, M.A. Hoon, J. Chandrashekar, I. Erlenbach, N.J.P. Ryba, and C. Zuker. 2003. The receptors for mammalian sweet and umami taste. *Cell* 115:255–266.

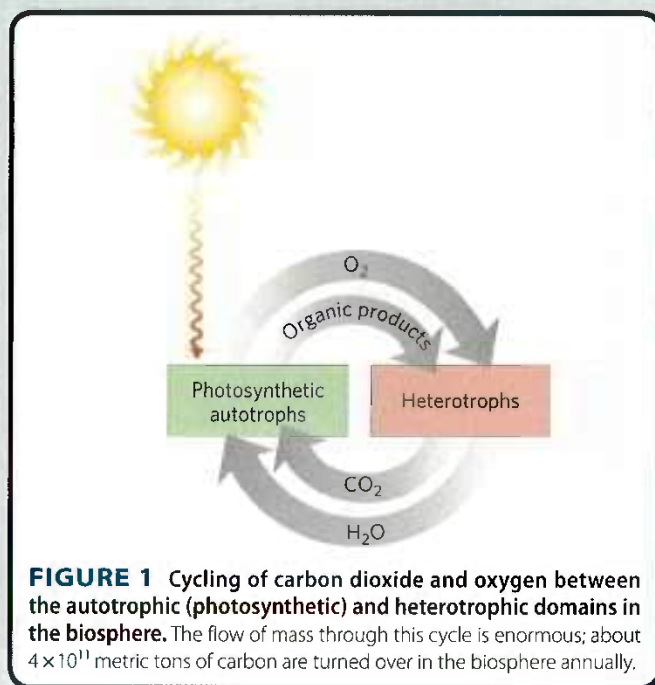
BIOENERGETICS AND METABOLISM

PART OUTLINE

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Every enzyme-catalyzed reaction and reaction sequence serves an important role in an organism's physiology: to (1) obtain chemical energy by capturing solar energy or degrading energy-rich nutrients from the environment; (2) convert nutrient molecules into the cell's own characteristic molecules, including precursors of macromolecules; (3) polymerize monomeric precursors into macromolecules: proteins, nucleic acids, and polysaccharides; and (4) synthesize and degrade biomolecules required for specialized cellular functions, such as membrane lipids, intracellular messengers, and pigments.

Although metabolism embraces many thousands of different enzyme-catalyzed reactions, our major concern in Part II is the central metabolic pathways, which are few in number and remarkably similar in all forms of life. Living organisms can be divided into two large groups according to the chemical form in which they obtain carbon from the environment. **Autotrophs** (such as photosynthetic bacteria, green algae, and vascular plants) can use carbon dioxide from the atmosphere as their sole source of carbon, from which they construct all their carbon-containing biomolecules (see Fig. 1). **Heterotrophs** cannot use atmospheric carbon dioxide and must obtain carbon from their environment in



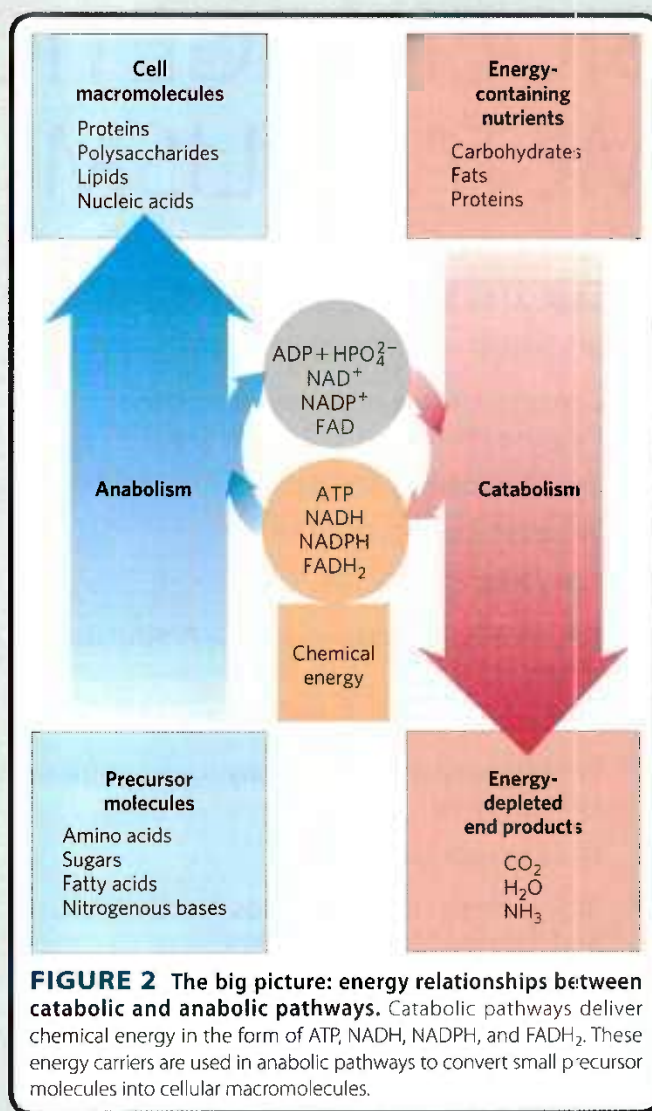
the form of relatively complex organic molecules such as glucose. Multicellular animals and most microorganisms are heterotrophic. Autotrophic cells and organisms are relatively self-sufficient, whereas heterotrophic cells and organisms, with their requirements for carbon in more complex forms, must subsist on the products of other organisms.

Many autotrophic organisms are photosynthetic and obtain their energy from sunlight, whereas heterotrophic organisms obtain their energy from the degradation of organic nutrients produced by autotrophs. In our biosphere, autotrophs and heterotrophs live together in a vast, interdependent cycle in which autotrophic organisms use atmospheric carbon dioxide to build their organic biomolecules, some of them generating oxygen from water in the process. Heterotrophs, in turn, use the organic products of autotrophs as nutrients, which they oxidize, releasing to the atmosphere the CO_2 produced by oxidation. Thus carbon, oxygen, and water are constantly cycled between the heterotrophic and autotrophic worlds, with solar energy as the driving force for this global process. A similarly complex global web connects molecular nitrogen (N_2) with nitrogen in its other oxidation states, including NH_3 (the form in which nitrogen enters metabolism). The cycling of carbon, oxygen, and nitrogen, which ultimately involves all species, depends on a proper balance between the activities of the producers (autotrophs) and consumers (heterotrophs) in our biosphere. Global warming by the greenhouse effect (the result of rising CO_2 concentrations in our atmosphere) is a biochemical phenomenon, occurring on a very large scale.

These cycles of matter are driven by an enormous flow of energy into and through the biosphere, beginning with the capture of solar energy by photosynthetic organisms and use of this energy to generate energy-rich carbohydrates

and other organic nutrients; these nutrients are then used as energy sources by heterotrophic organisms.

Precursors are converted into products through a series of intermediates called **metabolites**. The term **intermediary metabolism** is often applied to the combined activities of all the metabolic pathways that interconvert precursors, metabolites, and products of low molecular weight (generally, $M_r < 1,000$). **Catabolism** is the degradative phase of metabolism in which organic nutrient molecules (carbohydrates, fats, and proteins) are converted into smaller, simpler end products (such as lactic acid, CO_2 , and NH_3). Catabolic pathways release energy, some of which is conserved in the formation of ATP and reduced electron carriers (NADH or NADPH); the rest is lost as heat. In **anabolism**, also called biosynthesis, small, simple precursors are built up into larger and more complex molecules, including lipids, polysaccharides, proteins, and nucleic acids. Anabolic reactions require an input of energy, generally in the form of the phosphoryl group transfer potential of ATP and the reducing power of NADH and NADPH (Fig. 2).



Some metabolic pathways are linear, and some are branched, yielding multiple useful end products from a single precursor or converting several starting materials into a single product. In general, catabolic pathways are *convergent* and anabolic pathways are *divergent* (Fig. 3). Some pathways are cyclic: one starting component of the pathway is regenerated in a series of reactions that converts another starting component into a product. We shall see examples of each type of pathway in the following chapters.

Most cells have the enzymes to carry out both the degradation and the synthesis of the important categories of biomolecules—fatty acids, for example. The simultaneous synthesis and degradation of fatty acids would be wasteful, however. This is prevented by reciprocally regulating the anabolic and catabolic reaction sequences: when one sequence is active, the other is suppressed. Such regulation could not occur if anabolic and catabolic pathways were catalyzed by exactly the same set of enzymes, operating in one direction for anabolism, the opposite direction for catabolism: inhibition of an enzyme involved in catabolism would also inhibit the reaction sequence in the anabolic direction. Catabolic and anabolic pathways that connect the same two end points (glucose $\rightarrow$ pyruvate, and pyruvate $\rightarrow$ glucose, for example) may employ many of the same enzymes; but, invariably, at least one of the steps is catalyzed by different enzymes in the catabolic and anabolic directions, and these enzymes are the

sites of separate regulation. Moreover, for both anabolic and catabolic pathways to be essentially irreversible, the reactions unique to each direction must include at least one that is thermodynamically very favorable—in other words, a reaction for which the reverse reaction is very unfavorable.

As a further contribution to the separate regulation of catabolic and anabolic reaction sequences, paired catabolic and anabolic pathways commonly take place in different cellular compartments: for example, fatty acid catabolism occurs in animal mitochondria, whereas fatty acid synthesis occurs in the cytosol. The concentrations of intermediates, enzymes, and regulators can be maintained at different levels in these different compartments. Because metabolic pathways are subject to kinetic control by substrate concentration, separate pools of anabolic and catabolic intermediates also contribute to the control of metabolic rates. Devices that separate anabolic and catabolic processes will be of particular interest in our discussions of metabolism.

Metabolic pathways are regulated at several levels, from within the cell and from outside. A key enzyme in a pathway may be activated allosterically, or its amount may be changed by the rates of synthesis and breakdown of the enzyme. In multicellular organisms, the metabolic activities of different tissues are regulated and integrated by growth factors and hormones that act from outside the cell.

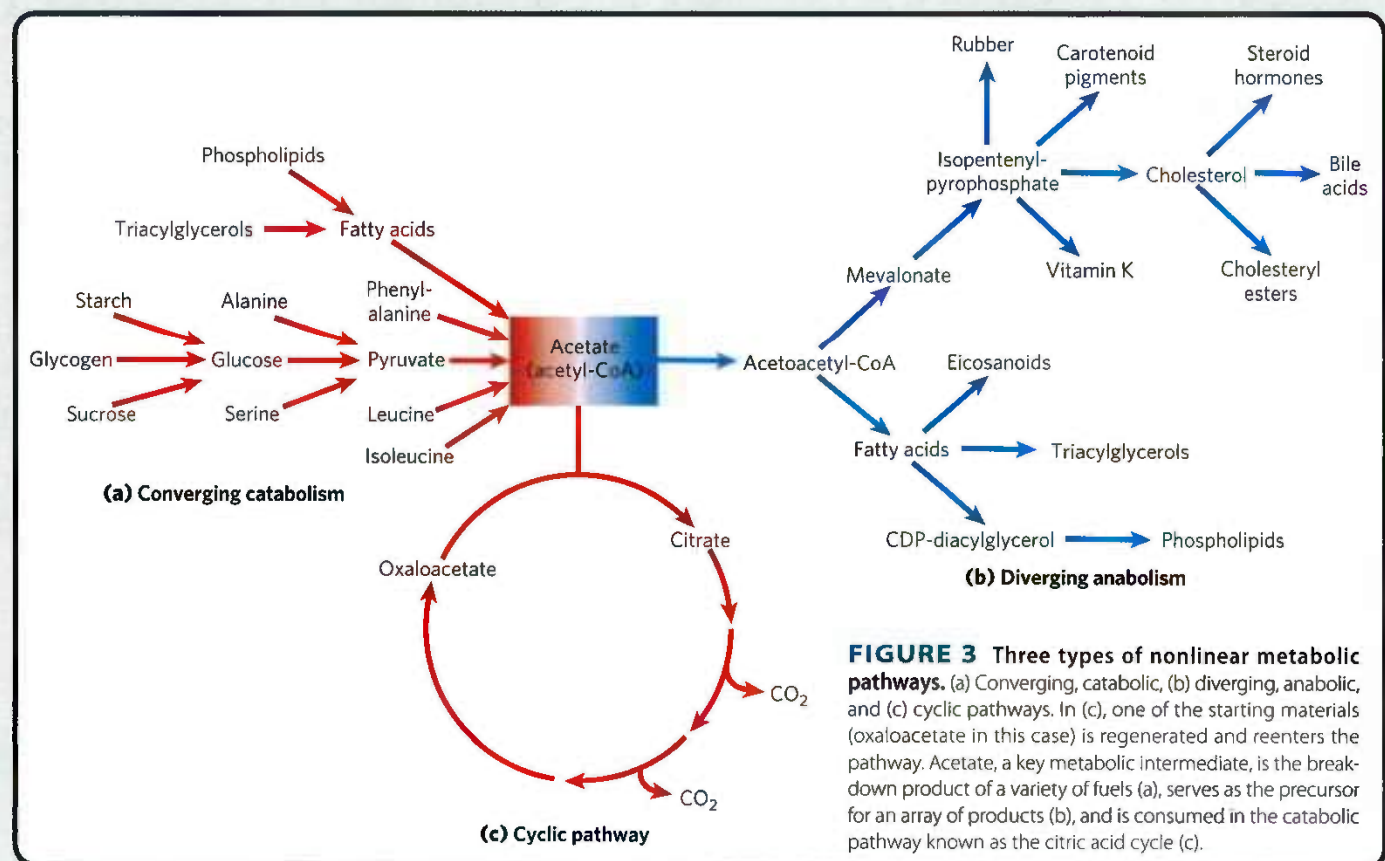


FIGURE 3 Three types of nonlinear metabolic pathways. (a) Converging, catabolic, (b) diverging, anabolic, and (c) cyclic pathways. In (c), one of the starting materials (oxaloacetate in this case) is regenerated and reenters the pathway. Acetyl-CoA, a key metabolic intermediate, is the breakdown product of a variety of fuels (a), serves as the precursor for an array of products (b), and is consumed in the catabolic pathway known as the citric acid cycle (c).

We begin Part II with a discussion of the basic energetic principles that govern all metabolism, as well as a refresher on the reactions from organic chemistry that we will see in metabolism (Chapter 13). We then consider the major catabolic pathways by which cells obtain energy from the oxidation of various fuels (Chapters 14 through 20). Chapters 19 and 20 represent the pivotal point of our discussion of metabolism; they concern chemiosmotic energy coupling, a universal mechanism in which a transmembrane electrochemical potential, produced either by substrate oxidation or by light absorption, drives the synthesis of ATP.

Chapters 20 through 22 describe the major anabolic pathways by which cells use the energy in ATP to produce carbohydrates, lipids, amino acids, and nucleotides from simpler precursors. In Chapter 23 we step back from our detailed look at the metabolic pathways—as they occur in

all organisms, from *E. coli* to humans—and consider how they are regulated and integrated in mammals by hormonal mechanisms.

As we undertake our study of intermediary metabolism, a final word. Keep in mind that the many reactions described in these pages take place in, and play crucial roles in, living organisms. As you encounter each reaction and each pathway, ask: Where does this piece fit in the big picture? What does this chemical transformation do for the organism? How does this pathway interconnect with the other pathways operating simultaneously in the same cell to produce the energy and products required for cell maintenance and growth? What would be the expected result of a defect in this enzyme or that pathway? Studied with this perspective, metabolism provides fascinating and revealing insights into life, with countless applications in medicine, agriculture, and biotechnology.

